

The University of Reading

*Knowledge construction in typography:
the case of legibility research and the
legibility of sans serif typefaces*

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Abstract

This thesis is first of all an epistemological study preoccupied with typographic knowledge construction. The thesis is also in its own right an idiographic historical study about legibility research and about the discourse on the legibility of sans serif typefaces.

Changing theoretical and operational definitions of legibility are discussed, an historical outline of legibility research is presented, and critiques of legibility research that from time to time have surfaced are analysed around recurring topics. Central to the thesis is a detailed and critical review of 28 typeface legibility studies based on a wide variety of rationales and operational methods, the first published in 1896, the last in 1997.

The thesis effectively reveals that nearly all of the 28 reviewed studies (of a surprising total of 72 identified studies) lack internal validity (the intra-paradigm *sine qua non* of experimental research). Other methodological flaws are also revealed. The detailed survey provides a thorough empirical substantiation of criticism that has been raised only in a sweeping manner at earlier occasions. The thesis shows that traditional experimental legibility research has provided a non-productive approach to typographic knowledge construction.

However, the thesis reveals that an increasing number of typeface legibility studies have been carried out during the last two decades. This stands in contrast to prevailing notions that legibility research to a large extent vanished in the early 1980s, or alternatively, that ‘too few’ legibility studies have been carried out during the last two decades. The thesis also shows that dubious and even seriously flawed legibility studies are frequently and indiscriminately cited in the field of information design.

Although the thesis primarily contributes knowledge to a self-reflective conversation in typography and information design, it also contributes knowledge of value to design history, design epistemology, reading research history and social science history.

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Author's declaration

The section 'Robinson, Abbamonte, and Evans 1971: Why serifs are (still) important', in chapter 5, 'A review of empirical studies', has previously, except for a few minor amendments, been published as 'Why serifs are (still) important', in *Typography Papers*, no. 3, 1997, pp. 91–104 (Lund 1997a).

The section 'Wheildon [1984] 1995', in chapter 5, 'A review of empirical studies', has previously, except for a few minor amendments, been published as a book review, in *Information Design Journal*, vol. 9, no. 1, 1997, pp. 74–77 (Lund 1997b).

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1 Introduction

This thesis is first of all an epistemological study preoccupied with typographic knowledge production. The thesis is also in its own right an idiographic¹ historical study about legibility research and about the history of a discourse, that is, the discourse about the relative legibility of serif and sans serif typefaces, while focusing on the marked category – sans serif typefaces.

Changing theoretical and operational definitions of legibility are discussed, and an historical outline of legibility research is presented. The main focus is on the hectic activity in the 1960s and 1970s, the relative loss of legitimacy and the changing paradigm from ‘legibility’ to ‘usability’ in the 1980s and 1990s, and furthermore, the status of legibility research today. Critiques of legibility research that from time to time have surfaced are analysed around recurring topics. Central to the thesis is a detailed and critical review of 28 typeface legibility studies based on a wide variety of rationales and operational methods, the first published in 1896, the last in 1997.

The thesis is in one respect a design historical study. However, the focus is not on designed artefacts, individual designers, period, style, movement, or for that matter, design in a broader socio-economic context, it is rather about a discourse about the legibility of a particular

1. The term ‘idiographic’ appears in historiography and science philosophy. It was coined by the German historian Wilhelm Windelband in 1884. It refers to individual matters and the unique and nonrecurrent, as opposed to the generalizing or lawful, described by the term ‘nomothetic’ (or ‘nomological’). ‘History’ is often referred to as ‘idiographic’, as opposed to the ‘nomothetic’ natural sciences, while social science (and especially positivist social science) has ‘nomothetic’ aspirations. However, these distinctions are certainly debatable. See for example Nagel 1961, pp. 547–551, and Baxandall 1985, pp. 12–15.

kind of designed artefact, sans serif typefaces. Thus, if the thesis is regarded as design history, then it is different from much design history with regard to subject matter, and it is also different in that, through the reviews, it engages in a critical dialogue with the empirical material which constitutes the discourse.

The primary source material that is reviewed consists of theses, dissertations, reports, papers in conference proceedings, and articles published in academic journals, in design journals, and in printing trade journals.² I am not implying that this material necessarily represents a 'complete', well defined and coherent historical discourse, if such a thing ever existed. Nevertheless, from the perspective of this thesis the dispersed texts in question all orbit around the same or similar questions, employ the same or similar concepts and methods, and furthermore, intersect each other in various ways.

My aim has been to combine the approach of a more traditional 'disinterested' historical study with an 'engaged' critical dialogue with the source material. However, the bias is towards the latter approach, by focusing more on 'worklike' aspects of the source-material than on 'documentary' aspects.³

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2. Sometimes the same piece of research appears in several guises. However, by using the expression 'in several guises' I do not imply that meaning relies on linguistic content alone and that concrete instantiations of a research 'paper' are arbitrary and without importance for interpretation. It makes of course a great difference whether a research paper is published or not, how it is published, when it is published, and by whom it is published. The source material's bibliographic history is therefore of great importance. Jerome McGann cogently makes this point in his writings on literary studies, textual criticism and bibliography. See for example McGann 1988a, as well as the condensed argument in his review essay 'Theory of texts' (McGann 1988b).
 3. For a stimulating advocacy of such a combined approach to historical writing – that is, especially in the context of intellectual history or history of ideas – see Dominick LaCapra's essay 'Rethinking intellectual history and reading texts' (1983, esp. pp. 27–33, 61–69). 'Documentary' refers to the referential function of historical sources with regard to empirical reality. 'Worklike' refers to the intellectual content of historical sources and questions of interpretation and dialogue. LaCapra points out the duality between the 'documentary' and the 'worklike' – both dimensions being present in parish registers as well as in philosophical texts. It is worth pointing out that it is exactly a combined approach LaCapra advocates; he does not advocate abolishment of documentary or empirical procedures. His position might be understood as a negotiation between more traditional history writing and intellectual history's more traditional interpretation-biased approach. There is however the possibility that focusing too much on 'worklike' aspects of the source material will

The thesis is idiographic, in that it approaches the ‘legibility discourse’ in question in its historical uniqueness. That is, by historically tracing, documenting, describing and discussing the socio-behavioural research in question, and further by contextualizing⁴ the research by looking into its dissemination and reception.

The thesis is also to some extent generalising,⁵ that is, by suggesting regularities based on a systematic dialogue with the primary source material in question – 28 typeface legibility studies. In these studies the typical mode of discourse is to refer briefly to the face results of a few pieces of previous research on the same topic, shown to be either heterogeneous or homogeneous, followed by a few critical remarks about technical aspects of the predecessors’ operational methods, before presenting the author’s own research on the same topic, which purports to straighten out the matter.

The systematic dialogue with the primary source material is performed not only by explanation and contextualisation but also by critically examining the research on its own terms with regard to construct validity, internal validity and external ecological validity,⁶ as well as by questioning its appropriateness or relevance from a perspective ‘outside’ the research paradigm. However, I am primarily focusing on the internal validity (the essential intra-paradigm condition for meaningful experimental research), by focusing on the typographic

yield something ahistoric, showing disregard for the source material’s ‘pastness’. For criticism of LaCapra, see for example Thompson 1993, where the author accuses LaCapra of conveying a confused concept of historicity and advocating techniques of interpretation that are not properly historical. In a critical assessment of what the author perceives as pros and cons of literary reception theory history (cf. Hans Robert Jauss) on one hand and the so-called ‘new’ history of political thought (cf. Quentin Skinner) on the other hand, Thompson identifies one of the cons of the reception theory history with LaCapra’s approach – reception theory history’s attempt to combine literary *history* with *literary* history. Nevertheless, in contrast to Thompson, Femia’s (1981) attack on the historiographical position of Quentin Skinner, can be read, although implicitly only, as a vigorous support of LaCapra’s position.

4. For interesting discussions on the role of ‘context’ in historical writing – pointing to difficulties posed by the textuality and constructedness of ‘context’, see Bal and Bryson 1991, pp. 176–180; Bryson 1994, as well as LaCapra 1983, pp. 26–27.
5. Note however that the thesis has no ambitions of synthesising arguments about typeface legibility or ‘findings’ from experimental research.
6. For definitions of these concepts, see the section ‘Terminology’, in chapter 5, ‘A review of empirical studies’.

‘stimulus material’. This strategy is based on an assumption that undetected confounding factors that pose a threat to the internal validity of the experiments easily reside in the ‘stimulus material’; for example typographic variables that un-intentionally vary systematically with the independent experimental variable. The reason for focusing on the stimulus material is that this is exactly where I can utilise my expertise as a typographer.

By questioning both the technical validity and the appropriateness of this legibility research, the thesis becomes an epistemological study about knowledge construction in typography. Thus – the question of ‘the legibility of sans serif typefaces’ becomes a manageable lens (or put another way: an extensive empirical and historical instrumental case study)⁷ for posing important epistemological questions about a certain kind of scientific knowledge, its claim to truth, and its interaction with quality judgements in the world of design. This knowledge is presupposed to be not only valid but also superior to craft-based and designerly domain knowledge, and it is very often produced with the explicit aim of guiding practical design activities within the fields of typography and information design.

It is somewhat unusual that a large part of a thesis is organised in such a way that the primary source material (the legibility studies) is reviewed as discrete units (one by one in a chronological order) and that the review is not organised as a category-based analysis around recurrent topics. The approach is not chosen as an easy way out, and should not be confused with something encumbering the reader with tedious ‘raw data’ or a tedious record of the researcher’s way through his source material. The intention has been to do justice to the intellectual content and possible ramifications of each piece of research in a dialogic way. Furthermore, the intention has also been to avoid on one hand a more or less critical but reductive ‘literature-review-like’ summarisation of research ‘findings’; and on the other hand to avoid a reductive treatment of this historical source material as mainly referential documentary evidence.

7. For a useful discussion on the characteristics of ‘case studies’, see Stake 1994.

The approach will admittedly to some extent benefit (or suffer) from the advantage of hindsight. Nevertheless, as numerous writers of both historiography and hermeneutics have repeatedly and forcefully pointed out: any writing based on material of the past necessarily projects values or knowledge of the present, and further, that historico-temporal distance does not necessarily imply myopic ‘presentism’, but rather represents a productive source of understanding. It is partly the intellectual content of the source material, partly its claim to truth and partly the fact that the source material is frequently treated as having more or less transhistorical validity, that demands the approach I have chosen (the prolific legibility researcher Miles Tinker once admitted that ‘the findings are not necessarily good for all time’ (1965, p. 122)). That means appearing judgemental in many instances; however, not without an awareness of the danger of displaying overt enthusiasm or righteous indignation in an anachronistic or ahistoric manner. Although entering into a ‘dialogue’ with the source material, my aim has been to avoid examining it as in a dehistoricised vacuum; independent of time and conditions of production.⁸

The thesis represents an attempt to contribute to a self-reflective conversation about certain aspects of the history of modern typography and information design, examining epistemological and theoretical pre-suppositions that may have relevance to design work or that even in some instances may have guided design work.

Legibility research has during the last few decades been rejected by several academics as well as by several designers, although most often in a sweeping manner only. The usefulness and adequacy of a construct like ‘legibility’ has also to a certain extent been questioned. However, this

8. Examples of an emerging body of historiographical writing on the method, philosophy, content, role and purpose of design history are: Dilnot [1984] 1989; Margolin 1988; Walker 1989; Margolin 1992 [1995], 1998; *Visible Language*, vol. 28, no. 3, 1994 (special issue on graphic design historiography); *Design Issues*, vol. 11, no. 1, 1995 (special issue on design historiography); as well as the introduction to Robin Kinross’ *Modern typography: an essay in critical history*, 1992, pp. 7–14. For a reasonably balanced but also vivid account of contemporary historiographical discourse, history writing and the so-called ‘linguistic turn’, see Samuel 1991/1992. Samuel’s account deals with problems of historical representation but nevertheless refuses to accept a ‘post-modern’ denial of the existence of a past independent of written representation or claims that reality is a product of discourse.

relative loss of legitimacy should not force researchers working on the history and theory of type and typography to exclude this phenomenon just in order to avoid guilt by association. It should be pointed out that although most legibility studies can be seen as atheoretical in their scope from a behavioural science perspective, legibility studies have, at least in quantitative terms, constituted an important if not major part of 20th-century attempts to bring some kind of theory to typography.⁹ So, there should be a place for a historical and critical involvement with the question of legibility – a question that researchers, designers, and perhaps not least design educators, have paid, and still pay, much attention to. Furthermore, the word ‘legibility’ today often surfaces in graphic design discourse, with a variety of notions and motives attached, often revealing a salient lack of historical knowledge. Aside from that – the past authority of legibility research and its methodology still lives on in some academic and other contexts. It is therefore necessary to place this research into a larger context in order to gain understanding of its past and current status, validity and function.

TIMES He that plays the king shall be welcome; his majesty shall have tribute of me; the adventurous knight shall use his foil and target; the lover shall not sigh gratis; the humorous man shall end his

BASKERVILLE He that plays the king shall be welcome; his majesty shall have tribute of me; the adventurous knight shall use his foil and target; the lover shall not sigh gratis; the humorous

ARIAL He that plays the king shall be welcome; his majesty shall have tribute of me; the adventurous knight shall use his foil and target; the lover shall not sigh gratis; the humorous man shall end

GILL SANS He that plays the king shall be welcome; his majesty shall have tribute of me; the adventurous knight shall use his foil and target; the lover shall not sigh gratis; the humorous man shall end

FIGURE 1. Text set in the serif (roman) typefaces Times and Baskerville (11 points nominal size), and the sans serif typefaces Arial (10 points nominal size) and Gill Sans (11 points nominal size).

9. Michael Macdonald-Ross even equates ‘legibility research’ with ‘typographic research’ (1994, p. 4691).

2 The construct of legibility

Terminology and theoretical definitions

Legibility has to do with the effect of design variables on reading performance, that is, ‘the effect of different typographical arrangements on the reader’s ability to carry out the reading task most easily, comfortably and effectively’ (Katzen 1977, p. 8). This seems as good a summary as any, and as wide in scope. However, before going any further, an attempt to clarify the sometimes confusing term ‘legibility’ will be made.

The term has been used when measuring (under a variety of conditions): the speed of reading of continuous text (combined with various kinds of comprehension tests), and the visibility or perceptibility of words or letters (either grouped or isolated, in either sense-making or non-sense combinations, at a variable distance, under variable illumination, or under variable time of exposure). Thus, when assessing the literature on the ‘legibility’ of sans serif typefaces we have to discriminate between studies where ‘legibility’ denotes the speed or ease of reading of continuous texts under ‘normal’ reading conditions; and where ‘legibility’ denotes the visibility or perceptibility of isolated displays of letters or words read at a distance or under other constrained reading conditions; and where ‘legibility’ denotes the comparative perceptibility of individual letters within the same typeface; and where ‘legibility’ denotes something else. R.L. Pyke, in his extensive review of early legibility research, suggested in 1926 that the term ‘legibility’ should be

reserved to the reading of continuous text. However, this point of view alone does not answer the question of what operational criteria to apply:

To read means to obtain meaning from written or printed symbols. But does ‘easily’ mean accurately, or rapidly, or both? Or does it mean with little effort, and if so, does that mean without eye-strain or without fatigue ... ? It does not follow, because you read accurately, or rapidly, or both, that you do so without effort ...; nor that you read accurately or fast because over a long period you read without eye-strain or fatigue. To decide which of these terms to retain as elements of our definition apply the test of *reductio ad absurdum*. (Pyke 1926, p. 25–26)

From the late 1930s the term ‘readability’ has to a certain extent been used more or less interchangeably with the term ‘legibility’.¹⁰ Around 1940 the influential legibility researcher Miles Tinker as well as the somewhat less influential legibility researcher Matthew Luckiesh preferred the term ‘readability’ to the term ‘legibility’ to describe their prolific work – even though they based their ‘readability’ construct on different theoretical and operational criteria. In correspondence between Tinker and Luckiesh in 1943, quoted by Sutherland (1989, pp. 77–81), both claimed to have used the term ‘readability’ first. In this argument Tinker explicitly pointed to his and Donald Paterson’s book *How to make type readable*, published in 1940.¹¹ Nevertheless, Luckiesh and Moss had already used the term ‘readability’ in the title of an article published in 1938 (i.e. Luckiesh and Moss 1938).

It is likely that Tinker wanted to use the term ‘readability’ as a narrower term as opposed to the then ambiguous and broadly embracing term ‘legibility’ – ‘readability’ denoting ease or speed of reading of continuous text under ‘normal reading conditions’ only (Tinker 1944, p. 385). The fact that Luckiesh and Moss used the term ‘readability’ as a broad embracing term complicated the picture (see Luckiesh and Moss 1942, pp. 4–7, 390, 393).

However, formulas based on lexical, syntactic and semantic variables for measuring *comprehensibility* of educational and

10. I am here disregarding occasional historical precedence, in literature and elsewhere, for various usages of the two terms, including interchangeable usage (cf. OED 1989).

11. In Tinker and Paterson’s book the terms ‘legibility’ and ‘readability’ were used interchangeably, both terms referring exclusively to the reading of continuous text (see Paterson and Tinker 1940, p. xvii).

instructional reading material in order to match the abilities of specific groups of readers, which first had appeared in the 1920s, and which from the mid 1930s more and more often were referred to as ‘readability formulas’,¹² had by the 1940s and 1950s become very influential. This made it difficult for Tinker and like-minded researchers to maintain the term ‘readability’ for their area of research (Tinker 1963, p. 4; Sutherland 1989, p. 108). In the summary of his oeuvre, *Legibility of print* (1963), Tinker acknowledged the confusion and stated that he had now confined himself to the term ‘legibility of print’. This acknowledgement was accompanied by the fact that both his *Legibility of print* and George Klare’s influential *The measurement of readability* was published the very same year in the same series by the same publisher. Tinker, who himself mainly relied on the speed of reading (of continuous text, combined with a comprehension test) in his legibility studies, in 1963 suggested that ‘legibility’ was best used as a broad comprehensive term and that ‘each of several research techniques contributes something of value to the study and understanding of legibility.’ Tinker suggested that legibility ‘is concerned with perceiving letters and words, and with the reading of continuous textual material’ (Tinker 1963, pp. 5, 7). Thus, in this respect, Tinker was back at square one, in exactly the ambiguous situation he tried to avoid some twenty years earlier.

In spite of all this, Herbert Spencer’s research unit established at the Royal College of Art in London in 1966 was named the ‘Readability of Print Research Unit’. However, Spencer did not refer to ‘readability’ in his book *The visible word* – where he exclusively used the term ‘legibility’, that is, as a broad embracing term (see Spencer 1969). Later, sometimes between 1977 and 1979, the research unit at the Royal College of Art changed its name to the ‘Graphic Information Research Unit’.

In 1965, the researcher Christopher Poulton seems to have preferred the term ‘readability’ to refer to the reading of continuous text (i.e. in accordance with Tinker’s earlier preferred usage) and ‘legibility’ to refer to the recognizability of individual characters (cf. Cheetham and

12. See Klare 1963, especially the extensive bibliography; and Zakaluk and Samuels 1988.

Grimbly 1965, p. 68; and Poulton 1965). However, only three years later, in 1968, Poulton seems to have changed his mind, and by then he used the term 'legibility' to refer to the reading of continuous text (i.e. in accordance with Pyke 1926), as well as for search tasks (Poulton 1968).

In 1968 Jeremy J. Foster pointed out that the terms 'legibility', 'visibility' and 'readability' needed to be standardised and that 'only recently have research workers bothered to make the distinction explicit (for example: Zachrisson, 1965).' (Foster 1968, pp. 279–280). Foster suggested that 'legibility' should be reserved for 'the ease with which running text matter can be understood under normal reading conditions' (i.e. in accordance with Pyke's (1926) and Poulton's (1968) suggestions; as well as Zachrisson 1965), and that 'visibility' should be reserved for the 'identifiability of a printed character or form' (Foster 1968, p. 279).

However, only five years later, still hanging on to his definition of 'visibility', Foster now, reflecting the then many new areas of investigation, uses the term 'legibility' as a very broad inclusive and more figurative term embracing not only 'running text matter' on paper but of information on all kinds of graphic displays; and not only embracing with regard to medium substrate, but also with regard to the various modes of symbolisation and configuration of various media types:

Legibility research is concerned with studying the effects of visual information format on the responses made to it by the reader. ... It therefore embraces not only typography, but also the use and design of signs, illustrations, maps, symbols, colour-coding systems' ... ('Visibility' is used in a more restricted sense, to refer to the ease with which a form, or character can be identified, and 'readability' usually denotes stylistic complexity of text). (Foster 1973, p. 20)

Foster acknowledges that the definition of 'legibility' is 'too' wide, and he therefore, in order 'to make sense of the field' tabulated a classification system to map legibility research: Here, either characteristics of the message (text, alphanumerics, or non-verbal marks) or characteristics of the environment (e.g. illumination or state of motion) represent the independent variables of most interest to the experimenter. Foster leaves out characteristics of the reader as an independent variable – and places research where characteristics of the reader are *the* independent variable in an even wider context than

‘legibility research’ which he calls ‘visual communication research’. Foster pointed out that ‘traditionally, most legibility research has compared typographic designs in terms of their effects on recognition of characters or reading of texts’ (p. 21). This kind of legibility research is what I hereafter will refer to as ‘traditional legibility research’. The studies into the relative legibility of sans serif typefaces as compared to serif (roman)¹³ typefaces, which are examined in this thesis, can all be described as ‘traditional legibility research’.

In 1980, Foster in his discursive bibliography of ‘legibility research 1972–1978’ still uses the term ‘legibility’ as a broad term – not only embracing the reading of continuous text and the visibility of words and letters, but also ‘reading’ and comprehension of symbols and signposting systems, illustrations in text, typographical factors in cartography, graphs, numerical tables, forms and algorithms. However, on page 8, where he equates ‘symbol identifiability’ with ‘visibility’, he equates ‘text reading performance’ with ‘legibility’ (Foster 1980).

As late as 1987, the International Standards Organisation in a proposal for an international standard for computer screen ergonomics, which included recommendations for ‘legibility testing of visual display screens’, used the terms *legibility* to refer to intrinsic characteristics of typefaces and *readability* to refer to the quality of the typographical arrangement on the page (Östberg *et al.* 1989, p. 146). This is in accord with the usage of the English typographer Geoffrey Dowding and the American typographer Marshall Lee (cf. Dowding 1957, pp. 1–8; Lee 1979, pp. 89–90).

An illegible type, set it how you will, cannot be made readable. But the most legible of types can be made unreadable if it is set to too wide a measure, or in too large or too small a size for a particular purpose.
(Dowding 1957, p. 5)

In accordance with the ISO 1987 (and Dowding and Lee’s) definition of *legibility*, but not in accordance with their definition of *readability*, type designer Walter Tracy with subtlety (and in accordance with Poulton’s 1965 definitions), refers to legibility as denoting the clarity of

13. The expressions ‘serif typeface’ and ‘roman typeface’ are used interchangeably in this thesis.

single letters of a typeface, whereas readability refers to the ‘visual comfort’ provided by the quality of typefaces when applied in long stretches of continuous text – what he refer to as ‘two different aspects of visual effectiveness’ (Tracy 1986, pp. 30–32). Tracy exemplifies his definitions the following way:

The difference in the two aspects of visual effectiveness is illustrated by the familiar argument on the suitability of sans-serif types for text setting. The characters in a particular sans-serif face may be perfectly legible in themselves, but no one would think of setting a popular novel in it because its readability is low. (Those typographers who specify a sans-serif for the text columns of a magazine may be running the risk of creating discomfort in the reader – to the ultimate benefit of a rival journal.) (Tracy 1986, p. 31)

To make things even more complicated: some typographers are using the term ‘readability’ to refer to how pleasurable and interesting (but also how comprehensible) the subject matter of a text is, while using ‘legibility’ as a broad embracing term like in Tinker’s 1963 usage (see for example Palmer 1979–1981; Rannem 1981). This usage of the term ‘readability’ is in accordance with lay everyday usage (think of the word ‘readable’), as well as historical precedence (see OED 1989).

As we can see from the exposition in this section, the use of the term ‘legibility’, and ‘readability’ for that matter, are far from straightforward and agreed upon.¹⁴

Nevertheless, it is not unique to legibility research to have problems with a mass of different definitions of what the object of study is, and how to measure it. There is a great diversity of theoretical definitions of constructs in the sociobehavioural sciences, and this applies not least to empirical operational definitions. It follows from this that different measures generate different results (see Pedhazur and Schmelkin 1991, p. 171).

14. Not to mention an ever more figurative use of the word ‘legibility’, particularly with regard to the built environment (paralleling the ever more figurative use of ‘to read’). Cf. also Abraham Moles’ modernist and socially concerned notion of graphic design as a means for making the man made world as accessible, understandable, and intelligible as possible, introducing his concern with the phrase ‘the legibility of the world’ (Moles 1989).

Ergonomics as a framework

In the above mentioned article (Foster 1973) – ‘Legibility research: the ergonomics of print’ – the author suggested that legibility research can be best described as research into the ergonomics of print. He also suggested that legibility research ‘may be seen as one branch of ergonomics’.

Since the mid 20th century the term ‘ergonomics’ (alternatively ‘human factors’; especially in the USA) has been used to describe research and other activities related to the design, adjustment, and organisation of tools, procedures, systems, and work environment – but also consumer products and any man made environment – in accordance with the biological characteristics of humans. Since legibility research is preoccupied with the interaction between the physical appearance of graphic displays and the perceptual capabilities of humans, the term ‘ergonomics’ seems appropriate to broadly describe and contextualize legibility research.

Another useful contextualization of legibility research is provided by Robert Waller. He borrows a framework of three ‘domains’ from educational theory related to the articulation of objectives for an educational outcome. While outlining typographic research undertaken by applied psychologists he suggests that legibility can be labelled as belonging to ‘the psycho-motor domain’ – as opposed to semantic and aesthetic dimensions of typography belonging to ‘the affective domain’, and typographic cueing belonging to ‘the cognitive domain’ (Waller 1987, pp. 28ff; Waller 1991, pp. 341ff).

Operational definitions

The existence of multiple operational criteria (methods) indicates that a construct (for example ‘legibility’) and its operational definition¹⁵ (the actual method employed) are not synonymous. Operational constructs

15. ‘Operational definitions’ are sometimes referred to as ‘empirical definitions’ in the research literature.

are by nature circular. A construct is defined in terms of the operations necessary to measure it, and the measurements are defined to be measures of the psychological construct.¹⁶ The emerging criticism of positivism in the social sciences in the 1960s focused among other things on easy going use in psychology of operational definitions divorced from theoretical definitions, and furthermore, that ‘operationalism’ limited psychology to be nothing but experimental psychology.¹⁷ Nevertheless, useful psychological experiments for evaluating legibility must in the last instance be based on convenient operational criteria, and a too persistent insistence on theoretical definitions of all constructs carries the seeds of infinite regress.¹⁸

Tinker (1963, pp. 9–31) and Zachrisson (1965, pp. 44–71) outline and thoroughly discuss various operational definitions, that is methods, techniques, or procedures applied by researchers to measure legibility. Shorter but useful overviews and discussions can be found in Spencer 1969 (pp. 21–24), in Reynolds 1984, and in Wendt 1994 (pp. 277–283, 291–293). Foster presents a discursive outline of a wide variety of operational methods employed by legibility researchers in the 1970s, including reading speed combined with comprehension tests (1980, pp. 4–8). A critical discussion on how the results of various legibility methods relate to each other, is given by Salcedo *et al.* (1972).

The various methods will be discussed or touched upon when necessary while examining each individual legibility study in the next section. However, a discursive overview of some of the most common methods of traditional legibility research, as well as some *ad hoc* ‘typeface topology’ methods almost exclusively used in typeface studies, is provided below in order to serve as a clarifying point of reference.

The various methods can be grouped as depending on either ‘outcome measures’ or ‘process measures’ (Schumacher and Waller 1985; Dillon 1994). However, I find these labels awkward to use in a meaningful and consistent manner when trying to classify typeface

16. For accounts of the historical role of ‘operationalism’ (or ‘operationism’) in psychology, see Rogers 1992; Bickhard 1992; Green 1992; and Koch 1992.

17. See Skjervheim 1963.

18. See Pedhazur and Schmelkin for a useful discussion on ‘theoretical’ and ‘empirical’ definitions (1991, pp. 164–179).

legibility studies. These labels are better used for describing a facet of individual methods. For a broad categorisation of operational methods and types of legibility studies, I find the following categories helpful: ‘Experimental performance studies’, ‘subjective preference studies’, and ‘typeface topology studies’.

Experimental performance studies

Continuous reading

- *Speed of reading.* This method (the quicker the better, the slower the worse) is probably the most frequently used method for measuring legibility and it was also the preferred method of Tinker and Paterson. ‘Reading speed is the best measure of type quality that has been found’ (Rubinstein 1988, p. 36). The method has been applied in various ways: with either time or amount of reading as invariants, and has often been combined with various forms of comprehension checks. Most often it has dealt with silent reading. Aside from criticism of Tinker’s methods by his ‘arch-rival’ Luckiesh, both Zachrisson and Poulton critically discuss various aspects of Tinker’s combined method based on the Chapman-Cook reading test. Zachrisson and Poulton are particularly critical of Tinker’s comprehension measure. Zachrisson thinks it is too much a measure of comprehension, while Poulton thinks that it is hardly a measure of comprehension at all, since comprehension is only required above a minimum level.¹⁹ Even if substantial differences in speed can be found in a legibility study, such differences may not decide the picture. For example: Don Bouwhuis suggests in an interesting study that elderly readers may need larger letters, but that they will read more slowly as a result (Bouwhuis 1993).

19. See Zachrisson 1965 (p. 46); Poulton 1960, 1968 (p. 72). Rayner and Pollatsek, although in the context of ‘readability formulas’, present a useful discussion on problems of what comprehension is and how possibly to measure it (1989, pp. 316–320). For a recent study on comprehension as a ‘psychophysical measure of reading performance’, see Legge *et al.* 1989. The authors conclude that comprehension is a poor ‘psychophysical measure’.

• *Eye movements.* Measurements of oculomotor behaviour have also been used in legibility studies. For the sake of order, eye movements refer here only to eyeball motion and not contraction and expansion of the lens. Several automatic unconscious eye movements during reading have been identified: *saccades* (frequent ballistic movements or jumps along the line), *fixations* (when the eye is nearly²⁰ immobile and visual information is being extracted by foveal vision – making up 90–95% of the total reading time), *regressions* (alternatively: *regressive saccades*; when the eye goes back and re-read characters or words), and *return sweeps* (when the eyes move from somewhere near the end of one line to somewhere near the beginning of the next line).

The *perceptual span* – according to Rayner and Pollatsek (1989) – refers to the effective visual field; that is, the amount of text or number of characters perceived in one fixation (5–20 characters, depending on contrast from low to high). In O'Regan's and Legge's alternative measure *the visual span*, the corresponding number of characters (that are recognised at each glance) are 2–10, depending on contrast from low to high. Nevertheless, the point is that the eye's acuity is increasingly diminished just outside the centre of fixation (that is, starting to diminish already within the effective visual field).²¹

Parafoveal information extraction refers to the repeated glimpses of text at a coarser scale ahead of *foveal* fixation. The psychologist Raymond Dodge already in 1907 theorised that 'It seems probable that the normal reading pause [i.e. the fixation] represents a comparatively late moment in the total process of perception of the fixated object' (Paterson and Tinker 1947, p. 388).

The eye movements in question were first identified and described in the latter part of the 19th century by the French ophthalmologist Emile Javal. His 'observation that eye movements in reading occurred in jumps ("per saccades") contrasted sharply with the prevailing wisdom of the time' (Venezky 1984, p. 7). Today, cognitive psychologists working in the field of reading psychology attempt to use eye movement data as

20. The eye oscillates continuously, something which is assumed to be necessary for seeing (Rubinstein 1988, p. 27).

21. See Morrison and Inhoff 1981; Rayner and Pollatsek 1989, pp. 113–152; O'Regan 1990; and Legge *et al.* 1997.

indices of the cognitive processes believed to be involved in reading, as well as for studying visual perception in general (O'Regan 1990). However, researchers are also aware of that since eye movements are sequential and cognition is likely not to be sequential, there can be no simple correspondence.²²

The length of saccades (the longer the better), the frequency of regressions (the fewer the better), the frequency and duration of fixations (the fewer and shorter the better), and the width of the perceptual span (the longer the better), have been viewed as indices of the relative ease or difficulty of discriminating characters and reading the test material. For example will a text of low contrast make the visual span shrink, reading speed will slow down because fewer characters are recognised at a glance, and the reader will advance in smaller but more frequent steps (Legge *et al.* 1997). In other words, the oculomotor measures mentioned are determined by the visual features of text.

Tinker, who extensively recorded eye movement data while performing his impressive range of legibility experiments, which involved more than 30.000 subjects, claimed 'that eye movement measures provide an excellent supplement to reading performance' (1963, p. 26). Paterson and Tinker actually photographed²³ the eye movements and recorded the number of fixations, the numbers of words per fixation, the duration of pauses, the perception time, and the number of regressions. They also published many papers explicitly on eye movement measures.²⁴ In a thorough survey of work on 'visual factors

22. Two collections of papers on eye movements in reading are Rayner 1983, and Kowler 1990. Recently the journal *Vision Research* hosted a heated debate on whether wordspaces guide eye movements or not – and for that matter, whether wordspaces are important in facilitating easy reading or not (see Epelboim *et al.* 1994; Rayner and Pollatsek 1996; Epelboim *et al.* 1996). Anyway, see Paul Saenger's interesting account of the historical role of wordspaces for the facilitation of easy reading (1997).

23. However, as Andrew Dillon points out, use of eye movement recording equipment is rarely non-intrusive (1994, p. 38). See for example the picture of a subject and the 'Minnesota eye movement camera' in Paterson and Tinker 1940a (inserted between pp. 4 and 5). However, current methods are less intrusive.

24. For example Paterson and Tinker 1940b ('Influence on line width on eye movements'); Paterson and Tinker 1947 ('The effect of typography upon the perceptual span in reading'); and Tinker and Paterson 1955 ('The effect of typographical variations upon eye movement in reading').

and eye movements in reading', published in 1981, Robert E. Morrison and Albrecht-Werner Inhoff conclude that:

Although comprehension requirements (e.g., skimming, reading for detail) and text difficulty may also have effect on eye movement measures within readers, and despite wide differences in eye behaviour between individuals, the oculomotor effect of the physical characteristics of text are undeniable and should not be ignored. (p. 143)

The explanatory and theoretical potential of focusing on eye movement measures in legibility studies becomes clear when the typographic variable line length is considered. Morrison and Inhoff, like researchers before them, focus on the important role of parafoveal information extraction in reading, and Tinker and Paterson accordingly theorised that short lines prevented the full use of parafoveal vision. This was supported by the observation that in the reading of lines with 'optimal line length' the fixation duration decreases over the last half of a line, presumably due to advance parafoveal information extraction. With line lengths beyond the 'optimum', regression frequency increases, and with extra long lines Tinker found problems with the execution of the return sweeps to next line.²⁵ Tinker also found that interlinear spacing expands the acceptable range of line widths. Mackworth (1965) theorised that extended interlinear spacing probably reduces lateral masking and widens the perceptual span. 'Lateral masking' refers to the decreased perceptibility of stimuli when surrounded by adjacent contours. This is supported by the observation that performance increases drastically when letters are presented in isolation within parafoveal vision, and decrease when not in isolation. (Morrison and Inhoff 1981, pp. 135–136).

Mary Dyson and Gary Kipping have recently carried out three studies on screen legibility.²⁶ These studies involved the variable paging versus scrolling and typographic variables such as number of columns and line length. Surprisingly, Dyson and Kipping found that reading was faster at longer line lengths.²⁷ They point out that this is in line with

25. See Paterson and Tinker 1940b; Tinker and Paterson 1955; and Morrison and Inhoff 1981.

26. Interestingly, these legibility studies are among several legibility studies that recently have been supported by the Microsoft Corporation.

27. See Dyson and Kipping 1997, 1998a, 1998b.

early findings on reading from screen (i.e. Duchnicky and Kolars 1983), but contradict traditional findings from print legibility research, such as Tinker's (the authors refer to Tinker 1963). It also contradicts common craft knowledge, as well as the theoretical arguments of Morrison and Inhoff, referred to above. While discussing their findings Dyson and Kipping suggest that it may be the screen medium that causes the surprising result; we tend to sit further away when we read on screen, thus allowing for a wider visual angle (1998b, p. 151–152).²⁸ Dyson and Kipping also suggest that the time spent in many more return sweeps for text set in short line lengths can partly explain their findings (1998a, p. 303). Nevertheless, in one of their studies they admit that very narrow line lengths have been compared with very long line lengths, and that none of them have been compared with moderate line lengths (1997, p. 711). Introducing moderate line lengths (as Paterson and Tinker did in their 1940b study) could very well have altered the results or put them in a different light. Interestingly, the authors point out that the longer lines were (subjectively) judged by the subjects as least easy to read.

Anyhow, it is not my intention to dwell on line lengths or eye movements. My intention is simply to foreground that eye movement recordings could seem to provide a basis for a more fertile and theory-inclined approach to legibility studies than for example theory-exempt speed of reading studies (in spite of contradictory findings).

- *Blink rate* (the rate of involuntary eyelid blinking). This method was used extensively by Luckiesh and Moss. Tinker, in numerous writings, as well as other researchers,²⁹ strongly contested the validity and reliability of this method. Foster seems to reflect a consensus when he refers to the method as 'the now-discredited blink-rate method used by Luckiesh and Moss' (1980, p. 8). The controversy between Tinker and Luckiesh has recently been described, although biased in Tinker's favour, by Sandra Sutherland who provides an interesting look into the personal and professional relationship between Tinker and Luckiesh. Their non-amicable and strained relationship was based on the exchange

28. See also Dillon 1994, p. 47, and further references there.

29. See for example English 1944, p. 219, and Carmichael and Dearborn 1947.

of letters and polemic articles, where they argued back and forth against each others' methods.³⁰

Nevertheless, a paper recently published in *Human Factors* (Stern *et al.* 1994), which includes a thorough review of the literature on blink rate as a measure of fatigue (47 items, old and new, typically related to either reading, driving, or aircraft pilot performance), contains a severe criticism of how Tinker as well as Carmichael and Dearborn (i.e. 1947), and others, reached their conclusions. The paper rehabilitates the possibility of linking increase in blink rate to fatigue and time on task. However, the authors also show that blink rate is a function of how demanding a task is: the more demanding the task, the lower the blink rate, and vice versa. Thus, the blink rate is significantly lower when reading than when not reading. In order to embrace both blink rate phenomena described here, the authors suggest that increase in blinking (as well as increase in blink closure duration and decrease in blink amplitude) during task performance, is a result of decrease in inhibitory control.

Threshold visibility

- *Variable time of exposure* (also referred to as 'speed of perception', 'brief exposure' and 'time-threshold'). By this threshold method, test material is exposed to the subjects for only very short periods of time – either incrementally increased (e.g. from 1/10 of a second), or by counting short successive exposures (e.g. each of 1/1000 of a second). This is performed with the help of specially designed instruments – tachistoscopes, also called t-scopes. This method is mainly used for 'investigating the legibility of individual letters and symbols, and in the field of word-perception research' (Rehe 1984, p. 18).
- *Variable distance* (also referred to as 'perceptibility at a distance' and 'distance threshold'. This method, too, has found application primarily in investigation into the legibility of individual letters and symbols, and

30. See Sutherland 1989 (pp. 72–87). The polemical articles in question appeared during the 1940s, most of them in the *Journal of Experimental Psychology* and the *Journal of Applied Psychology*.

more obviously, road signs. The method is certainly applicable for display purpose situations, such as road signs and instrument panels.

The results from visibility tests often contradict results from experiments where speed of reading is measured. 'Keeping the difference between these two measures in mind helps avoid apparent contradictions and paradoxes in reporting the quality of text' (Rubinstein 1988, p. 176).

Visibility is a function of visual angle, but within limits. Visibility raises steadily with visual angle, though the rate of rise diminishes with larger angles (Rubinstein 1988, p. 174ff). With sufficient magnification, interactions with the visual field will occur, that is, details of letterforms will start to fill up the visual field (Bouwhuis 1993).

Note however that traditional typeface design did not (necessarily) treat as linear the transformation of shape and proportions when scaling from size to size. That is, the different typeface sizes had individual designs (in traditional metal technology a 'font' equalled one typesize, as opposed to today when one 'font' may generate all applicable sizes). This design feature is referred to as *optical scaling*: a smaller type size results in heavier serifs, thicker strokes, bigger x-height with shorter ascenders and descenders, a more open form (bigger counters), wider letters, and larger default space between the letters. And the opposite applies for larger typesizes. This was done in order to compensate for side-effects of linear scaling; for example will a linear reduction of size into a smaller size create brittle, weak and anemic letterforms clogged together, posing technical reproduction problems, perception problems, and visual inconsistency. Based on research in visual perception, it has been theorised that 'such proportional changes are necessary because the human visual system has a non-linear sensitivity to visual features of different spatial frequencies' (Bigelow 1989, p. 77).³¹ Observe that the sophisticated knowledge of optical scaling is knowledge that is represented *in* or *by* artefacts, that is, typefaces. This 'within artefact

31. For good introductions to optical scaling in type design, and to contemporary approaches to optical scaling, see the following complementary papers: Harry Carter's classic paper [1937] 1984; Bigelow 1981; Seybold 1991; Benedek 1991; André and Vatton 1994; and Burke 1995/1996.

knowledge³² is not knowledge that could be invented and generated on the basis of experiments. It is of course possible that experiments could validate or falsify the underlying ‘theory’. However, to find procedures fine and accurate enough, or procedures that really measures the effect of optical scaling and not something else, would be difficult. Note also, that the concept of optical scaling suggests that to measure type size ‘objectively’ by referring to the visual angle subtended (as in some recent legibility research) is not necessarily as relevant as suggested.

‘Optical scaling’ can explain why in some comparative legibility studies (on proportional width typefaces like Times vs. fixed width or so-called mono-spaced ‘typewriter’ typefaces like Courier) the fixed width typefaces score best in very small sizes (see the results in Ariditi *et al.* 1970 and Mansfield *et al.* 1996). A fixed width typeface like Courier has exaggerated serifs, a more uniform stroke thickness, more space between the characters, as well as bigger default spaces between the words. These features of Courier parallels the features of the small size versions of typefaces generated on the basis of optical scaling, and Courier is therefore better suited than Times for linear scaling down to small sizes.

- *Other threshold visibility measures.* Based on a variety of unfavourable conditions, such as focus variation, presenting the stimulus material at an angle, poor illumination, with non-optimal brightness contrast between the text and the substrate on which it appears, and testing of peripheral vision.

Search task

- *Search task.* A measure very much directly related to the task of the reader, by measuring the time it takes to look up something in non-continuous text like dictionaries, timetables, telephone directories, bibliographies, or tables (Spencer 1969, p. 24). This method has not been used much in ‘traditional’ legibility performance studies, but became more common in the 1970s in studies that focused more on macro level aspects of typography.

32. See Carroll 1990, pp. 277–284; Carroll and Campbell 1989; and Cross 1999. See also the section ‘The epistemic alternative of “design-based theory”’, in chapter 6.

Subjective preference studies

Non-experimental preference studies constitute a particular genre of comparative typeface studies. Some of these studies are labelled as legibility studies, however, not all of them. Preference studies are most often individually published, but sometimes appear as subsidiary supplements to behavioural experimental legibility studies.³³

Paterson and Tinker published a typeface preference study in order to ‘determine the trustworthiness of readers’ opinions’ in their 1940 monograph *How to make type readable: a manual for typographers, printers and advertisers* (pp. 18–20). The ten typefaces employed were the same as had been employed in an earlier speed of reading study (Paterson and Tinker 1932b), and were now ranked according to ‘ease and speed of reading’. The results showed great differences among the typefaces. However, with one exception, the textura typeface Cloister Black, the results showed no agreement with ‘the objectively determined results’ from the 1932 speed of reading study. Paterson and Tinker actually suggested that ‘reader preference’ are of greater value for ‘the printing industry’ since legibility differences according to ‘speed of reading results’ are only slight for most ordinary typefaces. Thus, by choosing a preferred typeface among equally legible typefaces, the text ‘will give the impression of being easily and speedily read’. However, this argument only applies to ‘equally legible typefaces’, and the authors also concluded that

The failure of the students to recognize that American Typewriter is less legible than standard type faces in common use may be accepted as impressive evidence of the difficulty and probably the impossibility of deciding questions of legibility by mere inspection of printed material. ... The results of this special study should be accepted as evidence that mere opinions concerning matters of typography are unsafe guides (pp. 19–20)

In a paper published two years later, ‘Reader preferences and typography’, the authors reported some of the results on other typographic variables from the same series of preference studies as the 1940 typeface study cited above (Tinker and Paterson 1942). On these

33. See in chapter 5, ‘A review of empirical studies’.

other typographic variables, contrary to the results in the typeface preference study, the authors found that in many cases there was agreement between speed of reading studies and preference studies. What is more, here they also reported that they had obtained preference data on both ‘judged legibility’ (‘opinions on legibility’) and ‘judged pleasingness’ (‘aesthetic values’). They found ‘in all cases’ a close agreement between the two preference measures.

The typeface results reported in these studies by Tinker and Paterson have been misinterpreted by some writers, who claim that the preferred typefaces in Tinker and Paterson’s studies were also the most legible. They seem to have wrongly extrapolated to typefaces, the agreement between performance and preference reported on other variables in the 1942 article (see for example Kostelnick 1995, p. 187; and Schriver 1997, p. 302). Nevertheless, there seems to be near consensus in the literature that, regardless of typographic variable, user preference or perception of ease of reading tend to be inconsistent with user performance (see for example Spencer 1969, p. 23; Salcedo *et al.* 1972; and Dyson and Kipping 1998b). For empirical instances of typeface studies where preference and performance have been measured in the very same study, see several studies that include subsidiary preference studies and which are assessed in the next chapter of this thesis (for example Hvistendal and Kahl 1974; and Taylor 1990).

Typeface topology studies

By ‘typeface topology studies’ I refer to some scientific studies that with regard to method differ widely. These studies are not based on behavioural experiments, but rather on *a priori* theories of typeface legibility, based on properties of the shape of the glyphs³⁴ of individual

34. The common term ‘character’ is to some extent used interchangeably with the term ‘glyph’ in this thesis. However, the term ‘glyph’ has been introduced in digital typography to distinguish the physical instantiation (the mark on a substrate) from the abstract ‘character’. There is no one to one relationship between ‘glyph’ and ‘character’. The glyph A in a particular typeface can represent the character A in the Latin alphabet and the character A in the Greek alphabet. Accordingly, the character

typefaces (belonging to either the serif or sans serif category). These shapes are then measured, processed and compared by the researcher according to particular operational constructs.

For example, Legros and Grant's typeface legibility study from 1916, where the degree of similarity in the area covered by certain superimposed glyphs (based on physical measurements) constituted the legibility construct.³⁵

Another example is Robinson, Abbamonte and Evans' study from 1971, where they used a computer model of human vision, and where the computer processed input information about the shape of individual glyphs, and output resulting shapes. These shapes were then visually inspected by the researchers.³⁶

A third example is based on contemporary information processing theories about human vision and the sensitivity of the human eye to 'spatial frequencies' of typographic material. This interest has led to legibility studies and theories about how (the objective and measurable) spatial frequencies of typographic material determines its legibility.³⁷ The relative legibility of serif and sans serif typefaces has also been theorised on the basis of their 'spatial frequencies', albeit briefly and in a vague and not very convincing manner.³⁸

A fourth example is a recent typeface legibility study (van Rossum 1997). Van Rossum has devised an operational method based on an operational construct of legibility where images of text are blurred in a particular way. The resulting blurred images are not exposed to 'subjects', but visually inspected by the researcher in order to determine the relative legibility of the typefaces involved.³⁹

A in the Latin alphabet can be represented by different glyphs, that is, by As from different typefaces (see Bigelow and Holmes 1993). Compare with the similar linguistic distinctions between 'allograph' and 'grapheme' and between 'allophone' and 'phoneme'.

35. See the section 'Legros and Grant 1916', in chapter 5.

36. See the section 'Robinson, Abbamonte and Evans 1971', in chapter 5.

37. See the section 'Legibility research today', in chapter 3. See also Morris 1988; Rubinstein 1988; Clark 1989; Morris 1989; Morris, Berry and Hargreaves 1993; Hallberg 1992.

38. See the section 'Legibility research today', in chapter 3. See also Bigelow and Day 1983, p. 102; Morris 1988; Rubinstein 1988, pp. 43–47; Clark 1989; Hallberg 1992, pp. 106–108; Gelderman 1999, pp. 101–102.

39. See the section 'van Rossum 1997', in chapter 5.

3 A century of legibility research

Introduction

The history of legibility research goes back two hundred years and constitutes the oldest strand of reading research, although serious research first appeared in the late 19th century (Venezky 1984, pp. 22– 23). This large body of mostly experimental research most often seem to have been carried out by psychologists – under labels such as applied or educational psychology, and more recently, under the label ergonomics. Research has also been carried out by ophthalmologists, engineers, and since the 1940s under the labels ‘journalism’ and ‘mass communication studies’, and further, since the 1960s to some extent by typographers in co-operation with trained researchers.⁴⁰

Until the 1970s legibility research was to a large extent preoccupied with typography on a detailed level, and not the overall organisation of larger groups of elements on the page or within a document. Thus, research has been performed to answer questions about the relative legibility of: 1) single characters compared to each other regardless of typeface; 2) certain typefaces or typeface categories compared with each other; 3) type size, interlinear spacing, and line length – i.e. the influence of typographic micro and meso variables. However, the colour and qualities of paper, the colour of ink, and illumination, were also part of

40. Foster (1978) presents an overview which includes statistical information, over scholarly journals containing papers on legibility research in the 1970s. This overview gives an indication of the affiliations of legibility research then.

the repertoire of traditional legibility research. After World War II, especially in the United States, in addition to involvement with print, legibility research became more and more involved with, for example, road traffic signs and military applications like trans-illuminated cockpit displays (cf. the bibliography Cornog and Rose 1967), and since the 1960s new text-carrying media substrates like microforms and computer screens.

Early legibility research is admirably reviewed by Pyke (1926). The work of Miles Tinker and Donald Paterson, the most prolific legibility researchers ever, is reviewed by Tinker (1963). The whole area of legibility research is eclectically and lucidly surveyed by Spencer (1969). Legibility research carried out in the 1970s is reviewed by Foster (1980). Foster provides a comprehensive assessment of the vivid field of legibility research in the 1970s, although the 'field' by then was both moving astray in various directions and including many new approaches.

Legibility research seems to some extent to have grown, peaked and dwindled together with the behaviourist paradigm of psychology of this century. It is however not possible to link legibility research directly to behaviourist psychology. As Venezky points out while describing the development of reading research in the early part of the 20th century, legibility research existed before and was unaffected by the advent of behaviourism (1984, p. 23). However, one thing behaviourism and legibility research have in common is the focus on the controlled behavioural experiment.

Keith Rayner claims that the emergence of behaviourist psychology in the early 20th century unfortunately cut off an interesting strand within experimental psychology – a strand preoccupied with the process of reading, and memory and language processing – linked by Rayner to modern cognitive psychology. According to Rayner this activity reached its peak with Huey's acclaimed book *The psychology and pedagogy of reading*, published in 1908. 'Since cognitive processes involved in skilled reading cannot be observed and directly measured, interest in reading waned between 1920 and 1960' (Rayner and Pollatsek 1989, p. 6).⁴¹ Although Rayner acknowledges Tinker's work on eye movements in

41. See also Rayner 1981 and Venezky 1984.

reading, he dismisses most of Tinker's work, and thus legibility research, as dealing with 'purely peripheral components of reading'.

The rise and fall of legibility research as a 'research program'⁴² can be described as having gone through four phases. In the first phase, from around the turn of the century, legibility research was a visible part of reading research. Richard Venezky points out that around the turn of the century reading research (which then included legibility research) was prominent in the psychological literature, and that very few psychology journals existed – *Psychological Review* and *American Journal of Psychology* in the USA, *Mind* in England, and a few in Germany and France (1984, pp. 7, 23, 28). It is no coincidence that the first two experimental legibility studies involving serif and sans serif typefaces appeared in the journals *Psychological Review* and *American Journal of Psychology*.⁴³

The second phase is the 'Tinker and Paterson phase'; that is, the period from the 1920s until the 1960s. This phase is very much dominated by Tinker and Paterson's extremely prolific output of typography-oriented research papers in mainstream psychology journals like *Journal of Applied Psychology* and *Journal of Experimental Psychology*.⁴⁴ In addition; during and after the second world war legibility research expanded within new areas such as ergonomics and engineering, where studies of the legibility of instrument displays and road signs were typical research topics.⁴⁵

The third phase is represented by the hectic and expanding activity in the 1960s and 1970s, with a wider range, and at the same time also a more dedicated range, of journals functioning as outlets for the studies (described below).⁴⁶ The fourth phase is represented by the dwindling – but far from vanished – research activity since the 1970s and until today (this is also described below).

42. Lakatos's term 'research program' (1970), is here used in a loose way.

43. That is, Griffing and Franz 1896; and Roethlein 1912. Both papers are assessed in this thesis.

44. See for example Sandra Sutherland's biographical study of Tinker (1989).

45. See for example Cornog and Rose's bibliography (1967).

46. See Foster 1978.

The hectic activity in the 1960s and 1970s

Curiously, in the 1960s – at the same time as behaviourist psychology was in decline and cognitive psychology was emerging and reaching for its later dominant status, but independently of this shift – interest in legibility research grew rapidly. It was nevertheless a time with a spirit of optimism and when positivism not only as method but also philosophically still reigned in the social sciences. In the period from the mid 1960s to the early 1980s legibility research witnessed a culmination with a huge output of research papers, before it more or less vanished from the mainstream.

The hectic activity in this period may have several explanations. One is that many new type-carrying media appeared in this period, many with high constraints – like low resolution capitals-only displays and print-out devices, typewriter-based ‘typesetting’, nth generation photocopies, microfilm, etc. Another reason, pointed out by Foster, is that this period also coincided with the tremendous growth in the area of ergonomics (Foster 1973, p. 20). Nevertheless, in this climate, traditional typographic micro-variable legibility studies related to the print medium also flourished.

Interestingly, the culminating period of the 1960s and 1970s coincided with other comparable activities which may all be described as attempts to rationalise typography and graphic design. I am referring to the attempts to standardise and metricate typographic measurement.⁴⁷ I am also referring to the many efforts in the making of classification systems for typefaces.⁴⁸ Furthermore: attempts to theorise typography by applying metaphorical models from mathematical information theory to typography; see for example Gui Bonsiepe’s article bearing the suggestive title: ‘A method of quantifying order in typographic design’ (1968).⁴⁹ The article appeared in the same year in the journal *Ulm* and in the *Journal of Typographic Research*. This happened around the same

47. See Boag 1996 for a comprehensive introduction to the history of typographic measurement systems.

48. See for example Lund 1993.

49. For perceptive retrospective comments on Bonsiepe’s article, see Kinross 1985, and Waller 1987, pp. 23–24.

time when Abraham Moles and others developed aesthetic theories based on mathematical information theory in order to quantify style, originality and harmony. These highly imaginative, but nevertheless scientistic theories, used terms from information theory such as ‘redundancy’, ‘information’, ‘entropy’ and ‘noise’; terms that made more sense as a set of eye-opening metaphorical labels than the attempts to *quantify* style or originality.⁵⁰

It is perhaps not a coincidence that the 1960s also was a period when the style of the ‘cool and rational’ idiom of minimalist and high modernist ‘Swiss typography’ was influential and popular throughout the western world. And as well, the 1960s and 1970s were also the heyday of the ‘design methods’ movement,⁵¹ and furthermore, the heyday of the conception of design as a ‘science’, with Herbert Simon’s *The sciences of the artificial* (1969) as a basic and influential text.⁵²

And further, another relevant example could be mentioned: ‘Educational technology’ and ‘programmed learning’, were pedagogical methods and movements developed in the hey-day of positivism. ‘Programmed learning’ is an extremely goal oriented pedagogical theory, focusing on stringent predetermined goals of learning, ‘knowledge’ dissemination to the learner in small and measured doses (stimulus), answers (response), and instant reply (feedback) from the teacher (knowledge-administrator), and permission to progress to the next step granted only after mastering the current step (Kvernbekk 1995, p. 23). The basic idea behind ‘educational technology’ and the like is that all learning can be organised as an algorithmic pre-programmed production process with effective product evaluation and quality control during the process, not unlike the management theory of ‘Management By Objectives’, also developed in the 1950s and 1960s (Hellesnes 1975, pp. 141–157; 1998). Not only did these rationalistic and technocratic pedagogical methods appear and gain status in parallel to the other

50. See for example Abraham Moles’ *Information theory and esthetic perception* [1958] 1966.

51. For a balanced introduction to the ‘design methods’ movement, see Cross 1980.

52. For a critical assessment of the conception of design as ‘science’ – still alive in many quarters – see Cross, Naughton and Walker 1981. The authors suggested ‘technological activity’ as an alternative and more fertile conception of design.

‘rationalistic’ phenomena mentioned above, there were actually links to legibility research and the emerging field of ‘information design’. In England this had something to do with the establishment of the Open University in the early 1960s, and its need to develop effective procedures and methods for ‘distance learning’. It is an irony that Michael Macdonald-Ross and Robert Waller at the Institute of Educational Technology at the Open University published their post-positivistic ‘manifesto’, ‘Criticism, alternatives and tests: a conceptual framework for improving typography’, in exactly the journal *Programmed learning and educational technology* (1975).⁵³

The high level of activity related to legibility research was manifested in numerous ways: In 1967, ATypI (Association Typographique Internationale), set up a ‘Legibility research committee’ (Zachrisson 1968). The list of members included Prof. G. Willem Ovink in Amsterdam, Dr E.C. Poulton at the Medical Research Council’s Applied Psychology Unit in Cambridge, François Richaudeau in Paris, Alison Shaw at the Library Association in London, Herbert Spencer at the Royal College of Art in London, Dr Dirk Wendt at the department of psychology at the university of Hamburg, and the chairman of the committee, Dr Bror Zachrisson at ‘Grafiska Institutet’ in Stockholm.

In the same year as ATypI set up its legibility research committee, the direct predecessor of the scholarly journal *Visible Language* was established as *The Journal of Typographic Research*. Four of the five members on the editorial board listed in the first issue were well known ‘legibility researchers’: Dr Ovink from Holland, Dr Poulton from Great Britain, Dr Tinker from the United States and Dr Zachrisson from Sweden. This surely reflects the great optimism for legibility research of that time.

In addition to the large amount of research carried out by many researchers scattered around in several countries, some researchers or research teams stood out as especially prolific during this peak period.

Tinker published two books in the mid 1960s, mainly surveying his and Paterson’s work (i.e. Tinker 1963; Tinker 1965). In Germany Dirk

53. This paper is mentioned below, in the section ‘Post positivistic critique, and notions of tacit craft knowledge’, in chapter 4, ‘Critiques of legibility research’.

Wendt performed a series of studies. In Great Britain Christopher Poulton published many studies on legibility. In Sweden Bror Zachrisson was heavily engaged in legibility research; his doctoral thesis on the subject, written in English, was published in 1965 (i.e. Zachrisson 1965). In France, François Richaudeau was involved in legibility research (see Richaudeau 1984), and he published a facsimile edition of Emile Javal's *Physiologie de la lecture et de l'écriture* (Javal [1905] 1978).

Legibility studies related to the problems of certain categories of readers were published in the form of monographs, with topics like legibility for the partially sighted (i.e. Shaw 1969), and legibility in children's books (i.e. Watt and Nisbet 1974). Also several extensive bibliographies on legibility⁵⁴ and related⁵⁵ areas were published in the United States, Great Britain and elsewhere, during this relatively short period.

Legibility research did however not only exist within the boundaries of academe in this period. Research was also mediated to designers and to craftsmen in the typesetting and printing industry. Design magazines and printer's trade journals every now and then either summarised legibility research or published actual studies. This was certainly the case in the United States, Great Britain, Germany and the Scandinavian countries. What's more: The preoccupation with legibility research at educational institutions, like the Royal College of Art in London (see below) and 'Grafiska Institutet' in Stockholm, may have had an impact on the institutions' graphic design students. And in 1972, Rolf F. Rehe published the first of at least five editions in English and at least one edition in German of his compilation of findings from legibility research, *Typography: how to make it most legible* (see Rehe 1984). This book was distributed in Europe for its members by the International Association for Newspaper and Media Technology (IFRA), intended for use by editors, journalists, designers and compositors.⁵⁶

54. Cornog and Rose 1967; Foster 1971; Foster 1972; Tomaszewski 1973; Foster 1980.

55. Macdonald-Ross and Smith 1977; Felker 1980.

56. Rehe also published in American and German printing industry trade journals; e.g. Rehe 1970, and 1971.

Changing paradigms: from 'legibility' to 'usability'

In Great Britain the typographer Herbert Spencer published his acclaimed and influential review of legibility research, *The visible word*, in 1968 (Spencer 1969). For Spencer this book marked the start of a large number of legibility studies carried out together with Linda Reynolds and Brian Coe at the 'Readability of Print Research Unit' (established in 1966) at the Royal College of Art during the 1970s (see Spencer 1974; and Reynolds 1979a). Interestingly, the research carried out by this team moved away from legibility researchers' preoccupation with details in print, to the structuring and articulating of information segments on actual⁵⁷ pages, text more complex than continuous prose (for example bibliographies), and operational methods such as search tasks. It also moved further away, from print to microform and screen legibility (areas with a high degree of constraints), and even further to environmental wayfinding graphics (directional signing).

The typographer Peter Burnhill and the psychologist James Hartley, making up another prolific research team, published many of their papers in *Applied Ergonomics* and other journals in the 1970s. This team did not only include a typographer as was the case with the team at the Royal College of Art, but it also moved in the same direction as the RCA team – away from legibility researchers' preoccupation with details, to the structuring and articulating of information on actual pages. To use the framework suggested by Waller (1987), cited above⁵⁸ in this thesis: Both teams moved away from research primarily related to the 'psycho-motor domain' to research related to the 'cognitive domain'.

This move away from preoccupation with micro-features to macro-features parallels a contemporaneous development within linguistics, expanding from the earlier formalist preoccupation with elements within the sentence level as units of investigation, to contextualist macro-approaches such as discourse analysis, pragmatics and text linguistics.

Today, it might be said that the much broader field of 'information design' which emerged in the 1970s, with the *Information Design*

57. As opposed to the former common practice where the stimulus material often consisted of isolated letters, words, or isolated lines of text, or isolated chunks of text.

58. See the section 'Ergonomics as a framework', in chapter 2.

Journal launched in 1979 and the first ‘Information design conference’ held in Holland in 1979,⁵⁹ as important landmarks, has in one sense, but only in one sense,⁶⁰ replaced the former field of ‘legibility’ studies. Information design as it understands itself today⁶¹ is an expanded area with fuzzy borders having its roots not only in legibility research, but also in technical writing, graphic design, instructional technology, human–computer interaction, psychology, and linguistic areas of investigation like stylistics, rhetorics, pragmatics, discourse analysis and text linguistics.⁶²

Information design claims to consist of theory and academic research on the one hand and design practice on the other hand. Information design studies are to a large extent based on other methods and concerns than legibility studies. Where legibility research was preoccupied with typographic micro and meso variables and regarded research in the form of laboratory experiments as the sole method of generating valid (and universal) knowledge, information design takes quite another approach.

First: Information design is more concerned with the design of documents on a macro level, and it has a much greater scope with regard to media substrate and mode of graphic symbolisation and mode of graphic configuration.⁶³ Information design is not only preoccupied with verbal graphic media types and pictorial graphic media types, but also with ‘schematic’ graphic media types like diagrams and maps. Information design prefers to focus on relatively complex media interfaces (or systems) intended for use in relatively demanding situations, like static graphic artefacts such as forms, contracts, public

59. The papers presented at the conference was published under the title *Information design* in 1984 (i.e. Easterby and Zwaga 1984).

60. See the next section, ‘Legibility research today’.

61. For self-reflective contributions on the roots of ‘information design’ and what it is or might be, see Waller 1979; Kinross 1985; 1992, pp. 141–143; Schriver 1997, pp. 13–149; 1998; Sless 1994; 1998; Wright 1998; Horn 1998; 1999; Taylor 1998; and Burke 1998. For a highly relevant description of antecedents to the overlapping field of ‘human–computer interaction’, and its development during the last two decades, see Carroll 1997.

62. See for example Waller 1991; Campbell 1995; and Schriver 1997.

63. The concept-pair ‘mode of symbolisation’ and ‘mode of configuration’ appears in Twyman 1979.

transport timetables, warnings, user documentation and environmental wayfinding systems. And more vigorously,⁶⁴ on interactive graphic artefacts like the graphical user interfaces of computer software, multimedia applications and Internet web sites.

Second: Information design involves not only laboratory experiments as a behavioural methodology but strongly advocates the alternative epistemology of *iterative usability testing* (process oriented formative evaluation)⁶⁵ as a pragmatic methodology for giving ‘real-time’ support during actual design processes (Stiff 1995). In other words, focus is on the user’s behaviour at the interface, and not on traditional experimental research. Not only contextual usability testing, but also the advantage of focusing on the *conversational metaphor*,⁶⁶ as well as *user participation* in the design process,⁶⁷ has recently been forcefully advocated. ‘Usability’ and ‘the user’ are perhaps the two most central rhetorical concepts in ‘information design’; as they are in ‘human–computer interaction’.⁶⁸

Information design does not only advocate a multi-disciplinary approach with regard to research, but also advocates a multi-disciplinary participation in the activity of designing – while regarding authoring, editing, information structuring, and graphic design as an integral process.

64. Wright 1994, p. 1; Stiff 1995, p. 65.

65. For the sake of order: ‘Usability testing’ is not like hypothesis testing in science, but rather like fault diagnosis in technology and engineering. Although the ideal is *formative* evaluation (iterative testing) during the design process, most usability evaluations (in software design and development) are still primarily *summative* (Landauer 1997, p. 204). The concepts of ‘formative’ and ‘summative’ evaluation were introduced by Scriven (1967). For a widely used textbook on usability testing (in the context of the design and development of computer software), see Jacob Nielsen’s *Usability engineering* (Nielsen 1993). John Carroll, in the context of human–computer interaction, places usability evaluation in an historical and comparative context (1997).

66. See Sless 1996, 1998.

67. See for example Kyng and Greenbaum 1991 (in the context of ‘computer systems’); and Buur and Bagger 1999 (in the context of ‘solid user interfaces’).

68. For two papers that problematise the concepts ‘usability’, ‘user’, and ‘user interface’, see Grudin 1993; and Cooper and Bowers 1995.

Actually, as early as 1965 the researcher Christopher Poulton and co-authors, in connection with typeface design, suggested an iterative process oriented approach to legibility research:

Using such an experimental technique ... type designers could compare the high with the low scoring designs, attempt by insight to discover which factor in the designs had made them more or less readable, and redesign the high scoring faces to make them even better. A second round of precise experimental evaluation could then be undertaken. This process of scientific evaluation followed by intuitive design followed by scientific evaluation would continue until some extremely readable type faces had been developed. (Cheetham, Poulton, and Grimbley 1965, p. 51)

A similar idea (and empirical approach) had actually been pioneered by the famous industrial designer Henry Dreyfuss in the 1940's (see the chapter 'The importance of testing' in *Designing for people* (Dreyfuss 1955, pp. 62–69). John Carroll points out that Dreyfuss' approach incorporated four central ideas: early prototyping, the involvement of real users, introduction of new functions through familiar 'survival forms', and many cycles of design iteration (Carroll 1997, pp. 503–504).

Michael Macdonald-Ross and Robert Waller anticipated the usability testing trend in information design in their short but substantial article published in 1975 with the telling title 'Criticism, alternatives and tests: a conceptual framework for improving typography'.⁶⁹ Besides criticising legibility research as impotent, they forcefully advocated an approach for improving practice by moving away from the experimental paradigm of 'discovering universal truth'. This, by exchanging the experimental approach with iterative test procedures prefaced with informed criticism based on the tacit knowledge of typographers and designers. They also stressed the value of parallel testing of alternatives. They drew on Michael Polanyi's concept of 'tacit knowledge'⁷⁰ and on a particular feature of Karl Popper's philosophy of science stressing the fruitfulness of an approach of 'conjectures and refutations'.⁷¹ Popper's philosophy (and 'falsificationism') was based on a

69. Macdonald-Ross and Waller 1975. See also Schumacher & Waller (1985).

70. See Polanyi 1958, and 1966.

71. See Popper 1963.

deductive logic, where induction (generalisations or theory solely based on observations) was rejected as an appropriate scientific method.

A line of thought similar to Macdonald-Ross and Waller's was in the early 1970s advocated in theory on the architectural design process and design education, most notably in a substantial paper by Bill Hillier, John Musgrove and Pat O'Sullivan.⁷² They advocated the iterative 'conjecture-test' model of design, i.e. a *deductive* procedure, where form is generated first (based on a repertoire of already conceived solution types) and then checked against constraints. This model stands in contrast to the positivist principle of *inductive* analysis-synthesis and to what is claimed to be the modern movement's conception (or rationalisation) of design as an inductive process where *form* hardly has a relative autonomy and is believed to be primarily generated by the specific job's pre-existing requirements and constraints (purpose, specification, content, available materials and available production technology).⁷³ It is realised today that even highly formalised engineering design processes – with many participants and where scale and complexity demands that pre-articulated specifications to a large extent drives the problem solving (and problem definition) process – are deductive processes, characterised by frequent iterations and reformulations of aims and objectives.

Two prominent names in 20 century legibility research are linked to the (British) Medical Research Council; R.L. Pyke who published *The legibility of print* in 1926, and the researcher E.C. Poulton, at the Medical Research Council's Applied Psychology Unit in Cambridge, who was very active in the field of legibility research in the 1960s. One of the leading figures in the 'information design' movement for two decades, Patricia Wright, was for many years affiliated with the Medical Research

72. Hillier *et al.* [1972] 1984; Gelernter 1981, pp. 271–292; 1993. See also Cross *et al.* 1981; Buchanan 1992; and Jonas 1993. Hillier and his co-authors focus on the role of 'preconceptions' or 'prestructures' in designing (the designer's repertoire of solution types and his knowledge of genres). This resembles Heidegger's concept of 'fore-structure' and Gadamer's concept of 'prejudice'. However, no references to Gadamer's *Wahrheit und Methode* (1969) or the translation *Truth and method* (1975) is given in their [1972] 1984 article.

73. See also Jan Michl's relevant paper on the historical role of the inductivist, and impossible, slogan 'form follows function' as a designer-centred *carte blanche* for modernist aesthetic preferences (1995); as well as his paper: 'On the rumor of functional perfection' (Michl 1991).

Council's Applied Psychology Unit in Cambridge. I choose to interpret Patricia Wright's work as an index of the paradigm shift from 'legibility' to 'usability', and from legibility research to information design, that has gone on in the last few decades.

Legibility research today

Psychology

Interest in legibility research is today far removed from mainstream psychology. An indication of this can be found in the four-volume *Encyclopaedia of psychology* published by John Wiley in 1994. The encyclopedia does not contain any entry or information on 'legibility' or on Miles Albert Tinker who for four decades published prolifically in journals like *The Journal of Applied Psychology* and *The Journal of Experimental Psychology*. Tinker's fellow researcher Paterson was even the editor of the *Journal of Applied Psychology* for twelve years, as well as being the Secretary of the American Psychological Association for six years (Sutherland 1989, p. 131). Furthermore, in *The thesaurus of psychological index terms*, published by the American Psychological Association, 'legibility' appeared as an index term from 1978 until 1993.⁷⁴ The fact that interest in legibility studies is nearly non-existent in psychology today does not necessarily say anything about the prospects for or value of legibility studies, but probably more about the aversion to getting associated with such highly practical matters as the legibility of reading matter; in order not to be marginalized within psychology and cognitive science. Nevertheless, although legibility research has dwindled and lost status and legitimacy since its hey-day in the 1960s and 1970s,⁷⁵ it has certainly not vanished.

74. See the thesaurus, i.e. Walker 1994, pp. 120, 1282.

75. See the previous section, 'Changing paradigms: from "legibility" to "usability"', and chapter 4, 'Critiques of legibility research'.

Graphic design

Legibility research is today less visible in typographic and graphic design discourse than any time before. There are some scattered exceptions, for example a popular summary by Linda Reynolds about legibility and legibility research published in a fairly recent issue of the typography journal *Baseline* (Reynolds 1988). Legibility research is rarely mentioned in textbooks on typography or graphic design. One exception is Ruari McLean's widely read manual of typography (1980), of which the 1992 edition is still in print. However, McLean is sceptical about legibility research. Baird and Turnbull's widely read textbook on typography for journalism students, *The graphics of communication*, used to extensively report legibility research in earlier editions; however, in its sixth edition (1993) all references to legibility research are removed. A scholarly journal like *Visible Language*, which to some extent addresses the international scholarly graphic design community, might occasionally carry a legibility research paper (for example Wendt *et al.* 1997, and Dyson and Kipping 1998b).⁷⁶ However, the influential and leading graphic design journals *Eye* and *Print* hardly ever carry articles about legibility research. The same applies to printing industry trade journals; journals that earlier, and especially in the 1970s, every now and then carried articles about legibility research.⁷⁷ One conspicuous exception in the context of graphic design is a book on legibility research published by the Graphic College in Denmark a couple of years ago (Pedersen and Kidmose 1993).⁷⁸ Another exception is Colin Wheildon's *Type & layout: how typography and design can get your message across – or get in the way* (1995).⁷⁹

76. It is not often that legibility research is used in humanities texts; however, a recent example is Paul Saenger's historical study *Space between words: the origins of silent reading* (1997, pp. 26–29, 305–314).

77. This assessment is not based on a systematic quantitative study of any kind, but it is neither impressionistic; it is rather based on acquaintance with relevant primary sources and some searches in relevant bibliographical databases.

78. Recommended by the graphic designer Colin Banks in *Visible Language*, vol. 28, no. 1, 1994, p. 88; and reviewed by the writer of this thesis in *Information Design Journal* (Lund 1995).

79. Assessed in chapter 5; and reviewed by the writer of this thesis in *Information Design Journal* (Lund 1997).

Digital typography

There are three broadly defined areas where legibility research has legitimacy and appeal today. Within 'digital typography', within 'vision research', and within the 'paradigm' that in some sense took over from legibility research, 'ergonomics and information design'.

Legibility issues are a recurring concern within 'digital typography' (and to some extent within the wider area of 'computer graphics'). The canonical texts (e.g. Tinker 1963; and Zachrisson 1965) and individual empirical studies are every now and then discussed or briefly referred to,⁸⁰ or individual studies are published.⁸¹

A particular strand of legibility research within 'digital typography' is the evaluation of the legibility of 'greyscale' representations of fonts on computer screens.⁸² 'Greyscale bitmapping' is today performed by special applications or by the operating system. It represents one of several strategies for compensating the relative coarse resolution of the screen when representing high resolution outline fonts on the screen. That is, compensation beyond the enhancing rasterization managed by some fonts due to their format, or managed by special applications like Adobe Type Manager, or managed by the operating system.⁸³ Alternative methods to 'greyscaling' are: 'binary bitmapping' of particular size instances, like the complementary screenfonts to PostScript Type 1 outline fonts; the alternative or complementary method of 'hinting';⁸⁴ and not least, the design of typefaces where the primary font is the bitmap font dedicated for the screen, while the outline version for high resolution output is generated on a subsidiary basis; for example Apple's

80. See for example Rubinstein 1988; Clark 1989; Bowden and Brailsford 1989; Morris 1989; Naiman 1991; Farrell 1991; van Nes 1991; André 1993; Bigelow and Holmes 1993; Dyson 1993; Kahn and Lenk 1993; Karow 1997.

81. See for example Morris 1989; Morris *et al.* 1991; de Lange *et al.* 1993; Morris, Berry and Hargreaves 1993; Wendt 1994; Dyson and Kipping 1997, 1998a.

82. See for example Naiman 1991; Black and Boag 1992; Hersch *et al.* 1995; O'Regan *et al.* 1996; Morris *et al.* 1998; and Boyarski *et al.* 1998.

83. There is an ever growing literature on the www on this dynamic area of digital typography. A good starting point for information is the 'Microsoft Typography Home Page': <http://www.microsoft.com/typography/> (15 September, 1999).

84. See for example Deach 1992; Hersch 1993; and Stamm 1998.

system font Geneva,⁸⁵ Zuzana Licko's typeface Base (from Emigre),⁸⁶ and Matthew Carter's typeface Verdana (from Microsoft).⁸⁷ Interestingly, the research and development carried out in this area has involved first of all computer scientists and typographers. Furthermore, the efforts in this clearly critical area – of making typefaces as legible as possible on the relatively coarse screen – has primarily been based on type design domain knowledge,⁸⁸ advances in software engineering, and visual inspection, and less on behavioural experiments.⁸⁹

Another particular strand of legibility research affiliated with 'digital typography' is based on contemporary information processing theories about human vision and the sensitivity of the human eye to 'spatial frequencies' of typographic material. Here, one starting point was to investigate the necessary spatial resolution for output devices such as imagesetters, printers and computer screens in order to satisfy the requirements of human vision.⁹⁰ This interest has led to legibility studies and theories about how (and at least in one respect, the objective and measurable) spatial frequencies of typographic material determine its legibility.⁹¹ As mentioned above, the relative legibility of serif and sans serif typefaces has also been theorised on the basis of their 'spatial frequencies'.⁹²

85. See Bigelow and Holmes 1991.

86. See Licko 1996a.

87. See Boyarski *et al.* 1998.

88. For example: The (scalable) binary bitmap screen font Verdana is characterised by extra large x-height, extra space between glyphs, particularly even space between glyphs, special care with the design of glyphs such as l, I, l, i and J, and special curve treatment.

89. Exceptions are: Black and Boag 1992; O'Regan *et al.* 1996; Morris *et al.* 1998; and Boyarski *et al.* 1998.

90. See for example Bigelow 1982; Bigelow and Day 1983; Rubinstein 1988; and Naiman 1991. See Farrell 1991 for a good overview of basic sources.

91. See for example Legge *et al.* 1985a, 1985b, 1987; Morris 1988; Rubinstein 1988; Clark 1989; Morris 1989; Morris, Berry and Hargreaves 1993; Hallberg 1992.

92. See Bigelow and Day 1983, p. 102; Morris 1988; Rubinstein 1988, pp. 43–47; Clark 1989; Hallberg 1992, pp. 106–108; Gelderman 1999, pp. 101–102.

Ergonomics and information design

Importantly, legibility research is alive within the broadly designated area 'ergonomics and information design'. I here refer to areas such as ergonomics and human factors,⁹³ human–computer interaction (where an eager interest in computer screen legibility fits in),⁹⁴ technical documentation and information design,⁹⁵ instructional technology,⁹⁶ safety science and warning design research,⁹⁷ and transportation research.⁹⁸

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- 93. *Ergonomics at work*, a standard textbook, devotes a whole chapter to 'Man–man communication: words and symbols' (Oborne 1987). A subsection of this chapter is a presentation of findings from legibility research (pp. 73–80). In *Ergonomic Abstracts*, under the main category 'visual communication' several subcategories are today open to what can be described as traditional legibility studies (for example subcategory 8.1. 'design of alphanumeric characters', and sub-sub category 8.1.2. 'shape of characters').
 - 94. A comprehensive introduction to many studies of various aspects of screen ergonomics (including differences of reading from screen vs. paper) is found in Andrew Dillon's *Designing usable electronic text* (1994, pp. 28–58); earlier published in *Ergonomics* (Dillon 1992). For a comprehensive review of the 'legibility of continuous text on computer screens', see Frenckner 1990. For examples of recent individual studies of the legibility of various typographic formats on screen, see Dyson and Kipping 1997, 1998a, 1998b. For studies that compare serif and sans serif typefaces on computer screens, see Engman 1988; Smedshammar *et al.* 1990; Williams 1990; Holleran 1992; Garcia and Caldera 1996; Boyarski *et al.* 1998; and Stone *et al.* 1999. A typeface screen legibility study by Geske (1996) compares various sizes, weights and postures of sans serif typefaces only.
 - 95. See for example the prominent textbooks by Schriver (1997) and by Kostelnick and Roberts (1998); the annotated bibliography by Kempson and Moore (1994); Sidney Berger's *The design of bibliographies* (1991); overview articles like Benson 1985 in *Technical Communicator*, and Gribbons 1991 in *IEEE Transactions on Professional Communication*; and Gallagher and Jacobson's overview article (1993). For examples of brief references given to the legibility research literature in prominent 'information design' publications, see Kosslyn 1994 (pp. 90, 283); and Campbell 1995 (p. 4).
 - 96. See for example the PhD dissertations Stahl 1989; Taylor 1990; and Kravutske 1994.
 - 97. See for example the studies by Braun and Silver 1995; Silver 1993; Silver *et al.* 1994.
 - 98. See for example the comprehensive review by Zwalhen *et al.* 1995.

Vision research

And further, a vigorous strand of legibility research exists within the field of ‘vision research’ – where the interest is not only preoccupied with the understanding of human vision, but also with human vision’s role in the task of reading, with the design of optimal reading conditions, with the design of improved reading material, with developing aids for readers with impaired vision, with the aim to understand how visual disorders affect or hinder reading, and with the design of clinical reading acuity tests.⁹⁹ Examples of legibility research carried out within the field of vision research can be the typeface legibility study by Yager *et al.* (1998) published in *Vision Research*; a study by Roger Watt published in the collection *Computers and typography* (1993); Arnold Wilkins’ intriguing monograph *Visual stress* (1995);¹⁰⁰ as well as parts of the literature on ‘spatial frequencies’ referred to above in the context of ‘digital typography’.

And importantly; the impressive publishing program of influential articles under the series title ‘Psychophysics of reading’, published by Gordon Legge and various co-authors starting in 1985 and still running,¹⁰¹ with the eighteenth article published in 1998.¹⁰² Many of the articles in this series have appeared in the journal *Vision Research*, while some of them have appeared in other ophthalmology and vision research journals. Gordon Legge’s affiliation with Miles Tinker’s Department of Psychology at the University of Minnesota, is maybe not without some significance.

99. See for example Legge *et al.* 1985a; 1987, p. 1165; 1989, p. 51; Mansfield *et al.* 1996, p. 1492.

100. For mention of Wilkins’ book, see the section ‘Research that breaks with “received wisdom”’, in chapter 6, ‘Discussion: knowledge production and technical rationality’.

101. To a large extent based on ‘fastest reading rate’ as a measure of performance.

102. See for example Legge *et al.* 1985a (1: ‘Normal vision’); 1985b (2: ‘Low vision’); 1987 (5: ‘The role of contrast in normal vision’); 1989 (7: ‘Comprehension in low vision’); 1997 (16: ‘The visual span in normal and low vision’); Mansfield *et al.* 1996 (15: ‘Font effects in normal and low vision’); and Chung *et al.* 1998 (18: ‘The effect of print size on reading speed in normal peripheral vision’).

Many recent studies

Although the absolute number of ‘traditional’ legibility studies seems to be much lower today than in the 1960s and 1970s, and furthermore, that the relative number of these studies has diminished due to a much larger total research output today, a surprisingly large number of legibility studies are still carried out – at least a surprisingly large number of studies that compare serif and sans serif typefaces. These studies appear as PhD theses or as papers published in journals and conference proceedings. They are carried out under a variety of labels, and some of them are, as already mentioned, about computer screen legibility. Some of the studies appear in clusters, like three PhD theses under the label ‘instructional technology’ from Wayne State University,¹⁰³ a row of papers on ‘warning design’ by Clayton Silver and Curt Braun and co-authors published in journals like *Safety Science* and *Ergonomics* as well as in human factors conference proceedings.¹⁰⁴

A flurry of computer screen legibility studies and reviews were published by the Royal Institute of Technology in Stockholm in the late 1980s and early 1990s (for example the following studies and reviews, solely or in part on typeface legibility: Engman 1988; Smedshammar *et al.* 1990; Frenckner 1990; and Frenckner *et al.* 1991). Interestingly, Hans Smedshammar represents a link back to Bror Zachrisson. Together, they published papers and reports on legibility for the partially sighted in the early and mid 1970s (e.g. Zachrisson and Smedshammar 1971, 1973a, 1973b).

If the number of studies comparing serif and sans serif typefaces identified for this thesis (see the table on next page) is representative of the number of legibility studies carried out today, (although excluding computer screen studies) then Robert Waller was wrong in 1991 when he claimed that ‘since the late 1960s, research on ‘simple matters of legibility’ has tended to be undertaken only in special circumstances such as for new display technologies’ (p. 343). Likewise, Karen Schriver’s complaint that there ‘sadly ... has been very little empirical research on

103. Stahl 1989; Taylor 1990; and Kravutske 1994.

104. Two of these papers are reviewed in chapter 5 in this thesis (i.e. Silver and Braun 1993; and Silver, Kline and Braun 1994).

typography during the 1980s and 1990s' (1997, p. 276n) ought at least to be modified. And Susan Roth's plea in a recent issue of *Design Issues* (1999, p. 21), for more legibility studies to be performed, seems ill-informed. As many as two thirds of the 72 typeface legibility studies identified for this thesis have been carried out in the relative short period between 1970 and 1999. And the record number for any decade is the 20 studies carried out in the 1990s.¹⁰⁵

1890s	1900s	1910s	1920s	1930s	1940s	1950s	1960s	1970s	1980s	1990s
1	0	2	2	4	4	3	8	10	18	20

TABLE 1. Number (per decade) of legibility studies identified for this thesis on the relative legibility of serif and sans serif typefaces (or on the relative legibility of individual typefaces where both serif and sans serif typefaces are included). The total number is 72. The studies are predominantly experimental, except for 7 preference studies.

105. See the introduction to chapter 5, 'A review of empirical studies'.

4 Critiques of legibility research

Introduction

Traditional experimental legibility research has been attacked from various positions, especially during the last few decades. This, first of all is a reflection of the general reaction against positivism in the social sciences, but later also as a reflection of postmodernist thought. These criticisms have come from ‘internal’ positions (limited criticisms of certain methodological aspects only), and from ‘external’ positions (a rejection of legibility research, or a rejection of prevalent notions of legibility, or a rejection of the need for legibility *per se*). It is also worth pointing out that criticism of legibility research has not been confined to academe only, but has also been voiced in an articulate manner by type designers and typographers. The material below shows that serious criticism is not something new, and only made possible in a climate of post-modern problematising scepticism, but is in fact much older. Whether or not the current ‘low status’ of legibility research and its demise as a ‘research program’ is partly a reflection of these criticisms is difficult to establish.

Post-positivistic critique, and notions of tacit craft knowledge

The conventional view of ‘technical rationality’ (see Schön 1983) is well epitomised in the 1965 article ‘The case for research’ by Christopher Poulton and co-authors (of whom one is a typographer). While describing designers’ knowledge as mere ‘guesses’ as opposed to knowledge generated from research as ‘facts’, the authors state:

But how much is really known about the effectiveness of printed graphic design? There are plenty of opinions, plenty of guesses – but how much soundly based fact? ... there is no real body of knowledge about graphic design – slogans substitute for fact. (Cheetham, Poulton, and Grimby 1965, p. 48)

The overall validity of an approach to typography where professional knowledge, tacit craft experience, visual sensibility, and even careful common sense reasoning, is totally discarded in favour of a positivist belief in ‘science’ and where experiments are regarded as the only method of generating valid knowledge, has been questioned, most notably by Michael Macdonald-Ross, Robert Waller, and Joycelyn Chapman.¹⁰⁶

Reviewing the educational psychology literature on text design, it is quite common to find what at first sight appears to be a staggeringly naïve view of what counts as knowledge. In effect, a game is being played where a new ‘fact’ is admitted to the circle of those playing only when an experiment has appeared in the literature to support it. No other knowledge counts. The game is played in code: ‘nothing is known about ...’ or ‘we do not know ...’ means ‘no one has published an experiment about ...’ (Waller 1987, p. 73)

Macdonald-Ross and Waller partly based their arguments on Michael Polanyi’s concept of ‘tacit knowledge’, which he attached to the silent *un*-articulated and informal knowledge of experienced craftsmen and designers (see Polyanyi 1958; 1966).

106. See Macdonald-Ross and Waller 1975; Macdonald-Ross and Smith 1977; Macdonald-Ross 1978, 1989; Waller 1987, pp. 70–95; and Chapman 1978.

Notions of ‘tacit knowledge’¹⁰⁷ have, during the last three decades¹⁰⁸ gained considerable ground in social theory and philosophy, and have even opened up several discursive domains. These notions take part in contemporary epistemological discourse on a broad spectrum: As a part of the critique of conceptions about human knowledge in ‘artificial intelligence’ research and ‘expert system’ engineering,¹⁰⁹ within ‘Marxist labour process sociology’ preoccupied with the social power embedded in technology and the influence of skills and de-skilling on the degree of workers’ control of the labour process – versus various forms of managerial control – seen as important for collective bargaining strength and political influence;¹¹⁰ in the same vein, ‘action research’ for empowering vulnerable groups in the labour market, like the ‘working life’ sociology at ‘Arbetslivscentrum’ in Stockholm;¹¹¹ as a rhetorical aid in the professionalisation efforts of semi-professionalised occupations such as nursing (that is, in addition to the dual strategy of establishing a ‘nursing science’ and conquering management positions in the health services); in writings on the roles of skill versus self-expression and individuality, and on skill as bodily competence, in fine art, craft, and design;¹¹² in the ‘sociology of science and technology’ concerned with the construction of scientific knowledge;¹¹³ and not least within pedagogy related to the education, training, and practice of professionals.¹¹⁴

Critical voices have been raised,¹¹⁵ and disputes on defining what exactly ‘tacit knowledge’ is and is not, whether ‘tacit knowledge’ can be articulated or not, and what ‘tacit knowledge’ types there exist, are also

107. First of all relying on Michael Polanyi (see Rolf 1991), but also to some extent on Ludwig Wittgenstein (see Janik 1988, 1990), and on Gilbert Ryle (1949; cf. his distinction between ‘knowing how’ and ‘knowing that’).

108. Since Thomas Kuhn referred to Michael Polanyi’s ideas about tacit knowledge in the second and enlarged edition of *The structure of scientific revolutions* in 1970 (see Kuhn 1970, pp. 44, 191ff).

109. See e.g., Dreyfus and Dreyfus 1986; and Göranson and Josefson 1988.

110. See e.g., Manwaring and Wood 1984.

111. See e.g., Göranson and Florin 1990.

112. See e.g., Dormer 1994; and McCullough 1996.

113. See e.g., Collins 1987; and Cambrosio and Keating 1988.

114. See e.g., Schön’s extremely influential books (1983, 1987); Dreyfus and Dreyfus 1986; Polkinghorne 1992; and Molander 1996.

115. See e.g., Rolf 1991; and Fuller 1992.

present.¹¹⁶ A related concern can be found in writings on the role of visual or non-verbal thinking in the design process.¹¹⁷

The focus on tacit knowledge has meant that what counts as knowledge has undergone a kind of democratisation (for example: nurses' tacit knowledge should count as *knowledge*: nursing embodies a distinctive competence that nobody else has). However, one thing is to be aware of the tacit knowledge of, for example design practitioners, and its important role in the design process, and to be aware of the rich source of knowledge about process and product that it embodies. Another thing is the limitations in the explanatory potential of tacit knowledge. Tacit knowledge can be the practitioner's accumulated knowledge, but it can also be highly contingent and situated action-oriented knowledge. Furthermore, practitioners' tacit knowledge may differ even more than discursive knowledge among individual practitioners; tacit knowledge is by definition local knowledge. There are, for example, practitioners who find sans serif typefaces 'illegible', and practitioners who do not hesitate in using sans serif typefaces for whatever application, as well as practitioners who prefer to use sans serif typefaces only for specific purposes.

The philosopher Steve Fuller is a rare example of someone who vigorously questions the explanatory power of the epistemic concept 'tacit knowledge'. He points out that by relying too much on a concept like 'tacit knowledge' one can easily mistake *passable* for *optimal* performance.

The claim that a coal shoveler is an expert in what she does remains persuasive only as long as coal shovelling is treated as a *sui generis* activity with no basis for comparison or examination outside the vicinity in which the shovelling normally takes place. However, once the observer expands her horizons, even if it merely involves comparing shovelers at different locations, then the coal shoveler starts to lose the patina of expertise. The same may be said of the scientist, though several conceptual obstacles must be overcome before seeing the full implications. (Fuller 1992, p. 414)

116. See e.g., Janik 1988, 1990; Cambrosio and Keating 1988; Rolf 1991; Pleasants 1996; and Molander 1996.

117. See e.g., Ferguson 1992; Goldschmidt 1994; and Henderson 1995.

Accordingly, the philosopher Bertil Rolf claims that tacit knowledge has become a magic concept of honour that may function to conserve current practice. As he points out, it may be difficult to discriminate between ‘tacit knowledge’ and ‘tacit intolerance’ – or for that matter – between ‘tacit knowledge’ and ‘tacit stupidity’ (Rolf 1991, p. 53). Thus, does the apprentice internalise expert knowledge or deep-rooted inadequate patterns of behaviour? The German philosopher of pedagogy Erich Weniger (1990) has in a similar vein introduced the concept of ‘the tyranny of practice’ to describe educational theory where, according to Weniger, practice is held to be self-sufficient (Kvernbekk 1995).

Nevertheless, attempts to solicit design practitioners’ knowledge – on which they base their designing – through discursive reflection, (in addition to soliciting knowledge and theories from ‘within’ artefacts¹¹⁸), might possibly produce more relevant and interesting data than yet another of countless legibility experiments. Macdonald-Ross and Smith, and Macdonald-Ross and Waller, proposed that designers’ underestimated knowledge should become the starting point for ‘more fruitful typographic research’, thus far from advocating the abandonment of theory and research. I quote extensively:

How well ‘pre-scientific’ crafts solved difficult technical problems without the benefit of formal empirical tests! This is something of a mystery, but a mystery which can be dispelled by learning to use our eyes! The eye is a stupendous device for making comparisons; we must entertain the possibility that an expert craftsman can by the intelligent use of his eyes reach better solutions to particular problems than the researcher in his laboratory. (Macdonald-Ross and Smith 1977, p. 40)

Any casual acquaintance with typographers and designers will show that their skills are backed by tacit knowledge which is the product of experience rather than empirical tests or ‘book learning.’ Usually researchers dismiss such personal knowledge or even deny its existence simply because it has not been developed by orthodox experimental methods. But actually, as Polanyi (1962) has shown, all scientific knowledge (indeed all human knowledge) has its roots in the ‘tacit dimension.’ ... So we should value the personal skills of typographers and designers, and take them as the starting point for more fruitful typographic research. (Macdonald-Ross and Waller 1975, p. 77–78)

118. See Carroll and Campbell 1989. See also chapter 6, ‘Discussion: knowledge production and technical rationality’.

Tinker showed that too little or too much 'leading' (nowadays, inter-linear spacing) would affect legibility adversely. Fair enough. But this kind of knowledge has been the stock-in-trade of the master-printer for centuries! So is empirical research just a kind of 'aid to rhetoric' whereby educated people (who don't understand or won't accept the experiential basis of craft knowledge) will take heed of advice if it has a 'scientific' seal of approval? (Macdonald-Ross and Smith 1975, p. 42)

However, Macdonald-Ross do not want to throw the baby out with the bath water:

Tinker's work on the cumulative effect of suboptimal arrangements and his ideas about the 'hygienic reading situation' are both relevant and practical. Imagine a text full of conceptual muddle, unclear expression, visual confusion, and suboptimal setting being read by someone who is tired and disinterested, sitting in poor light, with the text not at right angles to the line of sight, and a noisy room full of interruptions and distractions. This gives you some idea of a real-life reading situation (we might get fewer no-difference results in conditions like these). Suboptimal settings may not produce large effects when tested one by one in clinically perfect reading conditions, but they do produce most significant effects when several occur together and the text is read under poor conditions. (Macdonald-Ross 1978, p. 77)

We see here two epistemologies (knowledge from controlled experiments vs. knowledge from personal trial and experience) and two tasks (universal knowledge vs. particular solutions). The question is, shall these differences continue to cause misunderstanding, or can they be mutually supportive? (Macdonald-Ross and Smith 1975, p. 43)

Not only the notion of expert practitioners' tacit knowledge but also a notion of careful common sense reasoning is central to Macdonald-Ross' argument.

Also, *reflection* (thinking) is a powerful research tool, much underestimated in my opinion. (Macdonald-Ross 1989, p. 147)

Thus, Macdonald-Ross is not advocating (as a platform for practice) the use of layman's 'contradictory words of wisdom', as in the caricature of non-scientific knowledge that often appears in introductory social science textbooks (see Furnham 1988, p. 24). Macdonald-Ross is advocating utilisation of professional expertise *and* 'common sense' in the sense of careful reflection, and then brought together with research methodologies for mutual benefit.

To put this in perspective: Aphoristic ‘words of wisdom’ can be viewed not firstmost as silly contradictory ‘folk wisdom’, but pragmatic ‘theories’ that are valid in certain circumstances. Similarly, theories, findings and guidelines of ‘scientific psychology’ are also often contradictory and only valid under certain circumstances (see Kelley 1992, p. 14). With regard to Macdonald-Ross’ insistence on the importance of careful common sense reasoning, several psychologists have during the last few decades voiced similar concerns, of which some have gone one step further and claimed that the technical language of scientific psychology is in fact often nothing but jargonized common sense.¹¹⁹ In 1974 psychologist R.B. Joynson cogently formulated this kind of criticism the following way:

it is altogether too easy to assume that, when the psychologist’s conclusions run counter to common sense, it is the psychologist’s conclusions which are correct and common sense which was wrong; and it is altogether too easy to assume that, when the psychologist’s conclusions agree with common sense, what was previously only guessed has now been reliably established. These assumptions are not necessarily justified. ... The layman’s understanding, though often imperfect, is not to be universally dismissed as intuitive guesswork, necessarily inferior to the special methods of the scientific psychologist. On the contrary, the layman’s conclusions may well be based on long and varied experience, frequently interpreted, of course by a highly trained experience. Experiment in psychology, by contrast, typically operates over short periods of time, in very restricted environments, and on narrow segments of behaviour. It would not be surprising if common sense often proved to be as reliable as experiment, and sometimes more reliable. So we should be prepared for the possibility that when the psychologist’s conclusions differ from common sense, it is common sense which is correct; and also for the possibility that, when the psychologist’s conclusions agree with common sense, it is the psychologist who has made the lucky guess. (Joynson 1974, pp. 8–9)

119. See Lee 1988, pp. 17–26; Furnham 1988, pp. 22–46; Kelley 1992; and Harré and Gillett 1994, pp. 5ff.

Lack of internal validity

Legibility research has also been accused – very much on its own terms – of inappropriately relying on single variable laboratory experiments. That is, on experiments where only one typographic variable¹²⁰ is manipulated as the independent experimental variable while other typographic variables are ‘controlled’ and treated as invariants – whereas in the real work of typographic design a large number of typographic variables are interdependent and interact simultaneously in a complex way.¹²¹ The experiments may thus become confounded by ‘nuisance variables’ which influence the experimental ‘dependent’ variable as much as – or more than – the experimental ‘independent’ variable (thus allowing for plausible alternative interpretations of the results). There is a strong case for claiming that most legibility researchers have simply been unaware¹²² of this specific threat to internal validity, residing in the ‘stimulus material’.

Take for example Pedersen and Kidmose’s experiment set up to measure the importance of the x-height for the relative legibility of typefaces (1993; reviewed by the writer of this thesis, i.e. Lund 1995). They measured reading speed of text set in ITC New Baskerville with its relatively large x-height and Monotype Baskerville with its relatively small x-height. However, the two Baskerville typefaces are, as is clearly shown by an illustration in their study,¹²³ two rather different typeface

120. Typographic ‘variables’ have often (perhaps most often) been referred to as ‘factors’ in the literature. Although the word ‘variable’ is ambiguous and used differently in various contexts, it seems to me that the connotations yielded by the ‘variable’ metaphor better convey and reveal the functional plasticity of interdependent typographic variables than the ‘factor’ metaphor.

121. See Buckingham’s early and poignant paper on this issue (1931, pp. 103–106). He also proposed, at least in principle, a solution to get around the problem. See also Carmichael and Dearborn 1947, pp. 113–116ff; Tarr 1949, p. 31; Cheetham, Poulton, and Grimby 1965, pp. 103–104; Wendt 1972; Watts and Nisbett 1974, pp. 38ff; Burnhill and Hartley 1975; Macdonald-Ross and Waller 1977; Wright 1978, p. 291; Rubinstein 1988, p. 176; Naiman 1991, pp. 14–15, 132–133; Waller 1991; Macdonald-Ross 1994, p. 4691).

122. ‘Hidden variables are ignored because they are not perceived.’ (Ions 1977, p. 4).

123. In some cases, researchers do not even bother to specify the values of important typographic variables (such as interlinear spacing); and illustrations that could at least have provided the opportunity for visual inspection of the stimulus material, are also omitted in the published material.

designs. It is thus difficult to say whether the reported differences in reading speed are caused by the different x-heights or by other design differences. Furthermore, since the same nominal interlinear space is used for both texts, it is hard to tell whether the differences in reading speed are caused by the different x-heights or by the different ratios of x-height to interlinear space (a possible optimal versus a possible non-optimal ratio). Thus, we have two typographic variables that unintentionally vary systematically with the independent experimental variable.

This example suggests that a solution is not as easy as suggested by Dirk Wendt (1969, p. 6; 1994, pp. 295–296). He suggested the creation of two nearly identical typefaces (by computer manipulation) where ‘under completely controlled conditions’ only the parameter to be tested vary while all other parameters are constant. If we look at Pedersen and Kidmose’s experiment, we realise that Wendt’s proposal eliminates one part of the problem: the x-height vs. other design differences. However, we also realise that we are still stuck with the confounding problem posed by the different ratios of x-height to interlinear space.

Recently, a Dutch student of psychology, in a dissertation on the merits of legibility research, poignantly suggested:

Perhaps researchers need to consider that the ultimate aim of typographic design is not to optimize the usability of every single typographic aspect. This is simply impossible, because all aspects need to be balanced and optimizing some will leave a choice of non-optimal options for the other aspects. (Wijnholds 1996, p. 84)

This thoughtful remark by Wijnholds has only one minor weakness: it implies that there really exist optimal options for ‘each aspect’, regardless of context. However, the whole point about interdependent interacting typographic variables is that isolated optimal options just do not exist. As Buckingham pointed out back in 1931: ‘No length of line, for example, can be said to be most desirable, independent of ... other characteristics of the type page.’ (p. 103).

Peripherality to the reading process

One of the most serious objections which has been raised against experimental legibility research, claims that ‘factors’ such as typeface are very much peripheral to the reading process, and thus that it will be very difficult to quantify any meaningful differences in legibility, for example by employing an operational method which measures the speed of reading. Put another way, that the typography of reading material on a micro-variable level (as long as it is *reasonably* legible) can only play a peripheral part in the whole complex process of reading.¹²⁴ This may explain the many results in legibility studies that shows little or no difference. David Sless puts it the following way:

It is therefore hardly surprising that experiments conducted to measure differences in typographic variables as they relate to reading should be so disappointing. The subjects are at best operating at the fringes of their normal awareness of the reading task as it is socially defined. Put another way, the reading schema is one which ignores typography. (Sless 1981, p. 170)

A somewhat different argument has been made that support this view: The reading researcher Ronald Carver points out that the reading rate (i.e. of ‘normal reading’, as opposed to other kinds of reading processes or what he calls ‘gears of reading’ like either ‘scanning’, ‘skimming’, ‘learning’ or ‘memorising’) is more or less constant for an individual and is only limited by their comprehension or ‘thinking rate’. He suggests that individuals move their eyes at the fastest rate they can think, what he refers to as their ‘cognitive speed’ (Carver 1992).¹²⁵

124. Although rarely seen, this point has been made by several researchers and typographers; among them, W.C. Ruediger as early as 1907; R.L. Pyke in 1926 (pp. 60–61); Tempte 1961, p. 6, while addressing an audience of prominent Scandinavian book designers, educators, and printing industry leaders; David Sless 1981 (p. 170); Morris, Berry and Hargreaves 1993; Wijnholds 1996, p. 32; Saenger 1997, p. 20; and not least, albeit obliquely and within a slightly different context, by Andrew Dillon in the chapter ‘Describing the reading process at an appropriate level’, in his *Designing usable electronic text*. However, this objection does not necessarily imply a rejection of the need carefully to design, modify, or choose, appropriate typefaces for specific applications (as well as the need carefully to manipulate other interdependent typographic variables; be it on paper, on screen or on any substrate), on the basis of aesthetic, craft-based, and ergonomic considerations.

125. See similar suggestions in Zachrisson 1965, p. 23; and in Nell 1988, pp. 88–91.

Patricia Wright formulates a similar concern the following way: 'The information processes concerned with the sensory information on the printed page are dominated by higher, conceptual and interpretative levels of analysis' (Wright 1978, p. 277). In fact, as early as 1907 the researcher W.C. Ruediger concluded that: 'reading rate is in the main determined centrally by the rapidity with which meaning is aroused after the words are seen' (Venezky 1984, p. 10).

Similarly, in a recent article in *Information Design Journal*, Paul Stiff, while discussing the German design company Metadesign's work for the public transport system in Berlin, points out that the 'legibility' (in the traditional sense) of the lettering of the transport system's information and wayfinding signs, has little to do with how effective these signs are and how well these signs answers the questions people ask when trying to navigate with the help of these signs (Stiff 1993b, pp. 44–45).

While pointing out that there is no easy way to speed up the reading rate, Carver nevertheless admits that there are many ways to slow down the reading process. He explicitly refers to an environment of high constraints and excessively low typographic quality, 'such as using dim lighting, dot matrix printers, poor handwriting, or poor screen contrast' (1992, p. 93). This last point is in line with a point made by Macdonald-Ross: 'We may well suspect that 'no difference' results would be less frequent in substandard reading conditions' (1977, p. 45), and Robin Kinross (but hardly expressed in the literature of legibility studies): 'Prospects for useful enquiry are greater than with ordinary printed texts in these areas of high constraint and low typographic quality'. (Kinross 1992, p. 142).

Also these points of view were anticipated by Pyke in 1926. I deliberately quote extensively:

The hypothesis is here put forward that extremely large typographical differences must be present before it is possible to say that there is any difference in the objective legibility of types. Of types more slightly differentiated it is impossible to say that one is objectively more legible than another. ... These two ideas – that within limits there is no objective optimum, and that only extremely violent changes in the stimulus can produce any significant variation in the reaction – are found in other psycho-physiological fields. ... Another factor which clogs

the influence of objective legibility is the ease with which reading habits are established. It is partly because this is operating without cease ... that the typographical difference must be large in order to invoke a change in the reaction. (Pyke 1926, pp. 60–61)

Some sixty years later, Richard Rubinstein, in his textbook *Digital typography*, while referring to the extensive work of Tinker, points out that Tinker's work 'supports the idea that individual parameters can be varied within ranges without great detriment to legibility, but that severe degradation sets in outside this range (1988, p. 177). However, Rubinstein also points out that typographic variables behave differently together than when viewed in isolation, and thus that several modest non-optimal factors may cause problems when they are added together (ibid. p. 176). The reading psychologists Keith Rayner and Alexander Pollatsek conclude that 'the fundamental conclusion to be drawn from the work on typography is that reading appears to proceed at about the same rate if the type font, size, and length of line are at all reasonable' (1989, p. 119). But again, this 'insight' is hardly surprising.

To round off, the 'peripherality of the reading process' perspective can also be tied to the often expressed complaints about the lack of 'significant' results and about the heterogeneous results of legibility research.¹²⁶ However, as Pedhazur and Schmelkin point out, heterogeneous results are not uncommon in sociobehavioural studies – 'the only consistent result is the inconsistency among results' (1991, p. 186).

Lack of theory

Several legibility researchers and commentators have pointed out what they perceive as a serious problem with legibility research – that it has been atheoretical, lacking theories¹²⁷ guiding research *and* failing to

126. For such complaints, see for example Whittemore 1948, and Schriver 1997.

127. Kantowitz, while pointing out that human factors research is conducted to answer pragmatic questions, nevertheless argues cogently for the usefulness of theories in human factors research (1992, pp. 389, 393).

generate post-hoc theories.¹²⁸ Foster, although hopeful for the future, pointed out that legibility research has been accused of neither having generated much of practical significance nor generated theories (1980, p. 77). Venezky describes the situation the following way:

Legibility studies ... have been and remain totally atheoretical. No conceptual framework for vision or reading has been invoked for guiding legibility research; no serious legibility models have been proposed, and consequently no theories of legibility exist. Yet careful and highly useful studies have been done on the relative legibilities of almost every printing variable from type style to page lay out. (1984, p. 4)

In this context, it is nevertheless important to point out that the focus of Tinker's and other legibility researchers' work was not, primarily to understand or explain human behaviour or cognition (or how human behaviour could be adapted to the environment) but how the environment should be adapted to the psycho-motor and cognitive characteristics of humans, and in that perspective one may say that Tinker's approach was to discover simple and applicable 'functional laws' based on empirical research (cf. Valentine 1982, p. 91–92). Tinker's concern was improved design of printed material and practical considerations of typographical variables in order to achieve the improved design. In an apologetic way, Sandra Sutherland (1989, p. 60) argues along the same lines as Valentine's general argument. She also clearly demonstrates that Tinker 'expected his work to be applied by typographical practitioners' (Sutherland 1989, p. 3). This point is certainly confirmed by the subtitle of Tinker and Paterson's first book summarising legibility research: *How to make type readable: a manual for typographers, printers and advertisers* (i.e. Paterson and Tinker 1940). Venezky concludes:

Research on legibility ... represents one of the most curious to render into psychological terms because its goal is not an improved understanding of human behavior but the improved design of printed materials. For this reason, legibility studies have been derived from practical considerations of typographical variables (type face, type size, line length, etc.) rather than from hypotheses about visual or psychological processes. Consequently, these studies have not produced theories (interesting or otherwise) about either reading or the effects of typographical variation on psychological or physiological abilities. (1984, p. 22)

128. Foster 1973, p. 23; 1980, p. 77; Burnhill and Hartley 1975, p. 72; Sutherland 1989, p. 108.

‘The hypothesis of habit’ – empiricism vs. rationalism

In the 1960s Dirk Wendt suggested the ‘hypothesis of habit’, or simply that we read best what we are most used to (Wendt 1970a). Many others before him (and after) have made similar proposals, among them the type designers Eric Gill and Adrian Frutiger, and legibility researchers like Pyke, Tinker, and Burt.¹²⁹ In fact, this ‘habituation’ point of view seems ubiquitous in contemporary graphic design discourse where legibility is mentioned.

As well as having recently been pointed out by the scholar and ‘traditionalist’ typographer John Dreyfus (1994, pp. 284–285), this point of view has recently also been proclaimed prominently by the ‘non-traditionalist’ type designer Zuzana Licko in the avant-garde-like graphic design magazine *Emigre* (Licko 1990, p. 13). The implications of this thesis are many and interesting and worth discussing – although the overall validity of the ‘hypothesis’ is far from clear. Licko’s statement has been frequently cited since it first appeared.¹³⁰

It’s the same with Blackletter, which was at one point more legible to people than humanist typefaces. That’s a shocker. I agree with the fact that if you are setting books and other things that just need to be read and understood easily, you need to use something other than Oakland Six. In those cases you need to use something that is not necessarily intrinsically more legible, but that people are used to seeing. This is what makes certain typestyles more legible or comfortable. *You read best what you read most.* However, those preferences for typefaces such as Times Roman exist by habit, because those typefaces have been around longest. When those typefaces first came out, they were not what people were used to, either. But because these faces were

129. See Gill 1936, p. 44, Eurographic Press 1962, p. 260; Pyke 1926, pp. 60–61; Paterson and Tinker 1932, p. 612; Burt 1959, p. xii; Burt 1960a, p. 278. See also Dreyfus 1964, p. 65; Lane 1966; Prince 1967, pp. 37f; Wendt 1969; Kinneir 1971, p. 8; Lee 1979, p. 72; Reynolds 1979, p. 317; Turnbull and Baird 1980, pp. 87–88; Reynolds 1984, pp. 197–198; Silver, Kline, and Braun 1994, p. 824; Schriver 1997, p. 301; Kostelnick and Roberts 1998, p. 142.

130. For example by Pedersen and Kidmose (1993, p. 52), MacKenzie (1994, p. 12), and Birkvig (1995, pp. 10–11).

frequently used, they have become extremely legible. Maybe some of my typefaces will eventually reach this point of acceptance, and therefore become more legible two hundred years from now, who knows? (Licko 1990, p. 13)

A similar statement has been repeated in a mantra-like manner, as a typeface specimen text, in multiples on almost every page in the many product and typeface catalogues issued by the Emigre company during the 1990s:

Typefaces are not intrinsically legible; rather, it is the reader's familiarity with faces that accounts for their legibility. Studies have shown that readers read best what they read most. Legibility is also a dynamic process, as readers' habit are ever changing.

While issuing a new typeface in 1996, Licko used the opportunity to once again reiterate her familiarity thesis. The typeface in question was Mrs Eaves, a free interpretation of one of the famous 18th century English typefounder and printer John Baskerville's typefaces; typefaces which in the 20th century have been copied or interpreted by many type foundries, while bearing the name its originator, Baskerville. The revived Baskerville typefaces have been, and still are, among the most successful text typefaces in the 20 century. In a promotional booklet for Mrs Eaves, Licko refers to the classic story – often told or alluded to in contemporary graphic design discourse – about the negative reception of Baskerville's typefaces. That is, that contemporaries of him declared his typefaces as illegible and damaging to their eyesight. Since the Baskerville typefaces has received such respectability in the 20th century, Licko sees this as proof of her stance and she explains:

This illustrates once again that readers' habits do change in time and are influenced by repeated exposure to particular typefaces, more so than by any measurable physical characteristics of the typefaces themselves. (Licko 1996b, p. 2.c).

However, in a review of the typeface family Mrs Eaves in *Print*, critic Paul Shaw points out that the reputed 'illegibility of Baskerville's types may be largely mythical'.¹³¹ Shaw's point of view is supported by the text in Daniel Berkeley Updike's classic *Printing types*, and the text

131. Shaw (1996, p. 28D); while referring to F.E. Pardoe's biography of John Baskerville (1975, p. 18).

in James Hutchinson's *Letters*, both which Licko quotes in her booklet. Neither sources do in fact say that Baskerville's types were generally received as 'illegible'.¹³²

The habituation point of view is in one respect a banal truism. It is obvious that the quality of text can vary considerably, from chunks of text set in barely decodable experimental alphabets and more or less illegible handwriting, to comfortably read text set in standard fonts or relatively legible handwriting. It is of course possible to familiarise oneself with experimental alphabets and 'illegible' handwriting (or to overcome the unease we temporarily feel when our favourite newspaper changes its layout). However, whether such familiarisations parallels the proclaimed effect of familiarisation among all reasonably standard and rather similar typefaces, is quite another question.

Maureen Mackenzie of the Communication Research Institute of Australia, makes Zuzana Licko's argument her own and adds that some 'recent testing of documents' at the institute (apparently including testing on serif vs. sans serif typefaces) suggests that legibility is not 'biologically determined' as 'earlier studies suggested' but 'culturally learned and sub-culturally developed', and that today, as contrary to a few decades ago, people are as accustomed to sans serif typefaces as they are to serif typefaces (MacKenzie 1994, p. 12). Birkvig, at the Craphic College in Copenhagen, although far more reserved, claims that this point of view is 'at least partly scientifically documented' (1995, p. 10).¹³³

However, in spite of Licko's claim that 'studies have shown that readers read best what they read most' and similar statements from MacKenzie, Birkvig, and others, it seems that the many statements about the role of habituation to typefaces and its effect on reading performance, seems to lack empirical verification in the literature. Maybe it is a question that cannot be solved empirically. The many statements proclaiming this view are as far as I can see *a priori* statements, or at best attempts to explain *a posteriori* aspects of research

132. See Updike [1922] 1962 (pp. 113–114); Hutchinson 1983 (p. 184); and Licko 1996 (pp. 2.b–2.c).

133. See also Twyman (1970), who claims that 'Research has shown that we ... read most easily, those types we are most familiar with' (p. 112); and Gribbons who claims that 'legibility studies tend to dispute the neurological hypothesis' (1991, p. 44).

findings that may point in such a direction.¹³⁴ The only apparently substantial exception I have come across is in one of Dirk Wendt's studies (published in 1969 and again in 1994) where on the basis of seven empirical studies he describes a trend that support his 'hypothesis of habit' with regard to typefaces. However, when we look into the details of the data he relies on, it turns out that his claim is unsubstantiated.¹³⁵

Mark Allen Williams represents another exception to the omnipresent *a priori* claim that legibility is merely a function of familiarity. He suggests that the question is something that warrants further investigation, and he simply suggests a replication of Paterson and Tinker's 1932 study in order to test their familiarity assumptions (1990, p. 11).

What can be said for sure on the familiarity question is that it is firmly rooted in a long-standing philosophical controversy between rationalism and empiricism. It is also expressed in other ways: nature vs. nurture; innateness vs. cognitive adaptation, or biologically determined factors vs. environmentally and culturally determined factors. The claim that aspects of normal vision are to a certain extent a function of visual exercise, and thus that activity shapes structures in the nervous system (cf. e.g. Kolars 1983, p. 54), is plausible. However, it is also plausible that there are certain physiological and psychological characteristics and limits in the psycho-motor and cognitive system of human beings. Where a possible 'correct answer' lies on the continuum between these two positions, and more important, whether a possible habituation to typefaces operates on the same physiological and psychological levels as this continuum or not, are still open questions.

134. Without going in detail: Prince 1967 (pp. 37f), and Watts and Nisbet 1974 (pp. 32–33; relying on Prince 1967), based on empirical research among 'visually handicapped' readers, present *a posteriori* explanations of research findings in favour of the familiarity thesis; while Coghill's empirical findings from a reading experiment among school children goes contrary to the familiarity thesis (1980).

135. See the section 'Wendt 1969', in chapter 5, 'A review of empirical studies'.

Critique from design practitioners

Many designers have also voiced their scepticism about legibility research; or alternatively, scepticism has been voiced by non-designers in design magazines. These voices may represent broad currents among reflective designers. A few of these are documented below, while Sandra Sutherland quotes some other voices of the same kind (1989, pp. 95–100). Recurrent topics are about lack of validity, the problem of habituation, and design practitioners' tacit knowledge. Other topics are that 'legibility' is an unstable concept, that attention to legibility research inhibits creativity, and that legibility research does not make a difference.

As early as 1942, Carl Purington Rollins, the Printer to Yale University, one of the central exponents of 'fine printing' in America, and a friend of leading lights such as Daniel Berkeley Updike, Bruce Rogers and William Addison Dwiggins, articulated his scepticism about legibility research the following way:

There has been a deal of nonsense written about the legibility of type, the proper length of line and the 'ideal' size. I once heard a psychological adept dogmatize on the subject of the proper type for children's books, asserting that ten point type and 88 mm. were the correct size and line. ... The trouble with the attempt to decide upon the most legible or readable type face is that there are too many factors involved to be controlled by any system of valuation so far devised. The trained eye of the skilled printer is more to be depended upon than all the investigations so far conducted. (Rollins 1942, pp. 28–29)

Only a few years later, a similar concern was voiced in the American graphic design magazine *Print*. There, [Dr] Irving C. Whittemore, in a belated review of Paterson and Tinker's 1940 *How to make type readable*, chose satire as a tool for questioning the relevance of legibility research. With a fine sense of humour he staged a fictitious conversation between Bruce Rolldike [*sic*] 'director of the University Press' and professor Tinkerson [*sic*] 'of the psych department'. Their conversation opens with Rolldike greeting Tinkerson: 'Ah there, Tinkerson, how's the psycho-ing?' However, although the conversation is satirical, it raises serious questions about the relevance of legibility research. In the latter part of his article, Whittemore leaves the satirical

conversation and focuses on the many different operational methods and the heterogeneous results of legibility research. He rhetorically asks what is meant by 'legibility':

Do you mean (1) easy to read fast, (2) easy to read at a distance, (3) easy to read in dim light, (4) easy to read when you haven't your glasses, (5) easy on the brain, (6) not tiring to the eyes, (7) possible to grasp in big gulps of meaning, (8) pleasant to read, (9) inviting to the eye, or (10) something else? (Whittemore 1948, p. 36)

Whittemore doesn't leave much honour to legibility research, and he suggests that an operational measure like 'speed of reading' does not have much relevance, when the many various contexts of reading, as well as aesthetic considerations and age old craft knowledge, are taken into consideration.

Soon after, Tinker and Paterson replied in the same journal to Whittemore's provocation. Whittemore's complaint is referred to as 'an excellent exposition of the scientific complexities of the problem of legibility' and 'laudable attempt to outline the complexity of the problems involved in legibility'. However, no concessions are given. Whittemore's complaint about legibility research is brushed off and rhetorically turned on its head as only confirming the complexity of the problem.

Unfortunately, insight concerning one's own ignorance of the complexity of the problem is rarely encountered in the practical printer or the type designer. For this reason, practical printers and type designers are likely to disregard findings from the research laboratory or at best to accept only those findings which happen to coincide with their own beliefs derived from experience or from erroneously designed laboratory studies. (Tinker and Paterson 1949, p. 61)

Tinker and Paterson also used this opportunity to re-iterate their dismissal of many of the operational methods that Whittemore mentions. They left it to those researchers who employ other methods than the ones they preferred¹³⁶ to answer for themselves:

We hope the visibility meter and eye blink boys will give their answer and that the printers who follow the manuals of style which have come down from the pre-scientific era of printing practice and the type

136. The methods preferred by Tinker and Paterson are: speed of reading, eye movement recordings, and reader preferences.

designers new and old will, if they can, likewise give answer. (Tinker and Paterson 1949, p. 61)

At the same time, on the other side of the Atlantic, John C. Tarr, who had directed the Monotype Corporation's training program during the 1920s, and having been a colleague of the influential Beatrice Warde,¹³⁷ in the article 'A critical discursus on type legibility' published in *Penrose Annual* of 1949, also voiced critical remarks on legibility research (Tarr 1949).¹³⁸ He acknowledged that 'The work already done by the scientist in the field of visibility and readability ... deserves closer recognition from the printing industry.' (p. 31). However, one of his points is that instead of narrowly focusing on the comparative legibility of different typefaces it would probably be better to focus on certain typeface variables such as x-height, typeface 'colour', a-z width, and typographic variables such as line length, and interlinear spacing ('leading'). Tarr, however, is clear in his opinion when it comes to sans serif typefaces: 'Experiments with sans serifs and egyptians have value only in relation to their use as display types, not for continuous text.' (p.31).

In 1962 the celebrated Swiss type designer Adrian Frutiger expressed his scepticism about legibility research, in an interview published in the trade journal *Print in Britain*:

As for the legibility of the character, this is to my mind a fictitious problem. I do not believe in tests of which one is more contradictory than the other. Certain statistical evidence shows, for example, that Bodoni is the most legible character [*sic*]; a little later, identical figures are produced to prove that Bodoni is the least legible of all the characters. Which are you to believe? I learnt to read with gothic characters and I never experienced the slightest difficulty. I think legibility is solely a matter of habit, and speed in reading depends not so much on the speed of the eye than on that of the mind. (Eurographic Press 1962, p. 260)

137. See Badaracco 1995, p. 82.

138. Another article by Tarr, published two years earlier in *Schweizer Graphische Mitteilungen*, carrying the title 'Legibility in printing', is (mainly) not about legibility research (Tarr 1947). This article offers craft-based *a priori* advice and reasoning about the physical properties of typefaces and legibility. The approach of this article is not unlike the approach taken by Alison Black (1990) in the chapter 'Legibility' in her *Typefaces for desktop publishing*. That is, in fact claiming the right to discursively use the term 'legibility' without marrying it to empirical experimental research.

In 1964 typographer John Dreyfus, in a letter to the editor of the British journal *Design* – while commenting upon a piece of legibility research carried out by Christopher Poulton and published in an earlier issue of the magazine¹³⁹ – expressed his scepticism about legibility research. Poulton's study involved several roman (serif) typefaces and the sans serif typefaces Gill Sans and Univers. Dreyfus had been heavily involved in launching and marketing Monotype's version of Univers, and it may be that he was offended by the result where Gill Sans (also a Monotype face) came out as more legible than Univers. However, his argument is not unreasonable, pointing out the conflict between single variable experimental designs and interacting typographic variables:

The legibility of a type depends not only upon the design, as supplied by the manufacturer, but on the relation of line length to size of type and degree of leading. These variables, added to the different habits of different nationalities, make it extremely difficult and costly to devise a valid set of tests. (*Design*, no. 188, 1964, p. 65)

A couple of years later, in the newly launched *Journal of Typographic Research*, the Belgian typographer and educator Fernand Baudin expressed scepticism about legibility research, while reviewing Tinker's latest book *Bases for effective reading*:

all typographers *en bloc*, whether expert or not, are presented merely as introspective aesthetes deserving, on the whole, of contempt (pp. 115, 125, 135, 136, 183). This is a pity. ... Expert typographers in fact do exist, and it is only fair to add on their behalf that there is little if anything in Professor Tinker's exposition that goes to counter their theories or practice. Indeed, it is a matter for regret that the author, who is also the typographic designer of his book, did not deem it worth his while to explain how it comes to be that his 'scientific' typography so much resembles many other 'merely' typographic designs in the current book production on the continent as well as in the U.S.A. This is not to say that all books are well designed or that Professor Tinker's book is not a good example of Aldine sobriety. (Baudin 1967a, pp. 204–205)

In a later issue the same year, Baudin simply claimed that legibility research had run into a blind alley (Baudin 1967b, p. 374).¹⁴⁰

139. See the section 'Poulton 1965', in chapter 5, 'A review of empirical studies'.

140. A counterargument by Jeremy Foster appeared in the consequent volume of the journal (Foster 1968).

Around the same time, the trade journal *Printing Technology* presented three articles on legibility in one issue. In addition to contributions from E.C. Poulton and Herbert Spencer, the issue contained an article by lettering artist and type designer David Kindersley on the legibility of typefaces. Part of this article is a reaction to Poulton's recently published test involving Gill Sans and Univers. Kindersley claimed that inter-character space is one of the most important variables determining the quality of a typeface, but that it is never taken into consideration by legibility researchers. According to Kindersley's introductory remarks much legibility research is of 'great interest' but needs 'interpretation in the light of typographical common sense if it is to be useful'. Later on in the article however, he is less diplomatic:

I am inclined to think that research into legibility has been of little use so far. Many designers have stated such feelings. Perhaps only in extremes of size – such as letters for road signs or labels on patent medicines – can it be of assistance, though it is generally ignored; but then it is more a question of visibility (Kindersley 1968, p. 70)

Kindersley, consciously of his role as a craftsman and artist, had also another concern which he stressed:

I am rather frightened by all this talk about legibility. I think we are trying to straighten out the problem at the wrong time. The alphabet must be allowed to develop. We may be wasting time comparing very similar objects without clear-cut results. In other arts we don't as a rule apply tests, because we know instinctively that standards could result. We don't want standards, because they prevent evolution. We need, above all, by continuous experiment, to avoid fossilization. (Kindersley 1968, p. 69)

Kindersley even anticipated Joel Roth who in 1969 pleaded for 'typography that makes the reader work' (Roth 1969, p. 193), and this idea's many later manifestations in typographic discourse as well as its manifestations in designed artefacts.¹⁴¹

It is often said that it takes time to get into a book. Why should it not take a page or two to get into the typography of book? It could be equally rewarding. (Kindersley 1968, p. 70)

141. For some critical comments on this idea, see Stiff 1993a, and Poynor 1999.

In 1979, in the American graphic design magazine *Communication Arts*, the designer George E. Mack¹⁴² not only voiced concerns similar to those above, but also clearly expressed what undoubtedly have become a widespread opinion among designers:

Personally, I think anyone who claims 'greater legibility' in more than a rough sense has a tough fight coming – educators and psychologists are still slugging out what this term means. The basic concept is so tangled up in decipherability, pattern recognition, reading speed, retention, familiarity, visual grouping, aesthetic response, and real life vs. test conditions that contradictory results can be obtained for the same type faces under different test conditions. And if that doesn't send you reeling, there is still the theory that if you make readers slow down and work a little harder they pay more attention and remember it longer. Research seems to find that people *can* read something (1) if they're familiar with the style, (2) if they want to, or have to, and (3) if the letterforms are even minimally recognizable – subject to your test conditions. (Mack 1979, p. 96)

Ruari McLean included a chapter on legibility in his widely read manual of typography (1980, pp. 41–48). The chapter first of all presents an informed common sense discussion on legibility, however not without strongly biased views. The author also explicitly discusses legibility *research*, and his scepticism is voiced the following way:

A great deal of research has been carried out during this century into the legibility of print, and also into the psychological and physiological processes of reading and comprehension. I believe that no research so far published has been seriously helpful to designers concerned with the design of straightforward reading matter for literate adults, except insofar as it has, in general, confirmed their practice. Research in legibility, even when carried out under the most 'scientific' conditions, has not yet come up with anything fundamental that typographic designers did not already know – or believe – with their inherited experience of five hundred years of printing history and their specialized observation of the civilization in which they live. For example, one authority, Tinker, had concluded that 'black print on a white background is over ten per cent more efficient than white on black'. ... I am glad to hear it, but after even ten years as a typographer it would never have occurred to me to think otherwise. (p. 47)

However, McLean does not reject research as such:

142. Not to be confused with the Scottish typographic designer Georgie Mackie.

It is in special cases – in which the typographer's work abounds – that research must be respected. Research into the legibility of lettering and print *has* been helpful in specialized areas, such as print for children and for those with poor vision or for handicapped or elderly people, the design of bibliographical entries, telephone directories, and motorway signing – where only research, and nobody's guessing, could discover how tall, and in what forms and colours, letters should be if they are to be legible at a safe distance by a car driver travelling at speed on a motorway. (pp. 47–48)

The voice of the British type designer and Linotype manager, Walter Tracy, echoes both Kindersley and McLean. The context is a critical discussion of reading research, orthographic and graphological reform proposals, and legibility research. Here is an extract:

The fact is, though, that so far from being suspicious of research, typographers are mostly indifferent to it. And with reason. Naturally, typographers admire the efforts of those who explore ways and means of helping children and the handicapped to read; and, too, they can see the case for research into specific matters like the legibility of directory entries, the details of military maps, the layout of official questionnaires, the organising of tabulations in a viewdata system, the design of alphabets for low-resolution print-out devices, and other particular tasks. ... A great deal of research, though, seems to be produced by academics for the interest not of designers but of other academics. Their motives are easy to understand: there is the need to add another title to their list of publications, that being the way to academic achievement. ... For them [designers], research and reform are occasionally interesting, frequently bewildering, but rarely of any consequence in the typographic scene as they know it. (Tracy 1988, pp. 84–85)

Accordingly, but this time from a psychologist; in a harsh review article in *Information Design Journal* in 1982 David Bartram encapsulated much of the then existing critique of legibility research and the emerging view of the way forward:

My own feeling is that any designer who picked up this book in order to see what psychology had to offer him, would come away from it very disappointed. ... for the most part, those situations in which psychologists manage to obtain so-called 'significant' effects are those which any self-respecting designer would avoid ... Perhaps it is time that psychology started asking different sorts of questions. Comprehension is a process involving an interaction between a reader who has a purpose, and a writer who has an intention – this interaction being mediated by some form of graphic display. ... we need to

understand design and designers in terms of process and function, and not treat design simply in structural terms as a static object of analysis. (Bartram 1982, pp. 75–76)

And in the same year, the British graphic designer Ken Garland described his initial enthusiasm and later disappointment with ‘ergonomics’ and legibility research:

In the late ‘fifties, inspired by the zeal of Michael Farr, then editor of *Design* magazine, I had been totally involved in the championing of ergonomics, or human factors studies. It had appeared to us that such a sensible and promising new aspect of the design process, and one in which we could happily collaborate with psychologists, anthropometrists and other applied scientists; we fully accepted that ergonomics would become a vital study for all design work, graphics included. But this didn’t happen. In one sense, ergonomics was a crusade that failed. (Garland [1982] 1996, p. 56)

Postmodernist critique

During the last decade or so the notion of legibility *per se* has come under attack from what might be called post-structuralist, anti-modern or even neo avant-garde modernist positions, exposed in influential graphic design magazines like *Emigre* and *Eye* (see for example Dauppe 1991)¹⁴³ – and not least in professional design practice. For some, traditional notions of legibility is simply rejected as irrelevant, accompanied with a few recurring themes such as stressing the importance of allowing artistic self-expression and rejecting the need for a ‘universally’ understood ‘language’ of typographic form. Another recurring theme is the claim that typography should literally reflect what is perceived as the fractured and fluxed condition of post-modernity. And furthermore – in a similar way as the idea behind Berthold Brecht’s ‘Verfremdungs’-technique – typography should take a liberating and critical stance against societal authority and thus problematise apparently smooth

143. For comments upon some such positions (projecting ideas about deconstruction on to the material world of typography), see Stiff 1993, and Robin Kinross’ pamphlet *Fellow readers* (1994a).

texts by disrupting them and making them cumbersome and less legible. This should be done in order to challenge their authority, and stop the reader to make him think and realise that the world is not as smooth as the smooth text purports it to be.

to engage the audience with the text, to make the audience 'work', and to emphasise the 'construction' of meaning. Radical typography might aim, not to flow seamlessly, legibly, but to halt and disrupt, to expose meaning and language as problematic. ... The cry of 'legibility' masks a reactionary attitude (Dauppe 1991, pp. 6–7)

5 A review of empirical studies

Approach

I have explained in the introduction to the thesis my purpose in reviewing the empirical studies. As I pointed out there, I have tried to avoid two possible approaches. Crudely put, one is to treat the empirical source material principally as referential documentary evidence of events or historic change. The other is to treat the empirical source material as simply a more or less valid source of ‘findings’. My approach has been to foreground the ‘combination approach’: to treat the source material as documentary evidence (or discursive events, for that matter), and at the same time critically focusing on the source material’s scientific validity. To extend the historiographical deliberations in my introduction to the context and reflexive vocabulary of social science philosophy: I am acknowledging that I to some extent, whether desirable or not, appear both as *participant* and *spectator*.

‘Participant and spectator’ is the translated title of an essay by the Norwegian philosopher Hans Skjervheim ([1957] 1996). However, while Skjervheim sees a crucial difference between ‘participant’ and ‘spectator’, the French social philosopher Pierre Bourdieu’s epistemological project is to suspend or go beyond the contradiction between an objectivist perspective and a subjectivist perspective (Solli 1998). In *An invitation to reflexive sociology* (Bourdieu and Wacquant 1992), Bourdieu introduces the apparently contradictory concept ‘participating objectification’, where

he suggests a methodological orientation that includes both ‘empathy’ and ‘rupture’.¹⁴⁴

The German social philosopher Jürgen Habermas advocates a position similar to Bourdieu’s in his critique of positivistic objectivism in *The theory of communicative action* ([1981] 1984, pp. 102–141). While relying on Skjervheim as a point of departure for his discourse on this matter, Habermas asserts that although the active presence of a ‘participant-observer’ ‘unavoidably alters the original scene’, a critical engagement (or ‘performative attitude’) that confronts knowledge and validity claims in the object domain and that makes accessible internal interrelations of meaning, is imperative in order to *understand* and not merely describe the object domain. This ‘performative attitude’ – or Bourdieu’s ‘participating objectification’ – might appropriately describe the ‘participant’ *and* ‘spectator’ approach taken in this thesis.

By critical engagement at least two approaches are possible. One approach (‘paradigm internal’) would be to focus on the studies very much on their own terms, e.g., by critically examining them with regard to technical aspects such as the quality of their experimental design or use of statistical methods. The other approach (‘paradigm external’) would be to apply a kind of discourse analysis and focus upon the use of rhetorical tropes and discursive strategies.¹⁴⁵ However, my approach is something in between. That is, by a relatively close reading, and by building on whatever criticism the unique individual pieces of source material invites to. This while trying to contextualise wherever possible, not least by including appropriate references to the target domains of printing, graphic design and information design.

Nevertheless, I *do* focus on one particular aspect of the legibility studies in question, the internal validity of the studies presented. One of the reasons for this, as explained in my introduction, is my assumption that problems of internal validity reside in the typographic stimulus

144. Bourdieu’s concept of ‘participating objectification’ is also brought to the fore (and further explained) in the penultimate chapter of this thesis; although in the context of legibility research in relation to its target domain of typography.

145. For a successful approach of this kind, in the context of human–computer interaction (and information design, for that matter), see Cooper and Bowers’ paper ‘Representing the user: notes on the disciplinary rhetoric of human–computer interaction’ (1995).

material, and that I can draw on my expertise as a typographer to do this. Thus, I have been attempting in my criticism to mediate between target domain (printing and graphic design) and the research domain (legibility studies and human factors).

I am avoiding any critical and thus rather technical examination of external population validity and statistical conclusion validity. Among the reasons for not assessing statistical conclusion validity are: statistical conclusion validity can only make sense if internal validity is present; and an *a priori* assumption that this part of the experiments in question are most often both elaborate, and adequate, if looked upon in isolation. 'Although researchers are expected to apply rigorous standards of statistical analysis to their data (and publicly demonstrate this analysis in their reports), their stimulus materials are rarely subjected to the same scrutiny.' (Schumacher and Waller 1985, p. 379).

Terminology

When using the term 'construct validity' I refer both to the theoretical definition of legibility, as well as its correspondence with the operational definition. That is, whether the theoretical definition of the legibility construct is reasonable or not, and whether methods and techniques employed really can be said to represent the construct or not.

When I occasionally refer to 'internal validity' and 'external validity', I do it, at least as a point of departure, in accordance with the use of Cook and Campbell (1979), discussed and lucidly laid out by Pedhazur and Schmelkin (1991). With respect to 'external validity' I also borrow from the exposition of Bracht and Glass (1968).

When using the term 'internal validity' I refer to whether or not there is a causal relationship between the manipulation of the independent experimental variable and the observed results. That is, whether alternative explanations of the observed result are plausible or not.¹⁴⁶

146. See also chapter 1, 'Introduction'; and the section 'Lack of internal validity', in chapter 4, 'Critiques of legibility research'.

When using the term ‘external validity’ I refer to whether or not the results of an experiment are generalisable to other people and situations (respectively ‘external population validity’ and ‘external ecological validity’), in other words: the coherence between the test situation and target situation. ‘Ecological validity’ in experimental psychology is sometimes called ‘setting representativeness’, and ‘operational fidelity’ in human factors research.

Berkowitz and Donnerstein (1982) and Kantowitz (1992) vigorously refute a common criticism of laboratory experiment’s lack of external validity. They argue that it is important to distinguish between realism and generalisability, and that the importance of realism (physical representativeness; i.e. the extent to which the test situation mirrors the physical target situation) ‘has been vastly overrated’ (Kantowitz 1992, p. 390). Thus, it is argued that external validity does not necessarily require ecological validity, and that whether it does or not is an empirical question in each case. Berkowitz and Donnerstein argue that laboratory experiments cannot be truly representative designs and that they are artificial by nature. Laboratory experiments, they argue, are about uncovering causal relations between variables while controlling irrelevant variables, and thus, that artificiality is the strength of experiments (1982, pp. 245, 255–256). Nevertheless, Bracht and Glass point out the obvious and banal fact that: ‘The intent (sometimes explicitly stated, sometimes not) of almost all experimenters is to generalise their findings to some group of subjects and set of conditions that are not included in the experiment.’ (1968, pp. 437–438).

With ‘overall validity’ I refer to the relevance or appropriateness of research when gazed from a reflexive standpoint ‘outside’ the realm of the discipline or paradigmatic framework within which the experiment has been performed (while not necessarily implying ‘neutrality’ or a privileged position).

Bibliographic sources

Some of the studies identified have been found through citations in similar studies, some as references in 'standard' texts like Tinker (1965), and some in review articles like Reynolds 1984, while some of the studies have been found in printed bibliographies.¹⁴⁷ More or less systematic searches¹⁴⁸ have been performed in on-line and off-line electronic bibliographic databases.¹⁴⁹ Complete runs of some journals have also been scanned.¹⁵⁰ Finally, some of the studies have been found by serendipity.

Selection criteria

Twenty-eight experimental behavioural studies on the relative legibility of serif and sans serif typefaces are examined in chronological order of appearance. The first of these studies was published in 1896, and the last was published in 1997. The 28 selected studies amount to almost half of the total of 72 legibility studies that I have uncovered on this

147. Among the printed bibliographies scanned were: Cornog and Rose 1967; Foster 1971; Foster 1972; Macdonald-Ross and Smith 1977; Foster 1980; Felker 1980; Kempson and Moore 1994; and the useful periodical *Ergonomic Abstracts*, unfortunately only for 1994, 1995, until June 1996. The following un-published bibliographies were also helpful: Brinch 1976; Rannem 1983; 'Information design database: authors A-K' 1992; and 'Information design database: authors L-Z' 1993. I have also scanned volumes of the following periodical, annual or irregular bibliographies: *Design and applied arts index* (1987–1990); *ABM ArtBibliographies Modern* (1970–1991); *PIRA Printing and publishing abstracts* (1975–1994); *Aslib Index to Theses* (predominantly British doctoral theses) (1980–1994).

148. The search terms have been: 'legibility', 'readability', 'typeface', 'font', 'sans serif', 'sanserif', and 'typography'.

149. Among the electronic databases scanned were: *ProQuest Dissertation Abstracts* (predominantly American doctoral theses); *OCLC World Cat* (American master dissertations are included in this 'world library catalogue'); *OCLC Article 1st*; *Medline*; *ERIC Education Resources Information Center* (1966–1998); *PsycLit* (1974–1994); *LLBA Linguistics and Language Behaviour Abstracts* (1973–1994); *ABM ArtBibliographies Modern* (1984–1993); and the *ISI Science Expanded*, *Social Science*, and *Art & Humanities Citation Indexes* (from 1987 to present).

150. *Journal of Typographic Research*; *Visible Language*; *Iconographic*; *Eye*; *Typos*; *Typographica*; *Information Design Journal*; and *Applied Ergonomics*.

topic. Most of them are experimental, a few of them are ‘typeface topology studies’ either with or without attached behavioural experiments, and a few of them are preference studies. Nevertheless, I am certain that a more systematic and persistent search would have uncovered several more relevant studies.

For the sake of order: Some of the 72 comparative studies are studies focusing on the legibility of individual typefaces (but which happens to include both serif and sans serif typefaces).¹⁵¹ Many of the 72 studies are univariate, focusing only on typefaces as the independent variable, while some of them are multivariate, including variables like line length and paper stock (as in Neman 1968), or typeface posture and typeface weight (as in Wendt 1969).

Most of the studies appear as articles in scientific journals, some appear as papers in conference proceedings, some appear as individually published reports, some appear as a chapter or subchapter within monographs on related topics, some appear as doctoral theses only, a few appear as master dissertations, while some appear in several guises (for example as a thesis, as an article in a scientific journal, and as a popular article in a printing trade journal).¹⁵²

The reason for assessing individual studies as discrete units in a chronological order has been explained in my introduction. The exercise of assessing as many as 28 studies has its background in the initial intention of assessing all the studies I could identify in order to document this research in a comprehensive empirical manner. However, as the number of detected studies grew, I realised that some sort of selection had to be undertaken so my task could stay manageable on the one hand, and not completely become a quagmire of tedious oversaturated empiricism on the other hand.

151. There exists other ‘genres’ of comparative typeface legibility studies: For example dichotomous studies on proportional set width vs. fixed set width typefaces (for example on Times vs. the ‘typewriter’ typeface Courier). Examples of such studies are: Payne 1967; Beldie *et al.* 1983; Ariditi *et al.* 1990; and Mansfield *et al.* 1996.

152. See for example de Lange 1992; de Lange 1993; and de Lange *et al.* 1993. It is interesting to observe that when a part or extract of a thesis or dissertation appears in a scientific journal, it happens that the supervisors are included as authors (compare for example Amachree 1975 with Amachree *et al.* 1977; de Lange 1993 with de Lange *et al.* 1993; and Stone 1997 with Stone *et al.* 1999).

Studies assessed

A list of the studies reviewed in this chapter appears on the content pages of this thesis (pp. 4–5).¹⁵³ Partly because my change in strategy, from assessing all of the relevant studies to assessing only a selection of the studies, happened en route, most of the omitted studies are from the period from the 1960s to and including the 1990s (with two exceptions: Ovink 1938, pp. 84–104; and Nolan 1959). However, the studies assessed are spread over the whole century, and while 15 of them are from the period from the 1890s to and including the 1950s, 12 are from the much shorter period from the 1960s to and including the 1990s.

The studies assessed also represent a wide variety of context rationale, operational methods, face results, and degree of formality of publication medium. Note also that as a criterion for inclusion I have given priority to studies that have been ‘published properly’ as opposed to ‘unpublished’ doctoral theses or master dissertations. The studies which I have not reviewed in this chapter have with very few exceptions all been inspected, and references to their content may appear elsewhere in the thesis. There is no reason to believe that another selection would have altered the picture that emerges and my conclusions in any substantial way. Observe that, except for the two studies mentioned below, the selection of studies for review has not been based on an advanced assessment (for example in order to pick out studies that could yield a more ‘productive’ result than others).

Note that two of the studies have on purpose received a more thorough and extended treatment than the others (Christie and Rutley 1961; and Robinson, Abbamonte, and Evans 1971).

The reason for the special treatment of Christie and Rutley’s study is that it was part of a unique instance of a public debate on legibility (of the new directional traffic signs for Britain’s new motorways and all-purpose roads in the early 1960s). Many people participated in this debate, including many designers. Furthermore, it was about alphabets that would become very prominent in the ‘visual landscape’ of Britain

153. For the sake of order: the reviewed study Crosland and Johnson 1928, is not counted among the 28 reviewed studies, nor among the total number of 72 studies. See the section ‘Crosland and Johnson 1928: the serifs that never were’, in chapter 5.

and elsewhere. And not least, this debate still surfaces every now and then, almost forty years later, and it relates to important questions about the nature of design.

The reason for the special treatment of Robinson, Abbamonte and Evans' study is the following. When it appeared in 1971 it was unusual in two respects: unlike most typeface legibility studies it proposed a *theory* for why serif typefaces are more legible than serif typefaces, and it also took a cognitive science approach, by employing an explanatory computer-based model that claimed to rely on how the human visual system works. Furthermore, the study is still cited, and several of these citations suggest that this study stands for something more scientific, objective, and conclusive than other typeface legibility studies. The study therefore deserves the extended treatment it has been given.

Studies not assessed

The studies of relative typeface legibility that have not been reviewed in this chapter can be categorised in the following way:

1. Practical legibility studies based on evaluation of image degradation of typefaces, for example created by repeated photocopying or by fax-transmission (Spencer *et al.* 1977; Bowden and Brailsford 1989; Adobe 1989; Birkvig 1990).¹⁵⁴

2. Experimental studies on typeface legibility for people with impaired vision (Nolan 1959; Prince 1967; Shaw 1969; Zachrisson and Smedshammar 1971; Jansen and Thomsen 1986; Plass and Yager 1995; Yager *et al.* 1998).¹⁵⁵

3. Experimental screen legibility typeface studies. The decision to omit screen legibility studies is partly arbitrary but also pragmatic: A line has to be drawn, and some of these studies were published too late for inclusion. The studies are: Engman 1988; Smedshammar *et al.* 1990;

154. These studies do not represent a new strand of research. See for example the (general) study by Bullington, 'Legibility determinations of multiple carbon copies' (1948).

155. The rationale behind these studies is certainly commendable. Most of these studies are based on subjects with impaired vision. However, some experiments are based on simulations of visual impairment, for example where subjects wear specially designed glasses.

Williams 1990; Garcia and Caldera 1996; Boyarski *et al.* 1998; and Stone *et al.* 1999.¹⁵⁶ Although the media substrate is different from paper, signboards, or various kinds of one-line displays (like LED displays), computer screens represent, at least in one respect ('the psycho-motor domain'), just another media substrate, and it is the same kind of questions that are asked as for print legibility studies, the studies have the same approach and similar operational methods are employed. And not least: The criticisms of traditional print legibility research reported elsewhere in this thesis, should to a large extent also apply to screen legibility research. The important difference to print (with regard to 'the psycho-motor domain') has of course to do with the fact that computer screens represent an environment of higher constraints (for example considerably lower image resolution, lower text/background contrast, disturbing flicker, and disturbing glare). Therefore, many typographers, who are indifferent to whether serif or sans serif typefaces are most legible on paper, clearly prefer sans serif typefaces for today's screen technologies. This is due to the relatively coarse image resolution of computer screens and problems with representing the fine serifs and the finely tuned stroke contrast of serif typefaces.

4. Some of the studies that appear as theses or dissertations only. For example, only one (Taylor 1990) of three PhD dissertations under the label 'instructional technology' from Wayne State University is included; the other two are Stahl 1989, and Kravutske 1994. The inclusion of Taylor 1990 happened because this was the first to arrive on my desk. Other PhD dissertations omitted are Neman 1968; and Kunst 1972. Two master theses that are omitted are: Sorg 1985 and Connelly 1998.

5. Three studies on 'projected' typefaces (from slides and overhead projection transparencies) are also omitted. These studies are: Adams *et al.* 1965; Grooters 1972; and Phillips 1976.

156. Stone *et al.* 1999 and Stone 1997 are basically the same study, only appearing in two guises. For the sake of order: A screen legibility study by Geske (1996), do not compare serif and sans serif typefaces, but only variants of a sans serif typeface.

6. A handful or two of other studies of various character are also omitted, i.e. Ovink 1938, pp. 84–104; Brachfeldt 1964;¹⁵⁷ Jha and Daftuar 1981; Moriarty and Scheiner 1984;¹⁵⁸ Hoffman 1987; Lenze 1989;¹⁵⁹ Misanchuk 1989a; Misanchuk 1989b; Regan and Hong 1994;¹⁶⁰ Smither and Braun 1994; Klitz *et al.* 1995;¹⁶¹ and Leat *et al.* 1999.¹⁶²

7. Finally, comparative non-experimental typeface ‘preference studies’ are also omitted. Some of these studies are labelled legibility studies; however, not all of them are. The studies in question are: Paterson and Tinker 1940, pp. 18–20 (about ‘legibility’ and ‘ease and speed of reading’);¹⁶³ Becker *et al.* 1970 (about perceived ‘appealingness’ and ‘attractiveness’); Amachree *et al.* 1977 (about ‘legibility’ and ‘easy to read type’); Cooper *et al.* 1979 (about ‘reader preference’); Bell and Sullivan 1981 (about ‘interest’ and ‘liking’); Holleran 1992 (about ‘favourite fonts’); Schriver 1997, pp. 288–303 (about ‘preference’, ‘legibility’, and ‘semantic fit’). Note however that several of the legibility studies included in the study-by-study assessment in this chapter contain preference studies as subsidiary supplements to their primary behavioural experiments.

A note on ‘semantic’ studies

One type of comparative ‘empirical’ typeface study involving both serif and sans serif typefaces has not been considered, as this type of study certainly does not relate to legibility: studies on semantic typeface connotations. That is, studies that relate to the more transient and culture-contingent ‘affective domain’, and not to the ‘psycho-motor

157. Brachfeldt 1964 is cited in the literature as if it is an empirical study, e.g. by Zachrisson (1965, pp. 37–38) and Wendt (1969). It is a short article – based on *a priori* arguments – that only contains a brief and vague mentioning of an experiment.

158. Primarily a study on ‘close-set type’.

159. Kravutske (1994) refers to this study, but her bibliographic details about the study are inadequate (pp. 16, 77).

160. Primarily a study about ‘texture-defined letters’.

161. This study appears only as a ‘conference poster’ (meeting abstract).

162. Primarily a study about ‘crowding in central and eccentric vision’.

163. This study has been described above, in the section ‘Subjective preference studies’, in chapter 2.

domain'.¹⁶⁴ Such empirical studies, dating back to the early 1920s (e.g., Berliner 1920,¹⁶⁵ and Poffenberger and Franken 1923), represent a particular strand of typographic research. Two of the 'standard' legibility research monographs, Ovink 1938, and Zachrisson 1965, deal not only with legibility, but also with typeface connotations (cf. the expression 'atmosphere-value' in the title of Ovink's book and Zachrisson's concern with the equivalent concept of 'congeniality'). Examples of contemporary comparative typeface studies of this kind are: Wendt 1968; Bartram 1982; Rowe 1982; Morrison 1986; Hoffman 1988; Lewis 1989; Lewis and Walker 1989; Pedersen and Kidmose 1993, pp. 17–23; Tantillo *et al.* 1995; Johnson 1995; Braun and Silver 1995).

Note that one of the legibility studies selected for assessment in this chapter is dual in its approach (Silver, Kline and Braun 1994). Although this study is about 'legibility', it can also be categorised as a 'semantic study' (as well as a 'preference study'), since it operates with variables like 'perceived hazardousness' in addition to less semantically charged variables.

Foregrounding connotative semantic aspects in typographic form is a central concern in the craft domain of typography and graphic design¹⁶⁶ (and much more so in advertising, packaging design, and corporate identity).¹⁶⁷ This concern has also been prominent throughout the history of graphic communication.¹⁶⁸ The wide variety of pluralistic-historicist display typefaces of the latter part of the 19th century represents a particular creative and rich instance of this concern.¹⁶⁹ The concern with semantic foregrounding (whether symbolic or iconic) applies more to 20th century eclectic-historicist 'traditionalist' typography,¹⁷⁰

164. See the section 'Ergonomics as a framework', in chapter 2.

165. For the sake of order: Anna Berliner used hand drawn letterforms, not proper typefaces.

166. See *Branding with type*, by Rögner, Pool and Packhauser (1995).

167. See for example Ehres and Lupton 1988. For a comprehensive annotated visual index of trademarks, see Møllerup 1998.

168. For a fascinating visual index, see Massin 1970. Cf. also the concerns of the semiotics journal *Word & Image*.

169. For a fascinating visual index and groundbreaking study, see Gray [1938] 1976.

170. See for example Vincent Steer's manual of typography from the 1930s (n.d.), and Crutchley 1986.

‘New York expressive’ typography,¹⁷¹ and ‘postmodern’ typography,¹⁷² than to 20th century ‘modernist’ typography. The concern is expressed in graphic artefacts and craft discourse, ranging from the obviously appropriate, via the subtle and intricate, via the speculative, via blatantly banal clichés, via the disturbingly interventionist and designer centred (colliding head on with authorial intention), to the absurd.¹⁷³ A spirited and witty instance of this ‘semantic concern’ in current craft discourse is represented by Eric Spiekermann and E.M. Ginger’s best-selling book *Stop stealing sheep & find out how type works* (1993).

This ‘semantic concern’ also appears as a joint interest between typographic studies and literary studies when the genre of (modernist) ‘visual’ or ‘concrete’ poetry is considered.¹⁷⁴ The concern is akin to certain strands within postmodern architecture as well as the semiotic movement in architectural design of the 1960s and 1970s, and the more current ‘product semantics’ movement in industrial design.¹⁷⁵ Not least, the concern is expressed through the prevalence of more or less appropriate visual metaphors and visual analogies in the graphical user interfaces of the operating systems, application software, multimedia applications and web sites of digital media. Finally, there is a kind of kinship between ‘typeface semantics’ and ‘graphology’, that is, graphology not in its linguistic sense, but graphology in the sense of handwriting analysis for revealing the psychological characteristics of the writer.

171. For a visual index, see Snyder and Peckolick 1985.

172. For visual indexes, see for example Aldersey-Williams *et al.* 1990; and Blackwell 1995.

173. See for example Baker 1985; and Stiff 1995a.

174. See for example Bohn 1986; and Levenston 1992.

175. On ‘product semantics’, see for example the collections of papers in Vihma 1990; and 1992. See also Richard Buchanan’s critical review of Vihma 1990 (Buchanan 1993).

Griffing and Franz 1896^{*}

The first experimental study ever published, which deals with the legibility of sans serif typefaces compared to serif ('roman') typefaces, was initially presented to the International Congress of Psychology in Munich in 1896, and subsequently published in the American journal *Psychological Review*.¹⁷⁶ The study is undertaken from the perspective of investigating the relationship between visual fatigue in reading and conditions causing this fatigue. The authors justify this approach by linking myopia and other eye disorders with visual fatigue; ultimately caused by the increased amount of reading in modern society. One of several conditions which they identify as contributing to visual fatigue is 'the quality of the type' (pp. 513, 522–525). By 'quality of type' they simply refer to typeface categories.

Myopia was a common concern in reading research and legibility literature around the turn of the century. For example; in an extensive article on legibility research that appeared in the leading Scandinavian printing trade journal *Nordisk Boktryckarekonst* in 1906, myopia was prominently pictured as caused by inadequate typography (see Schmidt 1906).¹⁷⁷ And not to mention Emile Javal, who as an ophthalmologist was preoccupied with myopia and its possible causes in typography.

The authors initially suggest that it can be theoretically assumed that legibility of typefaces is a function of 'complexity of structure'; i.e. the more simple the structure (e.g. sans serif typefaces) the more legible the typeface, and vice versa – the more complex the structure (e.g. fraktur typefaces) the less legible the typeface.

* Harold Griffing and Shepherd Ivory Franz. 1896. 'On the conditions of fatigue in reading'. *Psychological Review*, vol. 3, pp. 513–530.

176. Venezky points out that around the turn of the century reading research (which then included legibility research) was prominent in the psychological literature, and that very few psychology journals existed – *Psychological Review* and *American Journal of Psychology* in the USA, *Mind* in England, and a few in Germany and France (1984, pp. 7, 23, 28).

177. Interestingly this article is based on the work of Javal as well as a book written jointly by the psychologist Dr Prof. Herman Cohn and the printing ink factory director Dr Robert Rübencamp: *Wie sollen Bücher und Zeitungen gedruckt werden?*, published in Germany in 1903, and addressing the world of printing and typography. Both the article in the Scandinavian printing trade journal and the German book serve as an index of an early contact between research domain and target domain.

The operational criteria employed for measuring the relative legibility of six different typefaces are expressed through the ‘illumination threshold method’.¹⁷⁸ By the ‘illumination threshold method’ both time of exposure and the distance between the stimulus material and the subject’s eyes are constant. The luminous intensity of the light source is also constant. However, the distance between the light source and the stimulus material is varied.

The typefaces in question are: a roman typeface, a ‘German’ gothic typeface, a bold sans serif typeface of capitals only, a lighter sans serif typeface of capitals only, a modern fat face, and a normal sans serif typeface of normal weight of both small letters and capitals. The size of the letters is referred to as ‘1.5 mm. in height ... with slight variations, to .1 mm. in the individual letters’ (p. 523). The measure 1.5 mm. refers to the x-height of the typefaces that contains both small letters and capitals, and to the capital height of the capitals-only typefaces. Translated into nominal pica points, ‘1.5 mm in height’ should approximately equal a nominal typesize of 10 points (inferred from information on p. 514 and p. 518).

The results are not in line with the authors’ initial assumption that legibility is a function of the ‘complexity of structure’. The results show that there is only a slight difference of legibility between the roman typeface, the ‘German’ gothic typeface, the normal sans serif typeface and the light sans serif capitals-only typeface. However, the modern fat-face typeface and the bold sans serif capitals-only typeface are considerably more legible than the other typefaces. The authors point out that this phenomenon obviously makes legibility a function of stroke thickness.

That a rather fat typeface requires less light than a typeface of ‘normal’ width and per se becomes more legible, clearly illustrates a deficiency of the authors’ ‘illumination threshold method’. And the authors acknowledge that something might be wrong with their legibility-construct, and that their results therefore can be questioned. They also acknowledge that it is possible there is no direct connection

178. For the sake of order: The authors also employ the ‘speed of reading method’ and the ‘time of exposure method’ when doing experiments on other ‘conditions’.

between what is actually perceived at a specific threshold and the level of fatigue, i.e. that something perceived at a certain threshold may be either less or more fatiguing than something else perceived at another threshold.

There is nothing much left of the optimism after the authors have discussed their own results, and they conclude laconically: 'The form of the type is of less importance than the thickness of the letters.' (p. 530).

Roethlein 1912^{*}

The second experimental study ever published, which deals with the legibility of sans serif typefaces compared to roman typefaces, was published in the *American Journal of Psychology*.

The explicit aim of this large-scale experimental study was not to compare the legibility of sans serif typefaces to roman typefaces. It is an investigation into the relative legibility of up to 26 different typefaces, and including, for some of the typefaces, variants such as italic, bold and condensed. The relative legibility of each letter of the typefaces is measured, and the average legibility of the letters belonging to each typeface determines the legibility of the typeface. The study has also a more far-reaching aim: to identify the physical properties of typefaces which determines their degree of relative legibility – in order to improve legibility. The paper contains a relatively extensive bibliography (40 items) with references to Griffing and Franz' study (1896); Javal; Huey; and the American master printer Theodore Low De Vinne's authoritative excursion in type technology – *Plain printing types*. A breakdown of the bibliography into languages shows that it lists 20 English references, 17 German, and 3 French.

By the construct 'legibility' the author refers to the *visibility* of single letters – both isolated letters and letters appearing in awkward

^{*} Barbara Elisabeth Roethlein. 1912. The relative legibility of different faces of printing types'. *American Journal of Psychology*, vol. 23, no. 1, pp. 1–36. (Also published in the series Publications of the Clark University Library, vol. 3, no. 1. Worcester, Massachusetts: Clark University Press).

nonsense combinations like 'ksitugy'. The objective for investigating the legibility of letters in groups is to determine the role of adjacent letters when determining legibility. The operational criterion for measuring the legibility is expressed through the variable distance method. The time of exposure method was discarded by the researchers as too unreliable after some preliminary experiments – due to the interference between reaction-time and varying levels of concentration (p. 5).

The prints of the typefaces employed in the study were donated by American Type Founders Company. ATF also provided typographic expertise; Frank Berry, Linn Boyd Benton and Morris Fuller Benton contributed by suggesting typefaces and in the interpretation of the results. The investigation also took into consideration data on compositors' and proof-reader's errors.

The two sans serif typefaces included in the study are Franklin Gothic and News Gothic.¹⁷⁹ All letters were of ten points nominal size. The author explicitly acknowledges the problematics of nominal type size (p. 21), and its bearings for her interpretations.

The results of this extensive investigation are many and varied, and laid out in a large number of tables. About the comparative legibility of sans serif typefaces to roman typefaces the author concludes:

If legibility is to be our sole criterion of excellence of type-face, News Gothic [a sans serif typeface] must be regarded as our nearest approximation to an ideal face, in so far as the present investigation is able to decide this question. The aesthetic factor must always be taken into account, however, here as elsewhere. And the reader who prefers the appearance of Cushing Oldstyle or a Century face may gratify his aesthetic demands without any considerable sacrifice of legibility. (p. 29)

Roethlein is not satisfied with ranking typefaces. She points out that the letters of the typefaces in question differ not only in specific form, but also with regard to size and strokewidth (p. 22). She therefore measured strokewidth plus the height and width of the letters *m*, *o*, and *z* of 16 of the typefaces 'In order to obtain a clearer insight into the relative significance of each of these variable factors as determinants of

179. Although these typefaces carry different names, they are today frequently referred to as complementary typefaces, Franklin Gothic being a bold companion to the lighter News Gothic or vice versa.

legibility'. The results of these measurements lead her to suggest that the specific forms of the typefaces are less important than size and strokewidth as determinants of legibility (p. 24–25), (Franklin Gothic, the bolder relative of News Gothic, is however not included in this part of her study.)

Roethlein finally concludes that legibility of individual letters is less dependent on letter-form than: letter-size, letter strokewidth, letter spacing, position in a group, and the shape and size of adjacent letters (pp. 33–34). Roethlein's study appears to be thorough and well reasoned.

Legros and Grant 1916*

Chapter 11 is devoted to legibility in Lucien Alphonse Legros and John Cameron Grant's voluminous and impressive manual on type technology published in 1916: *Typographical printing surfaces: the technology and mechanism of their production* (pp. 156–192). The authors refer to four reading researchers, Sanford, Javal, Catell and Cohn; and in addition to Theodore Low De Vinne. There are however, no references to the work of Griffing and Franz (1896) or Roethlein (1912).

Legros and Grant are not only measuring legibility, they are also proposing a *theory* of typeface legibility. Their constructs 'legibility' and 'illegibility' refer to the degree of topological similarity between two single letters in certain pairs of similar looking and frequently occurring characters of the same typeface. They believe that similarity of form leads to confusion and misreading – with enormous consequences for the total amount of time millions of people spend reading. The study is not experimental, but a typeface topology study, based on measurements of the 'coincident' (common) and 'non-coincident' (peculiar) areas of the two individual letters of chosen characters-pairs when superimposed on each other. On the basis of such intra-typeface measurements, typefaces are compared. According to the theory, a high degree of non-coincident area

* Lucien Alphonse Legros and John Cameron Grant. 1916. *Typographical printing surfaces: the technology and mechanism of their production*. London: Longmans, Green, and Co.

suggests a high degree of legibility for that typeface. And on the contrary, a high degree of coincident area suggests 'illegibility'.

Practical measures were made possible by drawings where the characters had been enlarged 45 times. The study deals with three separate groups of characters – small letters, capitals, and figures – and present the result for each group independently. However, the emphasis is on the group of small letters because they are the most frequently used. The character-pairs compared are: for small letters: c/o, c/e, e/o, n/u, i/l, h/b and a/s; for capitals: C/G, O/Q, B/R and X/Z; and for figures: 3/5 and 8/6.

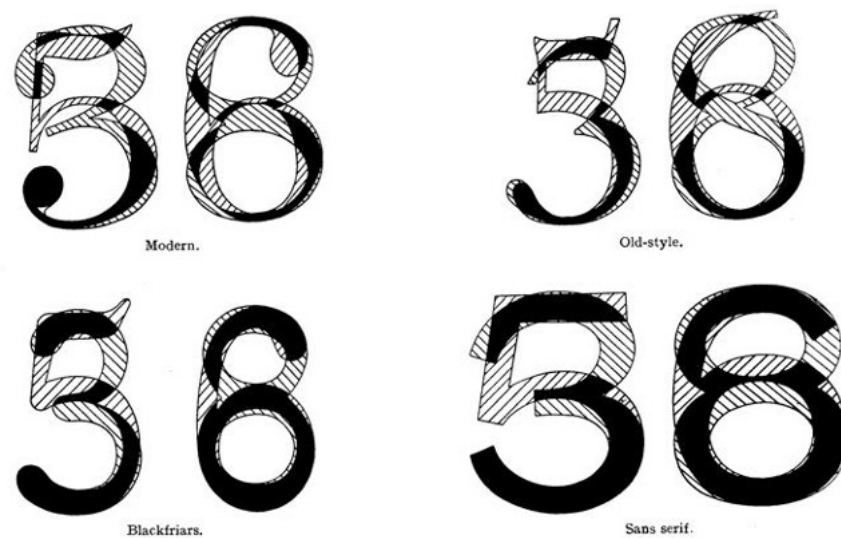


FIGURE 2. An illustration of Legros and Grant's concept of legibility. A high degree of non-coincident area suggests a high degree of legibility. A high degree of coincident area suggests 'illegibility'. (From Legros and Grant 1916, p. 172: figure 129.)

The ratio of the total peculiar area of both characters (of a pair) to the total area of the two characters is called the legibility coefficient (given as a percentage). If there is no common area, the peculiar area of the two characters would be as large as the total area of the two characters (peculiar area / total area = 1; i.e. 100%), and there would be 'perfect legibility'. The difference between 100% and the legibility

coefficient makes the illegibility coefficient (in percent). For determining the legibility of a typeface the authors take into consideration the occurrence 'given in the fount bill' of each of the characters of each pair, and then of each character of all the characters in question. Thus, the mean legibility coefficient and the mean illegibility coefficient of the typeface can be calculated (as percentages).

However, the authors realise that if the strokes of the characters of a typeface are very thin, the peculiar area will be larger than if the lines of the characters are very thick (imagine the character X of a light typeface, made of two very thin crossing lines, compared to the character X of an ultra bold typeface, made of two very thick crossing lines, while both Xs are of the same point size). To avoid that legibility/illegibility to a large extent becomes a function of the leanness/fatness ratio, and to avoid the threat this problem poses to their theory, the authors introduce two more concepts, 'blackness' and 'specific legibility'. The blackness is expressed by the ratio of the total area of the character to the area of a horizontal cross-section of the shank of the type (in per cent). Finally, the specific legibility (in per cent), is expressed by multiplying the mean legibility coefficient (in percent) with the mean blackness (in per cent). Specific legibility gives, according to the authors, the best comparative measure of the legibility of a typeface. The following example can serve to clarify the idea behind blackness and specific legibility: To avoid that the degree of fatness of a typeface unintendedly becomes the most important determinant for its legibility, the 'mean blackness' (expressed as a percentage) is multiplied with the 'mean legibility coefficient' (expressed as a percentage), resulting in the 'specific legibility' (expressed as a percentage), thus employed as a method to even out (reduce) the influence of the fatness when determining the legibility of a typeface. The following example can illustrate the relationship between the mean legibility coefficient and the specific legibility: for a light typeface with a high mean legibility coefficient of 25%, the coefficient is multiplied with its low mean blackness of only 10%, which makes the specific legibility 2.5%. Accordingly, for an ultra fat typeface with a low mean legibility coefficient of only 5%, the coefficient is multiplied with its high mean blackness of 50%, which actually results in the same specific legibility of 2.5%.

The authors' approach is applied to five more or less generic Latin-script typefaces: a modern typeface modified by the authors (given serifs with the thickness of old style serifs), an old-style typeface (with modified figures 'with a view to increased legibility'), a mannered typeface referred to as Blackfriars ('in which legibility of the lower-case was specially sought' and 'produced under the direction of one of the authors', with the main strokes thickening towards the thick serifs), a sans serif typeface, and a fraktur typeface.

It seems that the authors have problems in deciding whether it is the specific legibility or the mean legibility coefficient which should serve as the best measure of legibility. However, the results of the measurements and the calculations for both measures are laid out below.

The specific legibility of the sans serif typeface compared to the other four typefaces in question:

Small letters: the sans serif typeface comes out as the second most legible typeface (3.44%), only after Blackfriars (4.51%), before the modern typeface (3.33%), the old-style typeface (3.21%), and the fraktur typeface (2.45%).

Capitals: the sans serif typeface comes out as number three (2.60%), after Blackfriars (3.44%) and the modern typeface (2.88%), before the oldstyle (2.28%) and the fraktur typeface (0.89%).

Figures (the fraktur is not included): the sans serif typeface comes out as number three (4.82%), after the modified modern typeface (9.31%) and the oldstyle typeface (6.97%), but before Blackfriars (4.39%).

When only the mean legibility coefficient is considered (the 'blackness factor' not taken into consideration), the sans serif comes out far worse. For both small and capital letters it comes out penultimate only before the fraktur, and for the figures (fraktur not included) it comes out last. The mean legibility coefficients were:

Small letters: Blackfriars (27.3%), old-style (26.3%), modern (23.1%), sans serif (18.2%), fraktur (15.1%).

Capitals: modern (15.5%), Blackfriars (15.2%), old-style (15.1%), sans serif (10.2%), fraktur (3.8%).

Figures: modern (52.3%), old-style (46.8%), Blackfriars (20.4%), sans serif (18.0%).

The authors acknowledge that the blackness factor is ‘unavoidably’ larger for the sans serif typeface than for the other typefaces, something which gives the sans serif typeface a higher specific legibility than mean legibility coefficient. However, they also claim that the blackness factor does not help, because the specific legibility of the sans serif typeface is less than proportionally increased compared to the other typefaces when the blackness factor is included in the calculation (p. 164). The authors state that ‘Popular belief holds the sans serif, or, as it is popularly called, block letter, to be very legible, but to printers, and especially those who do much display work, this view is known to be erroneous.’ (p. 164). It is clear that the authors, independently of the results of their own investigation, view sans serif typefaces as inferior to roman typefaces. Their attitude is illustrated by the following somewhat witty comment:

It may be some comfort to motorists to know that the form of the characters and figures selected for car numbering by the governments of this and other countries is less legible than many others which might have been chosen; in fact, it would be difficult to improve them in the direction of greater illegibility except by combining German Fraktur capitals with the existing sans serif figures. (Legros and Grant 1916, pp. 169–170)

Legros 1922 *

In this publication, *A note on the legibility of printed matter*, Lucien Legros, now with the initials O.B.E. (Order of the British Empire) added after his name, maintains his and John Grant’s theory of typeface legibility. However, his claims are modified – he refers to the method employed in *Typographical printing surfaces* as a suggestion, and proposes that if the method is combined with a time of exposure method, ‘it is probable that a definite measure of legibility would be obtained’ (p. 6). The practical application of the method, as demonstrated in *Typographical printing surfaces*, is only briefly referred to. The mean

* Lucien Alphonse Legros. 1922. *A note on the legibility of printed matter*. Prepared for the information of the Committee on Type Faces. London: His Majesty’s Stationary Office.

legibility coefficient and the specific legibility of the group of small letters of the old style typeface from the 1916 investigation are referred to, respectively 26.3% and 3.21%. However, this information is followed by the following statement: 'A modern face of the same gauge and blackness would only have a specific legibility of about 2.8%, sans-serif about 2.2%, and German Fraktur 1.8%.' (p. 6). This tells us that the author in reality has discarded his formula for calculating specific legibility. He now arrives at a neat legibility hierarchy with old style on top, then modern, then sans serif, and then fraktur – in line with contemporary belief. He has discovered that the more narrow (condensed) a typeface is, the higher is its blackness, and vice versa for broad (expanded typefaces). Furthermore – the method is applied to an old style typeface family of eight weights, with the generic old style typeface referred to above as the starting point. The thickness of the main stroke has been gradually increased and decreased in each direction. Here we can see that the mean legibility coefficient and the mean blackness actually varies as a function of weight. Legros admits that 'this trial shows that the method gives the highest figure [both the mean legibility coefficient and the specific legibility] for type blacker than is generally used for reading matter' (p. 7). Since the results are obviously not in accordance with his preconceptions, and common typographic practice, he then suggests that 'it would be possible to fix a limit for the blackness and to require that the specific legibility should exceed say 2,5 per cent for any selected pair of letters. Such a stipulation would ensure that those characters that are almost commonly misread should be made to a definite percentage of dissimilarity as measured by the legibility coefficient.' (p. 7). This statement illustrates the deep troubles which the theory now has encountered; remember, this is not the first attempt to solve problems encountered; compare the introduction of the concepts of blackness and specific legibility.

However, the theory is dubious. In *Typographical printing surfaces* the authors tried to validate their theory by rhetorically referring to the form of the then abolished long s (identical form to the character f, except for the lack of the right-hand part of the f's crossbar), resulting in an extremely low legibility coefficient. It is not difficult to accept that two characters with such a degree of sameness easily can be confused and

misread. However, to generalise from this (and possibly other similar examples) into a theory as the above represents a step that is hardly defensible. Characters are definitely not read while superimposed upon each other.

Another fallacy of Legros and Grant is that they induce (without saying explicitly that they do) from specific typefaces (the ones actually employed in the investigation) with all their details and idiosyncrasies, to generic categories (old style, sans serif, etc.) – and then they judge the categories on this basis. Furthermore, the choice of character pairs is obviously not arbitrary, other character pairs could have been selected, and the results may have been different. Actually, in an application of their method in *A note on the legibility of printed matter* they have omitted some of the small-letter character-pairs where one of the eight typeface variants is identical with one of the typefaces employed in *Typographical printing surfaces*, and not unexpectedly – the figures differ – something which Legros acknowledges. There is not much left of the theory.

The typeface designer Walter Tracy commented upon this study in 1988:

The fundamental error ... in Legros and Grant's study of resemblances, is the assumption that a reader has to recognise each letter before he can recognise each word. But it has been shown, and experience confirms it, that the competent reader's eyes take in a text not by nibbling at it unit by unit but by gulping groups of words; and – such is the speed at which the perceptive faculty operates – even failing to notice a wrong letter in a word. (p. 83)

Pyke 1926 *

In 1923 the British Medical Research Council on the request of His Majesty's Stationery Office appointed the 'Committee upon the legibility of type'.¹⁸⁰ R.L. Pyke at the Psychological Laboratory at the University

* R.L. Pyke. 1926. *Report on the legibility of print*. Medical Research Council, Special report series, no. 110. London: His Majesty's Stationery Office.

180. Lucien Legros was one of six members; however, this committee should not be confused with the former 'Committee on Type Faces'; cf. Legros 1922).

of Cambridge was engaged by the committee, and in 1926 his *Report on the legibility of print* was published by HMSO as a 'Medical Research Council Special Report'. The report presents an unusually comprehensive discussion of previous legibility research. It also presents Pyke's own experimental work on the relative legibility of typefaces.

Pyke refers to Griffing and Franz (1896), Roethlein (1912), Legros and Grant (1916), and Legros (1922). He politely states that testing for legibility, in his opinion, can best be done by actual reading. Measuring legibility by involving real acts of reading 'springs from a more comprehensive aspect of the process of reading' than Legros' measure of 'specific legibility' (Pyke 1926, p. 30). Pyke discards what he calls *objective criteria* (e.g. 'specific legibility' – where 'objective' refers to the exact measures of typeface dimensions), *subjective criteria* (mental or physical states like fatigue, eye strain, and 'aesthetic satisfaction'), and *functional criteria* based on 'unconscious processes' (e.g. eye movements). What is left is *functional criteria* based on 'conscious processes'. By this he refers to the following five operational methods: distance threshold, illumination threshold, focus threshold, speed of reading, and errors. He then, after some discussion, discards distance threshold, illumination threshold, and focus threshold. Speed of reading is then, after some more discussion, also discarded: 'moderate changes of a realistic sort in the typographical stimulus are extremely unlikely to produce reactions in speed directly proportional, or even significant' (p. 31).

At last he settles with the rather odd 'number of errors' as the 'best available criterion' – although he is not completely satisfied with it. He then, after another elaborate discussion, decides that the tests must be based on reading aloud as opposed to silent reading, at maximum speed (as fast as the subjects can), and for four seconds at a time ('long enough for about one line of nonsense or two of sense to be read') (p. 34).

In most of his series of experiments he compares the legibility of eight typefaces: seven roman and one sans serif typeface. Three of the romans are described as 'standard' typefaces (Lanston Monotype Old

Style No. 2, L.M. Imprint Old Face No. 101, and Caslon Modern Series No. 23). Of the remaining four roman typefaces, one is 'narrow' (L.M. Modern Condensed No. 39), one is 'broad' (L.M. Modern Extended No. 7), one is 'thicked-limbed' (L.M. Old Style Antique No. 161, and one is 'thin' (L.M. Cushing Series No. 17). The sans serif is Stephenson & Blake, series no. 10, Lining Grotesque. All typefaces are illustrated as complete upper and lower case alphabets (p. 114–115).

On the face of Pyke's results, the standard Old Style is the most legible, 18 per cent better than the second – the sans serif typeface. However, the biggest difference between any typeface and the next on the ranking list is between the sans serif and the next (the standard Modern) which is 30 per cent less legible than the former. Pyke points out that the best type (the standard Old Style) is the most ordinary. However, he is not willing to attach any significant meaning to that, due to the fact that the second best, the sans serif typeface, 'is one of the most uncommon' (ibid. 52). Pyke suggests that the success of the sans serif might have something to do with the fact that its actual size (as opposed to its nominal size) was the second largest. However, he does not pursue this suggestion. He laconically states that the sans serif's 'relative legibility agrees with Roethlein's results, but disagrees with Legros' and Grant's' (p. 52).

In Pyke's final conclusion he expresses doubt and scepticism about his own results. He specifically questions the extent of the differences and 'whether the relative legibility of the types in the experiments holds in the ordinary world' (p. 60). He elaborates:

The hypothesis is here put forward that extremely large typographic differences must be present before it is possible to say that there is any difference in the objective legibility of types. Of types more slightly differentiated it is impossible to say that one is objectively more legible than another. Such types will be 'legible' according as they suit the psychological make-up of the reader. Hence the most legible type, in this subjective sense, is unlikely to be the same for all readers. ... These two ideas – that within limits there is no objective optimum, and that only extremely violent changes in the stimulus can produce any significant variation in the reaction – are found in other psychological fields. (p. 60)

Pyke's final conclusion comes like a deep sigh: 'The problem of legibility seemed simple at the outset; it is in fact complex and

elusive' (p. 61). Accordingly: The anonymous preface (by the Medical Research Council on behalf of the Committee) concludes that available methods are inadequate and that experimental work 'on the same lines' therefore should be discontinued and integrated into a more 'comprehensive scheme of research into the physiology of vision' conducted by a new committee (p. [3]).

Crosland and Johnson 1928: * the serifs that never were

This paper, an article published in the *Journal of Applied Psychology*, is every now and then cited in legibility studies. In 1959 Cyril Burt¹⁸¹ referred to Crosland and Johnson's article the following way:

Our conclusions are fully borne out by the results of H.R. Crosland and H. [sic] Johnson, who also found 'serified letters more legible than unserified' (*J. Appl. Psychol.* XII, 1928, p. 121). (Burt 1959, p. 9)

However, the compromising fact is that Burt cannot possibly have read anything but the short summary of the article (on p. 121) which states that 'seraphed letters are more legible than unseraphed letters'.¹⁸² James Hartley and Donald Rooum have pointed out that the only typeface involved in this study was Caslon (1983, p. 205). When Crosland and Johnson used the terms 'unseraphed' and 'seraphed' they clearly attached the terms to something other than serifs. They mean, respectively, small letters without ascenders or descenders, and small letters with ascenders or descenders.

A number of writers on legibility have since erroneously referred to this article. Such errors of disconnected intertextuality is something

* H.R. Crosland and Georgia Johnson. 1928. 'The range of apprehension as affected by inter-letter hair-spacing and by characteristics of individual letters'. *Journal of Applied Psychology*, vol. 12, pp. 82–124.

181. For information about Cyril Burt, see the section 'Burt, Cooper and Martin 1955 / Burt 1959' in this chapter.

182. Alternatively, he may have seen this quoted sentence in a secondary source which he refers to in his bibliography: Carmichael and Dearborn's *Reading and visual fatigue* (1947, p. 106).

anyone can fall victim to. Nevertheless, the sheer number of writers in question as well as the fact that they all imply that they cite Crosland and Johnson directly, without acknowledging that their source of information has been Burt or another faulty secondary one, is thought-provoking. The alternative explanation, that the writers in question, just like Burt, have not read the Crosland and Johnson article carefully, is also possible. Anyhow, they refer to the article the following way:

- Stanley Morison: ‘what other investigators had proved, that “serifed letters are more legible than unserifed” (in Burt 1959, p. xi)
- Miles Tinker: ‘Serifed letters were significantly more legible than unserifed letters.’ (Tinker 1963, p. 272)
- Dirk Wendt: ‘Crossland [*sic*] & Johnson 1928 fanden eine Unterlegenheit der Groteskschrift.’ (1969, p. 20)
- Sidney Berger: ‘Crosland and Johnson ... say that serifs increase legibility’ (1991, p. 7)
- Pedersen and Kidmose: ‘[Crosland and Johnson] disclosed that the subjects had more difficulty in reading sans serif typefaces than Roman ones’ (1993, p. 69)
- Dirk Wendt: ‘Crossland [*sic*] & Johnson (1928) reported an inferiority of a sans serif typeface’ (1994, p. 305)
- De Beaufort Wijnholds: ‘Among others ... Crosland and Johnson (1928) ... agree that serif typefaces are more legible’ (1996, p. 32)

Only two papers that I have come across erroneously refer to Crosland and Johnson without implying that the information is taken directly from the source. Zachrisson (1965, p. 37) and Jansen and Thomsen (1985, p. 58) refer to Crosland and Johnson’s ‘findings’ while acknowledging Burt as their source of information. Kunst (1972) reveals that he has actually read the paper, and he refers to it accordingly.

Moede 1932^{*}

In 1931/1932 the Bauer type foundry in Germany issued a promotional brochure called *Futuraheft* for the then new and innovative sans serif typeface Futura designed by Paul Renner. This brochure contained six pages of five dubious statements dressed in a self-important language about the excellence of Futura, while radiating some kind of scientific authority. The statements were made by a doctor, ‘a well known philosopher’, a medical doctor and ophthalmologist, another medical doctor, and a doctor and professor of psychology. At least three of the five statements are unsubstantiated claims where the only link between the content of the statements and science seems to be the scientific sounding vocabulary and the titles of the persons uttering them.

Two of the five statements are more elaborate than the others. One is a statement by Dr Med. G. Secker, an ophthalmologist in Hamburg. It includes a long, dubious and even referenced reasoning on the physiology of vision. The reasoning purports to scientifically ‘prove’ that a typeface like Futura, that is, a typeface without serifs and with a high degree of regularity, ‘obviously’ will cause less eye-fatigue than any other typeface.

The other elaborate statement, seemingly more substantial, and which avoids the pompous language of the other statements, is the only one which refers to experimental research and which thus is of special interest for this thesis. It is based on an unpublished report¹⁸³ of experimental work by Prof. Dr Walther Moede at the ‘Institut für industrielle Psychotechnik und Arbeitstechnik in Berlin-Charlottenburg’.

It is worth noting that Walther Moede (1888–1958) was a prominent and influential applied psychologist or ‘psycho-technician’ in the inter-

^{*} *Futuraheft*. [1931/1932]. Frankfurt am Main, Barcelona and New York: Bauersche Giesserei.

183. Ovink briefly refers to an ‘unpublished report by Moede’ (without giving further bibliographical information or mentioning *Futuraheft*) (1938, p. 111). The text in *Futuraheft* indicates that it is based on an independently existing report – as opposed to the other statements in the publication. However, I have made no effort in trying to track down the original report. A few comments on the content of *Futuraheft* can be found in Ovink 1938, p. 111; Zachrisson 1965, pp. 37, 68–69; and Burke 1998a, pp. 112–113.

war period. He was a teacher at Berlin Technische Hochschule, the founder of Germany's first institute for industrial 'psychotechnics' and for many years the president of the professional organisation 'Verbandes praktischer Psychologen'. He was an active researcher who published widely and he was the editor of numerous journals, among them *Praktische Psychologie* and *Industrielle Psychotechnik*. He was also the author of many books, among them several textbooks, and he was adviser to the German railways.¹⁸⁴

'Psychotechnik' was applied psychology in Germany in this period. Psychotechnics was preoccupied with applying psychology to industry, for example by psychological testing of individuals in order to match them to particular occupations and trades. It played a key role in German 'Arbeitswissenschaft', and was seen as an instrument for the 'rationalisation' of work, complementary to both 'Fordism' and Tayloristic 'scientific management'. However, it was also seen as an instrument for the 'humanisation' of work, by aiming at healthy and attractive work environments. Interestingly, psychotechnicians also showed an early interest in human beings as consumers, by carrying out quantitative research on the psychology of advertising and the consumption of commodities.¹⁸⁵

Moede's experiments, which involved Futura and a roman typeface, is described in *Futuraheft* under the title 'Welche Schrift ist leichter lesbar?' In the first experiment Moede measured the time it took subjects to read a text set in Futura and a roman typeface.¹⁸⁶ On average the text set in the roman typeface was read in 28.5 seconds and the text set in Futura in 27.8 seconds. In the second experiment the subjects had to read the text from a shaking apparatus. The average results were 57.6 seconds for the roman typeface and 52.4 seconds for Futura. The third experiment investigated the display effect ('die Reklamewirkung') of Futura. It was carried out with a tachistoscope, and here Futura came out as an even clearer winner: Futura was on average read after only 3.2

184. Campbell 1989, p. 142, 356; *Neue Deutsche Biographie* 1994, p. 611.

185. See Joan Campbell's fascinating study *Joy in work, German work: the national debate, 1800–1945* (1989, especially pp. 131–177).

186. Burke points out that 'Renner most certainly intended the typeface for use in [continuous] text.' (1998a, p. 113).

successive exposures of 1/6 of a second, while the average number for the roman typeface was 5.4. In addition, Moede refers very briefly to ‘distance’ and ‘low-illumination’ experiments, where, by now, to no surprise, Futura also came out as the superior. No wonder – Moede’s conclusion of Futura’s superiority is unambiguous. He even points out that the results were obtained in spite of the subjects’ habituation to roman typefaces.

Maybe I am too dismissive of Moede’s research. The fact that all the results, regardless of operational method, without exception were in Futura’s favour, and furthermore, that the research itself most likely was made to order, does not necessarily imply that the results were made to order. Nevertheless, as long as no details or illustrations of the ‘stimulus material’ is available, it is simply impossible to assess the validity of the research in a meaningful way.

It was not unusual for type manufacturers during the 1920s and 1930s to market typefaces by referring to scientific research on legibility, and not necessarily by such dubious statements as those in *Futuraheft*. American Type Founders promoted their hugely popular serif typeface Century Schoolbook, launched in the mid 1920s, by referring to a series of tests and experiments made by, among others, the British Association for the Advancement of Science and Clark University. The marketing material cleverly implied that Century Schoolbook was objectively far more legible than other typefaces, and claimed that the experimenters, who had worked independently of each other, all arrived at the same conclusions.¹⁸⁷

Linotype Mergenthaler extensively marketed their (serif) newspaper typefaces Ionic no. 5 (launched in 1925) and Excelsior (launched in 1931), and some other similar typefaces, as ‘The Legibility Group’ of typefaces. Mergenthaler Linotype hailed ‘The Legibility Group’ of newspaper typefaces as ‘a scientific solution to a typographic problem’ and Ionic no. 5 was claimed to have been designed to overcome ‘the

187. See for example the 1923 main typeface catalogue from American Type Founders (1923, p. 203). See also Shaw 1989.

handicap of eye fatigue and impaired vision' on the basis of 'a study of eye movements in reading'.¹⁸⁸

When Luckiesh and Moss published their book on legibility research, *Reading as a visual task*, in 1942, they emphasised that their book was printed in Texttype 'designed primarily for ... works requiring intensive study and prolonged reading' (p. [429]). Texttype is similar to Ionic no. 5 and belongs to Linotype's 'legibility group', but has considerably smaller x-height and was marketed for use in books and periodicals.

Linotype's competitor, the Intertype composing machine and typeface manufacturer, marketed their imitations of Linotype's newspaper typefaces, Ideal News and Regal, in a similar manner. Thus, Ideal News was marketed under the heading 'Readability, legibility, durability' while Regal was marketed under the heading 'A type face scientifically designed for easy reading'.¹⁸⁹

The strategy of selling a typeface, or any product, by procuring favourable statements from scientists (I am not necessarily including Moede's report as such) is not unique for the inter-war period. A similar strategy seems to have been employed in the 1980s for a typeface that later developed into one of the most successful typefaces of the 1990s, Erik Spiekermann's sans serif typeface Meta. In an essay by Spiekermann on the design process of the typeface (which initially was commissioned and intended to become the new corporate typeface for Deutsche Bundespost), he claims that before it was submitted (and subsequently turned down) 'a professor of applied psychology had written a very favourable report on it' (Spiekermann 1986). Paul Stiff, the editor of *Information Design Journal*, recently described a similar contemporary phenomenon the following way: 'some information design companies try to enlist the help of "researchers" for much the same

188. See for example the booklet *The legibility of type*, published by Mergenthaler Linotype in 1935 (p. 27); and the 1939 main typeface catalogue from Mergenthaler Linotype (1939).

189. See promotional type specimens for Ideal News and Regal from Intertype issued during the 1930s. For a general introduction to the history of dedicated newspaper typefaces, see Level 1989; and for a comprehensive visual index, see Gürtler 1988.

reasons that men in white lab coats appeared in the first toothpaste adverts on tv' (Stiff 1993, p. 43).

Parallels to the *Futura* kind of cynical, or naive, scientism – by vaguely referring to 'science' or by procuring favourable statements from medical doctors or psychologists – can be found in abundance in an area akin to typography. Exactly this kind of rhetorical strategy was ubiquitous in a long and heated debate on handwriting models for schoolchildren in post-war Norway. In an essay based on his research on the introduction and implementation of the Marion Richardson-like handwriting model 'formskrift' – which in the period from 1947 to 1962 monopolistically managed to take over for the older copperplate-like handwriting style – Trond Berg Eriksen discloses an astonishing amount of almost caricature-like scientific statements. This kind of rhetoric was applied by all sides of the debate, and the ergonomic arguments were often mixed with political, moral, social hygienic and aesthetic arguments – as well as more pragmatic arguments (Eriksen 1993).¹⁹⁰

Paterson and Tinker 1932*

The article 'Studies of typographical factors influencing speed of reading. 10. Style of typeface', was the tenth of thirteen numbered legibility studies published by Tinker and Paterson in the *Journal of Applied Psychology* in the period from 1928 to 1936. This study was also the first of four studies – three experimental and one non-experimental – carried out either by Tinker alone or with co-authors between 1932 and 1944, on typeface legibility, using the same typefaces but different methods each time.

190. For a description and survey of Marion Richardson's and other handwriting models, albeit in an English context, see Myers 1983. Gordon and Mock 1960 is brief, but still a comprehensive index of 20th century handwriting models. Sassoon 1999 is a recently published book on the same topic. For critiques of the methodology of several research papers on the ergonomics of handwriting and handwriting models, see Sassoon 1988.

* Donald G. Paterson and Miles A. Tinker. 1932b. 'Studies of typographical factors influencing speed of reading – 10: Style of type face'. *Journal of Applied Psychology*, vol. 16, pp. 605–613.

The authors open up by pointing out the tendency of ‘editorial, advertising and printing experts’ to show much greater interest in the legibility of various typefaces than for for example for line spacing and line width. They then go on, claiming that they will settle the question of the relative legibility of typefaces:

Absence of definite facts tends to perpetuate such a state of affairs. It is obvious that experiments on the relative legibility of type faces are needed in order to determine the merits of the claims advanced by partisan advocates. (p. 605)

They then continue by reviewing the literature, including Roethlein (1912) and Pyke (1926). They claim that, although Roethlein’s study has been quoted widely, and was a pioneering experiment, it must be discounted because it is based on stimulus material of isolated letters and meaningless groups of letters ‘and thus fails to give information regarding the relative legibility of typefaces as found in actual reading situations.’ (pp. 606, 612). They describe Pyke’s experiment as more ecologically valid than Roethlein’s because it represents ‘a closer approach to the normal situation’ (p. 607). However, they also discount Pyke’s experiment by referring to Pyke’s own doubts about his own experiments. Elsewhere in the paper they put it more bluntly, however without referring to Pyke explicitly: ‘... and the failure of any previous investigator to study the question under ordinary reading conditions led to the present study.’ (pp. 612–613).

Paterson and Tinker based their selection of typefaces on feedback from a large number of editors and publishers. On this background they selected the common typefaces Caslon Old Style, Scotch Roman, Garamont, Old Style, Bodoni, Cheltenham, and Antique. They added, in order to provide strong contrast, the typewriter-like typeface American Typewriter, the textura typeface Cloister Black, and the sans serif typeface Kabel light. All the typefaces are illustrated in the article (p. 608). The authors applied the Chapman-Cook speed of reading test (which includes a comprehension check), while displaying the typefaces in text set solid in 10 point nominal type size, at 19 pica line length, and read for 1 minute and 45 seconds.

Paterson and Tinker conclude that, except for the typewriter-like typeface and the textura typeface, all the typefaces in the experiment are

‘equally legible’ (pp. 609, 613). The tabulated results show that the sans serif typeface Kabel is 2.3 per cent less legible than the standard typeface in the experiment, Scotch Roman (p. 611). The authors emphasise the significance of the ‘findings’ for the printing industry (p. 612). They also point out that their results disagree with Roethlein’s results, but closely agree with Pyke’s conclusions.

The authors emphasise that the ‘ultra modern’ sans serif typeface Kabel came out ‘practically as legible as the other type faces in common use’ (‘however with a slight disadvantage’) (pp. 609, 612). To this they add that the sans serif typeface may have come out better if the subjects had been more used to read it.

The authors do however miss out on an obvious threat to the internal validity of their controlled single variable experimental design. The typographical variable ‘typeface’ is the independent experimental variable while the typographical variables linelength, interlinear spacing and (nominal) typesize are invariant. However, by applying the same interlinear spacing for each typeface, the results may very well be caused not by the difference in typeface, but by the differences in the ratio of the actual appearing typesize to interlinear spacing, or in the ratio of the x-height (the dominating dimension of a typeface) to interlinear spacing. At least, this will be a confounding factor.

Furthermore, the actual appearance size of Kabel light (as shown in an illustration) is very large in relation to the other typefaces (except for American Typewriter). Thus, the effective line space becomes comparatively small – to the possible disadvantage to Kabel light. Furthermore, among typographers in general, Kabel is not regarded as a sans serif typeface very well suited for continuous text, due to certain idiosyncratic design features such as the uneven distribution of internal space between the individual characters. It is certainly not a representative sans serif text typeface.

Webster and Tinker 1935 *

This article, 'The influence of type face on the legibility of print' by Helen A. Webster and Miles A. Tinker, was published three years after the previous article by Tinker that compares the relative legibility of various typefaces.¹⁹¹

The authors emphasise the study's potential usefulness to printing practice (p. 52). The rationale behind this experimental study is to compare the data generated by the speed of reading method (used in the previous study) with the variable distance method, by employing the same typefaces as last time. By the distance method the authors refer to the maximum distance from the eye which the stimulus material can be read. Five-letter words are arranged in four lines of four words each. The horizontal and vertical distances between the words are not given.

The authors conclude that the ranking order of the previous study hardly relates to the ranking order of the present study. This time, one of the least legible typefaces last time, American Typewriter, came first, 'read at a distance distinctly greater than the others.' (p. 46). This time Kabel light came out worse than last time, penultimate according to the ranking list, however not worse than 'slightly less legible than the Scotch-Roman standard' (against which all the typefaces were measured) (p. 46). The expression 'slightly less legible' refers to a difference in average distance between Kabel lite and Scotch Roman of 140.75 cm compared to 140.68 cm; i.e. less than one millimetre.

The authors point out that the theoretical and operational criteria differ between the two studies, and that the 'factors' which determine legibility may not be the same (or may even be the opposite) for reading at a distance as for reading continuous text under 'ordinary reading conditions'. They suggest that the relatively large (actual) size and the relatively large strokewidth of American Typewriter are probable determining factors for its good result in the present study. (The

* Helen A. Webster and Miles A. Tinker. 1935. 'The influence of type face on the legibility of print'. *Journal of Applied Psychology*, vol. 19, pp. 43–52.

191. Helen Webster did her M.A. thesis at Tinker's university, the University of Minnesota. The title of her thesis was 'The influence of type face and paper surface on the legibility of print' (Webster 1933).

relatively heavy typefaces Cheltenham and Antique also do well in this study). The authors also point to the ‘inadequate adjustment of spacing around the letters’ of American Typewriter¹⁹² as a determining factor for the poor result in the previous study. They explain this by pointing out that when text is carefully read as isolated words at a distance, the individual letters becomes more important, whereas when read rapidly as continuous prose under ‘normal conditions’, familiar word-forms and the context (surrounding word-forms) become more important. (pp. 49–51).

However, how can this explain the poorer result of Kabel light – another typeface with a large actual size – in the present study? By looking at the illustration that accompanies the previous study (by Paterson and Tinker), we can see that while Kabel light is rather large in actual size (but far from as large as American Typewriter, due to American Typewriter’s excessive x-height), it is one of the lightest, if not lightest, typefaces in the study. Thus, Kabel *medium*, nearer the weight of the other typefaces in the study, may have scored better.¹⁹³

Webster and Tinker are keen to stress that the clear differences produced by the two studies are not necessarily contradictory, but depend on the different methods employed. They further stress that this shows that different criteria of legibility cannot be used interchangeably.

The authors conclude ‘if ... perceptibility of words at the greatest possible distance from the reader is desired, and speed is less essential, American Typewriter type face is the best of the ten type faces used in this investigation’. However, they seem to have forgotten that American Typewriter had the largest *actual* size of the typefaces appearing in the study.

192. American Typewriter is a typewriter-like ‘monospaced’ typeface – not to be confused with the current ITC American Typewriter where the characters have individual set-widths.

193. However, whether Kabel medium existed in the USA at the time of the execution of the previous study (published in 1932) is not certain. According to Mac McGrew, Sans Serif Medium – the medium version of American Monotype’s series of copies of Gebrüder Klingspor’s foundry type Kabel – was designed sometime between 1930 and 1934, while the light version appeared in 1930 (McGrew 1994, p. 279). I assume that it was Monotype’s Sans Serif light which was employed in Tinker and his co-author’s studies, and not German foundry type. However, Tinker does not say anything conclusive about this.

Luckiesh and Moss 1937*

‘The visibility of various type faces’, published in the *Journal of the Franklin Institute*, by the engineers and prolific legibility researchers Matthew Luckiesh and Frank K. Moss at General Electric Company’s Lighting Research Laboratory, does not contain any references to the previous experiments described in this thesis. In their study they employ one of their methods, by measuring ‘the relative visibility’ with their Luckiesh-Moss Visibility Meter.

The article opens up with stating very much the same concerns as Griffing and Franz did in 1896, on impaired vision created by ‘visual tasks ... imposed by civilization’ (p. 77). They reason that such problems can be eased by correction of ocular deficiency, by proper lighting (not surprisingly, their affiliation considered), and by increasing visibility of reading matter (what the present study is about). ‘Since reading is a universal task in the home, schoolroom, and office, the visibility of the reading matter is of major importance.’ (p. 77).

Luckiesh and Moss acknowledge the validity of craft knowledge and the knowledge represented by existing artefacts to a larger extent than Tinker:

In general, an acceptable standard of type size has been more or less definitely indicated by the characteristics of good typography since these have been evolved through mass experience for generations. (p. 77)

They also stress that their results should not be taken too seriously and applied to all kinds of situations, for example where a less ‘visible’ italic typeface is used within continuous prose to catch attention, or where certain aesthetic considerations are important (p. 78).

The study employed 20 Monotype typefaces (of 8 typeface families) in 8 point size, including 3 variants of a sans serif typeface: Sans Serif light, medium, and bold (p. 79). The typeface Sans Serif light is probably identical to the Kabel ‘lite’ employed by Tinker. The serif typeface families included are Bodoni, Caslon, Cheltenham, Copperplate Gothic, Goudy, Cochon, and Garamond.

* Matthew Luckiesh and Frank K. Moss. 1937. ‘The visibility of various type faces’. *Journal of the Franklin Institute*, no. 223, pp. 77–82.

The 'Visibility Meter' employed is described in length by Luckiesh and Moss elsewhere.¹⁹⁴ It is criticised as inadequate (not measuring 'optimal' legibility, but only 'threshold' visibility) by Tinker (1963, pp. 9–10, 17). The main features of this apparatus are two circular filters where the density varies continuously when rotated in front of the eyes of the reading subjects, and it measures threshold visibility. Although the typefaces, the paper and the illumination is precisely described, the visual organisation of the 'stimulus material' is not. Due to the nature of the 'threshold' visibility experiment, the stimulus material is obviously not intended to be read continuously. However, illustrations and information elsewhere indicate that the 'stimulus material' may actually be continuous text.¹⁹⁵

The results show that the weight-variants of the Sans Serif typeface family are measured to have a better relative visibility than most of the equivalent weight variants of the roman typeface families (1937, p. 79). However, Luckiesh and Moss themselves do not discuss the results of the sans serif typeface variants to any extent. Of the four light typefaces (I am excluding the capital only typeface, Copperplate Gothic, from this and the next comparisons) Sans Serif light gets the second highest 'relative legibility' score – after Cochin Light (a mannered serif text typeface), but before Caslon Light and Goudy Light. Of the four medium weight typefaces, Sans Serif medium again gets the second highest legibility score – after Goudy Antique (a marginal difference), but before Cheltenham Wide and Bodoni Book. However, of the seven bold typefaces Sans Serif bold gets the lowest score.

Perhaps not unexpectedly, Luckiesh and Moss, working for General Electric as they did, throughout the paper emphasise that deficiencies with regard to visibility may be compensated by the level of illumination. Accordingly, they also express 'relative visibility' as 'footcandles required (for equal visibility)'. That is, a high score on 'relative visibility' matches a low number of 'footcandles required', and vice versa.

194. Luckiesh and Moss 1942, pp. 67–92, and inserted plate facing p. 67.

195. Luckiesh and Moss 1942, p. 398, insert facing p. 67, and insert facing p. 244.

Luckiesh and Moss 1942^{*}

Luckiesh and Moss' book *Reading as a visual task*, published in 1942, not only includes an exposition of their typeface visibility experiment from 1937 (see above), but also an exposition of another, until then unpublished visibility experiment (pp. 159–162). In this latter experiment (also based on the Luckiesh-Moss Visibility Meter) the typefaces employed were Caslon Old Face, Scotch No. 2, Texttype, the sans serif typeface Metrolite No. 2, Bookman, the newspaper typeface Excelsior, and the slab serif typeface Memphis Medium – all from Linotype. All typefaces were of ten points size, except for Caslon Old Face which was 11 points.

The sans serif typeface Metrolite No. 2 was ranked in the middle, 10.3 percent better than the one extreme Caslon Old Face, and 5.8 percent worse than the other extreme, Memphis Medium. By looking at the illustrated ranking (p. 161), it seems to be a clear correlation between weight and visibility – the rather bold Memphis Medium is ranked on top, and the very light Caslon Old Face on the bottom.

Tinker 1944^{**}

In 'Criteria for determining the readability of typefaces', published by Miles Tinker in *The Journal of Educational Psychology* in 1944, he tries to establish the superiority of speed of reading as a method for assessing legibility. It does however also introduce a fourth study by Tinker on typeface legibility, employing the same ten typefaces as in previous three studies (of which one is the preference study referred to in the section 'Subjective preference studies' in chapter 2). Tinker juxtaposes the results of the four studies – 'speed of reading',¹⁹⁶ 'perceptibility at a

* Matthew Luckiesh and Frank K. Moss. 1942. *Reading as a visual task*. New York: Van Nostrand.

** Miles A. Tinker. 1944. 'Criteria for determining the readability of typefaces'. *Journal of Educational Psychology*, vol. 35, no. 7, pp. 385–396.

196. Paterson and Tinker 1932. See the review in this chapter.

distance',¹⁹⁷ 'readers opinion',¹⁹⁸ and 'visibility' (the study in question here, based on the Luckiesh and Moss Visibility Meter) – and uses his juxtaposed interpretations to support in a well-articulated manner his advocacy for the speed of reading method (especially on p. 394).

In this fourth typeface study by Tinker, single five-letter words of ten points size were identified one at a time by each subject – through the rotating filters in the Luckiesh-Moss Visibility Meter. This time Kabel Light came penultimate, only 8.2 per cent better than the standard typeface in the test, Scotch Roman. Tinker points out that the typefaces with highest 'visibility' tended to be rather bold, thus Antique came first, 56.3 per cent better than Scotch Roman. Likewise, those typefaces with the lowest visibility tended to be rather light typefaces, cf. Kabel light.

These studies illustrate that the different operational methods employed produce incoherent results. Tinker claims that 'there is a slight tendency for the more visible faces to be read slower' and that a 'similar trend is found in comparing perceptibility and speed of reading.' (p. 391). Tinker points out that Scotch Roman with a very low score on both 'visibility' and 'perceptibility' 'is read as fast as all of the other commonly used typefaces.' (p. 392). Scotch Roman's score should however not come as a surprise. Its low score on 'visibility', and 'perceptibility' at a distance, may very well be due to the fact that it is the smallest of the ten typefaces with regard to actual size. In reading continuous text, a slight difference in size may not be so decisive, and may furthermore, due to the fact that all ten typefaces have the same nominal interlinear spacing, be counteracted by its relatively large and generous actual interlinear spacing.

197. Webster and Tinker 1935. See the review in this chapter.

198. Paterson and Tinker 1940a, pp. 18–20. See the section 'Subjective preference studies', in chapter 2.

English 1944^{*}

The year 1944 also saw another typeface legibility study published: ‘A study of the readability of four newspaper headline types’, by Earl English in *Journalism Quarterly*.¹⁹⁹ This article was based on a PhD dissertation at the department of psychology at the State University of Iowa. The article gives place to a thorough discussion of theoretical and operational criteria of legibility. The author refers to work on typeface legibility by Roethlein; Paterson, Tinker and Webster; and Luckiesh and Moss. What is more, the author has also corresponded with the type designer Frederic Goudy, and received material support and advice from the typeface expert Douglas McMurtrie of the Ludlow Typograph Company.

English measured the number of words read in three-line headlines of three different sizes (14, 24, and 30 points) during brief exposures through a tachistoscope. The typefaces employed were the serif typeface Bodoni bold, the slab serif typeface Karnak bold, and the sans serif typeface Tempo medium, all typefaces in common use.²⁰⁰ Specimens of the actual ‘stimulus material’ are not shown in the article.

Interestingly, this author acknowledges that by controlling all ‘factors’ (i.e. typographic and environmental variables) in order to isolate the effects of the independent experimental variable ‘one type may gain an advantage in the comparison by being better suited to the controlled conditions than another’ (p. 222). Thus, the line-spacing in this experiment was adjusted differently for the different typefaces: ‘The Ludlow type designing department considered spacing effects equalised

* Earl English. 1944. ‘A study of the readability of four newspaper headline types’. *Journalism Quarterly*, vol. 21, pp. 217–229.

199. This was the first of several legibility studies published in *Journalism Quarterly* in the post-war period. The journal carried its first mention of legibility research in 1941, with Douglas McMurtrie’s review of Paterson and Tinker’s *How to make type readable* (Neman 1968, p. 3).

200. The Karnak typeface family, designed by Robert Hunter Middleton at the Ludlow company, was derived from Linotype’s Memphis. The first versions of the Tempo family, by the same designer, was introduced in 1930 as an answer to the popular Futura and other Futura-like typefaces of that period (McGrew 1994, pp. 193, 303–304). Nevertheless, note that Tempo is distinctly different from Futura. It is not an imitation.

under this arrangement.’ (p. 223). This experimental design therefore represents an important but rare insight in legibility research. It also indicates an experiment with a high degree of internal validity, at least with regard to the stimulus material’s influence on the internal validity. In legibility research, the only way to control possible nuisance variables is not to hold them constant, but to adjust differently in an attempt to equalise ratios and relationships among the experimental typographic variable and other typographic variables. Thus, the design of English’s experiment contains (at least in principle) an answer to the criticism of legibility studies relying on single variable (univariate) laboratory tests where only one typographic variable is manipulated as the independent experimental variable while other typographic variables are constant, whereas in the real work of typographic design a large number of variables interact simultaneously in a complex way.

English also arranged a readers’ preference test, but discarded the results as accidental. And in another experiment, designed like the first, he tested family variants of Intertype’s serif typeface Cheltonian, in bold, bold condensed, and bold capitals only.²⁰¹

According to the results reported, Bodoni bold, Tempo bold, Cheltonian bold, and Cheltonian bold condensed was read ‘significantly’ faster than the Karnak medium and Cheltonian bold capitals only. The sans serif typeface Tempo bold got the highest score, however not ‘significantly’ higher than Bodoni and Cheltonian.

201. Cheltonian was a copy of American Typefounder’s Cheltenham. Except in a clarifying footnote, English consistently refers to Cheltonian as Cheltenham. ATF’s Cheltenham condensed was used for the size of 30 points.

Burt, Cooper, and Martin 1955^{*} / Burt 1959^{}**

Cyril Burt's and (fictitious?²⁰²) co-author's once acclaimed article 'A psychological study of typography' was published in 1955 in the journal Burt edited, *The British Journal of Statistical Psychology*. Burt's almost identical monograph, bearing the same title, with a foreword by Stanley Morison, was published by Cambridge University Press in 1959. The two publications' positive reception until the early 1980s, among both researchers and designers, have been thoroughly dealt with by Rooum (1981) and Hartley and Rooum (1983). They have convincingly shown that Burt's well known dubious if not fraudulent practices²⁰³ also extended into his work on legibility and typography.²⁰⁴

Neither version of 'A psychological study of typography' present original experimental research on the legibility of sans serif typefaces. Burt briefly discusses the legibility of sans serif typefaces in a footnote in the article (1955, p. 56n). In another footnote Burt mentions preliminary experiments involving four named typefaces 'found to be much inferior', including the sans serif typeface Gill Sans (1955, p. 32n). The former footnote is then expanded into four additional paragraphs plus a footnote in the monograph (1959, pp. 8–9).

Burt opens up his discussion by claiming that the 'four most usual arguments' in favour of sans serif typefaces are 'all unsupported by experimental evidence.' He then argues vehemently against the use of sans serif typefaces; while supporting his *a priori* arguments with subtle hints which might give the reader the impression that his views are based on scientific evidence. For example by employing the expression 'as psychologists have so often pointed out'. In order to substantiate his

* Cyril Burt, W.F. Cooper, and J.L. Martin. 1955. 'A psychological study of typography'. *The British Journal of Statistical Psychology*, vol. 8, pt. 1, pp. 29–57.

** Cyril Burt. 1960. *A psychological study of typography*. With an introduction by Stanley Morison. Cambridge: Cambridge University Press. (Reprinted by Bowker in 1974 for the College of Librarianship, Wales). See also Burt's article 'The readability of type' in *New Scientist* (1960a).

202. See Hartley and Rooum 1983, p. 209.

203. See for example Hearnshaw 1979.

204. Robert B. Joynson's long attempt to rehabilitate Burt (Joynson 1989), ignores Rooum and Hartley's articles. Rooum and Hartley's articles are not mentioned in Mackintosh's collection of articles (1995).

views, Burt cites two named sources (Kerr 1926, and Crosland and Johnson 1928), which when examined, turn out not to confirm his claims.²⁰⁵ Interestingly, Burt does not refer to Pyke, nor to the prolific writings of Tinker, a natural frame of reference for a study like *A psychological study of typography*. The picture of dubious rhetoric and dubious citation-practice is in stark contrast to Beatrice Warde's²⁰⁶ melodramatic review²⁰⁷ in *Penrose Annual* of Burt's article:

let me now tell you neighbours that the most reassuring and welcome document has come to the attention to the typographic fieldworkers this year (judging by the cries of 'At last!' with which they received it) is one which came fluttering down almost by accident from the very Other Hemisphere to that of the arts, namely that of the sciences. ... It was welcomed by practising typographers as the first scientific study of their subject ...[and] it was hailed as *surprisingly* helpful, coming as it did from the realm of science. (Warde 1956)

It is however worth noticing that Burt at least in one respect displays admirable sensitivity, compared to most previous legibility researchers – by pointing out that the x-height is the most dominant dimension of a typeface, and in addition, by pointing out that nominal typesize do not necessarily correspond to actual typesize (1955, p. 32; 1959, p. 6). It is reasonable to believe that Burt's frequent contact with typeface expert Beatrice Warde played a role here.

Burt's biased exposition is reinforced in Stanley Morison's foreword, where over several pages Morison proclaims the superiority of serif typefaces (pp. xi–xv). Morison appeals to the authority of scientific investigation, as well as appearing broadminded and unprejudiced:

Thus, while the investigators found that, in terms of word-recognition, the design known as sans serif 'was the worst of all', and confirmed what other investigators had proved, that 'serifed letters are more legible than unserifed' (p. 9), it does not follow that sans serif letters are never appropriate in any typographical composition, literary or otherwise. (p. xi)

205. See Hartley and Rooum 1983, p. 205; and the section 'Crosland and Johnson 1928: the serifs that never were' in this chapter.

206. See Hearnshaw for an exposition of the extraordinary relationship between Warde and Burt, exchanging hundreds if not thousands of letters (1979, pp. 199–203). Badaracco suggests that Beatrice Warde saw a potential in Burt's publications for the marketing of Monotype's typefaces (1995, pp. 78, 81).

207. Beatrice Warde also reviewed the book in the London *Times* (Badaracco 1995, p. 223).

Unfortunately for Morison both the ‘finding’ and the ‘proof’ in the above quote are the ones which in fact turned out not to confirm Burt’s claims. Given this background, the final pompous conclusion in Morison’s foreword becomes absurd:

No degree of consistency with an ideological, artistic, or intellectual theory^[208] is an acceptable substitute for conformity with verifiable paleographical, physiological and psychological facts. (p. xix)

Soon after the book was published, P.M. Handover in an extensive article on the history of sans serif typefaces, published in *Motif* and elsewhere,²⁰⁹ referred to Burt’s book and bluntly turned Burt’s claims into the following statement

The British psychologist, Sir Cyril Burt, has recently recalled and reaffirmed scientific findings that ‘for word recognition, a sans serif type face was the worst of all’. (Handover 1961, p. 81)

This statement by Handover was in turn used as evidence by Reynolds Stone, against the use of sans serif letterforms, in a public debate on the lettering on British traffic signs:

Fashionable or not, the use of sans meant ignoring experts like the psychologist Sir Cyril Burt, who ‘has recently recalled and reaffirmed scientific findings that “for word recognition a sans serif type face was the worst of all”.’ (see *Letters without serifs* by P.M. Handover in *Motif* 6). Presumably motorists, as well as *Motif* readers, are concerned with ‘word recognition’.²¹⁰

In modified versions of Handover’s article, that appeared in *Monotype News Letter* and *Revue suisse de l’Imprimerie*, Handover added yet another dubious piece of information, by paraphrasing from a then more recent article which Burt had published in *Yearbook of Education* (i.e., Burt 1960b):

Independent investigations in the U.S.A. support the result of British experiments. (Handover 1963, p. 9)

208. That is, modernist advocacy of common use of sans serif typefaces. My comment.

209. Of which in 1963 a slightly altered version was published both in *Monotype News Letter* (Handover 1963) and in *French in Revue suisse de l’Imprimerie* (the French language section of the Swiss journal *Typografische Monatsblätter*). The article appeared in other guises as well.

210. As an invited comment to Christie and Rutley 1961, p. 61 (appearing in the same issue).

Interestingly, more or less the same phrase also appeared a few years later in another context: in 'Map design and typography', an unsigned article in a special issue of *Monotype Recorder* entitled 'Precision in map making':

Scientific investigation has shown sans serif to be the worst of all type styles for word recognition. (*Monotype Recorder*, vol. 43, no. 1, p. 45)

As mentioned above, the reception history of *A psychological study of typography* has been thoroughly dealt with by Rooum (1981) and Hartley and Rooum (1983).²¹¹ However, in spite of their devastating critique in the early 1980s, Burt is still often cited in a positively or at least uncritical manner in contexts such as typography or information design.²¹² These citations are not necessarily hidden away in obscure privately published pamphlets or in printing industry trade journals. Examples can be found in recent issues of refereed journals such as *IEEE Transactions on Professional Communications* and *Safety Science*, and in the two most prominent and acclaimed textbooks on information design available today: Karen Schriver's *Dynamics in document design: creating texts for readers*, and Kostelnick and Robert's *Designing visual language: strategies for professional communicators*. These citations range from 'innocent' uncritical citations where Burt is treated as any other source of knowledge, but not necessarily with an explicit acceptance of his 'findings', as in Schriver's textbook; to positive endorsements as in Kostelnick and Robert's textbook, where Burt is described as an

211. I dare to mention two prominent publications that Rooum and Hartley overlooked. The American textbook *The graphics of communication*: 'Burt has offered evidence' (see Turnbull and Baird 1980, pp. 87–88). And the 1960 and 1979 editions (and 1996 reprint) of the well known *Glaister's glossary of the book*, also published as *Encyclopedia of the book*: 'Important results have been achieved' (Glaister 1979, p. 278).

212. See for example Gribbons 1991, pp. 44, 49; Kostelnick 1995, pp. 185–187; Kostelnick and Roberts 1998, pp. 142, 204; Pedersen and Kidmose 1993; Silver and Braun 1993, pp. 623f. For more such references, but specifically on the legibility of sans serif typefaces, see for example Gallagher and Jacobson 1993, p. 101; Lubell 1993; Schriver 1997, pp. 274, 276, 301. Badaracco represents an exception (she is aware of that Burt has been discredited). She nevertheless writes uncritically about the existence of experiments on the legibility of sans serif typefaces allegedly performed by Burt (Badaracco 1995 pp. 78–80, 223; cf. Hartley and Rooum 1983). Watts and Nisbet represent an early exception. They cite Burt, but add cautiously: 'Unfortunately, many of the opinions expressed are not substantiated by empirical evidence.' (1974, p. 32).

‘excellent source’. And similarly, in a recent paper in the *IEEE Transactions on Professional Communications*, Kostelnick states that

Studies in the legibility of typefaces, such as those conducted by Tinker and by Burt, represent the quantum physics of document design because they address basic performance questions about typefaces. (Kostelnick 1995, p. 187)

For a more complete picture of the reception of Burt’s book, why not include an annotation in a recent catalogue from a British antiquarian bookshop that specialises in books on typography and printing:

A landmark experimental survey of the mind’s capacity to take in a printed message. The first test of accuracy and speed of reading reveals the legibility of various styles of setting. The second test considers choice of typeface. (Wakeman 1999, p. 48)

To conclude: Burt’s suspect contribution to typography has been used for what it is worth by some, and is still alive in some quarters.

Christie and Rutley 1961: * ***The legibility of traffic signs and the public debate on Jock Kinneir’s Motorway alphabet***

In August 1961, two researchers at the Road Research Laboratory in Britain, published a paper on the ‘Relative effectiveness of some letter types designed for use on road traffic signs’ (Christie and Rutley 1961a). It appeared in *Roads and Road Construction*. A shorter version was published in *Design* the same month (Christie and Rutley 1961b). These two papers represented the culmination of a vigorous public debate on letterform legibility which had been going on since March 1959. The controversy and the Road Research Laboratory’s subsequent experiments happened in connection with the introduction of direction signs for Britain’s new high speed motorways.

* A.W. Christie and K.S. Rutley. 1961a. ‘Relative effectiveness of some letter types designed for use on road traffic signs. *Roads and Road Construction*, no. 39, August, pp. 239–244; and A.W. Christie and K.S. Rutley. 1961b. ‘[Research] on road signs’. *Design*, no. 151, August, pp. 59–60.

The design of these directional and other informational motorway signs represented the first phase of an overall development of a new coherent system of traffic signs in Britain between 1957 and 1963. The new system was a late British adaptation to (but not an adoption of) European practice and the UN Geneva protocol of 1949 (see Froshaug 1963). The new British system contained a large number of innovative direction signs of several categories for use on motorways (including direction signs on 'all-purpose' roads pointing to connected motorways), direction signs for use on major 'all-purpose' roads, and direction signs for use on local 'all-purpose' roads. The new British system also included basic categories of largely pictorial traffic signs (both iconic and symbolic), such as mandatory signs, prohibitory signs and warning signs. Although modified and redrawn, these signs were more directly adopted from the 1949 Geneva Protocol than the direction signs.²¹³

The Motorway alphabet

The Anderson advisory committee on traffic signs for motorways (1957– 1962), set up by the Ministry of Transport, consulted a large number of interested organisations and it also did something which was then groundbreaking.²¹⁴ in June 1958 it appointed a professional designer. Jock Kinneir – who by then had already designed the signing system for Gatwick Airport²¹⁵ – designed, on the basis of the committee's broad recommendations, the elegant and innovative motorway direction signs, as well as their accompanying letterforms.

213. For background information on this development, see Spencer 1961 (an illustrated article depicting the chaotic and insular traffic sign situation in Britain before this work started); Ministry of Transport 1962 (The Anderson report on traffic signs for motorways); Ministry of Transport 1963 (The Worboys report on traffic signs for 'all-purpose' roads); Froshaug 1963 (a profusely and systematically illustrated article on the historical development of traffic signs, leading up to the new British traffic sign system based on the recommendations of the Worboys committee); Krampen 1983 (a special issue of *Semiotica*, devoted to the origin and development of road sign systems in a broad international context), Department of Transport 1991 (a brief history of traffic signs in Britain); Department of Transport 1994 (a design-manual for British directional traffic signs); Department of Transport 1995 (a comprehensive all-colour exposition of the current British traffic signs).

214. Froshaug 1963, p. 50; also Krampen 1983, p. 110.

215. See Kinneir 1970, pp. 15–16.

Kinneir, who had rejected on ‘aesthetic grounds’ the committee’s initial wish²¹⁶ to employ the German DIN sans serif lettering of capitals and small letters,²¹⁷ designed a new sans serif alphabet. It resembles and was probably based on the typeface Berthold Grotesque from the German type foundry Berthold.²¹⁸ Like the DIN alphabet, Kinneir’s new design is characterised by obliquely cut terminals as well as relatively open semi-enclosed counters, of letters like a, c, e and s. However, the DIN alphabet suffers from having rather narrow and rectangular letterforms, and further, from inconsistent orientations with respect to how the terminals of its capitals with semi-enclosed counters are cut.²¹⁹

Kinneir’s new design got the committee’s full support, as well as ‘almost unanimous’ support from the consultative organisations.²²⁰

216. ‘We have as a committee got into the habit of accepting the general weight & appearance of the German alphabet as being the sort of things we need! I think therefore something on these lines is what the committee believes it wants.’ Letter from Colin Anderson to Jock Kinneir, dated 26 June, 1958. (Location: Margaret Calvert, London.)

217. Kinneir 1971, p. 6; 1984, p. 344.

218. Berthold Grotesque was issued in 1928. Nobel Grotesque (1929) from the Dutch type foundry Amsterdam is a similar typeface. Specimens of both typefaces are shown in Jaspert *et al.* 1970 (pp. 254, 312–313).

219. The numerals for use in route numbers only (and ditto letters A, B, and M) got a different design from the main numerals which were used for stating distances and later for the numbering of motorway exits (see illustrations in Ministry of Transport 1962, pp. 40–41, as well as illustrations in Department of Transport 1995, pp. 47–51). David Kindersley’s comment: ‘The road numbers, together with their letters, are even worse than the main alphabet, and do not conform with the simplest rules of legibility or differentiation.’ (Kindersley 1960, p. 465).

220. Ministry of Transport 1962, p. 4. Specimens of the alphabet appeared, without any name attached, in the Andersen committee’s report (i.e., in Ministry of Transport 1962, pp. 35–39). In more recent literature it is referred to as ‘the Motorway alphabet’ (Department of Transport 1994, p. 4). Jock Kinneir was subsequently engaged as designer to the Ministry of Transport’s Worboys committee (1961–1963) on traffic signs for ‘all-purpose’ roads. In the Worboys report specimens of two slightly modified variants of the same letterforms carried the names ‘Transport Medium’ (on a dark background) and ‘Transport Heavy’ (on a light background) (see Ministry of Transport 1963, pp. 97–102). However, an alternative style for route-numbering was not included for ‘all-purpose roads’, as for the motorways (see the previous footnote).

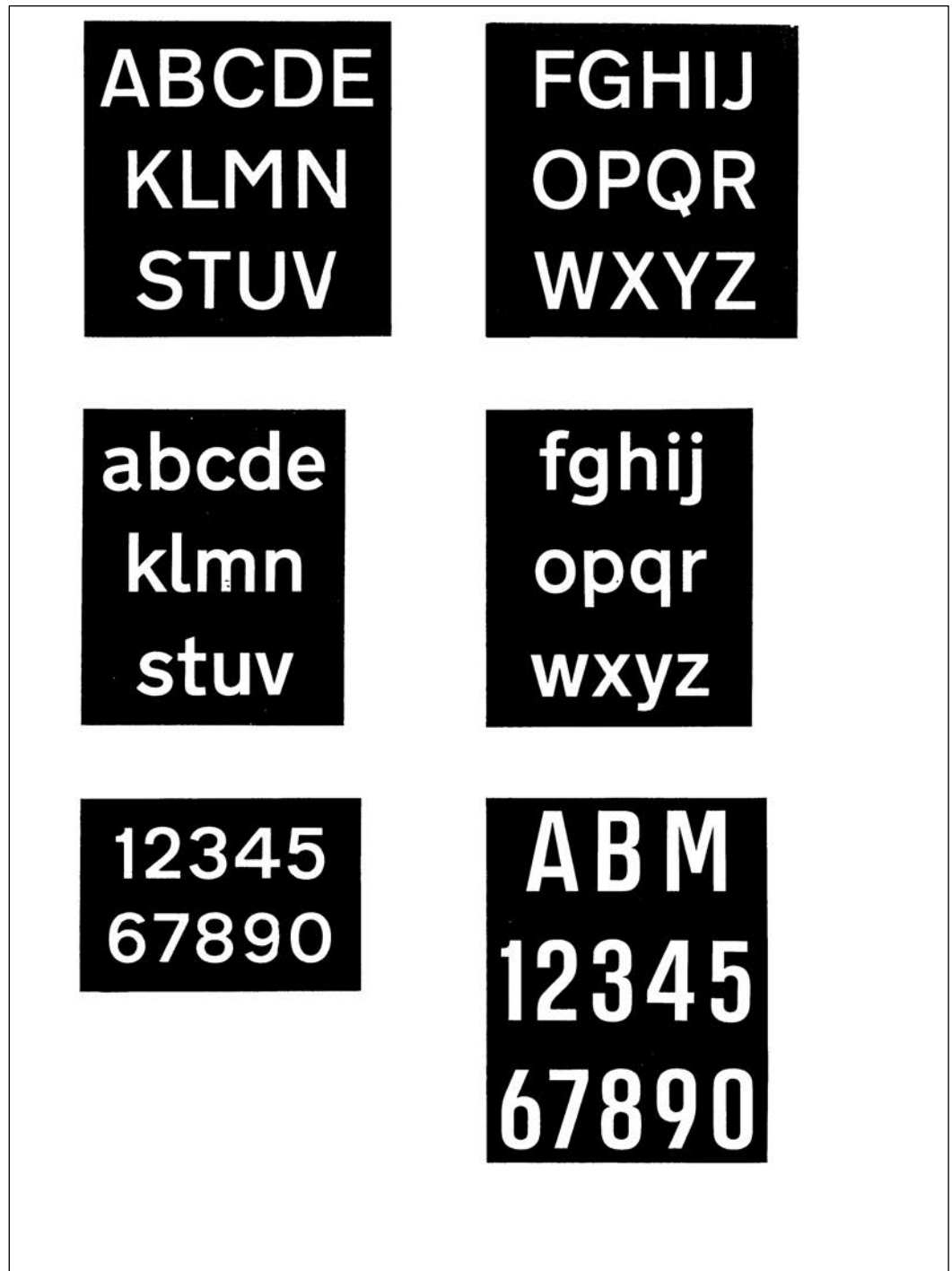


FIGURE 3. Jock Kinneir's Motorway alphabet. The letters ABM and numerals in the lower right corner are for route-numbers only. The original background colour is blue. (From: Ministry of Transport 1962, pp. 36–41: figures 1–4.)

When designing his sans serif alphabet, Kinneir performed informal ‘low-tech’ experiments: with reflective material in an underground garage in order to determine a sensible weight; in Hyde Park in London in order to determine sensible appearance-widths and a sensible x-height; and in addition, experiments in order to create an appropriate letterspacing system. Kinneir later commented on other aspects of the creation of the sans serif letterforms in question:

The basis of the letter design was the need for forms not to clog when viewed in headlights at a distance. For this reason counters ... had to be kept open and gaps prevented from closing. Also, as pointillist painting has shown, forms tend to merge when viewed from a distance, and this suggested a wider letter spacing than is usual in continuous text. (Kinneir 1984, p. 344)

The public controversy

It was exactly this lettering that – after appearing on the first experimental motorway signs put up in 1959 – provoked public controversy on letterform legibility. The controversy was reflected in the columns of publications diverse as *The New Scientist*, *Design*, *Roads and Road Construction*, *Traffic Engineering & Control*, *The Times*, *The Daily Telegraph*, *The Observer* and *Cambridge Daily News*.²²¹ In particular

221. The following chronological list does not claim to be exhaustive. Letters to the editor in *The Times* (1959): March 17, p. 11 (by Brooke Crutchley, printer to the University of Cambridge; attacking Kinneir’s work); March 20, p. 13 (by Noel Carrington, a designer and member of the advisory committee; supporting Kinneir); March 24, p. 13 (by J[ohn]. G. Dreyfus, and by G.S. Bagley, both attacking Kinneir’s work). Editorial note in *New Scientist*: vol. 5, no. 124, April 2, 1959, p. 731. Main editorial article, ‘Which signs for motorways?’, in *Design* no. 129, Sept., 1959, pp. 28–32 (*Design* 1959). This article reported a discussion organised by the journal, with contributors such as: a car manufacturer, a traffic sign manufacturer, the Ministry of Transport, the Road Research Laboratory, a landscape architect, Dr E.C. Poulton from the Medical Research Council’s Applied Psychology Research Unit in Cambridge, David Kindersley, Jock Kinneir, as well as other designers, typographers and Brooke Crutchley. Letters to the editor in *Design*, no. 132, Dec. 1959, p. 71 (by Herbert Spencer, and Aidron Duckworth); no. 133, Jan. 1960, pp. 75, 77 (by Ernest Hoch, and Norbert Dutton). An article by David Kindersley in *Traffic Engineering & Control*, Dec. 1960, pp. 463–465 (Kindersley 1960). A note in the ‘Peterborough column’ in the *Daily Telegraph*, 8 March, 1961, and a follow up note a week or so later. Kindersley also appeared in *Cambridge Daily News*, March 9, 1961. →

the radical solution of employing small letters with initial capitals – never before used on British road signs²²² – instead of using capitals only, was heatedly debated. Nevertheless, the use of sans serif lettering instead of serif was also debated in this unique instance of a public debate on letterform legibility.

The somewhat disgruntled²²³ opposition to Kinneir's solution was led by the 'traditionalist' letter cutter David Kindersley, and Brooke Crutchley, printer to Cambridge University, both linked to the British inter-war/mid-century typographic establishment (Stanley Morison and others). I can only guess that this 'establishment', as well as being suspicious of continental modernism, must have been seriously offended by the fact that a major national lettering project had been initiated and partly implemented without having been consulted. It was asserted that Kinneir's solution ignored specialised 'knowledge accumulated over the years' and that there had been some 'misguided work behind the present proposal'.²²⁴

It was argued that lettering for destination names on sign panels is more legible in capitals than in small letters,²²⁵ in spite of that words in small letters are more legible for continuous text in books. (It was argued that words in small letters make both irregular wordshapes *and* familiar word patterns in continuous text, and that this aids recognition.) The

In March 1961 BBC planned a debate between Kindersley and Kinneir in the 'Tonight' TV-program. However, the Ministry of Transport advised Kinneir not to participate, while reassuring Kinneir that he had the committee's full support. (Letter from Kinneir to Ministry of Transport, dated 17 March, 1961. Letter from Ministry of Transport to Jock Kinneir, dated 24 March, 1961. RTC.51/2/03.) (Location: Margaret Calvert, London.)

Subsequently, in August 1961, the two papers by Christie and Rutley at the Road Research Laboratory, was published in respectively *Roads and Road Construction* (1961a) and *Design* (1961b). To the article in *Design* were attached comments by the designers Herbert Spencer, Reynolds Stone and Colin Forbes. Letters to the editor of *Design* followed up the debate in subsequent issues: no. 154, October 1961, pp. 87, 89 (by Hans Schmoller, and by David Kindersley); no. 155, November 1961, p. 77 (by Colin Forbes); no. 156, December 1961, pp. 81, 83 (by A.G. Long).

222. Christie and Rutley 1961a, p. 239.

223. An unpleasant aspect of David Kindersley's argumentation was his public insults directed against Kinneir, alleging that Kinneir lacked competence and skill as a designer (see Kindersley 1960).

224. Brooke Crutchley in *The Times*, March 17, 1959, p. 11.

225. See especially the elaborate, but also occasionally dubious and rambling argument in Kindersley 1960.

reason for preferring all-capitals for sign panels, according to the same argument, is that horizontal eye movements is not an issue on sign panels, and that place names are not familiar word patterns.

It was also argued that capitals are intrinsically clearer than small letters, especially when compared in the same nominal size. Therefore, the argument went, all-capital lettering would allow for considerably smaller sign panels and therefore give large benefits with regard to the cost of production, as well as creating less impact on the landscape. The reasoning behind the size argument was that as long as the dominant dimension of capitals (baseline to top of capitals) is bigger than the dominant dimension of small letters (the x-height), big conspicuous all-capital lettering could be applied in a given area without the need to allocate space for ascenders and descenders.

It was further argued that serifs would strengthen terminals and thus define letters more clearly from a distance. David Kindersley actually proposed a theory on how serifs improve the legibility of letterforms in certain situations:

Try reading a page of sans-serif lower-case, and then a page of 'normal-face' and you will see at once that the normal one is more readable. The reason for the existence of the serif is clear, and is not just a meaningless tradition. In very small type, or in larger letters to be read at a great distance – in fact, wherever there is a question of distance in relation to size – there is always a loss of definition. The serif reinforces the individual character of the letter exactly where this loss is greatest. (Kindersley 1960, p. 465)

Against the arguments of the traditionalists, it was argued by supporters of Jock Kinneir that words with initial capitals and small letters with ascenders and descenders provided more differentiated as well as more familiar word-shapes, as opposed to the rectangular and monotonous shapes of all-capital words. Words would therefore be easier to recognise from a distance. It was also claimed that serifs and modulated strokes are not very well suited for reflective material.

The experiments

Comparative experimental research was recommended by several of the participants in the debate,²²⁶ and the experiments undertaken by Christie and Rutley for the Road Research Laboratory were, apparently, conducted as an answer to this demand. To put these experiments in perspective it should be taken into account that Kindersley's challenge was to a process that was already under way. It was not as if the job had been conceived as a competition between designers as in big architectural projects. The Anderson committee was fully in favour of Kinneir's proposal – based on parameters set by the committee itself. I think it is correct to say that the committee had no plans whatsoever abandon Kinneir's solution, regardless of the outcome of the Road Research Laboratory's experiment. The Kindersley 'challenge' was regarded with irritation in the Anderson Committee as well as in the Ministry of Transport. As pointed out in a letter to Jock Kinneir from the ministry:

You are already in a strong position vis-a-vis your detractors; it is you who were commissioned by the Department to do the job, it is your signs that have been erected on the motorways, and you can be sure of the solid support of the Committee for what you have done.²²⁷

In fact, Kinneir and the committee members tacitly regarded Kindersley as an indefatigable crank who carried out a campaign based on 'tendentious claims and half-truths' against both the committee's and Kinneir's work.²²⁸

226. Both parties claimed support from existing experimental research (mainly on the capitals versus small letters question, but also on the serif vs. sans serif question). See Noel Carrington in *The Times*, March 20, 1959, p. 13; John Dreyfus in *The Times*, March 24, 1959, p. 13; *Design* 1959, p. 30; and Kindersley 1960, p. 464. See also a reference to existing research in the Anderson report (Ministry of Transport 1962, p. 4), as well as a comment from Reynolds Stone in a comment to Christie and Rutley's paper in Christie and Rutley 1961b (p. 61); and a letter to the editor from David Kindersley, in *Design*, no. 154, 1961, pp. 87, 89.

227. Letter from the Ministry of Transport to Jock Kinneir, dated 24 March, 1961, Ref. RTC.51/2/03. (This is the letter where the ministry advised Kinneir not to participate in a TV-debate with Kindersley.) (Location: Margaret Calvert, London.)

228. See for example the letter from Jock Kinneir to the Ministry of Transport, dated 17 March, 1961; and a letter from the committee chairman Colin Anderson to Fred Salfield of the *Daily Telegraph*, dated 8 March, 1961, where he complained about the misinformed pro-Kindersley coverage in the newspaper's 'Peterborough' column the very same day. (Location: Margaret Calvert, London.)

Kindersley had stated optimistically that: ‘No decision should be finally and publicly announced on the M1 signs until the facts are established by the Road Research Laboratory’ (Kindersley 1960, p. 465). Similarly, Dr E.C. Poulton from the Applied Psychology Research Unit of the Medical Research Council, who participated in the discussion organised by *Design*, ‘was in no doubt that the facts could be established – providing the criteria could be agreed in the first place’ (*Design* 1959, p. 129).

Four different types of lettering were employed in the experiments:

- sans serif letters of capitals only, based on designs by Edward Johnston, commissioned by the Road Research Laboratory, from David Kindersley
- serif letters of capitals only, commissioned by the Road Research Laboratory, from the chief critic of Kinneir’s solution, letter cutter David Kindersley
- Jock Kinneir’s sans serif small letters (with initial capitals); by then already employed on the M1 motorway
- the same letters by Kinneir as above, but in a smaller size and applied with more interlinear space and more generous margins



FIGURE 4. One of the 24 basic signs that were used in the experiment (here shown in four alphabets). The total number of signs were 96, based on four alphabets and 24 basic signs (6 single-name destination signs, 6 two-name destination signs, 6 three-name destination signs, and 6 message signs like ‘Stop’ and ‘No entry’).

From the top: Kindersley’s ‘Edward Johnston’ alphabet; Kindersley’s own serif alphabet; Kinneir’s sans serif alphabet; and Kinneir’s sans serif alphabet with generous spacing. (From: Christie and Rutley 1961a, p. 240: figure 3c.)

The aim of the experiment was to find out which of these types of letterforms could be read at the greatest distance in order ‘to keep the angle between the driver’s line of sight and the road ahead as small as possible’ (Christie and Rutley 1961b, p. 59). However, the experiment also aimed at investigating the question of capitals versus small letters,

and in addition, investigating ‘the value of serifs ... because it has been suggested ... that serified lettering is more legible than sans-serif lettering’ (1961a, p. 240). The authors pointed out that the question of sans serif lettering versus serif lettering was only examined with respect to the two capital lettering styles employed (since no small letters with serifs were included in this multi-variable experiment).

Altogether 6000 reading distances were recorded. The experiments were conducted literally on a field. In order to speed up the experiments the signs were attached to a car moving towards stationary observers instead of the opposite natural way. Christie and Rutley sensibly pointed out that this reversing should not affect the relative order between the letterforms.²²⁹ The size of the letters on the test signs were around five times smaller than for Kinneir’s real signs already in use on the M1 motorway. However, absolute size and thus absolute distances, was not a question here, and had been dealt with experimentally earlier (see *Design*, no. 129, 1959). Sizes were probably anyway expected to vary for different applications and situations.

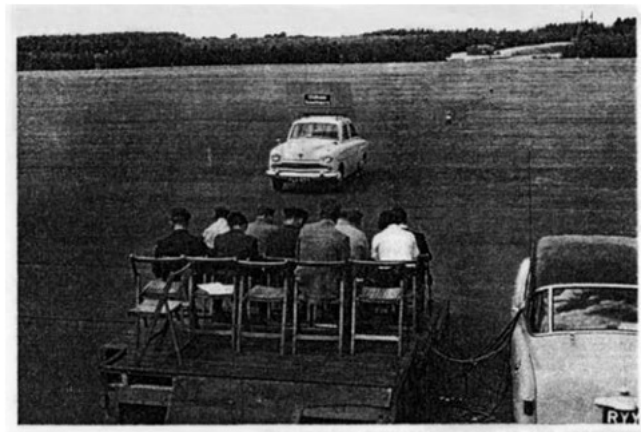


FIGURE 5. ‘The experiment was greatly speeded up by mounting the signs on a vehicle and driving them towards a group of 10–15 stationary observers seated on a tiered platform.’ (From: Christie and Rutley 1961a, p. 242: figure 4.)

229. Although Cohen (1981), according to Hughes and Cole (1986), demonstrated that eye movement behaviour is different in a laboratory situation from when actually driving on the road, Hughes and Cole claim that the pattern of eye movements is not a critical factor with regard to object conspicuity (Hughes and Cole 1986, pp. 1108–1109).

The mean reading distances were, in descending order:

- 247 ft. for David Kindersley's serif capitals
- 240 ft. for Jock Kinneir's sans serif small letters with initial capitals
- 239 ft. for the Edward Johnston-based sans serif capitals
- 212 ft. for Jock Kinneir's letterforms in a smaller size

Discussion

While discussing their results, Christie and Rutley pointed out that the difference in favour of Kindersley's serif capitals to the two sans serif lettering styles was statistically significant – 'about 3 per cent ... i.e. the difference is unlikely to be due to chance'. However, they added that spacing, layout, and width to height ratio of the letters, might could have been confounding factors. Nevertheless, the authors seem to have overlooked that – according to figures given in the text – the height of Kindersley's serif capital letters is at least 25 per cent larger than the dominating dimension of the largest version of Kinneir's sans serif letters, their x-height. Thus, since reading distance to a large extent is a function of size, adjusting upwards the size of Kinneir's sans serif letters, to a point somewhere on the 0 to 25 per cent range differential, while taking into consideration the size of the initial capitals in each destination name as well as the (relative short) ascenders and descenders, would undoubtedly produce results more in favour of Kinneir's letterforms.

However, the Road Research Laboratory, seemed taken in by Kindersley and his supporters' heavily promoted view that the competing letterforms should be compared while positioned tightly into areas of equal size hardly without surrounding space, with reference given to the production cost per square unit of a sign. Under this unrealistic condition, capitals (that is, letters without ascenders and descenders) will unavoidably create a more prominent visual image than small letters. This is because small letters will have to accommodate extra interlinear spacing for ascenders and descenders. Their visual size, expressed by their dominant dimension, the x-height, will under this condition have to be left smaller. Not only does this 'tightly crammed on

an equal area' argument rely on an unrealistic condition (both printed matter and sign panels usually rely on relatively large areas of space around the text), it also disregards a basic heuristic rule among designers: that interlinear spacing needs to be larger for text in capitals than for text in small letters. Furthermore, it also disregards the fact that capitals are wider than small letters and thus obviously need considerably more space width-wise, something which might become a critical factor on sign panels with long destination names. Some destination names even needed to be abbreviated or contracted on some sign-panels.²³⁰ Nevertheless, the Anderson report concluded this debate the following way:

We feel, however, that in designing a traffic sign regard must be paid to the space around the lettering as well as to the lettering itself, and that a sign that completely filled the space available would be so unattractive as to be quite unacceptable. (Ministry of Transport 1962, p. 4)

Kinneir reasonably pointed out that 'The criterion requiring an economic use of sign surface was to be largely overridden by the need to achieve clarity of layout on the more complex signs' (Kinneir 1971, p. 10). A similar point of view had also been present in the internal discussions of the committee, where it was pointed out that instead of filling the space available, a relatively large area of uninterrupted blue background against the landscape background was desirable in order to improve the 'target value' of the lettering.²³¹ Observe that the experimental signs employed by the Road Research Laboratory, probably of similar reasons ('target value'), were mounted on a *large* kaki-painted panel on top of the car that moved towards the stationary observers (see Christie and Rutley 1961a, pp. 241–242).

Christie and Rutley seemed to be fully aware that the technical concept of 'statistical significance' does not express meaningful

230. See the discussion in the Anderson report (Ministry of Transport 1962, p. 9). Also of relevance here is a discussion in the Worboys report on situations 'when site conditions restrict the width of signs' (Ministry of Transport 1963, p. 10).

231. See committee 'Notes', dated 29 February, 1960; and committee 'Minutes' of the 22nd meeting, dated 25 April, 1960; as well as a letter from Jock Kinneir to the Ministry of Transport, dated 17 March, 1961. (Location: Margaret Calvert, London.)

significance or substantial meaningfulness.²³² They concluded that ‘the most remarkable feature of the results for the three ... scripts is that the reading distances are so nearly equal ... the difference is so small that caution is necessary in interpreting its meaning’ (p. 243), and ‘the results do clearly indicate ... that none of the three scripts tested has any appreciable advantage over the others with regard to legibility’ (p. 60). It is thus reasonable to claim that no significant difference in legibility was found. In their conclusion Christie and Rutley also called attention to the fact that the small difference between the two capital lettering styles (one of them serif and one sans serif) was not necessarily based on the serifs or lack thereof, but might depend on other variables.

Christie and Rutley finally concluded: ‘Since there is little difference in legibility between the different types of lettering, it seems reasonable to make the choice on aesthetic grounds’ (1961b, p. 60). They went even further and perceptively suggested that ‘there are grounds for believing that aesthetic questions may be at the root of the controversy’ (1961a, p. 243). In the final Anderson report, it is admitted that ‘taste plays so important a part, as we believe it should’ (Ministry of Transport 1962, p. 5). Herbert Spencer in his comment in *Design*, stressed that aesthetic consideration – ‘taste, tradition, relevance and fashion’ – were of utmost importance. He clearly expressed his disapproval of Kindersley’s ‘partially’ seriffed letters, which he described as ‘clumsy’ and as ignoring ‘both taste and tradition’ (in Christie and Rutley 1961b, p. 61). Also Reynolds Stone seemed unhappy about Kindersley’s ‘unusually seriffed capitals’ and he complimented Kinneir’s sans serif lettering. Nevertheless, he suggested that if ‘good’ small serif letters had been included in the tests, they might have outdone the others.²³³

To no surprise: Cyril Burt’s name was called upon in the debate (by Reynolds Stone): ‘Fashionable or not, the use of sans meant ignoring experts like the psychologist Sir Cyril Burt, who “has recently recalled

232. For a brief but useful discussion on the technical concept of ‘statistical significance’ see Pedhazur and Schmelkin 1991, pp. 202–203.

233. Kinneir referred to Kindersley’s alternative lettering as ‘mis-serifs’ (although not in public): ‘As far as appearance goes I cannot imagine even the most obdurate philistine wanting to cover England with ‘mis-serifs!’ (Letter from Jock Kinneir to the Ministry of Transport, dated 17 march 1961.) (Location: Margaret Calvert, London.)



FIGURE 6. Davis Kindersley's capitals-only serif alphabet. (From: *Design* 1959, p. 35.)

and reaffirmed scientific findings that 'for word recognition a sans serif type face was the worst of all.' (in Christie and Rutley 1961b, p. 61).

David Kindersley defended his design. In a letter to the editor in a later issue he applauded the tests undertaken by the Road Research Laboratory. He however urged that 'the conclusions drawn from it are bad – really bad' (*Design*, no. 154, 1961, pp. 87, 89). He pointed out that the tests were not performed at 'real distances' (that is, they were performed at shorter distances). He claimed (with reference to a study by Paterson and Tinker on the legibility of newspaper headlines), that both capital legibility and serif letter legibility decrease at a lesser rate (assumingly with an increase in distance) than for both small letters and for sans serif letters. Furthermore, he claimed that his letters for motorway signs 'can be read from at least 175 ft further away than the existing lower case signs [i.e. Kinneir's] with equal areas' (p. 89). I can only guess that this figure is based on calculations where several quantities are included – for example the differences of the results between his and Kinneir's alphabets, the difference between test distance and a larger distance, and 'the decrease at a lesser rate' thesis referred to above.

Kinneir's supporter Herbert Spencer brought up a fundamental reservation about the value of experimental research, and cautioned:

Such tests of lettering as these are therefore useful in disposing of pseudo-scientific arguments, but in cases where the results strongly favour a particular design they must, to be of any practical use to designers, be elaborated upon so that we can clearly understand why one design functions more effectively than another. (In Christie and Rutley 1961b, p. 61)

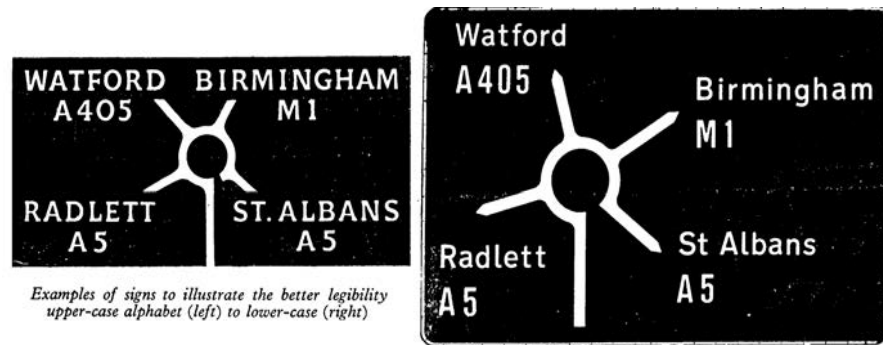


FIGURE 7. An illustration from Kindersley 1960 that shows a suggested sign by the author (left) and a Kinneir sign (right). Kindersley's caption reads: 'Examples of signs to illustrate the better legibility [of] upper-case alphabet (left) to lower-case (right)'. In his main text, Kindersley comments on Kinneir's sign: 'Apart from the ill-chosen type, the height of the sign is still further exaggerated by large areas of wasted space, resulting from the off-centre and asymmetrical contemporary typographical fashion.' (From: Kindersley 1960, p. 464.)

It is interesting to observe in retrospect that the public debate on these new direction signs only focused on one aspect of the new direction signs – their lettering. One of the letters to the editor of *Design* in the aftermath of the presentation of Christie and Rutley's research came from A. G. Long (assumably not a designer, perhaps a road engineer). He applied what today might be called a usability perspective and accused the research of suffering 'from an unnecessarily restricted consideration of some aspects and an unduly indiscriminate study of others'. He pointed out that the question of colour seemed to have been ignored, and the same applied to performance in bad weather, or in the dark while illuminated by different kinds of artificial lighting along the road and from the car. 'All these are surely more urgent problems of road design than a finicking survey of the effect of serifs on capital letters on large boards displayed in good conditions on the best roads.' (*Design*, no. 156, 1961, pp. 81, 83).

In retrospect

It was decided to use Kinneir's sans serif lettering of small letters and initial capitals (as well as his overall design-solution) for both motorways

and all-purpose roads.²³⁴ Some of the many ‘prominent’ ‘non-features’ of his directional motorway signs were: no boxes around destination names, no barbs on the heads of the arrows that symbolise the road ahead, and not least: no forced symmetrical or grid-based positioning of destination names. This represented an ‘exceedingly inferior layout’ and resulted in ‘large areas of wasted space’ according to Kindersley (1960, p. 464). Although the ‘no boxes feature’ adhered to an emerging modernist norm in graphic design at that time (groups were to be implied minimally by spatial relationships alone), it was also based on the wish of the Anderson committee, and it was in accordance with the 1949 Geneva protocol.²³⁵

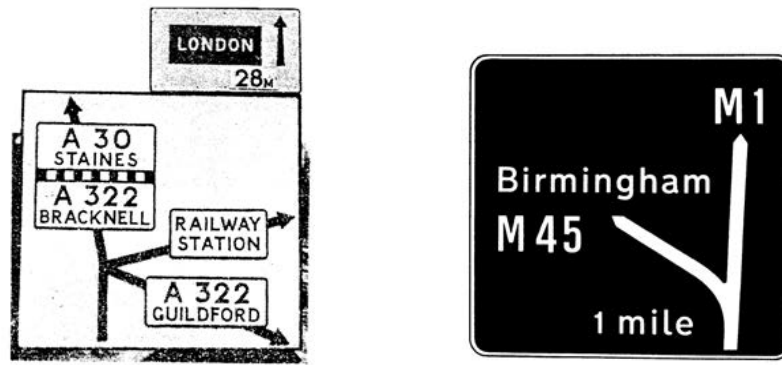


FIGURE 8. Left: a pre-Kinneir, pre-motorway, directional road sign. (From: Christie and Rutley 1961a, p. 239: figure 1.) Right: a Kinneir directional road sign. (From: Ministry of Transport 1962, p. 59: figure 38.)

Kinneir’s rather ‘neutral’ sans serif letterforms were undoubtedly, compared to Kindersley’s somewhat unusual serif capitals, more in line with contemporary aesthetic preferences among designers and taste trend-setters. To suggest that the final decision was taken already before the Road Research Laboratory performed their tests (with both explicit and implicit neutrality) is perhaps to overstate the issue, but nevertheless, a feeling that the experiments were some kind of play to

234. See Ministry of Transport, 1962, 1963; and subsequent regulations, referred to in Department of Transport 1991, p. 12.

235. See letter from Ministry of Transport and Civil Aviation to Jock Kinneir, signed R.L. Huddy, dated 12 June, 1958. RTC 53/4/024 Pt.4. (Location: Margaret Calvert, London.) See also the illustration on p. 46 in Froshaug 1963.

the gallery – only necessitated by the public debate and performed in order to shrug it off – is hard to avoid. Nevertheless, Kinneir's lettering solution corresponded more with the practice on the European continent; as well as with the direction signs on the American 'interstate highways'²³⁶ – an important imperative.

Kinneir and many other designers at that time strongly believed that sans serif letterforms – exactly in the combination of small letters and initial capitals – were intrinsically more legible for signing systems than serif capitals. In addition to the belief in the importance of the more distinctive word shapes of words in small letters with initial capitals, it was believed that sans serif letterforms were easier to handle and less 'aesthetically sensitive' and generally more 'forgiving' when actually produced; that is, in various modified forms for various applications, especially with the tools available at the time. Furthermore, if small letters was the preferred option: then sans serif small letters were undoubtedly more suited aesthetically than roman small letters to have relative short ascenders and descenders (i.e., to have a large x-height), and thus being more practical and less demanding of interlinear space on signboards (Mason 1994).

With the recommendations of the Worboys committee for all-purpose roads, words in all-capitals became to a large extent reserved for certain mandatory and prohibitory traffic signs. This distinction provided the means for a functional differentiation of certain important signs such as 'STOP' and 'GIVE WAY' (see Ministry of Transport 1963, pp. 10, 103).

It was also argued that since serif types are characterised by a high contrast between thick and thin strokes, they 'would almost certainly prove unsuitable when the letters have to consist of reflectionized material to catch the headlights'.²³⁷ To this, it has to be admitted that Kindersley's sturdy serif capitals were unusually low in contrast – not unlikely in order to solve exactly the problem suggested by Carrington. However, this characteristic of Kindersley's design, together with its highly idiosyncratic and unusual serifs, and not its serifness *per se*,

236. See *Design*, no. 129, September 1959, p. 31; *Design*, no. 132, 1959, p. 71; Ministry of Transport 1962, p. 4.

237. Letter to the editor from Noel Carrington, in *The Times*, March 20, 1959, p. 13.

might very well have created exactly the uneasiness people like Herbert Spencer felt towards it.

Interestingly, it seems that the leading figure of British typographic establishment, Stanley Morison, who did not participate in the public debate, undermined at least part of the argument of his fellow traditionalists, while addressing a continental public. In a postscript submitted in 1962, for a German language edition of his famous *First principles of typography*, published in Switzerland in 1966, Morison stressed the practical and pragmatic aspects of using sans serif letterforms for applications like traffic signs (see Morison 1996, pp. xiii, 39–40).

Sanserif type is ... quicker, easier and therefore cheaper to make. It is in fact the cheapest of all to make. Its forms can be mastered by the lowest category of draftsmen. Naturally, municipal architects and others to whom lettering is no more and no less than a necessary evil, gave the medium a cordial welcome, and with reason. That is to say with reason of a natural kind: of self-interest, which is the best – because a material and rational – basis for the choice of sanserif. It is not surprising that sanserif is superseding the serified style in all transport and street designations. Its economy of cost cannot but make sanserif the universal public medium of communication. (Morison 1996, p. 39)

The aftermath

Together with his associate Margaret Calvert, Kinneir came to dominate the design of public and official wayfinding and information signing systems and their various types of accompanying sans serif letterforms in Britain in the next few decades – for motorways and all-purpose roads, airports, the railways, the public hospitals, and the armed forces. His influence was also felt abroad,²³⁸ and his motorway signing system has ‘been called Britain’s true corporate identity’ (Rainford 1996, p. 13).

Although Kindersley lost, another serif capital letterform designed by him, a vigorous interpretation of the historic Trajan letterform, which was approved for street names by the Ministry of Transport in the early

238. For example (either on airports or on motorways) in Australia, in the Middle East, in Greece, on the Continent, and in Scandinavia.

1950s (Dreyfus 1957, p. 38), was later to become the most widely used letterform for street nameplates throughout Britain.²³⁹ Thus, both Kinneir and Kindersley came to put their decisive graphic marks on the public 'landscape' of post-war Britain.²⁴⁰

However, the dispute did not end in the 1960s. Since then it has come to the surface on many occasions. Robin Kinross appears prominently among the 'pro-Kinneir historians', celebrating Kinneir's work as a great achievement.²⁴¹

These signs were the first, in any country, in which 'visual' and 'functional' considerations were fused. They marked a new turn in British typography. And in the subtleties of their letterforms and of their rules of configuration, the signs showed a sophistication beyond the grasp of the title page- and inscription-bound traditionalism. (Kinross 1992, p. 167)

239. Nevertheless, in the road leading into the road where I used to live for a while, there is a still an old 'pre-Kindersley' street nameplate, which actually faces a 'Kindersley' name-plate on the other side of the road. Also: a remaining pre-Worboys directional signboard can be seen on a major all-purpose road nearby. The latest replacement date for pre-Worboys directional signs on all-purpose roads is 1st of January 2005 (Department of Transport 1994, p. 2).

Recently journalist, and since then author-of-fame, Helen Fielding wrote a spirited account of the current 'road sign madness' in Britain in *Independent on Sunday* (1994). It is not an attack on the British road signs in themselves or their design, but an attack on the current somewhat inadequate situation on British local roads and streets where signs too often are either lacking, hidden behind something, in the wrong place, or cluttered together in such a way that 'motorists of Britain just don't know which way to turn'. She perceptively states that: 'The trouble is, systems are usually set up by people who know the way anyway. They ought to be checked by people who are strangers to the area'.

Donald Norman comments in a similar but more polemic fashion on the signing of British (and American) local roads (as opposed to major national motorways) in his *The psychology of everyday things*. 'When making long journeys on the secondary roads of England, I learned to go around each roundabout two or three times, each time eliminating a different exit until I finally could select what appeared to be the best' (Norman 1988, pp. 226–227). The reason for why there are more problems on ordinary roads than on motorways has to do with the fact that it is less visual competition on motorways and fewer kinds of junctions. Information on all other roads are more dense and complex, and here there are more kinds of junctions and often more complex junctions (see Kinneir 1970, p. 18; and 1971, p. 10).

240. The ultimate winner of the dispute in question, Jock Kinneir, since wrote three substantial accounts of his work on signing systems: two on his road signs (one unpublished article in 1971, and one article published in 1984), and one article on several of his signing systems (published in Danish only, in 1970).

241. See Kinross 1984, 1989, 1992, 1994b.

Kinross is not only enthusiastic about the end result of the design process but also attaches great importance to the process itself: as an exemplary model for all design that aims to fulfil public needs. He sees it as an index of modernisation and public service democracy in Britain's post-war pre-privatisation era: a large-scale unglamorous planning process open to rational justification, in contradistinction to 'the recent cult of the designer, who reveals expensive master-creations to a boardroom, as a *fait accompli*' (Kinross 1989, p. 52). Kinross refers to the fact that the design process in question involved a broadly composed committee, an outward look towards continental Europe,²⁴² assessment of relevant research, consultation with a large number of interest groups, a public debate, an expert designer, and technical advisers who conducted experimental research. Furthermore, the committee was indeed not a passive body. The minutes from the committee's 27 meetings bear evidence of lively and constructive debates.

Another perspective that can be applied focuses on the 'system design' aspect. In a recent article, David Sless (as others have done before him), depicts changes in this century from a situation of small scale crafting of single artefacts (in a pre-designer era), via a situation of small-scale crafting and planning of individual artefacts for mass production (in the designer era), to a situation today where 'projects are concerned with designing whole systems rather than individual artefacts' and where the outcome may be a large number of unique artefacts produced by others (1998, p. 7).²⁴³ Kinneir's signing system is certainly an interesting and early example of 'system design' with a graphic designer playing the central role. First: the sheer number of signs: a whole range of different kinds of traffic signs related to each other, with separate solutions for motorways, major roads and local roads. Second: all the practical consideration on layout, lettering, size of

242. Not only did the 1949 UN Geneva protocol inform the work, the committee had also toured continental roads while taking colour slides (Kinneir 1971, p. 3).

243. Ironically, the emergence of computer assisted publishing (and the subsequent convergence of tools, production processes, and products) has also created a situation where many graphic designers today not only plan and specify, they themselves also often have to do the implementation that earlier was done by typographic craftsmen (see Stiff 1996b for an account of the historical rise and demise of specification as an important but hardly recognised part of typographic designing).

lettering, colour, background colour, reflection, illumination, mounting and siting, and not least, the content of the many different signs of each category. And third: the design of a system with prototypes and specifications which included a letter spacing system based on a limited number of tiles so the signs could 'design themselves', that is, so local sign manufacturers easily could space the letters consistently just by following the instructions in the manual.

Several contemporary writers are vigorous supporters of David Kindersley's position of the early 1960s. Montague Shaw describes in his book on David Kindersley Kinneir's lettering and Kindersley's involvement in the competition the following way:

It is one of the misfortunes of the creative person that his sensible work is, from time to time, set aside in favour of a vastly inferior article, by ignorant judges who are swayed by fashion and an uneducated taste. ... [David Kindersley's alphabet] were better in every single case. But the sanserif was used. (Shaw 1989, p. 19)

This account was recently accentuated in a more politicised manner by the designer James Souttar in a talk he gave at the Monotype Conference in Cambridge in 1992. In addition to quoting Shaw, he describes Kinneir's solution as reflecting 'a vision of shabby utopianism.' (Souttar 1992, p. 5).

However, not only opinionated accounts such as Kinross', Shaw's and Souttar's, but also unreliable accounts of what factually happened have been perpetuated:

A series of tests carried out by the Road Research Laboratory showed that in terms of recognition and legibility at speed, Kindersley's capitalized serif letters were greatly superior to the modernist sans serif, upper-and-lower-case letters. (Eason and Rookledge 1991, p. 98)

And similarly, in a recent article by Robert Long on David Kindersley and his oeuvre, in the American typographic journal *Serif*:

The all-caps alphabet that he designed used heavy, bracketed square serifs to promote legibility and intelligibility when seen from a rapidly moving vehicle. While it appears that practical tests clearly demonstrated its superiority to the Helvetica [*sic*] that was much in fashion at the time, Helvetica set in upper and lower case won. (Long 1995, p. 35)

However, the opposite misinterpretation is also present in the literature. The Dutch architect, information designer and professor,

Paul Mijksenaar in a recently published booklet, *Visual function: an introduction to information design*, claims that

1963 test models from the Road Research Laboratory in England ...
researchers found that place names in small letters with initial capitals
... were easier to recognize [than place names in capitals only]
(Mijksenaar 1997, p. 22)²⁴⁴

Poulton 1965 *

In 1965 Dr E.C. Poulton at the Applied Psychology Research Unit at the Medical Research Council, Cambridge, published a paper in the *Journal of Applied Psychology* – ‘Letter differentiation and rate of comprehension in reading’. The basic content of this paper had been presented the previous year in an article written by Dennis Cheetham and Brian Grimby in the journal *Design* (Cheetham and Grimby 1964). Poulton’s experiment was suggested by the Council of Industrial Design, the publisher of *Design*, and Cheetham and Grimby, who were affiliated with the council, had assisted Poulton in carrying it out.

The background for these writings and Poulton’s experiment can be found in contemporary craft discourse, and in marketing efforts, where the new and innovative sans serif typeface family Univers was hailed as a remarkable achievement.²⁴⁵ It was alleged that Univers’ subtle character shape made it an eminent and highly legible typeface. By 1965 – the heyday of high modernist design and ‘Swiss typography’ – the use of sans serif typefaces, for body text in both advertising, ephemeral

244. Elsewhere in his booklet Mijksenaar refers to Moore and Christie 1963 (a paper which I have not consulted), as the source for the accompanying illustrations of Kindersley’s and Kinneir’s lettering. The same paper is therefore possibly Mijksenaar’s source of misinformation about the outcome of the Road Research Laboratory’s research.

* E.C. Poulton. 1965. ‘Letter differentiation and rate of comprehension in reading’. *Journal of Applied Psychology*, vol. 49, no. 5, pp. 358–362. See also Dennis Cheetham and Brian Grimby. 1964. ‘Design analysis: typeface’. *Design*, no. 186, pp. 61–71.

245. See for example Ruder 1957, 1959, 1961; a special issue of *Typografische Monatsblätter*, no. 1, 1961; Handover 1960; Dreyfus 1961; Biemann 1961; Frutiger 1962; Eurographic Press Interview 1962.

printing, and design magazines, had become common, not only in Switzerland, on the continent and in the USA, but also in England. Many designers were enthusiastic about the use of sans serif typefaces, especially the new sans serifs adhering to the ‘Swiss idiom of sans serif typefaces’,²⁴⁶ for example Helvetica and Folio. ‘Many younger typographers have felt for some years now that sans serif faces are more suitable for text settings than has usually been acknowledged.’ (Cheetham and Grimbly 1964, p. 62).

However, this state of affairs was not without dissent – the ‘traditionalists’ also voiced their opinion – as in this letter from John Peters to *Monotype News Letter*:

News Letter 67 [set in Univers] is ingenious but unreadable. Having tried its acres of sans-serif on various friends, it is clear that I am not the only crank traditionalist. One busy friend declared that any advertisement he receives in sans-serif goes unread into the basket simply because the letter form slows down the reading process. (*Monotype News Letter*, 68, 1962, [p. 12])

More reasonable criticisms could also be heard. An example here is an article by R.S. Hutchings in *British Printer*, while attacking not only the new sans serif typefaces, but also the idiom of ‘Swiss typography’ (Hutchings 1965).

The contemporaneous enthusiasm for, and interest in, sans serif typefaces manifested itself not only in the widespread use of this kind of typefaces, but also in craft and academic discourse. For example, some substantial accounts, on various aspects of the history of sans serif letterforms and typefaces, were published in this period.²⁴⁷

The rationale behind Cheetham and Grimbly’s article in *Design* (1964), and Poulton’s research carried out on the request of *Design*, was to test the claims about Univers’ superior legibility and suitability for continuous text. The article in *Design* examines how Univers was

246. Sans serif typefaces adhering to the ‘Swiss idiom of sans serif typefaces’ were produced in the late 1950s and early 1960, primarily in Switzerland, but also by type foundries in Holland, Germany, France and Italy. They are characterised by horizontally cut terminals as well as relatively closed semi-enclosed counters of letters like a, c, e, and s. See Lund 1993, for a detailed exposition.

247. That is, Gray 1960; Handover 1961 (although somewhat speculative); Mosley 1965; and Falk 1965. Substantial contributions, from the same decade, on the form or topology of sans serif typefaces, are Gerstner 1963 and Schulz-Anker 1969.

designed. The issue of *Design* also contains an interview with its Swiss designer, Adrian Frutiger, carried out by the young British type designer Matthew Carter. Cheetham and Grimby state that their aim is ‘to examine how far it [Univers] meets the claims that have been made on its behalf.’ (p. 61).

Poulton’s experiments compared six different Monotype typefaces, three sans serif and three roman. This study does not include one sans serif typeface among several roman typefaces, but is unlike other studies evenly balanced. Thus, the probability of top ranking of at least one of the sans serif typefaces should be higher than in most of the previous experiments. The typefaces in question, *a priori* assumed to be reasonably readable, represented different subcategories or idioms of roman and sans serif typefaces. The typefaces, all of normal weight and appearance-width, were Bembo, Baskerville, Modern Extended, Grotesque 215, Gill Sans, and Univers. The line length was the same in all of the test material. Poulton explicitly acknowledges the possible problem of confounding due to differences not only in typeface design but also in actual size – posed by the discrepancy of nominal and actual appearance size of a typeface. In order to avoid this problem, the x-height (usually the dominant vertical dimension of a typeface) was equalised between the typefaces (as far as admitted by type size availability). Thus, the nominal sizes varied from 9.5 points (Univers) to 12 points (Bembo). However, Poulton does not explicitly acknowledge another possible source of causal confounding in that different results also can be interpreted as being caused by different ratios of size to interlinear spacing. Nevertheless, Poulton might accidentally have solved this problem – which he seems to be unaware of – while controlling ‘for the amount of paper covered by each typeface’ (p. 360). He achieves this aim by letting the sum of the eventual type size (nominal) and interlinear spacing become roughly the same for all the typefaces. Thus, the interlinear spacing varies between 1.5 points (for Bembo) and 3 points (for Univers). Thus, the ratio between the x-height and the interlinear spacing varies in accordance with ‘master craftsman’ advice – the larger the x-height the larger the interlinear spacing, and vice versa.²⁴⁸

248. See for example Dowding [1954] 1966, p. 14; 1957, p. 7.

Poulton makes the point that type size, interlinear spacing and line length are each as near optimal as possible – following Tinker’s recommendations – in order to make the typefaces in the experiment the only non-optimal variables.

Poulton’s operational criterion for measuring legibility is ‘rate of comprehension’. This criterion, developed by Poulton, reflects his scepticism towards Tinker’s comprehension checks deemed as insensitive because they are used merely as a means for checking that comprehension is above a minimum level, and thus to a large extent mainly relying on the speed of reading as a measure of legibility. Poulton juxtaposes two equal variables, comprehension and speed of reading, and the resulting measure is named ‘rate of comprehension’.

Earlier, Poulton had criticised Tinker and Paterson’s experiments both with regard to internal, external and statistical conclusion validity (Poulton 1960). Poulton strongly advocates the idea that by measuring the degree of comprehension in conjunction with speed of reading, legibility experiments would become more sensitive and thus yield differences that otherwise would go undetected (allegedly the reason for Tinker’s ‘negative’ results). A further exposition of his concept of ‘rate of comprehension’ can be found in Poulton (1968). The claim that rather subtle differences in the design of ‘ordinary’ typefaces can be reflected in the degree of comprehension measured by answering questions after having read a given passage for a limited time, seems to me to be implausible. It represents an extreme position in contradistinction to the view that typefaces are very much peripheral to the reading process. Furthermore, what *comprehension* is, is not easily answered: is Poulton measuring comprehension, or a facet of comprehension only, or something else?²⁴⁹ Nevertheless, Poulton claims that: ‘Using this method I have found statistically reliable differences between typefaces where previous research workers have failed to do so’ (Poulton 1968, p. 73).

The results of the experiment ranked a sans serif typeface on top – Gill Sans, not Univers. Gill Sans was found to be ‘reliably’ better than the two other sans serif typefaces, Grotesque 215 and Univers, which

249. For a brief but useful discussion on comprehension in the context of reading, see Rayner and Pollatsek 1989 (pp. 316–320).

came respectively penultimate and last on the ranking list. However, the study did not reveal 'reliable' differences between any of the sans serif typefaces and the roman typefaces.

Poulton argues – explicitly in opposition to Burt – that since a sans serif typeface produced the highest average rate of comprehension, it is likely that it is not the serifs (or lack of serifs) which determine the legibility of a typeface. In order to find a theoretical explanation, Poulton refers to his two typographic advisors, Cheetham and Grimbly (1964). They pointed out *character differentiation* as an answer, claiming that Gill Sans – 'with its geometrical approach allied to humanistic letterforms' – compared to the two other sans serif typefaces has a stronger character differentiation (the shape of individual characters) and thus less family resemblance. They also suggest the possibility that the rather uniform internal space ('counters') of the characters of Univers interact with the (assumably relatively small amount) of space between the letters, and thus creates confusion. Poulton does however not pursue these points of view any further. Cheetham and Grimbly conclude that:

The readability tests show that Univers does not in fact possess greater readability than its older rivals; but if it falls down on this claim, it still has abundant attraction for typographers. (1964, p. 71)

The exposition of this experiment in *Design*, provoked a heated discussion in subsequent issues, with prominent participants from the world of design – Ken Garland, John Dreyfus, Anthony Froshaug, Noel Carrington, Walter Tracy, and Jock Kinneir. The graphic designer Ken Garland – who had been the designer of the very same journal between 1956 and 1962 – opened up the discussion in the same issue as Cheetham and Grimbly's article:

a discovery has come out of the work on this article which should give great heart to all those who fear that the incursion of science means the death of whatever freedom designers have managed to wrest from their employers. For years they have been burdened with the traditional typographer's dogma that serif typefaces are much better for continuous reading than non-serif typefaces. Every time graphic designers attempted to use non-serif typefaces in any quantity they had this holy law slung at them. Now, glory, be, a scientist has proved that the difference in the degree of comprehension between commonly used serif

and non-serif typefaces is negligible; and that what small superiority does exist belongs to a non-serif face. Thus a great and venerable shibboleth is banished. Now I ask you, fellow designers, is this freedom or bondage? (*Design*, 1964, no. 186, p. 29)

John Dreyfus, who had been involved in launching the Monotype version of the Univers family (see Dreyfus 1961), aggressively expressed his doubts about the experiment's validity – and about Ken Garland's point of view. Dreyfus claimed that since

most people read most easily the types which they are most used to reading, it is not surprising that Gill Sans, popular in England since 1928, should come out of Dr Poulton's tests with such flying colours. (*Design*, 1964, no. 188, p. 65)

This last criticism was however vehemently repudiated by Cheetham and Grimby in a comment to Dreyfus and other letter-writers, while pointing out that Gill Sans was not at all a familiar typeface for continuous text (in the same issue of *Design*, p. 67).

In a follow-up article by Cheetham, Poulton and Grimby – while advocating an extensive programme of experimental research into graphic design – they showed that they certainly had picked up Garland's argument:

the fact that what we offered was conjecture rather than dogma (plus the fact that our results could free designers from a self-imposed restriction on the use of sans serif faces in text settings) seemed to suggest that scientific methods were both less rigid and more creative than had been thought. (Cheetham, Poulton, and Grimby 1965, pp. 48–49)

Thus, between the lines: science had a liberating potential and could become a weapon for young progressive non-dogmatic and creative designers in their battle with older traditionalist colleagues.

Shortly after, in 1968, letter cutter, type designer, and former apprentice of Eric Gill, David Kindersley, in a conference paper published in *Printing Technology*, discussed Poulton's experiment. Kindersley asked how Gill Sans, a sans serif typeface with geometric shapes and a very small x-height, could oust Univers, 'a beautifully tailored modern sans' with a very large x-height (Kindersley 1968, p. 71). Kindersley, who in the 1960s eagerly investigated and prescribed a

letter-spacing system (see Kindersley 1976), claimed that both Gill Sans and Univers were ‘wrongly’ spaced, but that the more generous internal and inter-letter spacing of Gill Sans (plus a higher degree of differentiation among characters) probably carried the answer. Kindersley pointed out both internal and inter-letter space as important variables which determines the qualities of a typeface, and he suggested that type manufacturer’s default inter-letter spacing of typefaces is not an intrinsic property of a typeface. He therefore concluded that inter-letter spacing must be controlled in experiments in order to avoid it as a confounding factor (1968, p. 71).

Zachrisson 1965 *

In 1965 Bror Zachrisson, the director of the educational institution the Graphic Institute in Stockholm, published his doctoral thesis, *Studies in the legibility of printed text*. The thesis was based on research carried out in 1954 and 1964, and was an extended version of his licentiate dissertation from 1957 – ‘Studies in the readability of printed text with special reference to type design and type size’.

Zachrisson, son of the famous Swedish printer Waldemar Zachrisson, had had a long experience with the printing industry and he was also one of the co-founders of the Graphic Institute. In his thesis, he investigated both ergonomic aspects (legibility) and semantic aspects (congeniality) of typefaces, and the problems found suitable for investigation had been chosen after consulting with ‘leading text-book publishers’.

A considerable part of Zachrisson’s study is devoted to investigate the relative legibility of sans serif typefaces and roman typefaces. Zachrisson refers to what he describes as an assumption often found in the printing trade: that sans serif typefaces are less legible than roman typefaces.

* Bror Zachrisson. 1965. *Studies in the legibility of printed text*. Stockholm: Almqvist & Wiksell.

While discussing research on reading and comprehension Zachrisson acknowledges that subtle differences in typeface design are peripheral to the reading process. He states that: 'Under normal circumstances, a reader is limited in speed only by his rate of comprehension' (p. 23). And he goes on: 'There is reason to assume that under normal conditions no significant differences exist between type faces in common use by adults for running text.' He makes the point that because children are slow readers struggling with comprehending the text, nuances of type design probably matter less to them than to adults (p. 36). However, these 'insights' do not deter him from performing many comparative experiments on the relative legibility of typefaces. His subjects are mainly schoolchildren, for whom – he states – 'the author is convinced that the question of type face design is one of the most pressing problems' (p. 93). Zachrisson claims that sans serif typefaces are widely used in school-books and juvenile literature, at least in Sweden and the United States, however not without controversy. It therefore 'seems worthwhile investigating whether there is any real difference between them' (p. 93).

Zachrisson refers to Pyke, Tinker and Paterson, Ovink, Burt, Moede, and Brachfeldt (pp. 36–38), and he employs a wide variety of operational criteria in his experiments:

- errors in oral reading at a normal rate (pp. 97ff)
- speed of silent reading at a normal rate plus comprehension check (pp. 109ff)
- instant perception of single words, through a tachistoscope (pp. 115ff)
- threshold from blur to focus, through a focal variator (pp. 121ff)
- indirect vision of single words, through a perimeter (pp. 124ff)
- ocular preference in a situation of binocular rivalry between typefaces, through a stereoscopic haploscope (pp. 128ff)
- readers opinion (pp. 131ff)

Speed and errors were the criteria used in the main experiments. The error criterion (oral reading) was used for 7–8 years old pupils in grade 1 for whom silent reading would prove difficult. A variety of sans serif and old style roman typefaces were employed in the various

experiments. In almost all of the experiments the result is described as ‘the null hypothesis may be accepted’, i.e. no significant difference was found.

Thus, Zachrisson’s conclusion is clear:

The empirical results show that *there is no significant difference, in objectively measured legibility, or subjective opinion regarding ease of reading between the OF [roman old style] and SS [sans serif] type faces.* (p. 132)

Wendt 1969*

In 1969, Dirk Wendt published an extensive study on typeface legibility. Translated to English, the title would read: *The influence of typeface category (Bodoni vs. Futura), typeface posture, and typeface weight, on the speed of reading.*²⁵⁰ This study was one of several in a series published by the Department of Psychology at the University of Hamburg under the series title ‘Untersuchungen zur Lesbarkeit von Druckschriften’ (‘Investigations into the legibility of typefaces’). Dirk Wendt, who later moved to the University of Kiel where he became a professor and since then has continued to work on legibility, published many articles on legibility and related topics in the late 1960s and the 1970s – in the *Journal of Typographic Research* and its successor *Visible*

* Dirk Wendt. 1969. *Einflüsse von Schriftart (Bodoni vs. Futura), Schriftneigung und Fettigkeit auf die erzielbare Lesegeschwindigkeit mit einer Druckschrift*. Bericht Nr. 5, Untersuchungen zur Lesbarkeit von Druckschriften. Hamburg: Psychologisches Institut der Universität Hamburg.

250. As late as 1994 the main content of this study was re-published as a substantial part of a 36 page article on legibility by Wendt (Wendt 1994), in Peter Karow’s anthology, *Font technology*, published by Springer-Verlag in both German and English editions. However, neither Wendt’s article, nor the general bibliography, contain any reference to the original study from 1969. Furthermore, certain clues and omissions in the text give the impression that the article is describing research carried out recently, and an illustration which claims to show ‘specimens of the typefaces actually used’ (pp. 296–297 in the English edition) most probably shows more recent (and somewhat different) digital versions of the typefaces.

Language, and not least in German printing trade journals, like *Druck-Print* and *Papier und Druck*.²⁵¹ He still publishes legibility studies.²⁵²

The aim of the study in question is to investigate the influence and possible interactions of *typeface category* (roman and sans serif), *typeface posture* (upright and cursive/slanted), and *typeface weight* (light, regular, bold, and extra bold), upon legibility. The ‘modern’ or neo-classical typeface Bodoni of the Berthold type foundry and the sans serif typeface Futura of the Bauer type foundry are chosen to represent respectively the roman category of typefaces and the sans serif category of typefaces. These typefaces, or more correctly these typeface families, were chosen because they have a large number of variants. The large range of family variants made it possible to investigate the relative legibility of eight variants (combinations of posture and weight) of each of the two typeface families – altogether 16 different typeforms. Wendt employed carefully adapted German versions of Tinker’s Chapman-Cook speed of reading tests, and generally seems to have invested heavily in designing sound experimental procedures – the study also contains an extensive introductory chapter on methodological considerations.

Wendt, points out that ‘The question of “roman or sans serif” has been controversially discussed among typographers for some time’, and then goes on discussing back and forth several arguments put forward in the defence of either of the two typeface categories:

- The characters of sans serif typefaces are better differentiated and therefore more legible, because they lack serifs which would have made them more similar to each other.
- The modulation of the stroke of roman typefaces are archaic remainders of earlier production methods.
- Serifs help to form more compact and typical word shapes.
- Serifs constitute a useful rail for the eye to slide along.

With respect to the last argument, Wendt refers to a preference study of typefaces and interlinear spacing, where the readers found that for a text set in a sans serif typeface to be appealing, it needed more

251. See for example Wendt 1970b, 1971, 1972.

252. See Wendt 1994; and Wendt, Groggel and Gutschmidt 1997.

interlinear spacing than for a roman typeface (Becker, Heinrich, von Sichowsky, and Wendt 1970). Wendt suggests that this may be so because of sans serif typefaces' missing rail of serifs (1994, pp. 294–295).

In Wendt's own study, each piece of test material was set in 8 point size, in five columns, unjustified (uneven right margin), and with slightly different line lengths so the same lines could be identical from one piece of test material to another. Two thousand subjects were involved in the experiment, which measured the number of words read in three minutes.

The results of interest to us here showed that the overall difference between the two typeface families (all the variants taken into consideration) was less than three words per minute (0.74%); in the favour of the sans serif typeface family Futura. Wendt's clear conclusion is that the difference 'cannot be considered significant' (1994, p. 305). Actually, if we compare only the regular variants of the two typeface families, which Wendt does not do, the difference was measured to less than half a word read per minute; in favour of the roman typeface Bodoni.

However, while discussing the overall result, Wendt surprisingly, but not without ulterior motive as we will later see, states that it

was a surprise in so far as Paterson and Tinker (1932), applying in principle the same experimental procedure, found a superiority of a roman typeface (Scotch Roman) over a sans serif (Kabel lite) of 2.3%, and Pyke (1926) even found a superiority of 18% for a roman typeface (Monotype No. 2 Old Style) over a sans serif (Stephenson & Blake No. 10 Lining Grotesque). (1994, p. 305)

Wendt seems to overlook the simple fact that while he himself has employed one typeface of each category (serif and sans serif), Paterson and Tinker employed only one sans serif typeface among a total of ten typefaces in their experiment, most of them roman. Likewise, Pyke employed one sans serif typeface together with seven roman typefaces in his experiment. Thus – the sheer statistical probability, of the sans serif typeface in Wendt's experiment to be ranked over or on a par with the roman is considerably higher than in Paterson & Tinker and Pyke's experiments. Furthermore, Paterson and Tinker's interpretation of their own results was that (except for the typewriter and textura typeface) all the typefaces (including the sans serif) were 'equally legible' (Paterson

and Tinker 1932, pp. 609, 613). Although Pyke ‘found a superiority of 18% for a roman typeface ... over a sans serif’ (Wendt), the sans serif was actually ranked second (of altogether eight typefaces, seven roman and one sans serif), and the biggest difference between any face and the next on the ranking list was between the sans serif and the next, which was 30 per cent less legible than the former. Pyke even emphasized that his second best typeface (the sans serif) was one of the most uncommon (Pyke 1926, p. 52).

While discussing his results Wendt also refers to Crosland & Johnson (1928) and Kerr (1926) (both from the same period of time as the two studies by Paterson & Tinker and Pyke) who ‘reported an inferiority of a sans serif typeface’. Wendt does this without acknowledging Cyril Burt as his flawed secondary source of information, and furthermore, without realising that the two publications in question in fact do not support either Burt’s claim²⁵³ or Wendt’s claim. Nevertheless, he also refers in his discussion to the more recent works of Brachfeldt (1964)²⁵⁴ and Zachrisson (1965), where very little difference in legibility between roman and sans serif typefaces was found.²⁵⁵

This leads on to Wendt suggesting that earlier reported differences between sans serif and serif typefaces in favour of serif typefaces (i.e., Paterson & Tinker, Pyke, Kerr, and Crosland & Johnson) have changed to no difference (Brachfeldt, and Zachrisson) due to increased habituation to sans serif typefaces, and consequently changed to an inverted situation:

253. See Hartley and Rooum 1983, p. 205. In fact, Kerr argued *a priori* that sans serif typefaces are less legible than serif typefaces due to irradiation (p. 552), while at the same time, on p. 554, referring to Roethlein’s study of 1912 where the sans serif typeface News Gothic ‘was found most legible’. However, this inconsistency is likely based on a confusion of Kerr – he refers to News Gothic as a serif typeface ‘similar to those on p. 554’, that is, a typical ‘modern’ 19th century serif typeface.

254. The article in the trade journal *Verlags-Praxis* by Professor Dr Oliver Brachfeldt is basically an *a priori* argument against the use of sans serif typefaces in books. Within the article there is a brief mentioning of what may have been an experimental empirical study.

255. However, Brachfeldt’s own brief interpretation of the study in question is that the relatively small difference of 3 second per page in favour of serif typefaces will be considerably larger when extrapolated to 20 or 30 pages of reading material.

Finally, in our own study^[256] we found even a slight superiority of a sans serif typeface over a roman one, though this finding does not yet reach statistical significance. If this impression is correct, the decrease in the difference of legibility between roman and sans serif typefaces may well be caused by the general increase in use of sans serifs in print. ([1969]/1994, p. [21]/306).

This argument, the ‘hypothesis of habit’, which Wendt also advocated elsewhere and in designer fora (Wendt 1970a), does not sound unreasonable. Wendt argues the following way further on in his discussion:

Therefore, familiarization with a typographic design is one of the main difficulties in legibility research: Based on experimental procedures, only very seldom do we discover whether a new typographic design is actually inferior to an established one due to intrinsic qualities, or simply rates lower because readers are less familiar with it. With respect to the perceptions concerning roman and sans serif typefaces, it may be the case that sans serifs were initially less legible because they were less familiar, but due to their increasing daily use in printed matter of all kinds they became more and more familiar, their inferiority to roman typefaces decreasing proportionally. (1994, p. 306)

This may very well be so. However, the problem is that the seven studies (from Pyke and onwards and including his own) – which Wendt painstakingly refers to in chronological order in order to show a trend which can substantiate his ‘hypothesis of habit’ – do not after all, according to what is demonstrated above, support his ‘hypothesis’. Paterson and Tinker only employed one sans serif typeface among a total of ten typefaces in their experiment. Pyke only employed one sans serif typeface together with seven roman typefaces in his experiment. And more importantly, the second best typeface in Pyke’s experiment was not only a sans serif typeface, but also the most uncommon. And furthermore, the biggest difference between any face and the next on Pyke’s ranking list was between the sans serif and the next. The studies by Crosland & Johnson and Kerr do not, after all, support Wendt’s claims. Whether the Brachfeldt article deserves status as an empirical study, and whether Wendt’s interpretation of its results is reasonable, are open questions.

256. That is, an even more recent study (my comment).

Furthermore: although Wendt qualifies his statement about ‘a slight superiority of a sans serif typeface over a roman one’ by stating that ‘though this finding does not yet reach statistical significance’, he still relates this ‘finding’ from his own study to a trend in order to support his familiarity thesis. What is more, it is not even nominally correct to say that the study showed ‘a slight superiority of a sans serif typeface over a roman one’. It was the sans serif typeface *family* that on average showed a ‘slight superiority’, while the opposite was the case for the regular variants of the two typeface families (the most common variant in legibility studies). And finally, the seven studies that are included in Wendt’s argument represent only a fraction of relevant studies available for trend comparison.

To conclude: Wendt’s study is an ambitious and large scale experimental study, and the results seem reasonable. However, Wendt’s trend reasoning is false, and his familiarity thesis therefore falls apart. Nevertheless, as a consequence of his trend reasoning, Wendt elsewhere suggests that the future, when sans serif typefaces have become more and more common, will possibly ‘demonstrate objectively a better legibility of sanserif types’ (Wendt 1970a, p. 43).

Robinson, Abbamonte, and Evans 1971: * ***Why serifs are (still) important***

In 1971, at the high point of ‘legibility research’, a study of the relative legibility of roman and sans serif typefaces was published in the journal *Visible Language*. The authors of ‘Why serifs are important: the perception of small print’ were David Owen Robinson, Michael Abbamonte, and Selby H. Evans. The legibility of roman and sans serif typefaces had been explored on many occasions beforehand, and although interest in ‘legibility research’ would pretty much evaporate only ten years later,²⁵⁷ experimental studies on this seemingly finicking question are even today produced at regular intervals. Robinson and his co-authors’ topic – which is more legible, typefaces with or without serifs? – is thus in one sense unremarkable. It seems to represent a neat dichotomous question which has apparently fascinated many researchers.

But their paper was unusual in two respects compared to most legibility studies. First: not only did its authors attempt to establish the

* David Owen Robinson, Michael Abbamonte, and Selby H. Evans. 1971. ‘Why serifs are important: the perception of small print’. *Visible Language*, vol. 5, no. 4, pp. 353–359.

257. For some references to criticism of ‘legibility research’ – a modest ‘research program’ that peaked in the 1960s and 1970s – see my review essay in *Information Design Journal* (Lund 1995). See also Richard Venezky’s (1984) article ‘The history of reading research’, where legibility studies are placed within a broader historical context of socio-behavioural reading research. See Tinker 1963, and Spencer 1969, for summaries of legibility research.

One of the most serious objections which repeatedly have been raised against experimental legibility research points out that ‘factors’ such as typeface are very much peripheral to the reading process, and thus that it will be extremely difficult to quantify any meaningful differences in legibility, for example by employing an operational method which measures the speed of reading. This point has been made by several researchers and typographers; among them, R.L. Pyke in 1926 (pp. 60–61), David Sless in 1981 (p. 170), and more recently, albeit obliquely and within a slightly different context, by Andrew Dillon in the chapter ‘Describing the reading process at an appropriate level’, in his *Designing usable electronic text* (1994). However, this objection does not imply a rejection of the need carefully to design, modify, or choose, appropriate typefaces for specific applications (as well as the need carefully to manipulate other interdependent typographic variables; be it on paper, on screen or on any substrate) – on the basis of aesthetic, craft-based, and ergonomic considerations.

superior legibility of roman over sans serif typefaces, they actually suggested a *theory* for ‘why serifs are important’. This was in striking contrast to the atheoretical approach of most previous legibility research. Second: the study was not based on behavioural experiments involving human ‘subjects’, but on a cognitive science approach, and it used a computer model of human visual perception.²⁵⁸ The published article has often been cited, especially in recent years.²⁵⁹ For all these reasons the article known in the literature of legibility studies as ‘Robinson, Abbamonte, and Evans 1971’ still deserves attention.

The content of ‘Why serifs are important’

The authors ask in a slightly ironic tone why serifs – which appear to be ‘at best only decorative and at worst merely superstitious’ – have not disappeared, as long as sans serif typefaces exist as an alternative to serifed typefaces (p. 353). After briefly referring to the research of the psychologists Christopher Poulton and Miles Tinker, and to the typographer Geoffrey Dowding’s discussion of sans serif typefaces, Robinson *et al.* put forward several arguments and counter-arguments about the usefulness of serifs. They bluntly discard as ‘unconvincing’ the suggestion that people’s preference for serifs is first of all an expression of aesthetic judgement. They sweepingly discard the possibility that preference is merely a question of conditioning or habituation. Furthermore, they discard the possibility that serifs enhance the eye’s horizontal movement along lines of typeset text. Here they argue that ‘since adults only make a few eye fixations in reading each average-length of print, it seems unlikely that the continuity from one letter to the next should be an important factor’ (p. 354). Finally, they discard the possibility that letters with serifs convey more information due to their more complex structure – with the counter-argument that serifs can just as well be regarded as unnecessary visual ‘noise’.

258. The research in question was ‘supported by the Department of Defense’ and the authors were affiliated to the ‘Institute for the study of cognitive systems’ at the Texas Christian University in Fort Worth.

259. For example in 1975, 1982, 1984, 1990, 1991, three times in 1993, 1994, and in 1997. See the subsection ‘The paper’s reception’ below.

Instead, the authors suggest that ‘the neurological structure of the human visual system benefits from serifs in the preservation of the main features of letters during neural processing’ (p. 353). They state that their theory depends on ‘the physiological structure of the human visual system’. As a tool for finding support for their theory, the authors employ a computer model which, they say, simulates human visual perception.

In the next few paragraphs I attempt to describe Robinson and his co-authors’ procedure. My attempt is based partly on the insufficient explanation given in their paper, and partly on a more detailed but nevertheless ambiguous and vague explanation given elsewhere (Evans *et al.* 1968).

Dot representations of four letters, each in two different sizes, are entered into a computer. Since the size of the dots are the same for the two sizes of letters, the relative representation of the smallest letters is far cruder than the representation of the larger letters. The letters are E, T, f, and h; from two IBM Selectric typewriter typefaces, the sans serif face Artisan and the seriffed face Courier (figure 9). After the input operation, the representations are detected and processed by the computer program’s ‘horizontal and vertical line operators’, which are sensitive respectively to horizontal and vertical lines. These line operators are meant to imitate ‘feature detectors’ which are assumed to be located in the brain’s striate cortex as part of the human visual system. The line operators can be described as matrices of 5×5 cells, while the dot representations of each letter are positioned within a matrix of 48×48 cells (Robinson *et al.* 1971).



T T E E

f f h h

FIGURE 9. The original letters used for the experiment, before they were turned into dot matrix representations ready for computer input. (From Robinson *et al.* 1971, p. 356: figure 1.)

The cells within each line operator carry differentiated values, such that the cells along the centre column of the vertical operator are assigned the highest values, the cells along the two adjacent parallel columns lower values, and the cells of the two outer columns zero value (Evans *et al.* 1968).

Each operator 'reads' the input pattern of the letters in such a way that it successively covers all 'distinct' locations of the input matrix in discrete overlapping steps. At each location the value of each cell of each operator is multiplied by either one or zero, depending on whether or not a 'dot' is assigned to the corresponding cell of the input matrix. Eventually the products of each cell are, in one way or another,²⁶⁰ added and then transferred to a new resulting matrix where eventually the cells with the largest sums will be constituted by dots in the print-out pattern (Evans *et al.* 1968).

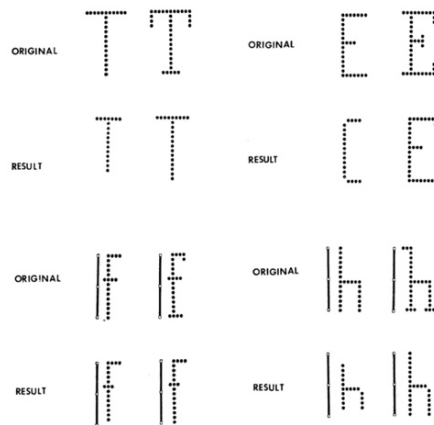


FIGURE 10. Dot matrix representations of the smaller-sized letters, before and after 'the computer model of the line detectors of the human visual system' was applied. (From Robinson *et al.* 1971, p. 357: figure 2.)

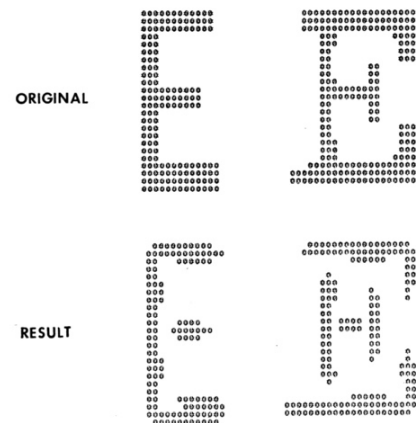


FIGURE 11. Dot matrix representations of the larger-sized letters, before and after 'the computer model of the line detectors of the human visual system' was applied. (From Robinson *et al.* 1971, p. 357: figure 3.)

260. The authors provide neither information about the exact values of the cells of the operators, nor details about the further mathematical operations.

The resulting print-out pattern shows that for relatively small sizes of letters (in ‘ordinary bookprint’ size), serifs help to preserve the original image (figure 10). For the capital E in particular, the serifs are crucial. However, for larger letters serifs are not found useful (figure 11). The authors explain the result for the larger letters by suggesting that such letters are probably perceived by different kinds of feature detectors – by edge detectors rather than by line detectors. However, they add that when larger letters are viewed from a distance, for example on billboards, the letters will be perceived by line detectors, and serifs will thus still be of importance.

The authors conclude:

If the computer model has any validity as an imitation of the human visual system, then one may conclude that serifs are important in preserving the image of small letters when they are represented in the neurological structure of the visual system. (p. 359)

The words ‘if’ and ‘may’ reflect their only detectable caution about the validity of their research. Elsewhere they are more categorical in their exposition.²⁶¹ However, as I shall argue below, the validity of this study does not rest, as its authors suggest, on the fidelity of the computer model alone, as if it is a matter of fact that ‘line detectors’ exist as part of the human visual system and that there is a simple and direct relationship between how the visual system works and the computer model.²⁶² First of all the study rests on a multi-level chain of theoretical constructs and assumptions about the human visual system.

A cognitive science approach

In order better to understand the claims which the authors make, I need to extend my discourse beyond the scope taken by Robinson, Abbamonte, and Evans.

261. See for example their claim on p. 354: ‘The explanation which this article proposes depends on the physiological structure of the human visual system.’

262. See for example the categorical references to ‘the computer model of the line detectors of the human visual system’ and ‘the model which imitates human visual feature detectors’ on p. 356. See also the abstract on p. 353.

On a fundamental level the authors rely on a cognitive science approach employing an explanatory computer model of human visual perception, implying that the brain *functions* like a digital computer that runs programs consisting of algorithms. This is in line with ideas which dominated cognitive science at the time of writing, and which continued to do so until the mid-1980s.

However, not only is it highly questionable to assume that the brain *operates* like a digital computer,²⁶³ it is also highly questionable to assume that the brain *functions* like a digital computer, that is, that it performs algorithmic manipulations of symbolic representations. Well-articulated criticisms of this widespread assumption – an assumption which is still vigorously defended – have been offered from various stances.²⁶⁴

For example, the philosophers Hubert Dreyfus and John Searle have argued against what is both the central conceptual belief of cognitive science and the basis for its central method of enquiry.²⁶⁵ that cognition has a computational information-processing character and can be treated as ‘device-independent’, and thus that cognitive processes, including visual perception, can be simulated or modelled compu-

263. Think of the physical hardware level with its wires, integrated circuits and magnetic discs; and the way particular operations are realised in registers.

264. For example, from the philosophers Hubert Dreyfus ([1972] 1992, pp. 163–188), John Searle (1990; 1992, pp. 197–226) and Peter Hacker (1987); from the neuroscientist Gerald Edelman (1994, pp. 211–252); the psychologist Thomas Hardy Leahey (1992, pp. 444–461); the sociologist Jeff Coulter (Button *et al.* 1995); and although more implicitly but not less forceful, from the computer scientist Terry Winograd together with Fernando Flores (1986).

265. I am concerned with Dreyfus’s and Searle’s critique of ‘cognitive reason’ or ‘cognitivism’ (the AI version of cognitive psychology, as opposed to applied AI engineering or cognitive psychology *per se*). For the sake of order: I am here not concerned with their rejection of the possibility of creating computer-based ‘general purpose’ *artificial intelligence*, where they basically argue that intelligence cannot be non-biological, disembodied, and independent of intentionality and of humans’ tacit and pragmatic background knowledge of the world. See Searle’s famous ‘Chinese room’ argument (‘syntax is not sufficient for semantics’); it appeared in a paper first published in *Behavioral and Brain Sciences* (Searle 1980). The paper has since then been republished on numerous occasions in anthologies and readers; e.g. together with Boden 1990 (pp. 67–88). It has also on numerous occasions been both vigorously contested (e.g. Boden 1990, pp. 89–104), vigorously defended (e.g. Kelly 1993, pp. 173–184), and vigorously yet conditionally defended (Button *et al.* 1995, pp. xi, 1–3, 15–22).

tationally by programs in order to generate relevant empirical data. To put it another way: that the best way to understand cognition is to study computer programs which try to reproduce cognition. To push it further: what the brain does may adequately be described by algorithms:

If a machine could be designed which could 'see', its computer program could constitute the implementation of a theory of how seeing is achieved by humans. (Bruce and Green 1990, p. 77)

These ideas imply that due to similarities between inputs of both human cognition and a computer model of human cognition, and similarities between outputs of both human cognition and a computer model of human cognition, there is a symmetrical relation between human cognition on the one hand and a computer model of human cognition on the other. This is as if minds are non-biological and disembodied computer programs only accidentally implemented in brains, and thus that computer programs and minds function on the same internal principles. John Searle offers the following polemical argument against such a view:

the word processing program simulates a typewriter better than any AI program I know of simulates the brain. But no sane person thinks: 'At long last we understand how typewriters work, they are implementations of word processing programs.' It is simply not the case in general that computational simulations provide causal explanations of the phenomena simulated. ... we do not suppose that because the computer simulates a typewriter, therefore the typewriter simulates a computer. (Searle 1992, p. 218)

And again:

The mistake [of cognitive science] is to suppose that in the sense in which computers are used to process information, brains also process information. (p. 223)

In the same vein, but more specifically on visual perception, the philosopher Peter Hacker attacks David Marr's currently very influential computational theory of both human visual perception and 'machine vision':

what Marr describes constitutes, as might be expected from a specialist in artificial intelligence, the outline of a novel approach to the problems of the theory of 'machine-vision', i.e. the theory underlying the design of machines that can identify what lies before

them as a consequence of light falling upon a sensor. But Marr's theory is neither an explanation nor even a description of animal or human vision. (Hacker 1991, p. 119)

[Marr's theory's] fruitfulness and coherence as a computational theory for artificial-intelligence research is a matter for evaluation for computer scientists. ... My sole concern is with the question of whether his theory is (as it purports to be) a theory of vision, in particular of human vision, or indeed whether it even provides a coherent framework for such a theory. (p. 121)

Similarly, in an issue of the *Annual Review of Psychology*, William P. Banks and David Kraljick spell out a point that is obvious but not often emphasised:

Does machine perception always tell us something about biological perception? The AI approach to perceptual theory is but one area of AI research. Many AI researchers in perception are simply trying to design machines that accomplish certain perceptual tasks, and they study human perception in the hope of learning something useful in that effort. However, it is unlikely that the constraints of biological evolution and the multiple competencies required of a living organism have produced solutions that apply directly to such single-purpose engineering problems. To attempt to implement the biological strategies of perception in a perceiving machine seems to some as foolish as designing airplanes with flapping wings. By the same token a machine model of perception may tell us surprisingly little about human perception. (Banks and Kraljick 1991, p. 310)

This scepticism points to the danger of invalid inference from computer models that only partially, or maybe even hardly, correspond to the natural phenomena being modelled, because of too many disanalogous elements. But that same scepticism does not deny the general usefulness of metaphorical models in science, in everyday thought and language, or in other contexts. However, the writers I have quoted so far point to a conceptual confusion and a too fuzzy boundary between the model and what is being modelled; a confusion which easily arises within a paradigm where cognitive models are viewed as 'functional' and thus 'device-independent'.²⁶⁶

266. It is worth pointing out that although the study in question (i.e. Robinson *et al.* 1971) is preoccupied *only* with human vision, the underlying context is not necessarily limited to vision proper. This is revealed by the authors' 'AI-like' approach, their institutional 'cognitive science' affiliation, and the implicit 'machine-vision' aim in one of the co-authors' former publications (Evans *et al.* 1968), on which the paper in

A multi-level chain of theory

With regard to broad theoretical positions on visual perception: there is a noticeable lack of consensus among both philosophers and psychologists. The most basic divide can be said to be between information processing theories on the one hand and ‘Gibsonian’ ecological theory on the other hand. Information processing theories emphasise the indirect, inferential and constructive nature of visual perception (whether mediated by *computational* cognitive processes or not). ‘Gibsonian’ ecological theory claims that visual perception has a more direct, unmediated nature; that it relies not on algorithmic processing of abstract symbolic representations of more or less static two-dimensional retinal ‘snapshots’, but on non-algorithmic detection, where a very information-rich surrounding optic array of light provides continuous input for the vision of locomotive humans.²⁶⁷

Robinson and his co-authors’ theory of ‘human visual processing’ is based on an information processing approach to visual perception. Their theory includes a ‘feature detection’ theory and a ‘degradation’ thesis, both of which I will discuss below. I will suggest that it is highly questionable whether their theory reflects the biological reality of human visual perception. The same doubt applies to the computer model of the theory, which is basically an algorithm, assumingly encoded in a high-level programming ‘language’ as a computer program. That doubt further applies to the actual computer implementation: that is, the running of the program on a computer, performing operations on representations of the input, and producing output as results of these operations.²⁶⁸

question is partially based. In fact much of the early work on pattern recognition in an AI context focused exactly on recognising alphanumeric characters, in order to develop useful applications, for example for automatic sorting of letters with hand-written postal codes (Bruce and Green 1990, p. 180).

267. See for example the exposition in Bruce and Green’s widely referred to textbook on visual perception (1990). Bruce and Green aim at an integration of aspects of both theoretical stances; as does Thompson (1995, pp. 215–250). For further expositions of ‘Gibsonian’ ecological theory, see Gibson 1979; Michaels and Carello 1981; Turvey *et al.* 1981; and Reed 1988.

268. The encoded program can of course be regarded as part of the implementation.

Feature detection theory

The authors' digital computer 'model of human visual processing' suggests that letterforms are recognised by 'feature detectors' in the visual cortex. The model is apparently based on a low-level 'feature analysis' theory, one among several projective 'feature analysis' theories popular among psychologists and computer scientists between the 1960s and early 1980s. Such theories in turn represent one of several kinds of approaches to 'object recognition'.²⁶⁹ More specifically, the computer model is based on a postulated 'feature detector' theory, which was inspired by David Hubel and Torsten Wiesel's classic neurophysiological research of the late 1950s and 1960s, in tracing the visual pathway in the brain from the retina to the striate ('visual') cortex. For this work Hubel and Wiesel shared half a Nobel prize in 1981. Hubel and Wiesel implanted micro-electrodes in single cells in the striate cortex in cats who were anaesthetised, had their eyes immobilised, and received artificial respiration. They recorded activity produced by relatively focused retinal stimulations, each of one-second duration (Hubel and Wiesel 1959, 1962).²⁷⁰

Hubel and Wiesel showed that many or most cortical cells reacted to specific geometrical features illuminating corresponding receptive fields (heavily overlapping areas of photo-receptors in the retina which feed single cortical cells). This applied especially to thin and long rectangular slits, edges and bars, with specific orientations, that is: vertical, horizontal, and all possible obliques. Hubel and Wiesel did not find any predominant orientation (Hubel and Wiesel 1962, pp. 110, 151; Hubel 1988, p. 71). The horizontal and vertical 'Hubel and Wiesel line detectors' applied by Robinson *et al.* (p. 355) apparently expresses

269. See for example Bruce and Green 1990, pp. 52–72, 175–202.

270. There is a broad consensus in the literature implying that it is reasonable to assume that the visual system of humans is not very different from the visual system of cats, the most popular non-human research species in vision research. However, Kolars questions the presupposition by arguing that *activity* shapes structures in the nervous system (Kolars 1983, p. 54–55). Nevertheless, for a thorough discussion on this question, see Crawford *et al.* 1990.

a ‘feature detector’ theory inspired by Hubel and Wiesel’s neuro-physiological findings.²⁷¹

It should be pointed out that since the 1970s, many psychologists of various theoretical stances have rejected feature detector theories in general – although admittedly these rejections have been most strongly directed against higher-level theories. Nevertheless, low-level feature-detector theory has also been seriously challenged. Moreover, Hubel and Wiesel’s experimental conditions and set-up have been questioned,²⁷² and doubts have been expressed about what has been described as a ‘reductive’ approach to visual perception.²⁷³

If low-level feature detector theory is accepted on its own terms, then possibly its most cogent rejection is based on what Bruce and Green refer to as the ‘problem of ambiguity in feature detection’. Although the responses of cortical cells to edges and lines and their orientation are prominent aspects of their activity, they are far from the only aspects. The response from a cortical cell is in fact a multivariate function: that is, it is interdependent on many variables, for example orientation, contrast, position, width, length, direction of movement, velocity of movement, binocular disparity, and vestibular and auditory stimulation.²⁷⁴ Thus Bruce and Green conclude that ‘we cannot identify single cells with feature detectors’ (1990, p. 65), and John and Schwartz conclude that ‘The very richness of the repertoire of trigger features seems to undermine the theoretical position that has been inferred from their existence’ (1978, p. 4). In the same vein, Wade and Swanston argue that although a particular orientation is optimal with regard to response from a single cell, other orientations, although decreasingly and within

271. These highly metaphorical and projective ‘feature detectors’, which are referred to in a literal and teleological manner by Robinson *et al.* (and elsewhere in the literature on visual perception), seem not to have originated as such with Hubel and Wiesel. No trace of the explicit term or concept ‘feature detector’ can be found in Hubel and Wiesel’s classic papers from 1959 and 1962, or in Hubel’s popular *Eye, brain and vision*, published in 1988.

272. See for example Kolars 1983, p. 54; Rock 1984, p. 141; and Wade and Swanston 1991, p. 193.

273. See for example Wade and Swanston 1991, p. 173; John and Schwartz 1978, pp. 24–25; Marr 1982, pp. 2–16, 340–341; and Michaels and Carello 1981, pp. 66–69, 174, 184.

274. Bruce and Green 1990, pp. 61–65; John and Schwartz 1978, pp. 3–4.

certain limits, will also produce a response. Accordingly: a whole range of 'orientation-selective neurons' will to a varying extent be excited by a particular line stimulus on the retina (1991, p. 84). Moreover, Michaels and Carello cogently demonstrate that perceiving higher-order properties does not entail detecting lower-order properties and doing computations (1981, pp. 66–69, relying on Runeson 1977).²⁷⁵

This brief exposition of the disagreements about whether or not 'feature detectors' can be said to exist suggests that Robinson and his co-authors' explanation, which claimed to simply rely 'on the physiological structure of the human visual system' (pp. 354, 356, 359), in fact relies on a specific and seriously contested theory or theoretical interpretation of some physiological findings.

It is a mystery why Robinson and his co-authors chose to include only horizontal and vertical line operators, and not also obliques at various orientations. The inclusion of obliques would be indicated by Hubel and Wiesel's findings, and also by the references made by Robinson *et al.* to an earlier study which one of them co-authored (Evans *et al.* 1968). The authors of this earlier study refer to 'straight lines of various slopes', and to *several* line operators that were 'successfully' employed in a similar application which to a limited extent managed to retrieve seriously distorted images of letter-like shapes (Evans *et al.* 1968, p. 404). So when Robinson *et al.* say that 'In Figure 2 it is clearly shown that serifs perform an important function in a perceptual system with horizontal and vertical line detectors', this points to a problem. What about line detectors at oblique orientations? Surely, this appli-

275. Keith Rayner and Alexander Pollatsek's widely read textbook *The psychology of reading* prominently gives credibility to 'feature detection theory' (1989, pp. 11–15). They conclude: 'While there are criticisms of feature detection as a theory of object perception in general, it provides a reasonably satisfactory model of how letters and words are processed' (p. 14). Rayner and Pollatsek refer to Hubel and Wiesel as literally discovering feature *detectors*, and claim in a circular way that Hubel and Wiesel's work is the 'best known *physiological evidence* in favour of feature detection theory' (p. 12, my italics). In fact it is the feature detection theory in question that is an interpretation of Hubel and Wiesel's findings. However, it should be noted that Rayner and Pollatsek's stance first of all seems to reflect their rejection of more implausible 'template matching' theories of human 'pattern' or 'object recognition' (described on pp. 11–15; also in Bruce and Green 1990, pp. 180–182).

cation of line operators at only two orientations could pose a damaging threat to the ecological validity of their research.

With regard to the widely acknowledged complexity²⁷⁶ of the ‘global’ human visual system it seems that the research in question by Robinson *et al.* is based on a very poor model – basically limited to only two kinds of ‘feature detectors’, and furthermore detectors from only one particular location in the global human visual system. This can only be amplified by a point made by Hubel and Wiesel in 1962 (p. 141), and which Hubel repeats in his *Eye, brain and vision*: ‘The striate cortex is just the first of over a dozen separate visual areas, each of which maps the whole visual field’ (Hubel 1988, p. 219). Hubel modestly adds:

We are far from understanding the perception of objects, even such comparatively simple ones as a circle, a triangle, or the letter A – indeed, we are far from even being able to come up with plausible hypotheses. (p. 220)

Degradation theory

A traditional view of visual perception since Descartes, still current in various influential guises,²⁷⁷ is that an impoverished and insufficient two-dimensional retinal image has to be reconstructed by cognitive operations into a rich three-dimensional image in the brain.²⁷⁸ Robinson and his co-authors offer an argument based on what seems to be a similar view.²⁷⁹ By referring to the fact that there are far fewer fibres in the optic nerve leading from the eye than there are photoreceptors (rods and cones) in the retina they imply that the transmission capacity of the

276. See for example Uttal 1990.

277. Also reflected in literature not primarily concerned with visual perception, for example in Richard Rubinstein’s admired *Digital typography: an introduction to type and composition for computer system design*. Writing about legibility and visual perception, he states: ‘In fact the “wiring” of the retina performs a great deal of feature detection and signal processing, creating a partially processed image to send on to the brain for higher-level analysis’ (1988, p. 28).

278. This view is vigorously contested by the ‘Gibsonian’ ecological school. See for example Gibson 1979; Michaels and Carello 1981; Turvey *et al.* 1981; and Reed 1988.

279. See Robinson *et al.* 1971, p. 354 and *passim*. See also Evans *et al.* 1968, *passim*.

optic nerve is too low.²⁸⁰ They further imply that, therefore, the ‘information’ (or nerve impulses) necessarily has to be enhanced when reaching the visual cortex, and that it is the ‘Hubel and Wiesel line detectors’ which are effective in ‘retrieving degraded patterns’ and preserving the main features of letters. Nevertheless, they still refer to the resulting ‘neural image’ as being degraded to a certain extent. The idea seems to be this: line detectors and serifs are naturally and especially adapted to each other in such a way that at the stage of a resulting ‘neural image’ in the brain, recovered letterforms with serifs will still be partly degraded (the serifs seem to have gone), but they will be degraded to a lesser extent than letterforms without serifs (where part of the main strokes and other structural elements such as the middle bar in the E have gone) (p. 358).

The authors’ explanation of why, after all, degraded ‘neural images’ of sans serif letters do not have a ‘disastrous’ effect on legibility is more fantastic than serious:

... the considerable influence of context. For example, if one erases one third of the letters in a sentence it is still readable:
An *xa*pl*of*se* te*ce*it* mi*si*gl*tt*rs. (p. 359)

I will limit myself to one further comment on the ‘degradation’ line of argument. With regard to the discrepancy in the ratio of photoreceptors in the retina to fibres in the optic nerve, David Hubel himself gives a plausible anatomical explanation for why this condition does not necessarily result in inadequate transmission capacity, and how detailed visual information can be preserved ‘without our having hopelessly crude vision’ (1988, p. 37–39). His explanation is simply that the details of connections between the retina and the optic nerve vary according to the distance from the fovea, ‘which corresponds to exactly where we are looking – our center of gaze, where our ability to make out fine detail is highest’. That is, that in the fovea a single receptor connects with a single nerve fibre, and that further out in the periphery of the fovea there is an escalating degree of convergence from many receptors to single nerve fibres.

280. There are about 1 million fibres in the optic nerve, and 125 million photoreceptors in the retina.

Implementation, interpretation, and reasoning – some remarks

The implementation of Robinson and co-authors' theory and model is problematic. Not only is the very crude dot matrix representation of the 'stimulus-material' dubious, but also the decision to represent serifed typefaces by a monoline and monospaced typewriter typeface with highly exaggerated serifs. One cannot possibly induce, as the authors do, from the specificity of IBM Selectric Courier (or similar typefaces) to serifed (i.e. roman) typefaces in general. Furthermore, it is an open question whether the letters T, E, f and h represent an appropriate sample of the 52 most common alphabetic glyphs.

While the authors state that 'In Figure 2 it is clearly shown that serifs perform an important function in preserving the original image of a small letter' (p. 358), the figure actually shows us that of the four letters T, E, f and h, it is only the E which is 'considerably degraded'. The E has lost its skeletal/structural form, by losing its original middle crossbar. The other three letters seem only to have contracted slightly (see figure 10).

Further, the authors' statement that 'Serifs are not useful when large letters are presented to a line detector system, as shown in Figure 3 ... [and] do not help to preserve the main features of a letter' (p. 358), is in fact an understatement. If we are to judge according to what Figure 3 actually shows, we have to conclude that serifs are not only 'not useful' and 'not helpful', but directly damaging (see figure 11). This conclusion is amplified by the additional verbal information given by the authors: that similar results occurred in the three other letter pairs.

Finally, the rationale of Robinson and co-authors' study seems to be based on a circular and slightly absurd argument. They claim that since serifed typefaces are more often preferred to sans serif typefaces, serifed typefaces must *necessarily* possess an intrinsic quality more easily adapted to how the human visual system works. Then, in their concluding discussion, the authors claim that readers' preference for serifed typefaces supports their theory (p. 359).

In fact, the paper's line of logical reasoning is not sound. It goes like this: serifed typefaces are, without any obvious reason, more popular

than sans serif typefaces. The theory (interchangeably referred to as an ‘explanation’ and proclaimed to ‘depend on the physiological structure of the human visual system’) says that serifs are important because they help to preserve the original shapes of letterforms ‘during neural processing’. The computer model of the theory is employed in order to test the theory. The computer simulation shows that serifs are useful. What is more, the popularity of seriffed typefaces gives further support for the theory. The theory is confirmed. However, simply to demonstrate that a model can work in a certain way does of course not demonstrate that what is being modelled works in the same way.

Theoretical assumptions or physiological facts?

It should by now be clear that Robinson and co-authors’ explanation relies first and foremost on an intervening multi-level chain of theoretical constructs and assumptions (right or wrong), and not as they claim: that it simply relies on ‘the neurological structure of the human visual system’. In short: what has been described above seems to be reasonably encapsulated by the following complaint:

In psychology the distinction between the perceptual algorithm and the biological hardware on which it hypothetically runs has not been so clearly maintained. Thus several theoretical terms are ambiguous: ‘Feature detectors’ [etc] ... suggests physiology but are in fact aspects of certain algorithms that hypothetically explain processing. ... *When theoretical terms masquerade as physiological ones*, speculation advances unchecked by reliable biological underpinnings. In such circumstances the purely theoretical components of research may not be properly tested, and the physiological ones fail to be verified by physiological experiment. (Banks and Krajicek 1991, p. 307, my italics)

Furthermore, not only does the authors’ explanation rely on a chain of theoretical assumptions while purporting to rely on physiological facts, but also on poor logical reasoning, a poor model, a crude implementation of the model, and a dubious interpretation of the results. Much seems to be wrong. To conclude: Robinson and co-authors’ paper demonstrates lack of ecological validity and is unconvincing.²⁸¹

281. In defence of the paper it can be pointed out that it is not based on a pure top-down model, frequently used in cognitive science (Wilkes 1990), where the theory (of what

The paper's reception

All the documents I have come across which cite 'Why serifs are important' accept it more or less uncritically.²⁸² The only somewhat cautious citation is one in (several editions of) Baird and Turnbull's textbook *The graphics of communication*. Although they treat it as an important paper, they do refer to it as being 'based on how the visual system is *believed* to function' (1975, pp. 64–65, my italics).

Rolf Rehe discusses the relative legibility of roman and sans serif typefaces in his book on legibility research, which has been published in at least five editions in English and one in German. He refers to 'Why serifs are important' in a positive way as 'an extensive study, [which] suggested that "the neurological structure of the human visual system benefits from serifs in the preservation of the main features of letters"' (1984, p. 32). A similar mention can be found in a fairly recent American master's thesis on computer screen legibility which compares roman and sans serif typefaces (Williams 1990, p. 12). The same applies to a recent American doctoral dissertation on the legibility of roman versus sans serif typefaces (Kravutske 1994, pp. 14–19).²⁸³

Pedersen and Kidmose are even more positive. They explain that Robinson *et al.* 'decided to *establish* their [i.e. serifs] importance' (1993, p. 70, my italics). And de Lange, Esterhuizen, and Beatty, in a paper in the journal *Electronic Publishing*, claim that this 'interesting study ...

must be going on) is the sole starting point (for making algorithmic descriptions of how), and where physiological data is regarded as irrelevant.

282. Admittedly, I do not take into account a possible tacit negative reception. On the other hand, I do not claim that my list of papers which uncritically cite 'Why serifs are important' is exhaustive. For the sake of order: Jeremy Foster's annotated *Legibility research abstracts 1971* contains a neutral description of 'Why serifs are important' (1972, p. 29). Michael Macdonald-Ross and Eleanor Smith's partially discursive bibliography *Graphics in text* contains a simple entry (1977, p. 46). 'Why serifs are important' was ignored by the cognitive psychologists who, with various aims, worked on 'letter perception' or 'letter recognition' in the 1970s and early 1980s. It may have been ignored because it was deemed to be improbable, or because it was seen as irrelevant (having too much of a machine-like optical character recognition approach). Nevertheless, the research on 'letter perception', largely based on various 'feature models', addressed a different agenda, and it also suffered from its own validity problems. For an example of such research, see McClelland and Rumelhart's well-known paper (1981).

283. Robinson *et al.*'s method of investigation is seriously misinterpreted by Kravutske.

determined the importance of serifs in the perception of individual letters' (1993, p. 242, my italics). Another research paper which refers to the Robinson *et al.* article attaches great importance to it (Gallagher and Jacobson 1993, p. 101).²⁸⁴

Although Karen Schriver, in her ambitious and well-publicised *Dynamics in document design: creating text for readers*, calls attention to some of the limitations of legibility research in general as well as its lack of homogeneous results (1997, pp. 276–277, 301), she nevertheless refers to 'Why serifs are important' in such a manner that it can only be perceived as an important paper (pp. 274, 276, 301).

The re-design of the journal *Radiology* back in 1982 provides an example where legibility research explicitly has guided, or at least post-rationalised, design decisions. The editors of the journal explained: 'Our decision [to use a typeface with serifs] was not arbitrary but based on evidence that serifs enhance legibility' (Eyler and Stewart 1982, p. 248, my italics). The 'evidence' referred to in the editorial is Robinson *et al.*'s paper.

It is tempting to ask why a study like the one in question can receive such uncritical acceptance. One of several possible answers is simply the seductive appeal, at least for outsiders, of the authority of scientific language, and in particular the language of cognitive science. Sidney Berger, himself a scholar in the humanities, puts it this way: 'Robinson ... gives us a *scientific reason* for preferring serifs: "the neurological structure of the human visual system benefits from serifs in the preservation of the main features of letters during neural processing"' (1991, p. 7, my italics). Another reason is simply the way 'research findings' often get summarised:

Unfortunately, what is served in the guise of a literature review is, in many instances, little more than a tiresome listing of studies and findings without even a hint that the author has been thinking when reading the literature, not to mention critically evaluating what he or she has been reading. (Pedhazur and Schmelkin 1991, p. 191)

284. Although Gallagher and Jacobson's paper is seemingly serious and thorough, it offers a remarkably flawed reading of Robinson *et al.*'s paper.

Finally: the paper's uncritical reception is obviously related to the fact that many of the underlying premises of the authors' approach, though seriously contested, are still very influential and even dominant.

Certainly, 'Why serifs are important', with its cognitive science approach, and its computer-modelling method of investigation, did not, after all, answer the question: which are more legible – typefaces with or without serifs? My investigation of this study and of its reception can only reinforce the truism that the translation of 'findings' from experimental research to the practice of typographic design has to be done with great caution. And, as I believe I have shown, that translation is even less assured when experimental validity cannot be taken for granted in the first place.

Harris 1973 *

In 1973 John Harris published the paper 'Confusions in letter recognition' in the trade journal *Professional Printer*. It was based on work carried out at the Computer Science Division of the National Physical Laboratory, and with support from Watford College of Technology.

Although the aim of this study was not primarily to investigate the relative legibility of serif and sans serif typefaces, it is of interest to us. Harris' starting point is that Poulton's research confirms that some typefaces are more legible than others. Harris is however more ambitious, and he wants to find out which features of a typeface contribute to its legibility. Although he acknowledges that adults do not read letter by letter, he nevertheless suggests that the key to the legibility of a typeface is found in the forms of its individual characters. He supports his view by pointing out that Poulton's experiments involved exactly the same pieces of text, only set in different typefaces. Therefore, he reasons, the differences in legibility found by Poulton must to a large extent be due to the form of the individual characters.

* J. Harris. 1973. 'Confusions in letter recognition'. *Professional Printer*, vol. 17, no. 2, pp. 29–34.

Thus, his study is based on the identification of individual letters of three of the typefaces employed by Poulton – the sans serif typefaces Gill Sans medium and Univers medium, and the roman typeface Baskerville. The operational criterion employed is the time of exposure method, performed with a ‘two-channel’ tachistoscope, on characters with equalised x-height.

Harris refers to evidence which shows that it is very likely that the time of exposure method as well as the variable distance method favour typefaces with relative large strokewidth. He points out that the employed variants of Gill Sans and Univers are almost of identical stroke width. He slightly increases the exposure time for Baskerville to compensate for its slightly thinner main strokes. After having performed the experiments, he then takes into consideration the frequencies in ‘English prose’ of the individual letters – in order to calculate the relative legibility among the three typefaces – when used in ‘text’.

With regard to Gill Sans, his results are in accordance with Poulton’s. Single letters in Gill Sans are more easily identified than those in the other two typefaces. Harris asserts that the superiority of Gill Sans over Univers is concentrated in the letters c, e, f, and t. The ranking order from Poulton’s experiment with regard to Univers and Baskerville is however not maintained. Harris calculates the ‘deficit in text’ of Univers compared to Gill to be 4.19% in Gill’s favour, and of Baskerville compared to Gill to be 11.13% in Gill’s favour. The author promptly gives an explanation: the apparent inferiority of the roman typeface Baskerville is caused by the ‘threshold effect’ on the thin strokes of certain letters – something which probably will not have an effect when read as continuous text. This explanation seems to oppose the rationale of the study.

Harris refers to Tinker, Paterson and Webster, who found neither significant differences in reading speed nor any correlation between distance legibility and speed of reading legibility, and he objects to their methodology. He suggests that the very short periods of reading employed by Tinker and Paterson in their experiments was not long enough to produce significant differences. He also suggests that differences in stroke width may have confounded the results in their distance experiments.

Harris also analyses the confusions of individual letters by his subjects, i.e. mistaking one letter for another. By doing this he arrives at some interesting conclusions (p. 32):

- serifs on adjacent verticals like h, n, and u, increase uncertainty about identity. ‘Confusions of b for h and n for u are significantly more likely to happen in Baskerville.’
- serifs on isolated verticals may have a beneficial function. ‘The letter i was correctly identified more often in Baskerville than in other faces, perhaps because the top serif serves in some way to emphasise the gap between the vertical and the dot. The scores for confusions of
- i with l and j with l, when combined, were significantly lower in Baskerville.’
- the gap size of the semi enclosed counters of c and e is important. ‘The letters c and e are less well-recognised in Univers than in Gill. ... In Univers this gap is much smaller’.

With regard to the perceptive last point made by the author, he nevertheless overlooks that it is not only the gap size which represents a considerable difference between the letters in question of Gill Sans and Univers – the curved end strokes of the same circumference are cut at different angles, respectively vertically (|) for Gill Sans and horizontally (–) for Univers.²⁸⁵

The author first concludes that ‘the evidence suggests that the legibility of single letters may be a major determinant of readability’. Then he more cautiously concludes that ‘Some evidence has been found to support the view that at least one of the variables underlying the readability of a typeface is the ease with which its individual characters can be recognised’ (p. 33). He supports this statement by pointing out that his results with regard to single letter identification are in accordance with Poulton’s results generated from reading continuous text. Nevertheless, he cautiously points out that other determining variables may exist, and that other experimental procedures (like visual scanning of arrays of letters) might give other results.

285. For an extended discussion on these morphological or topological properties of sans serif typefaces, see Lund 1993.

Hvistendahl and Kahl 1974 *

In 1974, professor of journalism at Iowa State University, J.K. Hvistendahl and Mary R. Kahl at Kansas State University published their study 'Roman v. sans serif body type: readability and reader preferences'. It was published in the *News Research Bulletin* of the American Newspaper Publishers Association, and later reprinted in the journal *Typographic*, published by the International Typographic Composition Association, in the United States.²⁸⁶

This experimental study was undertaken on request from the newspaper *Minneapolis Tribune*. It wanted advice on whether or not sans serif typefaces are suitable and legible for continuous newspaper text. This was at a time when many magazines in the United States had switched from roman type to sans serif type, and many newspapers had been experimenting with sans serif type. It is not unlikely that the possibility of switching to a sans serif type for 'body text' was seriously considered by many American newspapers in the mid 1970s – a period when radical re-design of newspapers became more and more usual.

Hvistendahl and Kahl's study, on the legibility of sans serif typefaces compared to roman typefaces in newspapers, was a two-part study – one experimental part measuring the speed of reading, and one part recording the reader's preferences.

The experimental study, employing altogether 200 subjects, measured the time it took the subjects to read 'at normal speed' newspaper articles of a certain number of words. No comprehension check was involved. Four different articles of different length were each

* Hvistendahl, J.K., and Mary R. Kahl. 1976. 'Roman v. sans serif body type; readability and reader preference', *Typographic*, vol. 8, no. 2, 4 pages. Published by the International Typographic Composition Association, in the USA. (According to the article in *Typographic*, and other sources, it was reprinted from *The ANPA [American Newspaper Publishers Association] News Research Bulletin*, no. 2, 1975, pp. 3–11. However, British Library's interlibrary loan services found no trace of the article in the 1975 volume (as well as in the 1974 and 1976 volumes) of the bulletin. The article is probably based on Mary Ruth Luna Kahl's M.S. thesis: 'A study of reading speed and reader preferences between roman and sans serif type', Iowa State University, 1974).

286. Not to be confused with the British journal *Typographic* of the Society of Typographic Designers.

printed in one roman version and one sans serif version. Altogether eight typefaces were employed – four dedicated newspaper typefaces with serifs,²⁸⁷ and four sans serif typefaces. However, each typeface from the one typeface category was compared to only one from the other typeface category. Thus,

- Intertype’s newspaper typeface Imperial, was compared to the sans serif typeface Helvetica
- Royal, Intertype’s version of Linotype’s newspaper typeface Corona, was compared with the sans serif typeface Futura
- The newspaper typeface News no. 2, was compared to the sans serif typeface Sans Heavy.
- The newspaper typeface News Bold, was compared with News Sans, a News Gothic look-alike

The two first pairs compared above, were set at a column width of 10.5 pica; while the two others were set at a column width of 14 pica. The ‘stimulus material’ was set by newspapers, and printed in specially made ‘experimental’ tabloid newspapers.

The result, as interpreted by the authors, is that the newspaper text set in roman type was read significantly faster in two of the four comparisons; and faster, but not significantly so, in one; and more or less the same to sans serif type in one. With regard to the preference recordings, a clear majority preferred roman to sans serif type. The authors also found that the widest columns gave the best results. Their conclusion is clear: ‘a well designed Roman type in combination with wider columns would produce optimum legibility’, and further that ‘Until more evidence is available, newspaper typographers would be advised to take a cautious approach toward using sans serif type for textual matter.’

Yet, it is worth noticing that both the ‘significantly faster’ (in two of the comparisons) and the ‘not significantly faster’ (in one of the comparisons) results express less than two seconds in difference of roughly around one minute reading time. It is therefore again appropriate to stress that *statistical significance* does not correspond to *practical*

287. For a general introduction to the history of dedicated newspaper typefaces, see Level 1989; and for a comprehensive visual index, see Gürtler 1988.

significance. The more or less identical result (in the fourth comparison) expresses 1/5 of a second in favour of the sans serif being compared.

There is one confounding factor which poses a damaging threat to the internal validity of the experiments. One of the ‘comparisons’ yielding ‘significant’ differences is illustrated in Hvistendahl and Kahl’s article – showing News no. 2 and News Sans. This illustration clearly shows that while the nominal line spacing, and probably the nominal point size as well, is the same in both examples, the x-height of the sans serif is considerably larger. Thus, the roman typeface looks generously linespaced, displaying distinctive articulated lines of text. The text set in the sans serif typeface looks badly in need of more interlinear spacing, or better, a smaller point size, which visually would make it match the text set in roman type.

To conclude: the internal validity of this ‘controlled readability experiment’ is flawed with respect to the integrity of its stimulus material. The typographic variables are ‘controlled’ with respect to nominal and arbitrary quantities but not to actual and visual quantities. It is thus impossible to decide whether the results are based on differences in typeface style or differences in the x-height to line-space ratio.

Eventually, the authors explicitly make a small but not unimportant one-word reservation about inducing from the specific typefaces employed in the experiment to typeface categories: *‘If the type faces used in this study are typical, it can be said that on the whole Roman type is more readable when used for textual matter than is sans serif type.’*²⁸⁸

In the mid-1980s, in a manual on *Typography and design for newspapers*, published by the IFRA, the International Association for Newspaper and Media Technology, Hvistendahl and Kahl’s study is the only legibility study referred to in the section on ‘text type’:

Sans serif type designs are not the best choices for newspaper text type. The lack of serifs causes an impersonal, monotonous look, resulting in lower reader appeal. Prof. Hvistendahl of the University of Iowa found, in a 1974 study, that readers prefer serif over sans serif type in newspaper text matter. (Rehe 1985)

288. On the first page (of this unpaginated article), my italics.

Vanderplas and Vanderplas 1980^{*}

Title of the paper: 'Some factors affecting legibility of printed material for older adults'. Published in the journal *Perceptual and Motor Skills*. Authors: James M. Vanderplas and Jean H. Vanderplas at Washington University, St. Louis.

The rationale provided for this study is reasonable. The authors display a good command of relevant literature, and they argue that almost all previous legibility studies have employed as subjects college students, young adults, children or military personnel. The many visual impairments that are common among older people suggest that recommendations based on studies of young adults may not be wholly applicable for older adults. The authors make a case about older people who often need to engage in crucial and demanding non-leisurely reading activities, for example of prescription labels, medicinal instructions, contracts, and application forms. They suggest that small type or inappropriate typefaces or particular layout-solutions may be ill suited for the visual needs of the elderly.

The study seems to be thorough with regard to its experimental design. Furthermore, the authors reflect in a mature manner on their object of study. The study consists of two experiments based on recordings of reading speed, as well as preference ratings. The first experiment includes the variables type size (several sizes), typeface 'style' ('Roman' and 'Gothic', i.e., serif and sans serif) and typeface 'font' (the serif typefaces Century Schoolbook, Times Roman and Bodoni; and the sans serif typefaces Helvetica, Trade Gothic and Spartan).²⁸⁹

* Vanderplas, James M., and Jean H. Vanderplas. 1980. 'Some factors affecting legibility of printed material for older adults'. *Perceptual and Motor Skills*, vol. 50, pp. 923–932.

289. Trade Gothic is equivalent to American Type Founders' well known sans serif typeface News Gothic. The Spartan typeface in question is an equivalent to the Bauer type foundry's well known sans serif typeface Futura (cf. Wheatley 1983).

The second experiment deals with line length and interlinear spacing.²⁹⁰ Twenty-eight subjects between the age of 60 and 83 participated in the first experiment (the one primarily of interest to us). The material presented to each subject was contained in a 90-page booklet with 30 combinations of type size, type style and typeface. Every three pages is occupied by one page with prose passages (i.e., the stimulus material), one page with a multiple choice comprehension test for control purposes only, and one page with a questionnaire for subjective preference ratings.

Among other things, the results

indicates that Roman styles are associated with higher over-all reading speeds than Gothic styles, with Century Schoolbook most consistently associated with the higher speed. (p. 927)

However, the authors add some important qualifications: two typeface/size combinations resulted in higher reading speeds than others, among them the sans serif typeface Helvetica in 16 point and the serif typeface Bodoni in 16 point. Increase in type size generally produced higher reading speed, except for the 14-point type which produced a decrease in reading speed.

The authors conclude:

If one were to use the present results as a basis for recommendations, one might not be ill-advised to choose Century Schoolbook or a similar Roman type style over Gothic styles, with size in the 12- to 14-point range, as opposed to the usual choice of 8- to 12-point type for younger persons. (p. 931)

The authors are however aware of that these style/size combinations also will interact with combinations of line length and inter linear spacing, and possibly produce additional variations in performance and preference. Furthermore, the authors admirably point out that this study only included a small fraction of available typefaces of which many

290. The article is not illustrated, but information is given about how to obtain photocopy or micro-fiche samples of the stimulus material. The authors' verbal description of the stimulus material suggests a wise elimination of typographic nuisance variables, for example by proportionalization of type size to line length from page to page of the stimulus material; in accordance with recommendations given by Tinker in his 1963 *Legibility of print*.

might be more legible or less legible than the ones employed. With these qualifications in mind, they suggest that instead of relying on existing empirical research results, one should rather be prepared to test for legibility any particular material that is being prepared for print. Interestingly, this idea (expressed in 1980) is similar to today's 'usability testing paradigm' in information design.

Suen and Komoda 1986 *

Title of the paper: 'Legibility of digital type-fonts and comprehension in reading'. Published in *Text processing and document manipulation*, edited by J.C. van Vliet, and published by Cambridge University press on behalf of The British Computer Society. Authors: C.Y. Suen and M.K. Komoda, at with Concordia University in Montreal, respectively at the Computer science department and the Psychology department. This paper, which is published as part of the proceedings from an international electronic publishing conference, presents two experiments.

The authors argue that advances in computer technology have brought many new methods of conducting research in typesetting and typography and that this especially benefits evaluation of 'type-font design'. Characters from typefaces can be input by an optical scanner and converted into digital images that can be easily reproduced by computers and matrix and laser printers (it seems that the authors are ignorant about the contemporaneous existence of high resolution digital typefaces). Therefore, the authors argue, legibility can be studied much more rigorously than before, because variables such as 'font style', character shape, format and spacing, exposure time, and presentation speed, can be controlled precisely by the computer. However, when it comes to the authors' own two experiments it is only exposure time and presentation speed that in the end is relevant. Furthermore, although

* C.Y. Suen and M.K. Komoda. 1986. 'Legibility of digital type-fonts and comprehension in reading'. In *Text processing and document manipulation*. Proceedings of the international conference, University of Nottingham, 14–16 April 1986. Edited by J.C. van Vliet, pp. 178–187. Cambridge: Cambridge University Press on behalf of The British Computer Society.

the text refers to an illustration of the typefaces employed, the illustration is not present. It is therefore impossible to judge visually whether potential nuisance variables are 'controlled precisely' or not.

Two 'typewriter' typefaces and one primitive printer typeface are studied, respectively Letter Gothic, Courier, and DECwriter. The Letter Gothic font represents sans serif typefaces and Courier represents serif typefaces, whereas DECwriter (with absence of real ascenders and descenders) represents alternative coarse 'dot-matrix'-typefaces. All the three fonts have been digitised by an OCR device, with resulting representations of each letter positioned within a matrix of 21×42 cells, presented on a 'high resolution' computer screen. All three fonts are typewriter/printer fonts, and all are probably monospaced. The fact that Courier, as opposed to most serif fonts, is monoline, and that its serifs are highly exaggerated (in order to compensate for it being monospaced) does not seem to bother the authors.

The first experiment investigated the legibility of individual letters (both capitals and small letters), while the second investigated the legibility ('readability') of continuous texts.

The first experiment was based on tachistoscope-like short exposures of individual letters on a computer screen followed with a masking stimulus after three different periods of time, 'the inter stimulus interval' (0, 16.7 and 33.3 ms). The approach is inspired by work by the cognitive psychologist Ulrich Neisser, and is referred to as 'the visual backward masking paradigm'.

the legibility of letters is increased due to the increased amount of time available for the processing of the letters before they are degraded by the mask. With respect to the present experiment, then, letters presented in inherently more legible fonts should lead to superior performance, especially at the shorter ISIs [inter stimulus intervals].
(p. 179)

Six subjects, with normal vision, participated in the experiment. The results show that with no interval (0 ms) between exposure and masking, performance was highest for Letter Gothic and poorest for DECwriter. With the longest interval (33.3 ms) performance were similar for all three fonts.

These results, then, suggest that the sans serif font Letter Gothic is the most legible among the three fonts and that the DECfont is the least legible among the three. (p. 180)

The second experiment, however, is based on an acknowledgement of that it is 'not clear' whether the results from the first experiment 'can be generalized to reading' (p. 184). The authors refer to the cognitive scientist David Rumelhart's suggestion that 'reading is accomplished through a set of interactive processes' where skilled readers constantly employ their pre-established knowledge of language to recognise individual words while reading. The authors here echo an insight that some commentators since the latter part of the 19 century have expressed in various ways: that 'factors' such as typeface are very much peripheral to the reading process, and thus that it will be extremely difficult to quantify any meaningful differences in legibility by various operational methods.²⁹¹ However, the authors do not seem to realise the implications of this 'insight', and go on devising yet another laboratory experiment, this time maybe one that has marginally more ecological validity.

In this second experiment, which basically is a comprehension test, the subjects read paragraphs taken from *Reader's Digest*. The words are presented on a computer screen one at a time at a rate of 600 words per minute, and questions for checking comprehension have to be answered. This method is called 'the rapid sequential visual presentation (RSVP) procedure for comprehension'. Thirty-six subjects with normal or corrected vision participated. Their reading proficiency was tested beforehand.

The results: As in the first experiment, the DECwriter font appears to be the least legible. The authors suggest that the explanation lies in the fact that the DECwriter font is a coarse dot-matrix font, and, that its small letters lack ascenders and descenders. They also point out that the results of the two experiments are not entirely consistent: whereas the sans serif font Letter Gothic was most legible in the first experiment, in the second experiment 'very little, if any, differences was observed in the reading performance of texts presented in' the serif and the sans serif

291. See the section 'Peripherality to the reading process', in chapter 4.

typeface. They therefore suggest that ‘the reading skills and strategies, brought to the reading situation can apparently attenuate the effects of the purely visual characteristics of the text being read’ (p. 186). Although this ‘reading situation’ certainly is an extreme laboratory situation, their suggestion is plausible: it is exactly the extreme unfamiliarity of the DECwriter font that has played a role in a ‘more’ real reading situation than in the first experiment.

Having in mind that this paper probably was written in the first half of the 1980s, one of the conclusions seems appropriate and timely: The authors conclude by questioning the adequacy of contemporaneous primitive dot-matrix printer fonts that lack proper ascenders and descenders – such as the DECwriter font.

Taylor 1990 *

Jeffrey Taylor is preoccupied with ‘the learning process and how effective [*sic*] the transfer of information takes place’ (p. 3). In the rationale provided for this study, Taylor describes a new situation after the advent of ‘desktop publishing’, where authors who produce manuals and technical documentation are making all the decisions, from content to printing style. Questions such as what typeface to use – which ‘have traditionally been the domain of publishing houses and printing companies’ – are now dealt with by the authors. However, authors and their organisations

do not have a great deal of research to guide them in making basic decisions about how to prepare effective and efficient technical manuals. (pp. 1–2)

* The title of this PhD thesis: ‘The effect of typeface on reading rates and the typeface preferences of individual readers’. Author: Jeffrey Lynn Taylor. Dissertation submitted to Wayne State University in Detroit. Major: ‘instructional technology’. This is the second of three recent PhD theses on the relative legibility of serif and sans serif typefaces, submitted under the label ‘instructional technology’ at Wayne State University (the other two are Stahl 1989, and Kravutske 1994).

And Taylor goes on:

There is a plethora of advice, but little in the way of empirical studies to help make the decision on serif versus sans-serif typeface. Which typeface, serif or sans-serif, is easiest to read? (p. 4) ... The reference material and literature, currently available, are filled with opinions, suggestions and inferences regarding the use of a serif or a sans serif typeface. Most of these opinions are intuitive based or based upon centuries of past practice by the printing and/or publishing industry. The past practice is usually provided as the rationale for choosing either the serif or sans-serif typeface. The printing and/or publishing industry has traditionally been responsible for the look of texts and technical documentation. They take the written material and apply their skills to make the printing of texts and textual material 'look' good from a page perspective. Little effort was taken to evaluate the design of these printed textual materials from the learner's perspective. (p. 15)

And, not unexpectedly, Taylor contrasts the situation he describes with the potential of his findings:

The significance of this study lies in its potential for providing a foundation for developing effective and efficient texts, documentation, and technical manuals. (p. 11; and in the abstract, p. 86)

Taylor's study is based on an unsubstantiated postulate and adjoining premise: that the most basic decision to be taken when designing text is whether or not to use a serif or a sans serif typeface:

The process of producing texts, technical documentation, or manuals should begin with the decision that is usually the last. What typeface should be chosen for the document, i.e. should it be a serif or sans-serif typeface? (p. 4)

There are many issues to resolve ... but they all must start with the same choice, typeface. (p. 11; and in the abstract, p. 86)

This explicit premise is far from the view of typography in which typefaces play a relatively peripheral role in the reading process.²⁹² What's more, Taylor point of view is carried to the extreme:

If a serif or sans-serif typeface will permit the reader to read faster, then the documenter can consciously choose a serif or a sans-serif typeface to accomplish the readability desired for either the entire document or various parts. (p. 5)

292. See the section 'Peripherality to the reading process', in chapter 4.

Using typefaces that can speed the reader up or slow the reader down can be important. (p. 22)

Accordingly, in his final conclusion Taylor claims that by changing between a serif and a sans serif typeface

designers and developers of texts ... can move the reader along with a serif typeface and then slow them down with a sans-serif area which they are drawn to by a preference for that typeface. (p. 75; and, prominently, in the abstract, p. 87)

This remarkable conclusion is based on the author's interpretation of his data: Text set in serif typefaces was read faster than text set in sans-serif typefaces, whereas the subjects preferred text set in sans serif typefaces. However, not only is the author's conclusion sensational and far from convincing. I will argue, further on, that also the author's interpretation of his data is far from convincing.

Nevertheless, elsewhere in his thesis – without fully acknowledging the implications – the author obliquely indicates that factors such as typeface might be peripheral to the reading process. While reasoning around his results he suggests that 'good' readers are the ones who can benefit the most by clever manipulation of typographic variables such as typeface, whereas less good readers (who assumingly struggle more with merely grasping the content) don't benefit.²⁹³ However, the author seems to forget that it is manuals and technical documentation that constitute the context-rationale for his study. The test-material actually used does not match this context-rationale. The test material – generating the data from which his conclusion stems – is not text taken from manuals and technical documentation, but relatively 'easy' subject-matter prose à la *Reader's Digest*. Manuals and technical documentation are typically read or consulted in less than ideal situations, and in such a 'reading for action'²⁹⁴ situation, the perceptual task of 'mere pick-up and registration' of the text – more than in other contexts – often comes second to trying to grasp a less than accessible written exposition while

293. In any case, the author also seems to contradict himself: 'The reader who must struggle with a typeface that slows the reading process down will also more likely be distracted by other elements contained in the material.' (p. 41).

294. For an interesting discussion on 'reading as goal-driven behaviour', see Bouwhuis 1990.

trying to solve an awkward problem.²⁹⁵ Thus, in such a situation, the typeface is probably even more peripheral than in most reading contexts. This means that the ‘peripherality of the typeface perspective’²⁹⁶ is even more relevant here than otherwise (because of the mismatch between context-rationale and test material in the study). Thus, when taking the context-rationale into account, then Taylor’s conclusion is not only sensational as suggested above, but also extreme.

Two methods were employed in this study: One ‘objective’, that is, by measuring the speed of reading, and one ‘subjective’, by recording the (implicit) preferences of the readers. The study seems in many respects to be well designed. It is based on a double blind design, where neither the subjects nor the administrators of the test knew precisely what was being tested and when. The tests were integrated with ordinary reading activities and regular periodic reading rate tests. The 74 subjects (the ones who participated all the time) were all high school students. High school students were chosen as population ‘because they are regularly reading textbooks and technical manuals’.

Only two typefaces were employed, Pacific Data Corporation’s version of the serif typeface Times Roman, and their version of the sans serif typeface Helvetica, both in 12 point size. Nominal interlinear spacing was 20% of the nominal point size. The text was excerpts from articles carrying titles such as ‘Amazing partnerships in nature’ and ‘Tropical forests’, mainly taken from a collection called *Be a better reader*. No illustration of the test material is shown in Taylor’s thesis.

The subjective study showed a clear preference for sans serif typefaces (pp. 58–62, 70). In the experimental study the mean serif reading rate was 260.3 words per minute, and the mean sans serif reading rate was 259.0 words per minute (computed by me, on the basis of a table on p. 50). The reading rate difference between serif and sans serif typefaces is thus negligible.

295. Nevertheless, I am not implying that the point size of the text does not matter when consulting an auto manual in a badly lit garage (this example is used in Krull 1997, p. 171). But Krull’s situational factor is not necessarily relevant to the point I have just made.

296. See the section ‘Peripherality to the reading process’, in chapter 4.

However, the reading rate result does not deter Taylor from trying to find information in his data that points to alternative conclusions. He does this by ranking the reading rates into four quartiles, and excluding the first and the fourth as 'extreme' quartiles. However, not only does he trim his data in order to obtain a central tendency measure, he also excludes the second quartile as below average. Eventually, Taylor is then left with the third quartile (the 'good readers'); which is used to test for significance.

By doing so Taylor found that 'students in the third quartile of reading rate rankings (the 'good' readers) do indeed read serif typeface text faster than they read sans-serif typeface material' (p. 62). For the third quartile the serif mean was 262.7 words per minute, while the sans serif mean was 254.5 words per minute (see p. 55). He adds (in order to support this finding) that 'for all students the overall mean reading rate for the sans-serif typeface material was *slightly lower* than the mean for the serif typeface reading rates (p. 63, my italics). The expression 'slightly lower' is certainly not an understatement. (It can easily be interpreted as conveying a bigger difference than the actual mean reading rate difference of 1.3 words per minute.) Taylor concludes that the difference for the third quartile readers was 'statistically significant' (p. 64). He attaches great importance to this finding:

The statistically significant difference between the serif reading rates over the sans-serif reading rates in the 3rd quartile of rankings is very important. This group is the good readers who can gain the most by the techniques and factors of manipulating texts, documentation, and textual material. ... These are the readers who can read above average and should be able to benefit from manipulations of the text, presentations, or other factors which may permit them to read faster. As evidenced by the statistically significant outcome of this group, the reading of a serif typeface permitted them to read faster than with the sans-serif typeface. (p. 66)

What is more, in the final summary, 'statistical significance' is confused with substantial or meaningful 'significance':

The speed of reading either the serif or the sans-serif typeface had a *statistically significant* outcome in this study. The preference study was *also* certainly *significant*. These high school students who were tested in the study had a strong preference for a sans-serif typeface. (p. 70; my italics)

The choice of a serif typeface permits [that] ‘good’ readers read *significantly* faster than [with] a sans-serif typeface. (p. 74, my italics)²⁹⁷

Moreover, one page further on in the final summary, the ‘good readers’ have become the ‘readers’:

Even though the *readers* could read faster with a serif typeface, they still preferred the sans-serif typeface. (p. 75, my italics)

I feel uneasy about the move constituted by ‘the third quartile argument’. Furthermore: The sudden and unacceptable (but almost invisible, when not reading the thesis from beginning to end) semantic transition in the final summary, from ‘statistical significance’ to ‘significance’, and from ‘good readers’ to ‘readers’, adds to this unease. In fact, the way the results are interpreted is not convincing, and only too convenient in relation to the final ‘interesting’ and sensational conclusion (on deliberately slowing down readers on occasions by using sans serif typefaces).

In addition, possibly confounding factors are over-looked. Because of the author’s non-adequate domain knowledge, the study lacks crucial awareness of confounding nuisance variables residing in the stimulus material; related to appearance size,²⁹⁸ x-height, and interlinear spacing. To compare text set in Times Roman with text set in Helvetica, with the same nominal attributes with regard to type size and interlinear spacing (and at the same time disregarding a most likely crucial difference in x-height), introduces a whole set of nuisance variables that undermines the internal validity of this study.

The irony in this respect is that the author has made it clear on pages 2 and 11 that ‘typeface’ may influence ‘the outcomes attributable to other aspects of text presentation’ (he refers to ‘leading’, ‘type styles’,

297. The fact that it has become common practice in some quarters to drop the word ‘statistical’ before ‘significant’ (see Pedhazur and Schmelkin 1991, p. 202), may explain the use of the expression ‘significance’ in the final summary. However, it does not explain the authors’ shift from ‘statistical significance’ to ‘significance’.

298. Without realising the implications, the author touches upon the problem of nominal size versus appearance size, while explaining basic typographic terms on pp. 10–11. Here, while explaining the (now obsolete) term ‘stamp’, he writes that ‘the printed letters may vary in size from specific type style to type style.’

‘page layout’, ‘cueing’, ‘bolding’ and ‘justification’, ‘etc’), and thus that ‘typeface contamination’ may have influenced the results of other studies on content, lay out, justification, leading, etc. Thus, the author here acknowledges that the typeface is only one of many typographic variables that may influence ‘the outcome’. However, the author only appreciates that the typeface variable may influence other variables. He does not appreciate that other typographic variables (both the micro and meso variables I mention above and the meso and macro variables the author mentions) *interact* with the typeface variable and thus may influence the effect of the particular typeface chosen. By isolating and privileging the typeface variable as a *basic* prerequisite, typeface is no longer a *variable*; it has become a one way *parameter*, able to influence, but not to be influenced.

Further lack of adequate domain knowledge is revealed on many occasions throughout the thesis. For example: Taylor wrongly claims that the point size incorporated on most desktop publishing software is ‘slightly less than the 1/72 ... inch’ (p. 10).²⁹⁹ He also uncritically circulates a piece of information (which certainly lacks empirical verification) that stems from the graphic design consultant Jan White, claiming that whereas the serif typeface is the default typeface in the United States, the sans serif typeface is the default in Europe (p. 72). More revealing: The author displays a (not un-common) confusion when it comes to the use of bold typefaces (see p. 73). ‘Bolding’ is wrongly understood as an alternative to regular type, and not as a contrastive navigational device or complementary articulatory device.³⁰⁰ I believe it is the same kind of confusion that (mis)-leads the author to his final core conclusion (about slowing readers down – when convenient – with a sans serif typeface, based on his ‘findings’ that readers both prefer and read more slowly, text set in sans serif typefaces). The author does not seem to realise that a contrast such as the one created by displaying with another typeface chunks of text within a larger text, in the first place is simply done in order to signal some kind of semantic or structural

299. For introductions to discourse on typographic measurement, see Boag 1993, and 1996.

300. For two papers on the use of bold typefaces for articulatory and navigational purposes, see Twyman 1986, and 1993.

difference, and not necessarily ‘important material’ that the reader will ‘prefer’ to read (see p. 70).

The lack of a more theoretically inclined domain knowledge is also revealed in Taylor’s thesis. In his final discussion, while discussing other areas that ‘need to be explored’, Taylor suggests ‘headings or guidelines used throughout the textual materials’. Certainly, such a high-level question on overall topic structure, and consequently the access structure of written text is, assumingly of great importance. However, when the author asks: ‘does the use of headings or guides within the textual materials give the reader a more structured technique and therefore is a more efficient and effective textual document?’ (p. 74), he reveals an understanding of text structure as something posterior and external to the text itself. Such an understanding is contradictory to the idea about ‘the textual function’,³⁰¹ that is, that any text in order to be a text is more than mere content; text is content and structure in the same instance. Text is thus material and not only a heap of data and loose or inter-related ideas. The concept of ‘the textual function’ suggests that structure is intrinsic to text. Content as text, becomes structured when it becomes text, that is, the textual function does not only give the ‘content’ a linear structure, but also an hierarchical semantic structure (on both the inter-sentential paragraph level and on the overall rhetorical discourse level). Whether or not to further structure the text (to reveal a latent structure or by re-arrangement) – by inserting headings that reflects, or generates, a semantic and hierarchical topic structure – is not a question that can be solved empirically by posing a dichotomous ‘either or’ question. Any text, to be a text, depends, more or less and on several levels, on linguistic and extra-linguistic devices that signals or indicates structure and thematic progression (such as for example backward and forward pointing cohesion cues for the bridging of gaps; and punctuation,

301. See Halliday 1970: ‘Finally, language has to provide for making links with itself and with features of the situation in which it is used. We may call this the textual function, since this is what enables the speaker or writer to construct “texts”, or connected passages of discourse that are situationally relevant; and enables the listener or reader to distinguish a text from a random set of sentences. One aspect of the textual function is the establishment of cohesive relations from one sentence to another in a discourse’ (p. 143).

paragraphing and headings for articulation of boundaries and labelling of segments).³⁰²

I point to these examples of inadequate domain knowledge, not in order to score cheap points, but because I suspect that the examples are symptomatic of many legibility studies. As I point out on several occasions elsewhere in this thesis, it is exactly the lack of adequate domain knowledge that seems to be responsible for the problems many legibility studies have struggled with. 'In practice, lack of insight and expertise in the domain of application can lead to naive or misleading conclusions and recommendations.' (Schumacher and Waller 1985, p. 379).

The concept of 'domain knowledge' is used in the field of 'human-computer interaction' and 'usability studies', while referring to 'task', 'target' or 'application' domains such as accounting – or for that matter – typography. It is not an unproblematic concept. There can be no sharp well-defined borders between 'domain knowledge' and 'non-domain knowledge'. In typography, domain knowledge can be argumentative, descriptive or prescriptive, and codified in manuals of typography. Domain knowledge is also represented by practitioners' practical knowledge (whether tacit or discursive). In addition, knowledge from a diversity of academic disciplines such as psychology, computer science, text preoccupied disciplines such as literary studies and linguistics, and 'text as physical object' preoccupied disciplines such as analytical bibliography and palaeography, have clearly contributed to establish a sophisticated typographic 'knowledge domain'.³⁰³ It follows from this that Taylor's thesis also contributes to this typographic knowledge domain; it is of course not separate from what it describes. However, it

302. This argument has been informed by the following works: Halliday 1970; Halliday and Hasan 1976; de Beaugrande 1980, 1984; Nash 1980; Pace 1982; Bernhard 1985; Southall 1989; Nunberg 1990; Waller 1985, 1991; Stiff 1994; and not least, Vagle 1995. For a fairly recent empirical study on the use of headings in text, see Williams and Spyridakis 1992. Nevertheless, Taylor must be excused if his concern is only with the problem of 'guidelines' and 'over-organised' text, as in some examples of distance-teaching materials; see Whalley 1993. For a relevant discussion – on text structures (in plural) and the ontological status of texts – see Renear, Mylonas and Durand 1996; Huitfeldt 1995; and Biggs and Huitfeldt 1997.

303. The references in the previous footnote may well serve as examples of such contributions.

does not follow from this that all knowledge claims belonging to or related to a knowledge domain are equally valid, plausible or persuasive.³⁰⁴

To sum up: the author's concern with 'the learning process' and his seemingly timely concern with providing 'desktop publishers' with empirically based knowledge sounds reasonable. However, much seems to be wrong with his thesis. It is based on an un-substantiated postulate and adjoining premise (by privileging the typeface 'factor' without presenting any sound reasons for that). The way the results are interpreted is not convincing. The final (sensational) conclusion is far from convincing. The way 'statistically significant' changes to 'significant' is dubious. And finally and most crucially – due to lack of adequate domain knowledge – the study lacks internal validity.

de Lange, Esterhuizen, and Beatty 1993 *

At the 'Raster imaging and digital typography' conference in Germany in April 1994 ('RIDT94'), Rudi de Lange from the Department of Graphic Design at Technikon OFS in Bloemfontein in South Africa presented the paper: 'Performance differences between Times and Helvetica in a reading task'. The paper has been published in the journal *Electronic Publishing*, and was based on de Lange's 1993 dissertation – 'The legibility of sans serif typefaces, an experimental and comparative study' (de Lange 1993), where his co-authors appeared as supervisors. The results of the study had been made public around a year earlier, in the

304. For a contribution to the question of what domain knowledge may or may not be – with references to the 'underdeveloped literature of typography' and 'a rich and diverse body of knowledge won from and tested by practice' (p. 150) – see Paul Stiff's instrumental case study 'The end of the line: a survey of unjustified typography' (1996).

* Rudi W. de Lange, Hendry L. Esterhuizen and Derek Beatty. 1993. 'Performance differences between Times and Helvetica in a reading task'. *Electronic Publishing*, vol. 6, no. 3, pp. 241–248. See also Rudi Wynand de Lange. 1993. 'The legibility of sans serif typefaces, an experimental and comparative study'. Master's Diploma dissertation. Bloemfontein: Technikon OFS; and [Lange, Rudi de]. 1992. 'The legibility of sans serif typefaces'. *Graphix*, December/January, 1993, pp. 15–16.

South-African printing trade magazine *Graphix* ([de Lange] 1992). For clarifications, I will on a few occasions refer to de Lange's unpublished dissertation. Otherwise I will only refer to the article in *Electronic Publishing*. No further references will be made to the article in *Graphix*.

The starting point for this study is a situation, as described by the authors, where typographers and printers even today regard roman typefaces as more legible for continuous text than sans serif typefaces. De Lange and his co-authors substantiate their claim by referring to a whole range of present-day American and British manuals and popular handbooks on graphic design, typography, and desk top publishing, including Richard Rubinstein's textbook *Digital typography*. In these sources they find that:

Authors argue that readers prefer serif typefaces, read them faster, recognize them easier and that there could possibly be a higher comprehension rate with material printed in these typefaces. (p. 241)

This is an interesting observation. Typographers today often reveal indifference towards the question of whether sans serif typefaces are less or more legible than roman typefaces or not, or alternatively see it as a false or dead dichotomy. Sans serif typefaces are used extensively for many different applications, including continuous text. However, it is still true that sans serif typefaces are used relatively little for continuous text in books and newspapers. De Lange and co-authors' suggestion that most manuals on typography promote the idea that sans serif typefaces are not as well suited for continuous text as roman typefaces are, is also worth noticing. Whether this is due to the aesthetic preferences of the authors of these books, it nevertheless, according to de Lange *et al.*, seems like this reluctance against sans serif typefaces is justified by referring to a notion of 'legibility'.

De Lange *et al.* contend that

Most of these opinions are, however, not based on any satisfactory empirical evidence, but appear to reflect the personal opinion of the authors. (p. 242).

Assumptions about the superior legibility of serif or roman typefaces appear to be untested generalizations. The author found no supporting evidence during an extensive literature study to confirm them. Many typographical practices are still based on the belief that romans are the most legible typefaces to use for text. This unsubstantiated belief, the

importance of legible instructional text, and the central part that typography plays in the graphic design process, motivated this study. (p. 243)

De Lange also refers to researchers who suggest equal legibility between sans serif and roman typefaces (Tinker 1963; Poulton 1965; Zachrisson 1965; and Moriarty & Scheiner 1984), and researchers who suggest roman superiority (Robinson, Abbamonte and Evans 1971) (pp. 242, 246). He also refers to the graphic design consultant and author of many books on typographic design, Jan White, who suggests equal legibility.

If we accept these research results as ‘empirical evidence’ (pointing as they do in another direction than most of the ‘untested’ ‘assumptions’ in manuals of typography), it is reasonable to ask why de Lange *et al.* felt it necessary to carry out yet another large scale experimental study on typeface legibility. No answer to this question is found in the *Electronic Publishing* article. However, an answer is present in the dissertation (pp. 7, 29–32). There de Lange refers to ‘errors’ in the research designs of the studies of Tinker, Poulton, and Zachrisson. According to de Lange, Tinker’s experimental design suffered from several errors, among them relying on nominal instead of actual typesize; relying on only one sans serif typeface among the ones being tested; and relying on the Chapman-Cook test which according to Zachrisson is too much an attention test and not a speed reading test. De Lange also points out that readers today are much more used to sans serif typefaces, and that there exist a much larger variety of sans serif typefaces, of which some are possibly more legible than the one employed by Tinker. Zachrisson’s experiments are criticised with regard to formal experimental procedure; that the ‘visual’ sizes of typefaces varied; and that he employed too small a sample size with regard to population. Poulton’s experiment measuring comprehension rate is implicitly criticised for measuring comprehension and not ‘legibility’. Robinson, Abbamonte and Evan’s seriously flawed ‘pro-serif’ study, which, in opposition to the above mentioned studies, is not only mentioned but also evaluated in the *Electronic Publishing* article, is referred to in an uncritical manner and only rejected by de Lange because it did not investigate the legibility of *continuous* text: ‘[an] interesting study ... [which] determined the importance of serifs in the

perception of individual letters and not the legibility of continuous text.’ (p. 242).

De Lange’s experiments³⁰⁵ were of four different kinds: a word recognition test (oral reading of discrete word units); a speed reading test (silent and subsequent repeated oral reading of discrete word units); a comprehension marathon test (answering questions after having read a text), and a scanning test (a search task in continuous text). These experiments involved 450 subjects, all primary school children. Method with regard to the selection of subjects, experimental procedures, and statistics, seems to be well considered and sound. However, the same can not necessarily be said of the overall experimental design.

Commendably, several sans serif and roman typefaces (cf. the dissertation) were employed to represent the two typeface categories in question, and in each experiment the two alternative typefaces were matched closely with regard to actual size, weight, letter spacing and line spacing. Times Roman and Helvetica (laser printed) were the typefaces employed in the scanning experiment, the only one of several experiments which are described in some detail in the article (hence its title). This experiment was regarded by the authors as ‘the most convincing experiment’. In fact, the other three experiments are, in the dissertation, but not explicitly in the article, more or less discarded as methodologically invalid.

The authors present the results of their study the following way:

The authors obtained sufficient evidence during the word recognition, speed reading, and comprehension test not to reject the research hypothesis of equal legibility between roman and sans serif typefaces. *Serifs do not appear to have a noticeable effect on legibility, as measured by all the tests employed in this study.* The subjects did not read and recognize words with serifs faster, their comprehension did not increase, and they were also not able to find a word in a portion of text more easily, when the text was set in a roman typeface. ... Sufficient evidence was also found during the scanning experiment not to reject the hypothesis that Times Roman and Helvetica are equally legible. (p. 243, my italics)

But exactly how ‘convincing’ is the scanning experiment for ‘determining’ the legibility of continuous text? The authors contend that since

305. That is, the ones reported in *Electronic Publishing*.

searching for a certain word does not require ‘verbal fluency’ and comprehension, the ‘physical structure’ of the text and typeface will cause the only interference with the scanning process. And since the ‘physical structure’ of the text is the same for both typefaces, the different typeface designs are what is left as possible causal factors in a otherwise carefully constructed experiment. Since de Lange found no significant difference between the roman typeface and the sans serif typeface as measured by the scanning procedure, he concludes that there is no significant difference between the legibility of the two typefaces (p. 246).

However, in the dissertation, and only there, de Lange wisely points out that the experiment might lack internal validity: it is possible that the experimental design is simply not sensitive enough to measure small differences in the design of two common typefaces. In order to verify the sensitivity of the experiment, he therefore devised a second scanning experiment comparing Times Roman with the ‘obviously less legible’ and mannered chancery cursive script typeface Zapf Chancery (reported in the dissertation only). Here he found a significant statistical difference in favour of the roman typeface, confirming the scanning approach as sensitive and valid.

Nonetheless, I am not convinced: the fact that a highly mannered and ‘obviously less legible’ typeface like Zapf Chancery produced the difference which de Lange found, does not rule out the possibility that a *crude* scanning procedure (which undoubtedly is appropriate when certain aspects of the visual organisation of documents are being tested), simply is not sensitive enough to detect possible differences of legibility, whether such differences exist or not, among well designed ‘common’ typefaces such as Times Roman and Helvetica.

Whether this study has fulfilled its aim – ‘in the absence of satisfactory empirical data ... to determine the comparative legibility of sans serif and roman typefaces’ – or not, is an open question.

As a conclusion, in order to justify their study, the authors state that ‘The results of this study differ from the opinion of most authors on the subjects of typography, legibility and printing.’ (p. 246,). Echoing Ken Garland, and Cheetham, Poulton and Grimbly back in 1965, de Lange continues:

These results can be interpreted as promising for designers and typographers, as it appears that legibility will not necessarily be sacrificed when sans serif typefaces are used for textual matter. (p. 246)

De Lange initially pointed out that 'Many typographical practices are still based on the belief that romans are the most legible typeface to use for text.' (p. 243). Nevertheless, even if we accept de Lange's conclusion at its face value (which with regard to its content ('equal legibility') is very reasonable), one problem still remains, the possibility that de Lange's notion of legibility (or any measurable legibility construct) simply does not correspond with the legibility notion of the 'many typographical practices' he refers to. That is, that the preference of many designers for roman typefaces in continuous text, although justified by a perhaps vague reference to 'legibility', is simply based on something else.

Noteworthy, in addition to the experiments, de Lange also includes an interesting and informed *pro et contra* discussion where he presents sensible counterarguments to many of the arguments which promote roman typefaces as more legible than sans serif typefaces – a discussion which by the way inevitably enforces the impression that his original motivation and starting point is as a fond advocate of sans serif typefaces. One of the pro-serif arguments which de Lange refutes through his counterarguments is the one which says that serifs assist the horizontal flow of reading and eye movements: de Lange points out that eye-movement research has shown that the eyes while reading continuous text do not follow lines of text in a strict horizontal flow, but in quick saccadic jumps.

Silver and Braun 1993 *

The authors of this paper are N. Clayton Silver and Curt C. Braun. At the time of publication, they were both affiliated with the University of Central Florida in Orlando. Together with various co-authors, they have published extensively in journals like *Safety Science* and *Ergonomics*, as well as in conference proceedings, during the 1990s on warning design and similar topics.

The rationale provided for this legibility study is reasonable. The context is the design of warning labels for potentially dangerous consumer products. The fact that ‘many consumer product related injuries occur despite warnings’ prompted this investigation of ‘factors influencing the willingness to read a warning’. Details of the study’s design as well the statistics indicate high standards.

The authors suggest that there are several attributes of warning labels that influence the willingness to read a warning. Among these factors are ‘perceived hazardness’, ‘attractiveness and understandability’, and ‘print conspicuity’. ‘Print conspicuity’ has – according to the authors – recently received increased attention, and they refer to a recent study that demonstrated that warnings set in 18 point Times with orange highlighting ‘was recalled better than’ warnings set in 12 point Helvetica with no highlighting.³⁰⁶ On this background Silver and Braun want to find out whether it is the typeface or the typesize that contributes the most to such an effect.

The authors refer to two comparative studies on sans serif versus serif typefaces which involved long passages of text: To Vanderplas and Vanderplas (1980), where text set in the serif typeface Century Schoolbook was read faster than text set in the sans serif typeface Helvetica, and to Moriarty and Scheiner (1984), who found no differences in reading speed between Helvetica and Times. Although the results of

* N. Clayton Silver and Curt C. Braun. 1993. ‘Perceived readability of warning labels with varied font sizes and styles’. *Safety Science*, vol. 16, no. 5/6, pp. 615–625, in a special issue on ‘Warning and risk communication’, edited by D.M. DeJoy and M.S. Wogalter.

306. The fact that a warning set in a large highlighted typeface ‘was recalled better’ can hardly be surprising.

the two cited studies apparently are heterogeneous (with ‘differing results’ according to Silver and Braun), and only one of the studies thus found a difference, Silver and Braun still want to find out if there are ‘similar differences’ (*sic*) on shorter warning labels. Nevertheless, an important premiss for the study is that Helvetica is considered to be the ‘standard’ typeface to use on warning labels, according to several manuals on the subject.³⁰⁷ The authors want to verify the rationale behind this ‘standardisation’.

Four type-related variables ‘associated with readability’ were examined: typeface (while focusing on the question of serif vs. sans serif typefaces), typeface weight, typeface size, and size contrast between the signal word and the main body of the warning. Three typefaces were included: The sans serif typeface Helvetica, and the two serif typefaces Times and Goudy. The warnings varied across the four typeface-related variables and were presented on 24 mock-up detergent labels. Forty undergraduate students and 22 elderly non-geriatric persons participated in the study.

The study can best be described as a ‘subjective legibility study’,³⁰⁸ rather than a ‘subjective preference study’, because it deals with perceptions of how *legible* the text is (the authors use the expression ‘readable’), and not the participants preferences for this or that alternative. Confusingly however, the authors use the term ‘readability’ in three different ways. First, they refer to readability as the lexical and semantic level of understanding of the text used in their material (‘the Kincaid readability index’ was used on the material in order to ascertain that the text could easily be understood by the participating subjects). Second, ‘readability’ is one of the three variables that the subjects should rate the material according to (the two other variables are: ‘how likely they would read the warning’ and ‘its salience’). Third, the three variables (because they showed a high degree of correlation) were ‘merged’ into the composite variable ‘perceived readability’.

307. The authors refer to FMC Corporation 1985; and Westinghouse 1981.

308. For a useful discussion on ‘subjective’ studies in this context, see the subsections ‘experimental studies’, ‘field studies’, and ‘subjective studies’ in Edworthy and Adams’ *Warning design* (1996, pp. 49–68).

Thus, the final and composite dependent variable ‘perceived readability’ was based on subjective ratings in response to a 24-page questionnaire with predetermined anchors of ‘how likely they would read the warning’, ‘its salience’, and ‘readability’. The authors point out that the construct ‘perceived readability’ should not be confused with ‘actual reading performance’, and they modestly admit that this might represent a limitation of the study.

Among the results: Helvetica was perceived to be more readable than Times and Goudy (‘a significant main effect’). And bold type was perceived as more readable than normal type.

The superior perceived readability of Helvetica bold when compared to the other font type and weight combinations conforms to the current standards (p. 622)

However, and of great consequence for the validity of the study: Out of the blue, in the middle of their paper,³⁰⁹ the authors almost by accident reveal that the Helvetica font used in the study is Helvetica Condensed (the two serif typefaces included in the study are not condensed). This information is not provided anywhere else in their paper, and their final conclusion reads accordingly:

The findings outlined in this study have implications for warning design. To increase the perceived readability of warnings, warning labels should be printed in a bold sans serif font such as Helvetica (p. 623)

Although Helvetica can easily be regarded as a more or less generic sans serif typeface, the condensed variant is certainly not, something which undermines the generalisability and thus external validity of this study (based on the authors’ own conclusion). Also: the condensed variant of Helvetica is not compared to condensed variants of the serif typefaces Times and Goudy (whether designed so, or electronically modified on the fly). This confounding condition, which introduces *appearance width* as a nuisance variable, alone undermines the internal validity of this non-experimental study: the authors’ conclusion could as well be that warning labels should be printed in condensed typefaces. The distinctive lesser area occupied by the text set in the condensed

309. In the main text and in a caption on p. 619.

typeface than for the two serif typefaces only amplifies this confounding condition.

Furthermore, the reason for including the serif typeface Goudy in the study (of all possible serif typefaces), is not exactly convincing: because it is a serif typeface ‘widely used in marketing applications’ that ‘has yet to be studied in terms of readability’. Nevertheless, to include two serif typefaces but only one sans serif typeface – when it is exactly generic differences which are focused upon – is unreasonable. There is always a chance that an additional sans serif typeface could have altered the mean result in either direction for the sans serif category.

Although in their introduction the authors are somewhat cautious about the validity of their approach (‘not measuring actual performance’), they are nonetheless far more categorical in their conclusion (‘warning labels should be printed in’). Systematic investigation of peoples’ subjective preferences ought to yield valuable information; humans are after all not automats. However, to ask if a bold variant of a typeface is more ‘salient’ than a normal variant of the same typeface, must necessarily produce a near unambiguous answer. This indicates another problem with this research: that the design of the questionnaire is not entirely sound. Nevertheless, the invalidating problem inherent to this investigation is not its main ‘subjective’ approach, but the authors’ lack of relevant domain knowledge about typographic variables that might function as ‘nuisance variable’ in studies like this.

This lack of domain knowledge also manifests itself in other ways than in the authors’ inability to perceive confounding typographic variables. The authors’ discussion on the use of bold type versus normal type is one instance where they display their lack of relevant (and crucial) typographic domain knowledge. Like Taylor (1990; see above), the authors display a not un-common confusion when it comes to the use of bold typefaces. The use of bold type is wrongly understood as in general being an alternative to regular type, and not a contrastive navigational device or a complementary articulatory device. So the authors inform the reader that: ‘Ralph (1982) suggested that boldface type should be used for stressing terms and ideas, rather than for all textual material’. This understanding is hardly groundbreaking,

reflecting, as it does, a practice of textual articulation which has existed since the mid 19th century.³¹⁰ Moreover, by citing the discredited researcher Cyril Burt without reservations (p. 623), the authors also display lack of sufficient knowledge of relevant research literature.³¹¹

There are interesting ‘findings’ in this paper (on size contrast between the signal word and the main text of a warning), but that is another story, and not of relevance to my investigation here. But at the same time, there are also some implausible suggestions. Among other things, the authors suggest that a possible reason why 10-point type is more readable than 8-point type, ‘is because the majority of type sizes used in general text range from 9-point to 11-point’ (*sic*). The ‘familiarity thesis’ of legibility is here stretched *in absurdum*.

The results of this research seem to be taken for granted by people in the human factors research community, eager to translate ‘research findings’ into guidelines for practical design. A recent paper on the development of design guidelines, published in a book on ‘human factors in consumer products’, simply states: ‘See Silver and Braun (1993) for readability issues of warning labels’ (Bonner 1998, p. 253). Bonner’s paper, which for the most part is a reflective and useful piece of writing, even promises that its aim is to deal with ‘broad issues at a high level of abstraction from the interaction process’ (p. 241). Nevertheless, Bonner’s reference to Silver and Braun’s low level and categorical – and problematic – research findings, has slipped through without reservation. Interestingly, the researchers Judy Edworthy and Austin Adams, in their recently published monograph *Warning design*, are more modest on behalf of the potential of research on ‘conspicuity’ and ‘attention getting’, while suggesting that

a good textbook on graphic design and typography, such as Baird *et al.* (1993), will be more informative than an attempt to use any of the available behavioural research data. (1996, p. 25)

310. For two papers on the use of bold typefaces for articulatory and navigational purposes, see Twyman 1986, and 1993.

311. See Rooum 1981; and Hartley and Rooum 1983; as well as the section ‘Burt, Cooper and Martin 1955 / Burt 1959’, above in this chapter.

Nevertheless, to conclude: Silver and Braun's categorical conclusion is inappropriate, and their study lacks validity, largely due to lack of adequate – and crucial – typographic domain knowledge.

Silver, Kline and Braun 1994 *

This study, 'Type form variables: differences in perceived readability and perceived hazardousness', published in the *Proceedings of the Human Factors and Ergonomics Society 38 annual meeting – 1994*, is another study by N. Clayton Silver, Curt C. Brown, and a co-author.

The primary object of interest in this study is the effective design of warning labels, as in Silver and Braun (1993). The rationale provided is similar: many injuries occur because people don't read warning labels. The authors suggest that this can be caused by many factors, like a high degree of familiarity with the product, underestimation of the risk, or inadequate labelling (whether lexical-semantically or graphically). While focusing on the labelling issue, the authors 'first purpose' is to 'determine if there is greater perceived readability' for Helvetica than for the two serif typefaces Century Schoolbook and Bookman.

Warning labels adapted from a common household insecticide are employed. The three typeface related variables in question are: Typeface; size of the signal word; and size contrast between the signal word and the main body of the warning. The resulting composite variable 'perceived readability' is averaged from two of the questionnaire variables: the variables 'salience of the warning' and 'warning readability'.

The results are generated on the basis of subjective preferences expressed by 50 undergraduate students who rated 36 insecticide warning labels which varied across the three type related variables ($3 \times 3 \times 4$ factorial design). The 'font variable' covered 3 typefaces: The sans serif typeface Helvetica and the two serif typefaces Bookman and

* N. Clayton Silver, Paul B. Kline, and Curt C. Braun. 1994. 'Type form variables: differences in perceived readability and perceived hazardousness'. *Proceedings of the Human Factors and Ergonomics Society 38th annual meeting – 1994*, pp. 821–825.

Century Schoolbook. One of the illustrations reveals that this time the Helvetica in question *is not* Helvetica Condensed (as in Silver and Braun 1993; see above), but an Helvetica variant of a ‘normal’ or ‘regular’ appearance width.

Among the results: the serif typeface Century Schoolbook is perceived as more ‘readable’ than the serif typeface Bookman, and both serif typefaces are more ‘readable’ than the sans serif typeface Helvetica. Thus, although compared to different serif typefaces than in Silver and Braun 1993, in this study Helvetica comes last, in contradiction to the former study.

Whether this result (and its striking contrast to the result in the former study) has something to do with individual differences between the serif typefaces employed in the two studies, or whether the answer simply is that condensed typefaces are the most readable (see my argument above, in the section ‘Silver and Braun 1993’), or whether the two groups of participants have different preferences, is an open question. However, the only possible explanation (implicitly) offered by the authors, suggests that the difference in stimulus material (domestic insecticide labels versus domestic detergent labels) has produced the different results (see below).

Although the authors discuss the results of this study in the light of heterogeneous results in previous studies conducted by themselves and other researchers, they still point out that

the superior perceived readability ... of Century Schoolbook as compared to Helvetica and Bookman corroborates with earlier research (p. 824)³¹²

By ‘earlier research’ the authors here refer to a paper ‘in press’ by one of the authors together with another co-author (Smither and Braun 1994) published in the *Journal of Clinical Psychology in Medical Settings*, and the paper by Vanderplas and Vanderplas (1980) (both papers are about ‘reading speed ... among older adults’). In this context they also refer to typographer Ruari McLean’s widely used textbook on typography in

312. However, on the first page in their paper, the authors refer to research that points in another direction (i.e. Moriarty and Scheiner 1984; Silver and Braun 1993; and an unpublished paper by Silver, Braun, Zeigler and Witt (‘Perceived readability of various fonts’, 1993).

order to substantiate their argument: ‘McLean (1980) pointed out that serif typefaces are easier to read when longer amounts of text are used.’ However, they refer to McLean as if McLean’s book is a research paper, without mentioning explicitly that it is a textbook on typography. The bewildering thing is that the authors, who only seems to trust ‘empirical evidence’, suddenly refer to typographer McLean’s point of view as if it carries the same authority as ‘empirical research’; but without mentioning explicitly that McLean’s point of view does not represent ‘empirical research’.³¹³ Accordingly, in Silver and Braun 1993, while doing a brief literature review, they referred to McLean in an even more ambiguous manner: the word ‘found’ in the expression ‘McLean (1980) *found* that serif typefaces are easier to read’ (p. 622, my italics), can denote ‘in his opinion’, but in the parlance of research papers like the one in question it denotes ‘research finding’.

The authors modestly deliver ‘obligatory’ qualifying statements about the validity of their research: First, the limitation that the research is not about ‘objective’ performance measurements. And second, that the stimulus material was insecticide labels: ‘It is equivocal as to whether these results will generalise to other products that are perceived as more or less hazardous’. I doubt that these reservations are the most urgent.

Except for the single qualifying word ‘consideration’, the authors conclude categorically on the choice of a typeface for warnings:

These results have important implications for warning design. To increase the perceived readability (or perceived hazardousness), consideration should be given to printing warning labels in Century Schoolbook. (p. 824)

The recommendation is even more low level than the recommendation in Silver and Braun (1993). This time, not a typeface category, but a particular typeface, is recommended. And furthermore, it even belongs to the opposite category than the category recommended last time. Furthermore, the fact that a particular typeface has been recommended and that that typeface has only been compared to two

313. McLean carries forward a reasonably plausible argument in favour of using serif typefaces for running text; but that is not my point here.

other typefaces does not seem to bother the authors, while ignoring the large number of readily available typefaces that was not included in the study.

The authors are apparently unaware about the inherent problems connected with the translation of low level and possibly chance results like the one in question to general recommendations. In addition: the heterogeneity of the results between this study and Silver and Braun's 1993 study suggests on its own that translations of this research into recommendations for practical designing is hardly feasible.

The other composite dependent variable in this study, 'perceived hazardousness', and especially its correlation with 'perceived readability' is problematic (see pp. 823–824).

the labels printed in Century Schoolbook were perceived as more hazardous than those printed in either Bookman or Helvetica (p. 823)

There was a significant linear relationship between perceived hazardousness and perceived readability ... Hence, the greater perceived readability, the greater the perceived hazard. (p. 824)

First: To subjectively rate or 'perceive' a typeface as more or less readable (and salient!) is more about utilitarian aspects that possibly has something to do with the experience of mediated effectiveness, or some kind of experience of ease of seeing. However, to rate typefaces on 'perceived hazardousness' is more about semantic connotations. Second: the fact that 'perceived readability' and 'perceived hazardousness' strongly correlate should get the alarm bells ringing: Do the participants really know what they are rating? Are there any confounding 'nuisance' variables? What if other (control) variables were introduced, and they correlated as well? Maybe then one conclusion that could be drawn is that the ratings simply have to do with 'saliency' (or something else, for that matter)?

That a typeface perceived as the most 'readable' also is associated most with hazard, is puzzling. It sounds reasonable that a typeface that is perceived as the most 'readable' can also be the most *effective* (in a utilitarian way) in conveying hazardousness. However, it is doubtful that such an explanation is in accordance with a reasonable interpretation of Silver, Kline and Braun's exposition, where they recommend that

To increase the perceived readability (or perceived hazardousness), consideration should be given to printing warning labels in Century Schoolbook. (p. 824)

That the (relative) subtle design differences between, say two rather anonymous bread and butter typefaces like Century Schoolbook and Bookman, on its own should have impact regarding ‘hazardousness’ is nothing but fantastic. My suggestion is that the overall design and validity of this study is seriously flawed.

It is admirable that the authors in this paper not only cite the research literature and manuals on warning design, but also Ruari McLean’s manual of typography (i.e. Mclean 1980). However, it should be pointed out that there is a puzzling lack of references to instances of highly relevant research literature. Although there are references to Paterson and Tinker, Reynolds, and two typeface legibility studies performed by someone else (i.e. Vanderplas and Vanderplas 1980, and Moriarty and Scheiner 1984), there are no references to any of the many existing typeface preference studies,³¹⁴ or to any existing typeface connotation studies.³¹⁵ Accordingly, there are no references to papers where the correlation or non-correlation of various operational constructs are critically discussed. What is more, Salcedo *et al.*’s (1972) extensive and fairly well known legibility study that explicitly discusses³¹⁶ (and discards) the correlation of operational constructs (including speed of reading, comprehension, and reader preference), which is based on the same concerns as Silver *et al.*’s study, is not mentioned. The rationale is the same (people don’t read the warnings), the topic is similar (pesticide labels), and a similar factorial design is employed (three typeface variables).

A final note: In a later published *semantic* typeface study (about ‘perception of product hazard’) published in 1995 (Braun and Silver

314. See the section ‘Subjective preference studies’, in chapter 2.

315. See the subsection ‘A note on “semantic” studies’ above in this chapter.

316. Where the authors, in contradiction to Silver *et al.*, discard such correlation.

1995),³¹⁷ the authors acknowledge that the results of their legibility studies and the other few studies they have cited, are contradictory (pp. 984–985). Furthermore, although the authors believe that the presence or absence of serifs affects legibility, they after all seem to have reached a new understanding, while suggesting that serifs or not may not affect semantic connotations of hazardousness:

the underlying hazard continuum for typefaces might not be adequately described by the presence or absence of serifs [*sic*] (p. 985)

Wheildon [1984] 1995 *

Back in 1984 the Australian newspaper and magazine editor Colin Wheildon published a booklet with the tickling title *Communicating, or just making pretty shapes?* (Wheildon 1984). The third edition of this booklet came out in 1990, reaching a total of 20,000 printed copies (Wheildon 1995, p. 14). According to the preface in the original booklet, the study had been accepted as a ‘mass communication research paper’ at the Virginia Commonwealth University in the USA, and had also been presented at various seminars, among these at the Poynter Institute for Media Studies in Florida. The aim of the study was to subject ‘some of typography’s maxims to research’. One of the questions Wheildon carried out research on was the relative ‘comprehensibility’, and ease of reading, of continuous text set in either roman or sans serif type.

317. ‘Legibility’ appears in this study only as an independent variable in a factorial design. That is, there are two levels of legibility, one represented by capitals set in the ‘high legibility’ sans serif typeface Helvetica, and the other by capitals set in the ‘low legibility’ decorative art nouveau display typeface Arabia. (Arabia is an imitation of the typeface Arnold Böcklin, supplied with the illustration software package CorelDRAW).

* Colin Wheildon. 1995. *Type & layout: how typography and design can get your message across – or get in the way*. Edited and with an introduction by Mal Warwick. Berkeley, California: Strathmoor Press. See also Colin Wheildon. 1984. *Communicating, or just making pretty shapes? a study of validity – or otherwise – of some elements of typographic design*. Sydney: Newspaper Advertising Bureau of Australia. This section of the thesis has, except for a few minor alterations, appeared as a book review in *Information Design Journal* (Lund 1998).

This research on typeface ‘comprehensibility’ was based on a test which involved 224 subjects (in two groups, one of which was a control group) who had read magazine- or newspaper-like articles with content of ‘direct interest’ to themselves. In addition, answers and comments from the readers to ‘leading’ questions about their attitudes ‘were collected for anecdotal rather than scientific value’ (1984, pp. 8–9).

Texts set in the newspaper typeface Corona and the sans serif typeface Helvetica were compared (pp. 18–20). The sensational results showed that 67 per cent of the readers reached a high comprehension level and 14 per cent a low comprehension level when reading the material set in Corona. By contrast, only 12 per cent of the readers reached a high comprehension level, and 65 per cent a low comprehension level, when reading the material set in the sans serif typeface Helvetica.

In addition, of the 67 readers who had reached only a low level of comprehension when faced with the sans serif type: ‘53 complained strongly about the difficulty of reading the type’, ‘11 said the task caused them physical discomfort’, ‘32 said the type was merely hard to read’, ‘10 said they found they had to backtrack continually to try to maintain concentration’, ‘5 said when they had to backtrack to recall points made in the article they gave up trying to concentrate’, ‘22 said they had difficulty in focusing on the type after having read a dozen or so lines’. Wheildon stressed that when the same group afterwards read an article in the serif typeface ‘they reported no physical difficulties, and no necessity to recapitulate to maintain concentration’. In order to stress the implications of his ‘findings’ Wheildon pointed out that this means that if the body text of an advertisement in a magazine with one million readers is set in a sans serif typeface, then ‘the message will be comprehended thoroughly by only 120,000 of our readers’ (1984, p. 4).

Wheildon did not leave the question with this ‘empirical evidence’. He extended his argument against the use of sans serif type for continuous text, by referring to the (postulated) optical phenomenon of *irradiation* which allegedly interferes harmfully with the perception of

print.³¹⁸ He did this by including an oblique reference to R.L. Pyke's *Report on the legibility of print* prepared for the British Medical Research Council in 1926; but without mentioning the author's name or including any bibliographical information:

In research collated by the British Medical Council [*sic*] in 1926, it was asserted that the absence of serifs in sans serif body type permitted what the council referred to as irradiation, an optical effect in which space between lines of type intruded into the letters, setting up a form of light vibration, which militated against comfortable reading. Serifs the research said, prevented this irradiation; thus serif types were easier to read. (1984, p. 18)

It was not the Medical Research Council or Pyke who 'asserted' or 'said' this about serifs and irradiation. What Pyke did, in his very comprehensive review of legibility studies, was simply to include, in a duty-bound and non-approving manner, coverage of six papers where the authors on a non-experimental basis ascribed their preference for serif typefaces by arguing that serifs helped to prevent harmful irradiation (see Pyke 1926, pp. 21, 76, 99–101).

After this rather dubious appeal to the authority of past research, Wheildon showed a reproduction of a magazine page set in a sans serif typeface. He commented: 'An attempt to read this, and comprehend it thoroughly, will lend a measure of credence to the irradiation theory.' (1984, p. 19).

Wheildon also vigorously, amusingly, and earnestly, argued against the idea that the legibility of sans serif typefaces is a question of habit and of people getting conditioned to them:

This is nonsense. It's analogous to saying that instead of feeding your children wheatie pops, you should feed them wood shavings. They'll get used to them and in term will learn to love them. (1984, p. 20)

No wonder Wheildon's conclusion was unambiguous – 'body type must be set in serif type if the designer intends it to be read and understood' (1984, p. 20).

318. "Irradiation" is the apparent extension of edges of an illuminated object as seen against a dark ground. A bright point appears bigger than reality, while a dark point appears smaller' (Watts and Nisbet 1974, p. 32).

Then, in 1995, Wheildon published *Type & layout: how typography and design can get your message across – or get in the way*. It is a collection of research material formerly published in three separate booklets. The content of his 1984 booklet constitutes a substantial part of this new book, which claims to be ‘an international benchmark study on the effects of typographic elements on the reader’ (p. 244).³¹⁹

If we are to believe Wheildon, his research has been ‘adopted by the New South Wales State Government, and influences design standards for all new state legislation aimed at improving comprehension in Australia’ (p. 244).

It is an interesting publication, not only for its dubious content, but also for its persistent appeal for external praise and legitimacy. In fact, this becomes excessive. The eight pages of praise at the front of the book, as well as a foreword and the editor’s introduction, are nothing short of hilarious. Among the persons praising the book on these pages are the famous advertising specialist David Ogilvy, the distinguished graphic designer Milton Glaser, the renowned graphic design consultant Jan White, a university professor, an ex-university professor, the director of design at the University of California Press, and the editor-in-chief and director of magazine development at Hearst Magazines. Here are some samples:

Now there’s nothing left to argue about! At last, conjecture, tradition, and hearsay have been replaced by the irrefutable logic of numbers and scientific survey results.

It’s reassuring to find that the typographic truths I have practised for years have some basis in hard evidence.

Hitherto designers have to rely on their guesses as to what works best in choosing the typography and lay out. ... Thanks to Colin Wheildon, they no longer have to guess. No guesswork here. Only facts.

Unfortunately, many of the ‘rules’ of effective design and typography I’ve offered have been based primarily on age-old truisms or minor research studies. But Colin Wheildon has changed all that. For the first time there is a body of statistically significant research to show which techniques achieve the objective of all printed material: maximum communication with readers.

It scientifically proves some long-held assumptions and makes other surprising and important discoveries.

319. The material on sans serif versus roman typefaces is found in the book on pp. 19–24, 53–60.

This book reports the results of nine years of his hard-nosed, rigorous research.

Wheildon himself contrasts his research with the former state of affairs in the following way:

Texts on typography frequently allude to research into some of the elements to be examined, but, regrettably, discussion of this research is usually anecdotal rather than empirical. (1995, p. 186)³²⁰

Moreover, the author elsewhere in the book lists the name of four advisors, of whom three are university professors and one is a Dr and director of research (pp. 190–192). In addition Wheildon claims to have

sought the advice of research consultants and academics in the United States, Britain, and Australia, and [I] submitted my proposed methodology, and later the results, to them for reservation, comment, or dissent. The consensus: that the study was both valid and valuable (p. 35)

Among the people from whom Wheildon claims to have sought advice are professor David Sless (the information design specialist), David Ogilvy, professor Rolf Rehe (who once published a book on legibility research), members of the academic staff at the University of Reading, and members of the academic staff at the Royal College of Art in London (p. 192). My guess is that it is highly doubtful that these persons were involved in Wheildon's work in the way that he implies, and that they would not authorise such use of their names.

Interestingly, Wheildon's original 1984 publication, together with one of the two other booklets on which *Type & layout* is based, recently received approval in a book from the Policy Studies Institute in London: *Designing public documents: a review of research* (Kempson and Moore 1994). In this book, commissioned by the Department of Social Security and partly based on work funded by the British Library, Wheildon's booklets are the most frequently cited sources of recent research on typography. At least twelve references are made to Wheildon's research: on sans serif versus roman type, justified versus unjustified text, lower case versus upper case, the use of bold type, and the use of italic type (Kempson and Moore 1994, pp. 51–56, 283–286). Statements like: 'Most

320. Also in Wheildon 1984, p. 8.

readers found it difficult to hold their concentration when reading the sans serif text' are offered without reservation. The authors' statement 'We hope ... that we have covered all the significant research' (p. 1), is depressing in light of what a close reading of Wheildon's work reveals.

It is worth asking whether this lack of discrimination is unique to Kempson and Moore's book, or is symptomatic of how 'research findings' often get summarised. Karen Schriver, assessing 'the research literature on typography', in her impressive *Dynamics in document design*, more or less uncritically includes two references to Wheildon's book (Schriver 1997, pp. 274–278). One is to Wheildon's research on roman and sans serif typefaces. Furthermore, in a review of the book in the journal *Technical Communication*, it is recommended

to all technical communicators who are interested in making sure that their content is delivered as comprehensibly as possibly. Furthermore, I encourage them to use it as a starting point for further research and consideration of typographic design. (Skarzenski 1996, p. 426)

At the time of its publication Wheildon's book received coverage in the American typography magazine *x-height* (1995). The irony is: whereas this unpretentious non-scholarly magazine admirably treats Wheildon with tongue in cheek, two books on information design, as well as the journal *Technical Communication*, reveal difficulties in discriminating between seriousness and blatant outrageousness.³²¹

Van Rossum 1997 *

In 1997, a typeface legibility study appeared in the Dutch/Belgian scholarly journal *Quaerendo* (a leading journal within the field of analytical bibliography and printing history). The author, Mark van Rossum, a scientist in theoretical physics, with a doctorate from the

321. Wheildon's book has produced many more reactions, and among the ones I have come across, only positive, except for a contra and pro exchange (of rather unsubstantial) letters to the editor in the American typography magazine *Serif* (no. 5, 1997, p. 7; no. 6, 1998, p. 4).

* Mark van Rossum. 1997. 'A new test of legibility'. *Quaerendo*, vol. 27, no. 2, pp. 141–147.

University of Utrecht,³²² claimed to have developed a brand new and easy to perform method for measuring legibility; a method ‘more accurate and more objective than traditional tests’ that also ‘clarifies the ... the function of serifs’. What’s more, according to the author, the method is based on nothing less than ‘the reading process and the properties of the human eye’. In other words, bold claims, put forward by a scientist in a highly regarded scholarly humanities journal.

Van Rossum acknowledges that an operational method like ‘speed of reading’ has seldom demonstrated any significant differences in the legibility between typefaces in general or between serif and sans serif typefaces. However, he states that it is ‘still very important to know which typeface is the most legible, even if the differences are slight’. The reason for this, according to van Rossum, is that the more legible typefaces can be used in smaller sizes, and thus yield ‘massive savings in printing and paper costs’.

Van Rossum has devised an operational method where images of text scanned into a computer are blurred by a Gaussian function. The blurred images are inspected visually by the researcher in order to determine the relative legibility of the typefaces involved.³²³ The theoretical construct behind his method is based on the fact that when we read, in each fixation only a limited number of characters are seen sharply in the visual field, and that the eye’s acuity is somehow increasingly diminished outside the centre of fixation (both within and outside the visual field). On this background van Rossum theorises that letters have to stay legible also at the edges of the visual field (where they are increasingly blurred), in order to allow a perceptual span as large as possible. Van Rossum’s blurred text images are thus claimed to simulate text as it is perceived around the edges of the visual field.

The typeface that withstands the detrimental effect of the blurring with respect to the number of recognised letters will be the most legible.

322. *Quærendo*, vol. 27, no. 2, ‘Notes on contributors’.

323. The details about the practical implementation of van Rossum’s test procedure do not come through clearly. This is partly due to inconsistencies between what the illustrations show, what the captions say, what the main text say and what examples in the main text say.

Note that all typeface sizes are normalised according to the x-height, the dominant dimension of a typeface.

Interestingly, this legibility study does not only involve ‘standard’ serif and sans serif typefaces, but also the new serif typeface Gulliver, designed by the author’s fellow countryman, typeface designer Gerard Unger. The study was actually ‘prompted by the arrival of Gulliver, a new typeface that has been designed to be more legible than the common typefaces, meaning that it can be set smaller’.³²⁴

The results: Serif typefaces were more legible than sans serif typefaces. Gulliver was deemed to be 5.7% more legible than the serif typeface Times, and respectively 7.7% and 8.9% more legible than the two sans serif typefaces Helvetica and Argo. A cynic would claim that this research – whether intentionally or not – can easily be mistaken for a not too subtle marketing ploy for the typeface that best fit the test parameters, namely Gulliver.

Nevertheless, not only does van Rossum’s study resemble a marketing ploy, while being immersed in over-confident rhetoric. His test and test-results are based entirely on unsubstantiated assumptions. It is correct that the eyes only fixate on a limited number of letters during a fixational pause, and that the surrounding letters are somehow looking increasingly diffuse outside the centre of fixation. However, van Rossum provides no evidence that suggests that the letters around and outside the edge of the visual field appear blurred in the same – or even similar – smeary way in real human vision. And even if there to some extent should exist a similar blurring effect, then there is no guarantee that another similar mathematical blurring function (or an optical filter of one kind or another) would produce the same results.³²⁵

Subsequently the author comes up with something which at first sight promises to qualify the bold claims about his method’s excellence: He suggests that the test he has demonstrated is only ‘a first step on the way to a wholly objective test of the legibility of different typefaces’. However, it is soon revealed that by this statement van Rossum does not

324. Gulliver has large counters and a relatively large x-height.

325. Compare for example, that although the author states that Helvetica’s ascenders ‘soon disappear in the “mist”’, he also admits that Helvetica ‘benefits from its clean style’ in some letter combinations.

refer to the basic quality of his method, but rather to an aspect of its technical implementation. He suggests that in the future, computers using neural networks for executing the visual inspection of his test procedure 'will make the test truly objective'. And this is not all; at this stage the author suggests that type designers, in order to ensure maximum legibility of their end product, can use his easily implemented test during the design process.

To conclude: This study, based on unsubstantiated assumptions, with categorical and over-confident formulations, while published in a highly respected scholarly journal, is 'truly' a suspect piece of legibility research.

6 Discussion: knowledge production and technical rationality

A comment on the reviewed studies

As suggested in the introduction, to synthesise or summarise the face results of the reviewed studies (on the relative legibility of serif and sans serif typefaces) would represent a betrayal of the intention and approach of this thesis. This thesis has never been meant as yet another legibility study, only in a comprehensive review format. On the contrary: the thesis is preoccupied with the production and validity of knowledge that is intended to be applied in typography and information design. The thesis is meant as an epistemological study about the history of legibility research, all the time while documenting one ‘genre’ of legibility research and using that ‘genre’ as a lens.

However, one thing can be said for sure: The face results are not conclusive. Some studies found serif typefaces most legible, some studies found sans serif typefaces most legible, while some studies did not find a difference. The tremendous effort that has gone into producing the 72 typeface legibility studies,³²⁶ based on a wide variety of rationales and of operational methods, has not resulted in an incremental body of

326. Note that the actual number of publications is bigger, since many of these studies appear in two or three guises; for example as a thesis, in an academic journal, and in a printing trade journal.

knowledge. Nor has it resulted in any clarifying theories. By some arguments, we are no further than we were in 1886.

While assessing the empirical studies one by one, focusing in particular on one assessment criterion – internal validity, the *sine qua non* of experimental research – it has been demonstrated in the previous chapter that the overall state of past as well as contemporary comparative research on typeface legibility is not too impressive. However, the quality is uneven. On one end of the scale we find Tinker and Paterson's relatively sober studies (and impressive research program). Their studies certainly have faults, but can nevertheless be situated in an historical context dominated by positivist views of social science. Their research program seemed timely and it represented a laudable effort by social scientists to provide a scientific basis for typographic designing. It is therefore ahistoric to dismiss the work of Tinker in the way that David Sless does: 'His [Tinker's] work consistently reflects a choice of inappropriate units of analysis.' (Sless 1981, p. 165).

On the same end of the scale as Tinker we also find in many ways interesting studies, with modest claims, like English 1944, Zachrisson 1965, Harris 1973, Vanderplas and Vanderplas 1980, and de Lange *et al.* 1993, which are all relatively sober (but nevertheless not without faults and inherent problems).

On the other end of the scale we find studies of dubious value, but which are all the same categorical and self-confident, like Robinson, Abbamonte and Evans 1971, Taylor 1990, Silver and Braun 1993, and van Rossum 1997. At the extremity we find studies that are nothing but outrageous, like Burt 1959 and Wheildon 1984. Nevertheless, as I have shown above, these are frequently and indiscriminately cited in the contemporary discursive realms of 'document design' and 'information design'.

Domain knowledge

It has become clear that the lack of internal validity reflects inappropriate manipulation (or lack of manipulation) of the typographic stimulus material employed. This, in turn, is to a large extent based on lack of adequate domain knowledge (of typographic design).³²⁷ However, to speculate that ‘more’ domain knowledge could have weeded out all serious problems with regard to internal validity is probably too far stretched. There are many possible pitfalls and many more validity problems and reliability problems than the internal validity problems posed by the quality of the typographic stimulus material. This last point is indicated by the three studies mentioned below.

Some of the authors have in their studies actually demonstrated crucial domain knowledge. That is, an awareness of the necessity to manipulate typographic variables that might function as confounding nuisance variables – so that they do not vary systematically with the independent experimental variable – with potential detrimental effect on the internal validity of the studies in question. I refer specifically to English 1944, Poulton 1965, and surprisingly, Burt 1959. Interestingly, this ‘employment’ of target domain knowledge seems to correlate with extensive contact between researcher and typographic expertise.

Although the study by English (1944) has its limitations (a t-scope tests of newspaper headlines), it is in many ways a sober study and it contains among other things a good literature review and a thorough discussion of theoretical and operational criteria of legibility. But more importantly, the author has been in close contact with typeface experts. That is, the famous type designer Frederic Goudy, and Douglas McMurtrie, director of the Ludlow Typograph Company. I assume that this contact has had something to do with English’s careful manipulation of interacting typographic variables, in order to avoid confounding of the results.³²⁸

Although Poulton’s study (1965) suffers from severe weaknesses, it is balanced (as many sans serif as serif typefaces are included, and each

327. See for example my review of Taylor 1990, in chapter 5.

328. See the section ‘English 1944’, in chapter 5.

category is represented by three stylistic idioms), and he adjusts for the dominant x-height dimension by varying the nominal point size from typeface to typeface, and he also intervenes on the ratio of size to interlinear spacing, by varying the interlinear space. And like English, Poulton has closely co-operated with ‘typographic advisors’, of whom at least one is a trained typographer.³²⁹

Burt’s (1959) otherwise discredited publication reflects a similar sensitivity by emphasising the lack of correspondence between nominal type size and actual appearance size, and he also emphasises the importance of the x-height. It is reasonable to believe that Burt’s extensive contact with the typeface expert Beatrice Warde of the Monotype Corporation played a role so that he could articulate these insights of his target domain.³³⁰

An extreme position

The perhaps most serious ‘external’ objection to legibility studies is expressed by the ‘peripherality of the reading process’ perspective.³³¹ My review has revealed instances of an *extreme* opposite position (explained in the next paragraphs). This position is represented by Poulton (1965), Wheildon (1984), and Taylor (1990). To these three, at least one of the studies that have not been reviewed could be added, that is, Kunst (1972).

In order to determine the relative legibility of typefaces Poulton bases his study on the speed of reading and comprehension. However, Poulton’s comprehension measure is not merely a check, but equally weighted with the speed of reading measure. Poulton argues vehemently for the appropriateness of his operational method and refers to it as ‘rate of comprehension’. However, as pointed out above, to claim that rather subtle differences in the design of ‘ordinary’ typefaces can be reflected in the degree of *comprehension* (measured by answering questions after

329. See the section ‘Poulton 1965’, in chapter 5.

330. See the section ‘Burt, Cooper and Martin 1955 / Burt 1959’, in chapter 5.

331. See the section ‘Peripherality to the reading process’, in chapter 4.

having read a passage for a limited time), seems to this writer to be wholly implausible. It certainly represents an extreme position compared to the view that the impact of typefaces is very much peripheral to the reading process.³³²

Taylor (1990) concluded that readers could be moved along with a serif typeface and slowed down with a sans serif typeface. This (sensational) conclusion was based on his interpretation of his face results, that readers prefer sans serif typefaces while they perform better with serif typefaces. Taylor's conclusion certainly represents, from a perspective where typefaces are regarded as relatively peripheral to the reading process, an extreme position.³³³

Accordingly, the typeface legibility study by Kunst (1972), which relied solely on a comprehension measure, expresses an extreme position on a par with Poulton's and Taylor's positions.

Observe that the introduction of the 'peripherality perspective' makes it much more difficult than before to have confidence in the operational constructs and conclusions of the four studies just mentioned.

Negative knowledge

What can be said is that all the typeface studies assessed in this thesis, except for occasional insights, at least represent a body of *negative knowledge*. Many pitfalls have been revealed in my assessment, and inappropriate approaches to typographic knowledge production have been made clear.

The escalating number of typeface legibility studies produced since the 1980s seems not to have taken notice of the many criticisms of legibility studies. Furthermore, the insights reached, and warnings uttered by people like R.L. Pyke (as early as 1926), Michael Macdonald-Ross and Robert Waller (1977), David Sless (1981), as well as many design practitioners,³³⁴ seem not to have had the deserved impact.

332. See the section 'Poulton 1965', in chapter 5.

333. See the section 'Taylor 1990', in chapter 5.

334. See chapter 4, 'Critiques of legibility research'.

Research that breaks with ‘received wisdom’

However, this does not mean that all legibility studies (of all ‘genres’) are of the same doubtful value. Practical legibility studies that resemble contextual usability tests are often less dubious. These studies might suffer from problems, for example with regard to internal validity due to the configuration of the stimulus material. However, external validity is less of a problem here because these studies do not claim general validity; and furthermore, construct validity is less of a problem, because there is a rather direct relationship between theoretical and operational definitions. I am thinking of studies like Christie and Rutley’s practical letterform legibility study on distance visibility for the British Road Research Laboratory,³³⁵ and practical typeface legibility studies on the image degradation of typefaces created by fax-transmission, like Birkvig 1990 (not reviewed in the previous chapter).

Furthermore, as suggested in the material on eye movement measures,³³⁶ there are approaches and studies that bring our understanding forward, whether these studies express ‘truth’ or not. There are also studies that go contrary to received wisdom. Such studies can provide insights of great value; like Emile Javal’s finding that the eyes move in saccadic jumps when we read. Of similar value, if not scale, is for example the insight of Don Bouwhuis that elderly people with partially impaired eyesight may need relatively large type to read with comfort, in spite of the fact that this large type may actually slow down the speed of their reading. In some studies, it seems that the researcher has stumbled over valuable or at least provocative insights which are secondary to their main aim. For example in a comparative typeface legibility study by Prince (1967) on people with impaired vision, where perhaps the most interesting and important result is the recommendation that it is necessary to enlarge the punctuation marks in texts for people with impaired vision.

Similarly, Arnold Wilkins’ intriguing monograph *Visual stress* focuses on pattern-sensitive epilepsy, headaches and discomfort. He has

335. See the section ‘Christie and Rutley 1961’, in chapter 5.

336. In the section ‘Experimental performance studies’, in chapter 2.

discovered through his work that pattern-sensitivity, triggered by the omnipresent stripes in our modern urban environment (think of clothes, furnishing fabrics, grills, and computer screen flicker), is more prevalent than earlier thought. Wilkins has also discovered that the stripy pattern of text has the same negative effect on predisposed persons. He suggests that one of the strategies that can be employed to reduce this problem is to rely on liberal amounts of interlinear spacing and liberal amounts of (inter) word spacing. The suggestion of employing liberal amounts of interlinear spacing resonates (within limits) with expert typographic advice; if not with the idiom of high modernist Swiss typography, where the even 'grey colour' of text areas on a page could seem to be more important than the visual articulation of each line of text (with 'white channels' of interlinear spacing). However, the second suggestion, of liberal (inter) word spacing does not resonate with expert advice. The rationale behind such an expert consideration is not only based on aesthetic sensitivity, but also on ideas inspired by, or at least in accord with, gestalt psychology about perceptual grouping (the distance between lines should be markedly bigger than the distance between words, etc).³³⁷ As long as only the interlinear spacing and (inter) word spacing increases but not the type size, the pattern created by the individual words of the text will visually 'fall apart'. If the interlinear spacing is kept relatively narrow while only the (inter) word spacing is extended, as in an illustration of a preferred text in Wilkins' book (pp. 74–75), the result will be 'rivers' in the text, something that is regarded as amateurish, ugly, and bad for legibility by typographic designers.³³⁸ I am not saying this in an attempt to repudiate Wilkins' argument. On the contrary; I want to point out that research can sometimes come up with surprising results or suggestions that go contrary to expert domain knowledge, and that such results may represent the better solution (at least for special applications). What I am trying to say (regardless of the validity of Wilkins' suggestions), is that it is exactly this role that research should play – to generate

337. For a study on perceptual grouping and typography, see Rivlin 1987.

338. For discussions on 'rivers' in typeset text, see Dowding 1966, pp. 4–7; and Rubinstein 1988, pp. 182–184.

knowledge that breaks with 'received wisdom' (cf. Bourdieu and Wacquant 1992).³³⁹

There is of course no guarantee that studies that go contrary to received wisdom, are free of validity problems. The contrary is probably most often the case. One such contrary example appears in a chapter on 'words and symbols' in David Osborne's prominent textbook *Ergonomics at work* (1987). On 'line spacing', while citing a study by Herman Bouma from 1980, Osborne claims that 'within reason, therefore, for long lines of text the interline spacing should be reduced' and further, that 'with very long text lines ... the interline spacing would have to be extremely small' (pp. 77, 78). This sensational claim, whether it derives from Bouma or not, is certainly contrary to received wisdom. However, it is not only contrary to received target domain knowledge, but also contrary to research findings (on the typographic variables line length and interlinear spacing). My *a priori* position is that Osborne's (and Bouma's?) claim is nonsense. The fact that Osborne does not reveal any signs of awareness of how sensational this claim is, enforces my suspicion that there is some kind of misunderstanding about, or alternatively, that Bouma's study, or conclusions, are somehow invalid. Nevertheless, I chose to interpret Osborne's recommendations on appropriate interlinear spacing as yet another instance of inadequate domain knowledge (as well as inadequate knowledge of the relevant research literature). In addition: it is an expression of how unreliable recommendations can be; recommendations based on uncritical and indiscriminate citations of whichever research 'finding' the collector stumbles over.

Passing by other points of comparisons, Mary Dyson and Gary Kipping's findings in the context of computer screen typography (mentioned above),³⁴⁰ that long line lengths are better, can also pass as a finding which is contrary to received wisdom. However, as pointed out earlier, the fact that Dyson and Kipping only compared long lines to

339. I choose deliberately not to describe the domain knowledge in question as 'everyday common sense knowledge', because I think expert domain knowledge is at least one level removed from 'everyday common sense knowledge'.

340. See the section 'Experimental performance studies', in chapter 2, 'The construct of legibility'.

short lines and not with something in between (a point that easily can be missed), suggests that caution should be exercised in interpreting their results.

Nevertheless, communication between research domain and target domain is helpful. This is exactly what Pierre Bourdieu, in *An invitation to reflexive sociology* (1992), refers to as ‘participating objectification’ (briefly referred to above while expounding on my own methodological strategy in the chapter ‘A review of empirical studies’). ‘Participating objectification’ is what Bourdieu promotes as the most appropriate methodological approach in social science research. Bourdieu not only emphasises that it is impossible to establish watertight walls between research on the one hand and everyday language and intimacy with the ‘subject’ on the other hand, he also emphasises that it is not desirable. Bourdieu stresses the necessity of maintaining both intimacy with and distance from the subject in order to construct a fertile object of study. Bourdieu is thus eager to transgress the contradiction between ‘participant’ and ‘spectator’, or phrased with Bourdieu’s own words, between ‘reductive subjectivism’ and ‘reductive objectivism’. Nevertheless, he also stresses that what is interesting from a research viewpoint is the epistemological rupture in relation to the immediate conceptions or *doxa* of the subject domain (as well as research reflexivity, that is, the necessity to reflect on and aim at an epistemological rupture with the *doxa* of the researchers’ own domain as well). Bourdieu sees the *systematic* aspect of research as a major contribution to achieve satisfying results (see Bourdieu and Wacquant 1992; as well as the exposition in Solli 1998).

Conceptions similar to Bourdieu’s ‘participating objectification’ are expressed by the British sociologist Anthony Giddens’ concept of a ‘double hermeneutics’ (1976, p. 158; 1987). Giddens emphasises that concepts and insights flow back and forth between research and the social sphere, and furthermore, that the social scientist depends on the subjects’ own concepts and motives in order to construct an object of study.

The question of contact and flow of information *in both directions* between the domain of experimental psychology and the target domain of typography and information design has been dealt with by Patricia

Wright (1978) and by Robert Waller (1979). In the context of language comprehension Patricia Wright focuses on the contact between 'pure' and 'applied' research (while acknowledging that it is an oversimplification to classify research as one or the other). However, she is also concerned with the contact between research domain and the domain of designing written communication (information design). Both Wright and Waller emphasise the importance of mutual respect and the importance (and difficulties) of bi-directional contact. Wright identifies three existing viewpoints on the relationship between the domains in question. The first viewpoint sees successful application as dependent upon advances in basic knowledge; the second sees applied research as the ideal starting point for advances in basic theories; while the third considers basic and applied research as separate but equal. Wright proposes a fourth viewpoint where 'a flow of information from applied to pure and back to applied could be a very fruitful pattern of interaction' (p. 253). In addition she also points in the direction of the alternative epistemological strategy that has become known as usability testing: 'There will never be a substitute for pretesting written material empirically' (p. 297). Furthermore, in a way similar to Michael Macdonald-Ross (1978, p. 24), Wright also focuses on the problem of extracting applicable recommendations from research, not least because research findings are often based on comparisons of incompetently designed materials: 'Knowing that flow charts are an improvement on legal language is no guarantee that they are better than good prose' (p. 297). However, Wright also acknowledges the problem of not necessarily knowing beforehand what is better or worse. Nevertheless, although there is no safe way out of this dilemma, Wright emphasises that researchers should

be concerned to optimise within formats. It is logically necessary to know how to write good prose or how to design a good flow chart before one can ask questions about the circumstances in which one of these formats is more useful than the other. (p. 297)

and she also echoes Hillier *et al.*'s ([1972] 1984) and Macdonald-Ross and Waller's (1975) papers inspired by Karl Popper:³⁴¹

In advancing an information flow from applied to pure and back to applied it has been suggested that perhaps the most worthwhile starting point is with applied *solutions*, rather than with applied *problems*. (p. 302)

Similar to the positions of Bourdieu and Habermas on social science research, Wright is recognising the importance of taking into account the 'everyday' or 'expert' knowledge of the target domain before undertaking research meant for application in the very same target domain in the next round. However, she also acknowledges the problem that

The researcher who starts with a real problem, but extracts only some of the variables to take back for examination in the laboratory, may very well find difficulties in moving from the laboratory findings back into the problem domain. (p. 252)

Thus, intimate contact and knowledge of the target domain is necessary in attempting to avoid many of the pitfalls of applied experimental research. This acknowledgement resonates with the 'findings' of this thesis, that lack of adequate (target) domain knowledge is responsible for faulty stimulus material, which leads to lack of internal validity, the basic requirement of experimental research.

Shadow positivism

It is now several decades ago since positivism³⁴² was repudiated as an adequate philosophy for the social sciences. However, the empirical approach of this thesis has provided substantiation of that positivism still lives on as a methodological platform for legibility research. This 'finding' resonates with recent observations on the state of sociology as well as the state of psychology (see for example Bickhard 1992;

341. See the section 'Changing paradigms: from legibility to usability', in chapter 3, 'A century of legibility research'.

342. For a good exposition of positivism, criticism of positivism, and post-positivist theory, see Outhwaite 1987.

Wacquant 1993; Giddens 1996). The Norwegian sociologist Ragnvald Kalleberg has suggested 'shadow positivism' as a description of this state of affairs; that is, when positivism is left behind as a philosophical basis, but still implicitly informs and permeates practical research (Kalleberg 1996; 1997). This 'shadow positivism' influences what are regarded as valid questions to ask, what are regarded as valid empirical data, what are legitimate methods to employ (controlled experiments and quantitative measures), and is described by 'technical rationality' and a prevalent 'nomologism'. 'Shadow positivism' is realised through the 'conventional view' that only experimental research can generate valid knowledge and that quantitative data are better than qualitative. As Robert Waller poignantly put it in 1987 (cited above in chapter 4, 'Critiques of legibility research'.): The expression 'nothing is known about' in the research literature means that no experiment has been performed on this question (Waller 1987, p. 73).

Kalleberg also suggests that the tendency of 'shadow positivism' thrives best in immature 'hyphen areas' of the social sciences. This last suggestion may very well fit the findings of this thesis. Most of the typeface legibility studies performed during the second half of the twentieth century, do not belong to dominant or major areas of social science such as psychology, sociology or economics, but rather modest and minor areas such as educational technology, journalism, and ergonomics. Other reasons suggested by Kalleberg for this 'shadow positivism' is a lack of a minimum of adequate competence in the theory and philosophy of science and social science, in the social sciences, and furthermore, the rather abstract and non-accessible character of much of science philosophy.

For the sake of order, note that a rejection of positivism does not imply a general rejection of quantitative methods or statistical analyses. Furthermore, opposition to narrow positivist conceptions of science, and the insight that knowledge is constructed and contingent of culture and social interest, does not imply that truth and knowledge does not exist or can not be achieved. Christopher Norris, while paraphrasing the British ('realist') philosopher of science Roy Bhaskar (1986), asserts that

Where the relativists err is in confusing *ontological* with *epistemological* issues. Thus they take the sheer variety of truth-claims advanced (and very often subsequently abandoned) down through the history of scientific thought as evidence that no truth is to be had (Norris 1995, p. 111)

Operationalism

The ‘shadow positivism’ suggested above is also realised through the prevalent *operationalism* that has been indirectly demonstrated in my review of empirical studies: Find something that can be measured and *voilà*, ‘legibility’ is not only operationalised, but also defined.³⁴³ The old saying about positivist science, ‘if you can’t count it it doesn’t count’, certainly describes prevalent notions exposed in this thesis. Furthermore, it may very well be that exactly the easily invented operational constructs within the frame of ‘legibility’ has distorted questions of design which *are* of importance to designers and readers alike.

Interestingly, operationalism has in fact not only been criticised by philosophers of science, but also by critics of legibility research from the domain of typography and graphic design. The typeface designer and retired director of British Linotype, Walter Tracy, perceptively hit the (easy going operationalism)-nail on its head in his 1988 *The typographic scene*:

Psychologists, and engineers like Lucien Legros, need to *measure* things: cognition, parts of letters, anything that can be identified and then evaluated by time or dimension. They make up a theory, devise a testing procedure, tabulate the results, and draw a conclusion. (Tracy 1988, p. 83)

And R.L. Pyke’s complaint from 1926, in his extensive review of legibility research, reminds us that much of early legibility research did not only lack theoretical definitions but even operational definitions:

343. The last study reviewed in the previous chapter (Van Rossum 1997), can stand as an example of an interesting and original idea about a particular measure of legibility – that changes to an uncritical reification of the same measure.

four times as many writers have measured legibility as have defined it. Three out of every four writers have been attempting to measure something the exact nature of which they have not paused to examine. (Pyke 1926, p. 10)

Joycelyn Chapman in an article about ‘re-thinking research into visual communication’ published in the late 1970s, in an issue of the design journal *Iconographic*, expressed similar concerns as Walter Tracy:

In complex ‘real’ life situations, speed of reading is only one, rather unimportant factor amongst a mass of other influences. Yet it is the main measuring tool used in legibility studies. It is the one human factor of concern. Why? *Partly because you can easily measure it.* Time can be quantified in minutes or seconds. How do you quantify interest? How do you discover deep motivations affecting reactions to print? (Chapman 1978, p. 28, my italics)

And Donald Norman, in the context of human–computer interaction, recently formulated the same concern the following way:

Unfortunately, things we can measure and quantify, if not selected with care, can be things that don’t have great practical value. (Norman 1995, p. 35)

Nevertheless, easy-going ‘operationalism’ is not confined to legibility research and human–computer interaction. Experimental psychology in general has also been described in a similar manner. For example by the psychologist Paul Kline (in a polemical book-length essay on the state of psychology):

the demand for precision has led empirical psychologists to choose variables *because they can be so measured* and not for reasons of theoretical psychology. This has led to research into trivial issues. (Kline 1988, p. 222, my italics)

And furthermore, as indicated above,³⁴⁴ the emerging criticism of positivism in the 1960s focused among other things exactly on easy-going use of operational definitions in social science research, including psychology. So, although Kalleberg’s ‘shadow positivism’ may fit as a descriptive label of contemporary legibility research, it would be wrong historically to regard legibility research as a ‘shadow’-exception, that is,

344. See the section ‘Operational definitions’, in chapter 2.

as a poor and peripheral relative to ‘flawless’ and dominant social science research domains (or, for that matter, natural science research domains like vision research and ophthalmology).

Translation of ‘findings’

It goes without saying that the ultimate aim of applied research on legibility (or, for example, human–computer interaction research) is to improve the design of informational artefacts. However, as pointed out above,³⁴⁵ the problem is not only to translate with great care ‘findings’ from experimental research to the practice of typographic design (or ‘information design’ or ‘human–computer interaction’), especially not when experimental validity cannot be taken for granted in the first place. There are frequent instances in the information design literature (and even more specifically in the ‘document design’ literature) where the application of findings of experimental research is reduced to a problem of translation and access for design practitioners. See for example Mary Beth Debs’ article ‘A history of advice: what experts have to tell us’ in the collection *Effective documentation: what we have learned from research* (Debs 1988, pp. 11–23); Paul Buckley in the article ‘Expressing research findings to have a practical influence on design’ in the collection *Cognitive ergonomics and human–computer interaction* (Buckley 1989); and Robert Krull’s thoughtful and thorough article ‘What practitioners need to know to evaluate research’ recently published in a special issue on the topic of ‘Bringing communication science to technical communication – advancing the profession’ in *IEEE Transactions on Professional Communication* (Krull 1997).³⁴⁶

Interestingly, Krull employs Miles Tinker’s legibility research as example-material in his article. Krull’s paper is an interesting and

345. See the section ‘Robinson, Abbamonte and Evans 1971’, in chapter 5.

346. In the introduction to this special issue, it is revealed that ‘Technical communicators who base their recommendations on established research are able to support their recommendations better and over time, establish better credibility with the scientists, engineers, and technical specialists with whom they work.’ (Grove and Zimmerman 1997, p. 158).

thorough exposition of all precautions that ought to be taken when applied research is to be interpreted and utilised by practitioners. Nevertheless, Krull's underlying epistemological assumption is less pragmatic. As an example he asks (in the opening paragraph): 'Should practitioners set body copy in serif type?'. He continues: 'Practitioners could answer such questions without themselves conducting research if they had methods of finding, evaluating, and applying existing social science research' (p. 168). Thus, in Krull's universe, there exists only one epistemology, that is, one based on experimental research. Either you do the experimental research yourself,³⁴⁷ or you find the right answers by (critical) reading of research papers.

Certainly, if the research described in this thesis – on the relative legibility of serif and sans serif typefaces – is representative of legibility research, we may polemically conclude that the existence of the 'problem' of translation and access represents an advantage rather than a problem.

Parallels between legibility research and human–computer interaction research

There are parallels between the critiques of legibility research that have been described in this thesis (as well as the critique constituted by this thesis), and a similar critique from the last fifteen years of psychological human–computer interaction research. The growing field of human–computer interaction (HCI) is, like information design, a research *and* practice domain, and information design and HCI overlap each other. HCI has 'inherited from software psychology and human factors both a laboratory-based, experimental program and a strong belief that such a program could alone answer whatever question – applied and basic – computer and human interaction might arise' (Nyce and Löwgren 1995, p. 37). Critiques of this belief has been articulated by several

347. Note that Krull is not talking about pragmatic 'on-site' usability testing, but 'applied' research on less context-dependent questions.

researchers, most notably by John Carroll and John Landauer.³⁴⁸

Landauer expresses a well-articulated scepticism of borrowings from psychology, and among these borrowings

we are sceptical of the practical value of classical comparative experiments to determine the relative merits of two or more features or systems and conventional testing of significance. (Landauer 1997, p. 224)

and the reasoning behind this scepticism echoes earlier critiques of legibility research

Because the parts can interact in subtle and unpredictable ways, it is not sufficient to study them in isolation. (Landauer 1997, p. 224)

And not unexpectedly, Landauer points to ‘human performance analysis’ and ‘formative evaluation’ as epistemic alternatives to experimental laboratory research. John Long, who in an article is preoccupied with specifying relations between research and the design of human–computer interactions, concludes that ‘in spite of that [human–computer interaction] research would be expected to support design’ and ‘the optimism that pervades the proceedings of the CHI-conferences’, ‘little of the knowledge has been incremental’ and ‘effectiveness in supporting design has not been generally demonstrated.’ (Long 1996, pp. 878–880).

Epistemic alternatives to legibility research

Donald Schön describes the ‘conventional view’ of ‘technical rationality’ as prevalent within most professions. That is, researchers are perceived as producers of valid knowledge that should be applied by practitioners (Schön 1983). Schön is extremely sceptical of this view, and he claims that it creates a wrong picture of the rich knowledge of practitioners.

The view of ‘technical rationality’ is most often implicit, but sometimes it is made explicit. Frank Smith, the editor of *Technical Communication*, the journal of the Society of Technical Communicators,

348. See for example Carroll and Campbell 1986, 1989; Carroll 1990; Landauer 1991; 1997. For valuable meta-commentaries on HCI-research and HCI-rhetoric, see Nyce and Löwgren 1995, and Cooper and Bowers 1995.

has explicitly expressed this view the following way (which also reveals an instrumental second reason for why practitioners should utilise research: in order not to be challenged by clients on the choices taken):

I submit that we must change our habitual approach to our jobs. Typically we work on the basis of intuition and folklore, and when a client asks us why we want to change his expression or his table or his organization, our only answer is that we think it's more effective our way. The client is perfectly justified in that case to say that he thinks it isn't. We need to be able to say that experimental research has proven *conclusively* that our recommended approach is superior. And if we are to do that, we must learn what has been *proven* and who is doing the work and where it is being published. And those of us who have the proper training and bent of mind and circumstances must begin or accelerate controlled experiments designed to test the old saws and establish *new truths* (Frank Smith in an editorial of *Technical Communication*, p. 5, 4th quarter, 1985; quoted by Doheny-Farina 1988, pp. 2–3, Doheny-Farina's italics)

And similarly, from T. Brooks of IBM a couple of years later, in the same journal, *Technical Communication*:

most [technical communicators] would probably agree that text set in all uppercase letters is harder to read than mixed-case text. That a well-designed serif type is easier to read than sans serif. But are you really sure why, or do you just *know* that? If you are challenged on a question like that, it helps to be able to back up your opinion with published research results or studies. (Brooks 1991, p. 183; quote taken from Campbell 1995, p. 4)

However, this thesis has so far substantiated that research on the relative legibility of serif and sans serif typefaces, in spite of the large number of studies carried out, has not been incremental or conclusive. It has also been indicated that traditional legibility research, regardless of genre, has been a blind alley that probably should be abandoned the sooner the better. If that is the case, it might be appropriate to ask if there exist alternative and more promising epistemologies. That is, alternative epistemologies for research that are meant to, or can, guide practice.

As suggested in chapter 4, 'Critiques of legibility research': an epistemic alternative that has been advocated by Michael Macdonald-Ross and others is to focus on knowledge in the design domain, that is, the 'tacit knowledge' of expert practitioners and to admit designers'

‘informed reasoning’ higher status. However, my exposition in the same chapter has shown that the explanatory potential of the concept of ‘tacit knowledge’ has its limits. But again, design domain knowledge is more than the personal tacit knowledge and contextual informed reasoning of expert practitioners, it is also ‘the rich and diverse body of knowledge ... [and] current of critical reflection on practice’ that typography conceals ‘not far below the surface’ of its ‘underdeveloped literature’. This last point has been demonstrated by Paul Stiff in his instrumental case study ‘The end of line: a survey of unjustified typography’ (1996a).

Also suggested in chapter 4: usability testing (and other contextual evaluation methods) represent pragmatic and powerful epistemic alternatives to experimental laboratory research. Observe that usability testing has seen a tremendous growth within the areas of ‘information design’ and ‘human–computer interaction’ during the last two decades.

As suggested above in this chapter (in the section ‘Domain knowledge’), there is also a certain potential for improving experimental research. This is suggested by the argument that inadequate domain knowledge is in the last instance responsible for the lack of validity in much existing research. Thus, adequate domain knowledge will yield better research. A potential for such improvement is also suggested by Patricia Wright’s emphasis of bi-directional contact between research domain and design domain. Michael Macdonald-Ross takes for granted the potential for improved research based on adequate domain knowledge:

There have been few empirical studies of total designs. Such studies require researchers who are prepared to grasp the basics of typographic design and the manufacturing processes. (Macdonald-Ross 1994, p. 4691)

The epistemic alternative of ‘design-based theory’

John Carroll, who in the contexts of document design and human–computer interaction has developed a powerful critique of much psychological research (including information-processing theories) meant to service the practical development of user interfaces (e.g. Carroll and

Campbell 1986), has suggested ‘design-based theory’ as an epistemic alternative to the technical rationality represented by ‘the conventional view’ of ‘theory-based design’ – which ‘doesn’t ... seem to work’ (see Carroll 1990, pp. 277–284; and Carroll and Campbell 1989).

Carroll’s quest for ‘design-based theory’ is partly informed by Patricia Wright’s focus on the need for a bi-directional flow of information (see Carroll 1990, p. 283). However, the quest for ‘design-based theory’ goes beyond Wright’s proposal. Carroll claims that

There is a conventional view of the relationship of scientific research and the invention, design, and development of practical artefacts. The idea is that basic science provides understanding of nature that can be applied deductively in practical contexts. This idea has been rather thoroughly absorbed, at least by scientists, and is familiar to anyone who have studied science. (Carroll 1990, p. 277)

Against this ‘conventional view’ Carroll points out that technological inventions often vastly predate their own scientific analysis.³⁴⁹ As one of several examples from the history of technology and from the history of human–computer interaction techniques, he mentions that

current practice in designing texts far outstrips what can be grounded in the basic psychology of text comprehension (Carroll 1990, p. 281)

and he continues

... Such inversions of theory-based design cannot be understood in the conventional view. Their resolution lies in a different view of the relation between science and design, one that takes these artifacts and the process of invention and development that produces designed artifacts more seriously. We refer to this view as design-based theory. (Carroll 1990, p. 281)

Carroll stresses the potential of focusing on, and extracting knowledge from, existing artefacts. This is possible because designed artefacts embody implicit theoretical claims. However, artefacts develop in a ‘task-artifact cycle’ and user interface design depends on understanding this ‘ecology of tasks and artifacts’. In order to reach such understanding, several things are needed: a competent understanding of domain details; scientists who can provide conceptual guidance in

349. ‘Optical scaling’ in typeface design may represent such an invention. See my excursus on ‘optical scaling’ in the section ‘Experimental performance studies’, in chapter 2.

design; and tools that expose the theoretical and psychological claims embodied in designed artefacts.

Once HCI is understood in terms of an ecology of tasks and artifacts, it becomes clear that the conventional views have been looking for psychology in the wrong place. The psychology in HCI is not to be found, for the most part, in laboratory methods and information-processing theories; it is to be found in the user interfaces, the artifacts, that HCI researchers build and evaluate. (Carroll and Campbell 1989, p. 254)

Carroll's position may be brushed off as wishful thinking, but if we transfer Carroll's concern to text design and typography (ranging from concerns about comprehension to traditional legibility concerns), we may very well realise that there already exist examples of systematic inquiry where design and artefacts are taken seriously on their own terms (as opposed to instances of design enquiry where design in its own right is made invisible and is seen through the lenses of more dominant and colonising fields of enquiry like experimental psychology, art history, sociology, cultural studies, or literary theory).

I can think of several examples of highly relevant and systematic scholarly investigation on the function and usability of typographic artefacts where the focus is on the relationship between 'content' and 'form', that is, investigations that focus on artefacts on the terms of design. Take for example a row of papers by book-historian Margaret Smith on early printed books where usability and the needs of the reader are strongly present, although in an implied manner (with 'textual articulation' as an important concept). In contradiction to many book-historians or analytical bibliographers, Smith is consistently focusing on the relationship between content and form as well as the highly relevant role of navigational and other information-technological devices of the book (see for example Smith 1983, 1987, 1993, 1994). Richard Southall has in the context of 'electronic publishing' produced a series of highly relevant theoretical papers that consistently focus on the relationship between content and form, and thus the artefactual interface between 'content' and the reader (see for example Southall 1988, 1989, 1992). Michael Twyman has produced a row of historical and theoretical papers on typography that has a similar inclination (see for example Twyman 1979, 1982, 1986, 1993). And Robert Waller has produced a row of papers

where the focus is on ‘the typographic contribution to language’, that is, the focus is on the relationship and border-zone between writing as a linguistic medium and writing as a graphic medium (see for example Waller 1980, 1982, 1987, 1991). What all these authors have in common is a will to employ a combined empirical and interpretive approach that very much focuses on design in its own right, and not through the lens of for example art history or cultural studies. Furthermore, not only are insights produced by these studies, but also a conceptual vocabulary, that in the last instance has a potential for guidance in practical design.

It is important to realise that although such preoccupation with the artefact may signal a turn away from reader-centred design and back to an earlier traditional preoccupation of design (and design education) with the artefact and its formal visual structures, this need not be the case. Rather, the artefact must be understood not as a *thing*, but rather as an *interface* that negotiates between content and ‘message’ on one hand and the reader and his/her needs, capacities and intentions on the other hand.³⁵⁰ If we look at informational artefacts in the way suggested by Carroll, we realise that by focusing on the artefact, the reader need not to be forgotten, but rather, that concern for the reader can be taken care of implicitly instead of by explicit rhetoric.

Similar ideas as Carroll’s have been tentatively expressed by Nigel Cross (1999), and obliquely expressed by Clive Dilnot (1999), on the question of what ‘design research’ could or should be. Nigel Cross echoes Bill Hillier *et al.* ([1972] 1984), and Michael Macdonald-Ross and Robert Waller (1975),³⁵¹ when he stresses that design knowledge resides in ‘products themselves’:

Much everyday design work entails the use of precedents or previous exemplars – not because of laziness by the designer but because the exemplars actually contain knowledge of what the product should be ... knowledge implicit within the object itself of how best to shape, make,

350. An abstract ‘logical’ text structure may suggest six chapter levels; and similarly, an orderly ‘logical’ organisation of content for a web-site may suggest a very deep hierarchical structure. However, the artefactual and visual instantiation has to take into account that it may be difficult for readers to discriminate in a meaningful way between six chapter levels, and unnecessary or even devastating to have to navigate through an excess of hierarchical levels in a website.

351. See the section ‘Changing paradigms: from legibility to usability’, in chapter 3.

and use it. ... we would be foolish to disregard or overlook this informal product knowledge simply because it has not been made explicit yet; that is a task for design research. (Cross 1999, p. 6)

And Clive Dilnot advocates the importance of focusing on the artefact, especially in a climate of 'artefact denial'. As he argues: artefacts are seen everywhere, but remain 'invisible', not least in the university. However, Dilnot's emphasis on the artefact is not made without at the same time expressing a warning about 'the fetishistic "distraction" [of the artefact itself] that on occasion so blinds design to its own consequences and implications' (Dilnot 1999, p. 89).

7 Conclusion

The thesis effectively reveals that nearly all of the 28 studies which have been reviewed (of a surprising total of 72 identified studies) lack internal validity (the intra-paradigm *sine qua non* of experimental research). It is shown that this lack of internal validity is largely due to confounding factors that resides in the stimulus material, in the last instance caused by the researchers' inadequate domain knowledge (about typography). Other methodological flaws are also revealed.

The tremendous effort that has gone into producing all these typeface legibility studies, based on a wide variety of rationales and operational methods, has not resulted in an incremental body of knowledge. Nor has it resulted in any clarifying theories. The reviews of the individual studies provide a thorough empirical substantiation of criticism that has been raised only in a sweeping manner at earlier occasions. The thesis shows that traditional experimental legibility research has provided a non-productive approach to typographic knowledge production.

However, the thesis has revealed that an increasing number of comparative typeface legibility studies have been carried out during the last two decades. This stands in contrast to prevailing notions that legibility research pretty much vanished in the early 1980s, or alternatively, that 'too few' legibility studies have been carried out during the last two decades. What is more, the thesis has also shown that dubious and even seriously flawed legibility studies are frequently and indiscriminately cited in the contemporary discursive realms of 'document design' and 'information design'.

It is suggested that much of contemporary legibility research is naive (far from representing 'superior knowledge') and that it can be

described as an expression of ‘shadow positivism’ and easy-going ‘operationalism’, thriving in the ‘hyphen areas’ of the social sciences, where advances in the philosophy of science made decades ago seem not yet to have been fully recognised. Epistemic alternatives to traditional legibility research have also been discussed.

As well as being an idiographic historical study in its own right, and primarily contributing knowledge to a self-reflective conversation in typography and information design, the thesis also contributes knowledge of value to design history, design epistemology, reading research history and social science history. The thesis has, by its empirical approach, acknowledged the idea that epistemological studies (or science history for that matter) should focus on what researchers actually do, and the works that are actually delivered, rather than idealised conceptions of what research is.

References

Bibliographical details about cited letters to the editor of newspapers, magazines and journals are most often given in footnotes only, and thus do not appear in this bibliography. The same applies to cited correspondence, cited archival documents, and electronic bibliographic databases. URLs were correct at the time of writing.

The format '[Year] Year', as in '[1949] 1973', indicates that the document was first published in 1949, but that the edition cited was published in 1973.

A

- Adams, Sarah, R. Rosemier, and P. Sleeman. 1965. 'Readable letter size and visibility for overhead projection transparencies'. *Audiovisual Communications Review*, vol. 13, pp. 412–417
- Adobe. 1989. 'Face fax'. *Font & Function*, Fall 1989, p. 12
- Aldersey-Williams, Hugh, Lorraine Wild, Daralice Boles, Katherine McCoy, Michael McCoy, Roy Slade, and Niels Diffrient. 1990. *Cranbrook design: the new discourse*. New York: Rizzoli
- Amachree, Tomlinson K.P., Hubertus L. Bloemer, and Bob J. Walter. 1977. 'Typographic legibility on maps: a comparative study'. *Bulletin of the Society of University Cartographers*, 11, pp. 27–39. (Probably based on Tomlinson Krakrayemo P. Amachree. 1975. 'Typographic legibility on maps: a comparison between Sans-serif (Gill) and Serif (Times Roman) type'. MA thesis. Ohio University)
- André, Jacques. 1993. *Contribution à la création de fontes en typographie numérique*. Rennes: l'université de Rennes 1, Institut de Formation Supérieure en Informatique et en Communication. (On the cover it says: Jacques André. *Création de fontes en typographie numérique*. Documents d'habilitation. Rennes: IRISA and IFSIC)
- André, Jacques, and Irène Vatton. 1994. 'Dynamic optical scaling and variable-sized characters'. *Electronic Publishing*, vol. 7, no. 4, pp. 231–250
- Arditi, A., K. Knoblauch, and I. Grünwald. 1990. 'Reading with fixed and variable pitch'. *Journal of the Ophthalmological Society of America*, vol. 7, pp. 2011–2015

B

- Badaracco, Claire Hoertz. 1995. *Trading words: poetry, typography, and illustrated books in the modern literary economy*. Baltimore and London: The Johns Hopkins University Press
- Baker, Steve. 1985. 'The hell of connotation'. *Word & Image*, vol. 1, no. 2, pp. 164–175
- Bal, Mieke, and Norman Bryson. 1991. 'Semiotics and art history'. *The Art Bulletin*, vol. 73, no. 2, pp. 174–208
- Baird, Russel N., and Arthur T. Turnbull. 1975. *The graphics of communication: typography, layout, design*. 3rd ed. New York: Holt, Rinehart and Winston
- Baird, Russel, N., Duncan McDonald, Ronald K. Pittman, and the late Arthur T. Turnbull. 1993. *The graphics of communication: methods, media and technology*. 6th edition. Fort Worth: Harcourt Brace Jovanovich College Publishers
- Banks, William P., and David Krajicek. 1991. 'Perception'. *Annual Review of Psychology*, vol. 42, pp. 305–331
- Bartram, David. 1982. 'The perception of semantic quality in type: differences between designers and non-designers'. *Information Design Journal*, vol. 3, no. 1, pp. 38–50
- Bartram, David. 1982. 'Jeremy J. Foster, *Legibility research 1972–1978: a summary*'. Review article. *Information Design Journal*, vol. 3, no. 1, pp. 75–76
- Baudin, Fernand. 1967a. 'Miles A. Tinker, *Bases for effective reading*'. Review article. *Journal of Typographic Research*, vol. 1, no. 2, pp. 204–207
- Baudin, Fernand. 1967b. 'Typography: evolution + revolution'. *Journal of Typographic Research*, vol. 1, no. 4, pp. 373–386
- Baxandall, Michael. 1985. *Patterns of intention: on the historical explanation of pictures*. New Haven and London: Yale University Press
- Beaugrande, Robert de. 1980. *Text, discourse, and process: toward a multidisciplinary science of text*. Norwood, New Jersey: Ablex
- Beaugrande, Robert de. 1984. *Text production: toward a science of composition*. Norwood, New Jersey: Ablex
- Becker, D., J. Heinrich, R. von Sichowsky, and D. Wendt. 1970. 'Reader preferences for typeface and leading'. *Journal of Typographic Research*, vol. 4, no. 1, pp. 61–66
- Beldie, I.P., S. Pastoor, and E. Schwartz. 1983. 'Fixed vs. variable letter width for televised text'. *Human Factors*, vol. 25, pp. 273–277
- Bell, Richard C., and James L.F. Sullivan. 1981. 'Student preferences in typography'. *Programmed Learning and Educational Technology*, vol. 18, no. 2, pp. 57–61
- Benedek, Andy. 1991. 'The craft of digital type'. *Eye*, vol. 1, no. 2, pp. 86–87
- Benson, Philippa. 1985. 'Writing visually: design considerations in technical publications'. *Technical Communication*, no. 32, pp. 35–39

- Berger, Sidney E. 1991. *The design of bibliographies: observations, references and examples*. No. 6 of Bibliographies and indexes in library and information science. London: Mansell, a Cassell imprint; Westport, Connecticut: Greenwood Press (1992)
- Berkowitz, Leonard, and Edward Donnerstein. 1982. 'External validity is more than skin deep: some answers to criticisms of laboratory experiments'. *American Psychologist*, vol. 37, no. 3, pp. 245–257
- Berliner, Anna. 1920. 'Atmosphärenwert von Drucktypen'. *Zeitschrift für Angewandte Psychologie*, 17, p. 165
- Bernhardt, Stephen A. 1985. 'Text structure and graphic design: the visible design'. In *Systemic perspectives on discourse: volume 2*. Selected applied papers from the 9th international systemic workshop. Edited by James D. Benson and William S. Greaves, pp. 18–38. Norwood, New Jersey: Ablex
- Bhaskar, Roy. 1986. *Scientific realism and human emancipation*. London: Verso
- Bickhard, M.H. 1992. 'Myths of science'. *Theory and Psychology*, vol. 2, no. 3, pp. 321–337
- Biemann, Emil O. 1961. 'Univers: a new concept in European type design'. *Print*, vol. 15, no. 1, February, pp. 32–36
- Bigelow, Charles A. 1981. 'Technology and the aesthetics of type: maintaining the "tradition" in the age of electronics'. *The Seybold Report*, vol. 10, no. 24, pp. [1–2], 3–16 (This is part 1 of a series of 3 articles by Bigelow in this journal, under the series title: 'Aesthetics vs. technology')
- Bigelow, Charles A. 1982. 'The principles of digital type: quality type for low, medium and high resolution printers'. *The Seybold Report on Publishing Systems*, vol. 11, no. 11, pp. [1], 3–23. (This is part 2 of a series of 3 articles by Bigelow in this journal, under the series title: 'Aesthetics vs. technology')
- Bigelow, Charles. 1989. 'On type: form, pattern, & texture in the typographic image'. *Fine Print*, vol. 15, no. 2, pp. 75–82
- Bigelow, Charles, and Donald Day. 1983. 'Digital typography'. *Scientific American*, no. 8, pp. 106–110, 112, 114–119
- Bigelow, Charles, and Kris Holmes. 1991. 'Notes on Apple 4 fonts'. *Electronic Publishing*, vol. 4, no. 3, pp. 171–181
- Bigelow, Charles, and Kris Holmes. 1993. 'The design of a Unicode font'. *Electronic Publishing*, vol. 6, no. 3, pp. 289–305
- Biggs, Michael, and Claus Huitfeldt. 1997. 'Philosophy and electronic publishing: theory and metatheory in the development of text encoding'. *The Monist*, vol. 80, no. 3, pp. 348–367
- Birkvig, Henrik. 1990. 'Frutiger: en sikker vinner'. *Bogtrykkerbladet*, no. 9, pp. 11–14
- Bi[rvig], H[enrik]. 1995. 'Typografiens ekstrapolation'. *Rubrik*, no. 37, pp. 10–11
- Black, Alison. 1990. *Typefaces for desktop publishing: a user guide*. London: Architecture Design and Technology Press
- Black, Alison, and Andrew Boag. 1992. 'Choosing binary or grayscale bitmaps: some consequences for users'. In *EP 92: Proceedings of Electronic Publishing 1992*, edited by C. Vanoirbeek and G. Coray, pp. 247–260. Cambridge: Cambridge University Press

- Blackwell, Lewis. 1995. *The end of print: the graphic design of David Carson*. London: Laurence King
- Boag, Andrew. 1993. 'What is the point?' *Print*, vol. 47, no. 2, pp. 109–110
- Boag, Andrew. 1996. 'Typographic measurement: a chronology'. *Typography Papers*, no. 1, pp. 105–121
- Boden, Margaret A. 1990. 'Escaping the Chinese room'. In *The philosophy of artificial intelligence*, edited by Margaret A. Boden, pp. 89–104. Oxford: Oxford University Press
- Bohn, Willard. 1986. *The aesthetics of visual poetry: 1914–1928*. Cambridge: Cambridge University Press
- Bonner, John V.H. 1998. 'Towards consumer product interface design guidelines'. In *Human factors in consumer products*, edited by Neville Stanton, pp. 239–258. London: Taylor & Francis
- Bonsiepe, Gui. 1968. 'A method of quantifying order in typographic design'. *Ulm*, no. 21, pp. 24–31. (Also published in *Journal of Typographic Research*, vol. 2, no. 3, 1968, pp. 203–220. Republished in *Readings from Ulm: selected articles from the journal of the Ulm school of Design*, edited by Kirti Trivedi, pp. 249–258. Bombay: Industrial Design Centre, at the Indian Institute of Technology, 1989)
- Bourdieu, Pierre, and Loïc J.D. Wacquant. 1992. *An invitation to reflexive sociology*. Cambridge: Polity Press
- Bouwhuis, Don G. 1989. 'Reading as goal-driven behaviour'. In *Working models of human perception*, edited by Ben A.G. Elsendoorn and Herman Bouma, pp. 341–362. London: Academic Press
- Bouwhuis, D.G. 1993. 'Reading rate and letter size'. *IPO Annual Progress Report*, no. 28, pp. 30–36
- Bowden, Paul R., and David F. Brailsford. 1989. 'On the noise immunity and legibility of Lucida fonts'. In *Raster imaging and digital typography*, edited by Jacques André and Roger Hersch, pp. 205–212. Cambridge: Cambridge University Press
- Boyarski, Dan, Christine Neuwirth, Jodi Forlizzi, and Susan Harkness Regli. 1998. 'A study of fonts designed for screen display'. *CHI 98*, pp. 87–94
- Brachfeld, Oliver. 1964. 'Zuviel Grotesk'. *Verlags-Praxis*, no. 9, September, pp. 245–246
- Bracht, Glenn H., and Gene V. Glass. 1968. 'The external validity of experiments'. *American Educational Research Journal*, vol. 5, no. 4, pp. 437–474
- [Brinch, Ole]. 1976. 'Bibliografi over kommunikationsteori, kommunikationsmidler, læselighetsforskning o.l.' Copenhagen: The graphic College of Denmark. (To a large extent based on bibliographic information from FOGRA Literaturdienst, Deutsche Gesellschaft für Forschung im graphischen Gewerbe e.V., München, under the index term 'Schriften: lesbarkeit'; and ZIID-Referatkartei Grafische Technik, Institut f. graf. Technik, Leipzig)

- Braun, Curt C., and N. Clayton Silver. 1995. 'Interaction of warning label features: determining the contributions of three warning characteristics'. In *Designing for the global village: proceedings of the Human Factors and Ergonomics Society 39th annual meeting – 1995*, volume 2, pp. 984–988. Santa Monica, California: Human Factors and Ergonomics Society
- Brooks, T. 1991. 'Career development: filling the usability gap'. *Technical Communication*, vol. 38, no. 2, pp. 180–184
- Bruce, Vicki, and Patrick Green. 1990. *Visual perception: physiology, psychology and ecology*. 2nd edn. Hove, East Sussex; and Hillsdale, New Jersey: Lawrence Erlbaum. (3rd edition, with Mark A. Georgeson as co-author, was published in 1996 by Psychology Press, an imprint of Erlbaum (UK) Taylor Francis. Hove, East Sussex)
- Bryson, Norman. 1994. 'Art in context'. In *The point of theory: practices of cultural analysis*, edited by Mieke Bal and Inge E. Boer, pp. 66–78. Amsterdam: Amsterdam University Press
- Buchanan, Richard. 1992. 'Wicked problems in design thinking'. *Design Issues*, vol. 8, no. 2, pp. 5–21
- Buchanan, Richard. 1993. 'Semiotics and design'. *Semiotica*, vol. 97, no. 1/2, pp. 189–197
- Buckingham, B.R. 1931. 'New data on the typography of textbooks'. *Yearbook of the National Society for the Study of Education*, vol. 30, pp. 93–125
- Buckley, Paul. 1989. 'Expressing research findings to have a practical influence on design'. In *Cognitive ergonomics and human-computer interaction*, edited by J. Long and A. Whitefield, pp. 166–190. Cambridge: Cambridge University Press
- Bullington, Edward Weeks. 1948. 'Legibility determinations of multiple carbon copies'. MS thesis. Virginia Polytechnic Institute
- Burke, Christopher. 1995/96. 'Typeface review: ITC Bodoni'. *Printing Historical Society Bulletin*, no. 40, pp. 36–37
- Burke, Christopher. 1998a. *Paul Renner: the art of typography*. London: Hyphen Press; New York: Princeton Architectural Press
- Burke, Christopher. 1998b. 'From functionalism to information design: a review of Jan Tschichold's The new typography'. *Information Design Journal*, vol. 9, no. 1, pp. 51–58
- Burnhill, P., and J. Hartley. 1975. 'Psychology and textbook design: a research critique'. In *Communication and learning*, edited by Jon Baggaley, G. Harry Jamieson, and Harry Marchant, pp. 65–78. Volume 8 of Aspects of educational technology. Pitman Publishing
- Burt, Cyril. 1959. *A psychological study of typography*. With an introduction by Stanley Morison. Cambridge: Cambridge University Press. (Reprinted in 1974 for the College of Librarianship, in Wales, by Bowker)
- Burt, Cyril. 1960a. 'The readability of type'. *New Scientist*, vol. 7, no. 168, pp. 227–229
- Burt, Cyril. 1960b. 'The typography of children's books: a record of research in the U.K.'. In *Yearbook of Education*, edited by G.Z.F. Bereday and J.A. Lauwerys. London: Evans

- Burt, Cyril, W.F. Cooper, and J.L. Martin. 1955. 'A psychological study of typography'. *The British Journal of Statistical Psychology*, vol. 8, pt. 1, pp. 29–57
- Button, Graham, Jeff Coulter, John R.E. Lee, and Wes Sharrock. 1995. *Computers, minds and conduct*. Cambridge: Polity Press
- Buur, Jacob, and Kirsten Bagger. 1999. 'Replacing usability testing with user dialogue'. *Communications of the ACM*, vol. 42, no. 5, pp. 63–66

C

- Cambrosio, Alberto, and Peter Keating. 1988. '“Going monoclonal”: art, science, and magic in the day-to-day use of hybridoma technology'. *Social Problems*, vol. 35, no. 3, pp. 244–260
- Campbell, Joan. 1989. *Joy in work: German work, the national debate 1800–1945*. Princeton, New Jersey: Princeton University Press
- Campbell, Kim Sydow. 1995. *Coherence, continuity, and cohesion: theoretical foundations for document design*. Hillsdale, New Jersey: Lawrence Erlbaum
- Carmichael, Leonard, and Walter F. Dearborn. 1947. *Reading and visual fatigue*. Boston: Houghton Mifflin.
- Carroll, John M. 1990. *The Nurnberg funnel: designing minimalist instruction for practical computer skill*. Cambridge, Massachusetts: The MIT Press
- Carroll, John M. 1997. 'Human–computer interaction: psychology as a science of design'. *International Journal of Human–Computer Studies*, vol. 46, pp. 501–522
- Carroll, John M., and Robert L. Campbell. 1986. 'Softening up hard science: reply to Newell and Card'. *Human–Computer Interaction*, vol. 2, pp. 227–249
- Carroll, John M., and Robert L. Campbell. 1989. 'Artifacts as psychological theories: the case of human–computer interaction'. *Behaviour and Information Technology*, vol. 8, no. 4, pp. 247–256
- Carter, Harry. [1937] 1984. 'Optical scale in type founding'. *Printing Historical Society Bulletin*, no. 13, 1984, pp. 144–148. (Originally published in *Typography*, no. 4, 1937, pp. 2–6)
- Carver, Ronald P. 1992. 'Reading rate: theory, research, and practical implications'. *Journal of Reading*, vol. 36, no. 2, pp. 84–95
- Catalogue 43: the book arts*. [1999]. Oxford: Frances Wakeman Books
- Chapman, Jocelyn. 1978. 'Re-thinking research into visual communication'. *Icographic*, no. 13, pp. 28–32
- Cheetham, Dennis, and Brian Grimby. 1964. 'Design analysis: typeface'. *Design*, no. 186, pp. 61–71
- Cheetham, Dennis, Christopher Poulton, and Brian Grimby. 1965. 'The case for research'. *Design*, no. 195, pp. 48–51
- Christie, A.W., and K.S. Rutley. 1961a. 'Relative effectiveness of some letter types designed for use on road traffic signs'. *Roads and Road Construction*, no. 39, August, pp. 239–244
- Christie, A.W., and K.S. Rutley. 1961b. '[Research] on road signs'. *Design*, no. 152, August, pp. 59–60

- Chung, S.T.L., J.S. Mansfield, and G.E. Legge. 1998. 'Psychophysics of reading – 18: The effect of print size on reading speed in normal peripheral vision'. *Vision Research*, vol. 38, no. 19, pp. 2949–2962
- Clark, Malcolm. 1989. 'Book review: *Digital typography: an introduction to type and composition for computer system design*, Richard Rubinstein'. *TEXline*, no. 8, January, pp. 32–33
- Coghill, Vera. 1980. 'Can children read familiar words in unfamiliar type?' *Information Design Journal*, vol. 1, no. 4, pp. 254–260
- Cohen, A.S. 1981. 'Car drivers' pattern of eye fixations on the road and in the laboratory'. *Perceptual and Motor Skills*, vol. 52, pp. 515–522
- Cohn, Herman, and Robert Rübencamp. 1903. *Wie sollen Bücher und Zeitungen gedruckt werden?* Braunschweig: F. Vieweg & Sohn
- Collins, H.M. 1987. 'Expert systems and the science of knowledge'. In *The social construction of technological systems: new directions in the sociology and history of technology*, edited by Wiebe E. Bijker, Thomas P. Hughes, and Trevor J. Pinch, pp. 329–348. Cambridge, Massachusetts: The MIT Press
- Connolly, Kevin G. 1998. 'Legibility and readability of small print: effects of font, observer age and spatial vision'. MSc thesis. University of Calgary
- Cook, Thomas D., and Donald T. Campbell. 1979. 'Validity'. Chapter 2 in *Quasi-experimentation: design & analysis issues for field settings*, pp. 37–94. Chicago: Rand McNally
- Cooper, Geoff, and John Bowers. 1995. 'Representing the user: notes on the disciplinary rhetoric of human–computer interaction'. In *The social and interactional dimensions of human–computer interfaces*, edited by Peter J. Thomas, pp. 48–65. Cambridge: Cambridge University Press
- Cooper, M.B., H.N. Daglish, and J.A. Adams. 1979. 'Reader preferences for report typefaces'. *Applied Ergonomics*, vol. 10, no. 2, pp. 66–70
- Cornog, D.Y., and F.C. Rose. 1967. *Legibility of alphanumeric characters and other symbols. Vol. 2: a reference handbook*. Miscellaneous publication 262-2. Washington, D.C.: National Bureau of Standards
- Crawford, M.L.J., Richard A. Andersen, Randolph Blake, Gerald H. Jakobs, and Christa Neumeyer. 1990. 'Interspecies comparisons in the understanding of human visual perception'. In *Visual perception: the neurophysiological foundations*, edited by Lothar Spillmann and John S. Werner, pp. 23–52. San Diego: Academic Press
- Crosland H.R., and Georgia Johnson. 1928. 'The range of apprehension as affected by inter-letter hair-spacing and by characteristics of individual letters'. *Journal of Applied Psychology*, vol. 12, February, pp. 82–124
- Cross, Nigel. 1980. 'An introduction to design methods'. *Information Design Journal*, vol. 1, no. 4, pp. 242–253
- Cross, Nigel. 1999. 'Design research: a disciplined conversation'. *Design Issues*, vol. 15, no. 2, pp. 5–10
- Cross, Nigel, John Naughton, and David Walker. 1981. 'Design method and scientific method'. In *Design: science: method. Proceedings of the 1980 Design Research Society Conference*. Edited by Robin Jacques and James A. Powell, pp. 18–29. Guildford: Westbury House. (Also published, with a few minor alterations, in *Design Studies*, vol. 2, no. 4, 1981, pp. 195–201)

- Crutchley, Brooke. 1986. '“Congeniality” in the typography of books'. *Matrix*, no. 6, pp. 134–139, plus 8 inserted pages with illustrations

D

- Dauppe, Michèle-Anne. 1991. 'Get the message?'. *Eye*, no. 3, pp. 4–7
- Deach, Stephen. 1992. 'Outline font hints and rasterization: a technology primer'. *The Seybold Report on Desktop Publishing*, vol. 6, no. 7, pp. 21–32
- Dearborn, W.F. 1906. *The psychology of reading*. New York: Columbia University contribution to philosophy and psychology, vol. 14, no. 7
- Debs, Mary Beth. 1988. 'A history of advice: what experts have to tell us'. In *Effective communication: what we have learned from research*, edited by Stephen Doheny-Farina, pp. 11–23. Cambridge, MA: The MIT Press
- Department of Transport. 1991. *The history of traffic signs*. London: Department of Transport, Traffic Signs Branch, Network Management and Driver Information Division. (Photocopied leaflet of 35 pages)
- Department of Transport. 1994. *The design and use of directional informatory signs*. Local transport note, no. 1/94. London: HMSO
- Department of Transport. 1995. *Know your traffic signs*. Fourth edition. London: HMSO. (First edition 1975).
- Design*. 1959. 'Which signs for motorways'. No. 129, pp. 28–30, 32, 35
- Design Issues*, vol. 11, no. 1, 1995. (Special issue on design historiography)
- Dillon, Andrew. 1992. 'Reading from paper versus screens: a critical review of the empirical literature'. *Ergonomics*, vol. 35, no. 10, pp. 1297–1326
- Dillon, Andrew. 1994. *Designing usable electronic text: ergonomic aspects of human information usage*. London: Taylor & Francis
- Dilnot, Clive. [1984] 1989. 'The state of design history, part 1: mapping the field', pp. 213–232; and 'The state of design history, part 2: problems and possibilities', pp. 233–262. In *Design discourse: history, theory, criticism*, edited by Victor Margolin, Chicago and London: University of Chicago Press, 1989. (Reprinted facsimile-like page for page from *Design Issues*, vol. 1, no. 1, 1984, pp. 4–23; and vol. 1, no. 2, 1984, pp. 3–20)
- Dilnot, Clive. 1999. 'The science of uncertainty: the potential contribution of design to knowledge'. In *Doctoral education in design*. Proceedings of the Ohio conference, October 8–11, 1998. Edited by Richard Buchanan, Dennis Doordan, Lorraine Justice, and Victor Margolin, pp. 65–97. Pittsburgh: The School of Design, Carnegie Mellon University
- Doheny-Farina, Stephen. 1988. 'Methods and results: a range of possibilities'. Introduction to *Effective documentation: what we have learned from research*, edited by Stephen Doheny-Farina, pp. 1–7. Cambridge, Massachusetts: The MIT Press
- Dormer, Peter. 1994. *The art of the maker*. London: Thames and Hudson
- Dowding, Geoffrey. 1957. *Factors in the choice of type faces*. London: Wace & Company

- Dowding, Geoffrey. 1966. *Finer points in the spacing & arrangement of type*. 3rd edition. London: Wace & Company. (A new edition, based on the 3rd edition, with a foreword by Crispin Elsted, was published in 1995, by Hartley & Marks, Vancouver.)
- Dreyfus, John. 1957. 'David Kindersley's contribution to street lettering'. *Penrose Annual*, vol. 51, pp. 38–41
- Dreyfus, John. 1961. 'Univers in action'. *Penrose Annual*, vol. 55, pp. 15–19
- Dreyfus, John. 1994. '“Who is to design books now that computer [sic] are making books?”'. In *Into print: selected writings on printing history, typography and book production*, pp. 283–297. London: The British Library. (Based on a lecture given in 1988. The essay was first published in *Classical typography in the computer age*, by Hermann Zapf and John Dreyfus. Los Angeles: William Andrews Clark Library, 1991)
- Dreyfus, Hubert L. 1992. *What computers still can't do: a critique of artificial reason*. 3rd edition. Cambridge, Massachusetts: The MIT Press. (*What computers can't do: a critique of artificial reason*, 1st edition 1972; *What computers can't do: the limits of artificial reason*, 2nd edition 1979)
- Dreyfus, Hubert L., and Stuart E. Dreyfus. 1986. *Mind over machine: the power of human intuition and expertise in the era of the computer*. New York: Free Press
- Dreyfuss, Henry. 1955. *Designing for people*. New York: Simon & Shuster
- Duchnick, Robert L., and Paul A. Kolers. 1983. 'Readability of text scrolled on visual display terminals as a function of window size'. *Human Factors*, vol. 25, no. 6, pp. 683–692
- Dyson, Mary C. 1993. 'Improving discrimination of symbols for display at low resolution'. *Electronic Publishing*, vol. 6, no. 3, pp. 231–239
- Dyson, Mary C., and Gary J. Kipping. 1997. 'The legibility of screen formats: are three columns better than one?'. *Computers and Graphics*, vol. 21, no. 6, pp. 703–712
- Dyson, Mary C., and Gary J. Kipping. 1998a. 'Exploring the effect of layout on reading from screen'. In *Electronic publishing, artistic imaging and digital typography*, edited by Roger D. Hersch, Jacques André, and Heather Brown, pp. 294–304. Berlin: Springer-Verlag
- Dyson, Mary C., and Gary J. Kipping. 1998b. 'The effects of line length and method of movement on patterns of reading from screen'. *Visible Language*, vol. 32, no. 2, pp. 150–181

E

- Eason, Ron, and Sarah Rookledge (edited by Phil Baines and Gordon Rookledge). 1991. *Rookledge's international handbook of type designers: a biographical directory*. Carshalton Beeches, Surrey: Sarema Press
- Easterby, Ronald, and Harm Zwaga (editors). 1984. *Information design: the design and evaluation of signs and technical information*. Chichester: John Wiley
- Edelman, Gerald M. 1992. *Bright air, brilliant fire: on the matter of the mind*. London: Penguin, 1994

- Edworthy, Judy, and Austin Adams. 1996. *Warning design: a research prospective*. London: Taylor & Francis
- Ehses, Hanno, and Ellen Lupton. 1988. *Rhetorical handbook: an illustrated manual for graphic designers*. Design Papers no. 5. Halifax, Nova Scotia: Design Division, Nova Scotia College of Art and Design
- Emigre. 1996. Catalog (of various merchandise and typefaces). Sacramento: Emigre
- English, Earl. 1944. 'A study of the readability of four newspaper headline types'. *Journalism Quarterly*, vol. 21, September, pp. 217–229
- Engman, Petra. 1988. 'Fontexperiment'. Stockholm: Department of numerical analysis and computing science, Royal Institute of Technology
- Epelboim, Julie, James R. Booth, and Robert M. Steinman. 1994. 'Reading unspaced text: implications for theories of reading eye movements'. *Vision Research*, vol. 34, no. 13, pp. 1735–1766
- Epelboim, Julie, James R. Booth, and Robert M. Steinman. 1996. 'Much ado about nothing: the place of space in text'. *Vision Research*, vol. 36, no. 3, pp. 465–470
- Eriksen, Trond Berg. 1993. 'Form-skrift: en skriftreform'. In *Bokspor: norske bøker gjennom 350 år*, edited by Per Strømholm, pp. 233–242. Oslo: Universitetsforlaget
- Eurographic Press Interview. 1962. 'Designer's profile: Adrian Frutiger'. *Print in Britain*, no. 9, vol. 9, January, pp. 258–262
- Evans, S.H., A.A.J. Hoffman, M.D. Arnoult, and O. Zinser. 1968. 'Pattern enhancement with schematic operators'. *Behavioral Science*, vol. 13, no. 5, pp. 402–404
- Eyler, William R., and Donald A Stewart. 1982. 'Notes on our new format'. Editorial. *Radiology*, vol. 142, no. 1, p. 248

F

- Falk, Valter. 1965. 'Grotesk-revy: den serifflösa bokstavsformen under 150 år'. *Biblis*, pp. 71–105
- Farrell, Joyce E. 1991. 'Fitting physical screen parameters to the human eye'. In *The man-machine interface*, edited by J.A.J. Roufs, pp. 7–23. Basingstoke: Macmillan Press
- Felker, Daniel B. (ed). 1980. *Document design: a review of the relevant research*. Washington DC: American Institutes for Research
- Femia, Joseph V. 1981. 'An historicist critique of "revisionist" methods for studying the history of ideas'. *History and Theory*, vol. 20, no. 2, pp. 113–134
- Ferguson, Eugene S. 1992. *Engineering and the mind's eye*. Cambridge, Massachusetts: The MIT Press
- Fielding, Helen. 1994. 'Life in the lost lane: motorists of Britain just don't know which way to turn: Helen Fielding on road sign madness'. *Independent on Sunday*, 2 October
- FMC Corporation. 1985. *Product safety sign and label system*. Santa Clara, California: FMC Corporation

- Foster, Jeremy J. 1968. 'Commentary: psychological research into legibility'. *Journal of Typographic Research*, vol. 2, no. 3, pp. 279–282
- Foster, Jeremy J. (ed). 1971. *Legibility research abstracts 1970*. London: Lund Humphries
- Foster, Jeremy J. (ed). 1972. *Legibility research abstracts 1971*. London: Lund Humphries
- Foster, Jeremy J. 1973. 'Legibility research: the ergonomics of print'. *Icographic*, no. 6, pp. 20–24
- Foster, Jeremy J. 1978. 'Locating legibility research: a guide for the graphic designer'. *Visible Language*, vol. 7, no. 2, pp. 201–205
- Foster, Jeremy J. 1980. *Legibility research 1972–1978: a summary*. [London]: Graphic Information Research Unit, Royal College of Art
- Frenckner, Kerstin. 1990. *Legibility of continuous text on computer screens: a guide to the literature*. TRITA-NA P9010. Stockholm: Department of numerical analysis and computing science, Royal Institute of Technology
- Frenckner, Kerstin, Caroline Nordquist, Staffan Romberger, and Hans Smedshammar. 1991. *Legibility of continuous text on computer screens: a series of experiments*. Research report TRITA-NA-P9033. Stockholm: Department of numerical analysis and computing science, Royal Institute of Technology
- Froshaug, Anthony. 1963. 'Roadside traffic signs'. *Design*, no. 178, October, pp. 37–50
- Frutiger, Adrian. 1962. 'How I came to design Univers'. *Print in Britain*, no. 9, vol. 9, January, pp. 263–263–266
- Fuller, Steve. 1992. 'Social epistemology and the research agenda of science studies'. In *Science as practice and culture*, edited by Andrew Pickering, pp. 390–428. Chicago: University of Chicago Press
- Furnham, Adrian F. 1988. *Lay theories: everyday understanding of problems in the social sciences*. Oxford: Pergamon Press
- Futura*heft. [1931/1932]. Frankfurt am Main, Barcelona, and New York: Bauersche Giesserei

G

- Gadamer, Hans-Georg. [1960] 1993. *Truth and method*. Second, revised edition. London: Sheed & Ward (first English language edition, 1975) [*Wahrheit und Methode: Grundzüge einer philosophischen Hermeneutik*]
- Gallagher, Thomas J., and Wendy S. Jakobson. 1993. 'The typography of environmental impact statements: criteria, evaluation, and public participation'. *Environmental Management*, vol. 17, no. 1, pp. 99–109
- Garcia, Marelys L, and Cesar I. Caldera. 1996. 'The effect of color and typeface on the readability of on-line text'. *Computers and Industrial Engineering*, vol. 31, no. 1/2, pp. 519–524

- Garland, Ken. [1982] 1996. 'Some thoughts. From Ken Garland and Associates Designers: 20 years work and play, London, 1982'. In Ken Garland, *A word in your eye: opinions, observations and conjectures on design, from 1960 to the present*, pp. 53–57. Reading: The University of Reading, Department of Typography and Graphic Communication
- Gelderman, Maarten. 1999. 'A short introduction to font characteristics'. *TUGboat*, vol. 20, no. 2, pp. 96–104
- Gelernter, Mark George. 1981. 'The subject-object problem in design theory and education'. PhD thesis. Bartlett School of Architectural Planning, University College of London, University of London
- Gelernter, Mark. 1993. 'The concept of design and its application to architecture'. In *Companion to contemporary architectural thought*, edited by Ben Farmer and Hentie Louw, pp. 399–403. London and New York: Routledge
- Gerstner, Karl. 1963. 'The old Berthold sans-serif on a new basis'. *Druckspiegel*, no. 6. (Also published in Karl Gerstner, *Designing programmes*, pp. 29–48. London: Alec Tiranti, 1964)
- Geske, Joel. 1996. 'Legibility of sans serif type for use as body copy in computer mediated communication'. AEJMC Visual Communication Division, 22 pp. (Paper presented at the 79th annual meeting of the Association for Education in Journalism and Mass Communication. Anaheim, California, 9–3 August, 1996)
- Gibson, James J. 1979. *The ecological approach to visual perception*. Boston and London: Houghton-Mifflin. (Republished by Lawrence Erlbaum. Hillsdale, New Jersey, 1986)
- Giddens, Anthony. 1976. *New rules of sociological method: a positive critique of interpretive sociologies*. London: Hutchinson
- Giddens, Anthony. 1987. 'Nine theses on the future of sociology'. In Anthony Giddens, *Social theory and modern sociology*. Cambridge: Polity Press
- Giddens, Anthony. 1996. 'What is social science?'. In Anthony Giddens, *In defense of sociology*, pp. 65–77. Cambridge: Polity Press
- Gill, Eric. 1936. *An essay on typography*. 2nd edition. Sheed and Ward
- Glaister, Geoffrey Ashall. 1979. *Glaister's glossary of the book*. 2nd edition. London: George Allen & Unwin. (The second edition was republished Oak Knoll Press and The British Library in 1996, with a new introduction by Donald Farren, under the title *Encyclopedia of the book*, a title which had been used for the first American edition in 1960.)
- Goldschmidt, Gabriela. 1994. 'On visual design thinking: the vis [sic] kids of architecture.' *Design Studies*, vol. 15, no. 2, pp. 158–174
- Gordon, V.E.C., and Ruth Mock. 1960. *Twentieth-century handwriting*. London: Methuen
- Göranzon, Bo, and Ingela Josefson (eds). 1988. *Knowledge, skill and artificial intelligence*. London: Springer-Verlag
- Göranzon, Bo, and Magnus Florin (eds). 1990. *Artificial intelligence, culture and language: on education and work*. London: Springer-Verlag
- Gray, Nicolette. 1960. 'Sans serif and other experimental inscribed lettering of the early renaissance'. *Motif*, no. 5, pp. 66–76

- Gray, Nicolette. [1938] 1976. *Nineteenth century ornamented typefaces*. New edition, and with a chapter on ornamented types in America by Ray Nash. London: Faber and Faber (Originally published as *XIXth century ornamented types & title pages* in 1938)
- Green, C.D. 1992. 'Of immortal mythological beasts'. *Theory and Psychology*, vol. 2, no. 3, pp. 291–320
- Gribbons, William M. 1991. 'Visual literacy in corporate communication: some implications for information design'. *IEEE Transactions on Professional Communication*, vol. 34, no. 1, pp. 42–50
- Griffing, Harold, and Shepherd Ivory Franz. 1896. 'On the conditions of fatigue in reading'. *Psychological Review*, vol. 3, pp. 513–530
- Grooters, Lyle E. 1972. 'The relationship of letter style, letter size, and viewing distance to the readability of transparent visuals'. PhD dissertation. University of Oklahoma
- Grove, Laurel K., and Donald E. Zimmerman. 1997. 'Introduction: bringing communication science to technical communication – advancing the profession'. *IEEE Transactions on Professional Communication*, vol. 40, no. 3, pp. 157–167
- Grudin, Jonathan. 1993. 'Interface: an evolving concept'. *Communications of the ACM*, vol. 36, no. 4, pp. 110–119
- Gürtler, André. 1988. 'History of the development of the newspaper: a survey of newspaper text faces of the 20th century'. *Typografische Monatsblätter*, no. 1, pp. 1–36. (Tri-lingual text in German, French and English.)

H

- Habermas, Jürgen. [1981] 1984. *The theory of communicative action, volume 1: reason and the rationalization of society*. Translated by Thomas McCarthy. London: Heinemann [*Theorie des kommunikativen Handelns, Band 1, Handlungsrationality und gesellschaftliche Rationalisierung*. Frankfurt am Main: Suhrkamp Verlag]
- Hacker, Peter. 1987. 'Languages, minds and brains'. In *Mindwaves: thoughts on intelligence, identity and consciousness*, edited by Colin Blakemore and Susan Greenfield, pp. 484–505. Oxford: Basil Blackwell
- Hacker, Peter. 1991. 'Seeing, representing and describing: an examination of David Marr's computational theory of vision'. In *Investigating psychology: sciences of the mind after Wittgenstein*, edited by John Hyman, pp. 119–154. London and New York: Routledge
- Hallberg, Åke. 1992. *Typografin och läsprocessen: grafisk kommunikation med text och bild*. Halmstad: Bokförlaget Spectra
- Halliday, M.A.K. 1970. 'Language structure and language function'. In *New horizons in linguistics*, edited by John Lyons, pp. 140–165. Harmondsworth: Penguin
- Halliday, M.A.K., and Ruqaiya Hasan. 1976. *Cohesion in English*. London: Longman
- Handover, P.M. 1960. 'Palette for printers'. *Motif*, no. 5, pp. 94–95
- Handover, P.M. 1961. 'Letters without serifs'. *Motif*, no. 6, pp. 66–81

- Handover, P.M. 1963. 'Grotesque letters: a history of unseriffed type faces from 1816 to the present day'. *Monotype News Letter*, no. 69, pp. 2–9
- Harré, Rom, and Grant Gillett. 1994. *The discursive mind*. Thousand Oaks, California: Sage
- Harris, J. 1973. 'Confusions in letter recognition'. *Professional Printer*, vol. 17, no. 2, pp. 29–34
- Hartley, James, and Peter Burnhill. 1977. 'Understanding instructional text: typography, layout and design'. In *Adult learning: psychological research and applications*, edited by Michael J.A. Howe. London: John Wiley
- Hartley, James, and Donald Rooum. 1983. 'Sir Cyril Burt and typography: a re-evaluation'. *British Journal of Psychology*, vol. 74, pt. 1, pp. 203–212
- Hearnshaw, L.S. 1979. *Cyril Burt: psychologist*. London: Hodder and Stoughton
- Hellesnes, Jon. 1975. *Sosialisering og teknokrati: ein sosialfilosofisk studie med særleg vekt på pedagogikkens problem*. Oslo: Gyldendal Norsk Forlag
- Hellesnes, Jon. 1998. 'Skulen og den teknokratiske freistinga'. *Syn og Segn*, vol. 104, no. 3, pp. 210–215
- Henderson, Kathryn. 1995. 'The visual culture of engineers'. In *The cultures of computing*, edited by Susan Leigh Star, pp. 196–218. Oxford: Blackwell and Sociological Review
- Hersch, Roger D. 1993. 'Font rasterization: the state of the art'. In *Visual and technical aspects of type*, edited by Roger D. Hersch, pp. 78–109. Cambridge: Cambridge University Press
- Hersch, Roger D., Claude Bétrisey, Justin Bur, and André Gürtler. 1995. 'Perceptually tuned generation of grayscale fonts'. *IEEE Computer Graphics and Applications*, vol. 15, no. 6, pp. 78–89
- Hillier, Bill, John Musgrove, and Pat O'Sullivan. [1972] 1984. 'Knowledge and design'. In *Developments in design methodology*, edited by Nigel Cross, pp. 245–264. Chichester: John Wiley. (Originally published in 1972, in *Environmental design: research and practice*, edited by W.J. Mitchell. Los Angeles: University of California)
- Hoffman, Valerie. 1987. 'Effect of typeface on iconic storage capacity'. Paper presented at the Annual meeting of the Western Psychological Association, Long Beach, California, April 23–26, 25 pp.
- Hoffman, Veronica M. 1988. 'An investigation of the affects of typefaces upon reader's perception of the meanings of messages using the semantic differential testing technique'. MS thesis. Rochester Institute of Technology
- Holleran, Patrick. 1992. 'An assessment of font preferences for screen-based text display'. In *People and computers 7: proceedings of HCI 92, York, September 1992*, British Computer Society Conference Series 5, edited by A. Monk, D. Diaper, and M.D. Harrison, pp. 447–459
- Horn, Robert E. 1998. 'Comments on "Building the bridge across the years and the disciplines"'. *Information Design Journal*, vol. 9, no. 1, pp. 21–23
- Horn, Robert E. 1999. 'Information design: emergence of a new profession'. In *Information design*, edited by Robert Jakobson, pp. 15–33. Cambridge, Massachusetts: The MIT Press
- Hubel, David H. 1988. *Eye, brain, and vision*. New York: Scientific American Library

- Hubel, D.H., and T.N. Wiesel. 1959. 'Receptive fields of single neurones in the cat's striate cortex'. *Journal of Physiology*, vol. 148, pp. 574–591
- Hubel, D.H., and T.N. Wiesel. 1962. 'Receptive fields, binocular interaction and functional architecture in the cat's visual cortex'. *Journal of Physiology*, vol. 160, pp. 106–154
- Huey, Edmund B. 1908. *The psychology and pedagogy of reading*. New York: Macmillan. (Republished by MIT Press, Cambridge, Massachusetts, 1968.)
- Hughes, Philip K., and Barry L. Cole. 1986. 'Can the conspicuity of objects be predicted from laboratory experiments?'. *Ergonomics*, vol. 29, no. 9, pp. 1097–1111
- Huitfeldt, Claus. 1995. 'Multidimensional texts in a one-dimensional medium'. *Computers and the Humanities*, vol. 28, pp. 235–241
- Hutchings, R.S. 1966. 'Neo-grot types and the new Swiss typography'. *British Printer*, vol. 79, no. 3, pp. 116–130
- Hutchinson, James. 1983. *Letters*. London: The Herbert Press
- Hvistendal, J.K., and Mary R. Kahl. 1976. 'Roman v. sans serif body type: readability and reader preference', *Typographic*, vol. 8, no. 2, 4 pages. Published by the International Typographic Composition Association, in the USA. (Allegedly reprinted from *The ANPA* [American Newspaper Publishers Association] *News Research Bulletin*, no. 2, 1975, pp. 3–11; however British Library's interlibrary service did not find it in the 1976 volume of the bulletin. The article is probably based on Mary Ruth Luna Kahl's M.S. thesis: 'A study of reading speed and reader preferences between roman and sans serif type', Iowa State University, 1974)

I

- 'Information design database: authors A–K'. 1992. Reading: Department of Typography & Graphic Communication (unpublished bibliography)
- 'Information design database: authors L–Z'. 1993. Reading: Department of Typography & Graphic Communication (unpublished bibliography)
- Ingvar, David H.; and Åke Hallberg. 1989. *Hjärnan, bokstaven, ordet*. Halmstad: Bokförlaget Spektra
- Ions, Edmund. 1977. *Against behaviouralism: a critique of behavioural science*. Oxford: Basil Blackwell

J

- Janik, A. 1988. 'Tacit knowledge, working life and scientific method'. In *Knowledge, skill and artificial intelligence*, edited by Bo Göranson and Ingela Josefson, pp. 53–63. London: Springer-Verlag
- Janik, A. 1990. 'Tacit knowledge, rule following and learning'. In *Artificial intelligence, culture and language: on education and work*, edited by Bo Göranson and Magnus Florin, pp. 45–55. London: Springer-Verlag

- Jansen, Linda, and Pauli Thomsen. 1986. *Typografi for synshæmmende: en undersøgelse af mulige sammenhænge mellom nogle typografiske faktorer og læselighed for synshæmmende*. Copenhagen: Instituttet for blinde og svagsynede
- Janson, Marius A., and Stephen J. Morrissey. 1991. 'Legibility of video display units: one more look'. *Behaviour & Information Technology*, vol. 10, no. 6, pp. 525–542
- Jaspert, W. Pincus, W. Turner Berry, and A.F. Johnson. 1970. *The encyclopaedia of type faces*. Third edition. Poole: Blandford Press
- Javal, Emile. [1905] 1978. *Physiologie de la lecture et de l'écriture*. [Paris: Félix Alcan]. Reprinted as facsimile, with a new preface by François Richaudeau, in the series *Les classiques des sciences humaines*. Paris: Retz-CEPL
- Jha, S.S., and C.N. Daftuar. 1981. 'Legibility of typefaces'. *Journal of Psychological Researches*, vol. 25, no. 2, pp. 108–110
- John, E. Roy, and Eric L. Schwartz. 1978. 'The neurophysiology of information processing and cognition'. *Annual Review of Psychology*, vol. 29, pp. 1–29
- Jonas, Wolfgang. 1993. 'Design as problem-solving? or: here is the solution – what was the problem?'. *Design Studies*, vol. 14, no. 2, pp. 157–170
- Johnson, Edward Allan. 1996. 'Meanings of typography in trademarks'. PhD dissertation. The University of Alabama
- Joynson, R.B. 1974. *Psychology and common sense*. London: Routledge & Kegan Paul
- Joynson, Robert B. 1989. *The Burt affair*. London: Routledge

K

- Kahn, Paul, and Krzysztof Lenk. 1993. 'Typography for the computer screen: applying the lessons of print to electronic documents'. *The Seybold Report on Desktop Publishing*, vol. 7, no. 11, pp. 3–16
- Kalleberg, Ragnvald. 1996. 'Forskningsopplegget og samfunnsforskningens dobbeltdialog'. In *Kvalitative metoder i samfunnsforskningen*, edited by Harriet Holter and Ragnvald Kalleberg, second edition, pp. 26–72. Oslo: Universitetsforlaget
- Kalleberg, Ragnvald. 1997. 'Skjervheim, sosiologi, filosofi og vitenskapsstudier'. In *Regime under kritikk: om Hans Skjervheim i norsk filosofi og samfunnsdebatt*, edited by Hermund Slaattelid, pp. 239–266. Oslo: Aschehoug
- Kantowitz, Barry H. 1992. 'Selecting measures for human factors research'. *Human Factors*, vol. 34, no. 4, pp. 387–398
- Karow, Peter. 1994. *Font technology: methods and tools*. Foreword by Gerard Unger. Berlin: Springer-Verlag. (Original title: *Schrifttechnologie*). (On the cover the title reads: *Font technology: description and tools*)
- Karow, Peter. 1997. 'The quality display of text on computer screens'. *Type*, vol. 1, no. 1, pp. 47–64
- Katzen, May. 1977. *The visual impact of scholarly journal articles*. Primary Communications Research Centre, University of Leicester
- Kelley, Harold H. 1992. 'Common-sense psychology and scientific psychology'. *Annual Review of Psychology*, vol. 43, pp. 1–23

- Kelly, John. 1993. *Artificial intelligence: a modern myth*. Ellis Horwood series in artificial intelligence. New York and London: Ellis Horwood
- Kempson, Elaine, and Nick Moore. 1994. *Designing public documents: a review of research*. London: Policy Studies Institute
- Kerr, J. 1926. *The fundamentals of school health*. London: Allen & Unwin
- Kindersley, David. 1960. 'Motorway sign lettering'. *Traffic Engineering & Control*, December, pp. 463–465.
- Kindersley, David. 1968. 'Legibility'. *Printing Technology*, vol. 12, no. 2, pp. 69–71
- Kindersley, David. 1976. *Optical letter spacing: for new printing systems*. London: Wynkyn de Worde Society.
- Kinneir, Jock. 1970. 'Det offentlige skilt'. In *Offentlig design*, edited by Christian Ejlers, Erik Ellegaard Fredriksen and Niels Kryger, pp. 15–23. Copenhagen: Christian Ejlers Forlag
- Kinneir, Jock. 1971. 'Technical notes on the redesign of the United Kingdom road signs'. Unpublished machine-written manuscript, 15 + 7 pp. Written for a Russian monthly journal, but never published. (Location: Margaret Calvert, London)
- Kinneir, Jock. 1984. 'The practical and graphic problems of road sign design'. In *Information design: the design and evaluation of signs and technical information*, edited by Ronald Easterby and Harm Zwaga, pp. 341–358. Chichester: John Wiley
- Kinross, Robin. 1984. 'Kinneir, Jock'. In *Contemporary designers*, edited by Ann Lee Morgan and Colin Naylor, p. 324. London: Macmillan Publishers
- Kinross, Robin. 1985. 'The rethoric of neutrality'. *Design Issues*, vol. 2, no. 2, pp. 18–30. (Reprinted facsimile-like page for page, in *Design discourse: history, theory, criticism*, edited by Victor Margolin, pp. 131–143. Chicago and London: University of Chicago Press. 1989. Originally presented as a paper to the first Information Design Conference in the UK, in 1984)
- Kinross, Robin. 1989. 'Road signs: wrong turning?' *Blueprint*, October, pp. 50–51
- Kinross, Robin. 1992. *Modern typography: an essay in critical history*. London: Hyphen Press
- Kinross, Robin. 1994a. *Fellow readers: an essay on multiplied language*. London: Hyphen Press
- Kinross, Robin. 1994b. 'Obituary: Richard Jock Kinneir'. *Directions: newsletter of the Sign Design Society*, vol. 1, no. 7, pp. [2–3]. (Extended and revised version of obituary first published in *The Guardian*, August 30, 1994)
- Klare, George R. 1963. *The measurement of readability*. Ames, Iowa: Iowa State University Press
- Kline, Paul. 1988. *Psychology exposed or the emperor's new clothes*. London and New York: Routledge
- Klitz, T.S., J.S. Mansfield, and G.E. Legge. 1995. 'Reading speed is affected by font transitions'. (Poster presentation / meeting abstract). *Investigative Ophthalmology & Visual Science*, vol. 36, no. 4, p. S670
- Koch, S. 1992. 'Psychology's Bridgman vs. Bridgman's Bridgman'. *Theory and Psychology*, vol. 2, no. 3, pp. 261–290

- Kolers, Paul A. 1983. 'Locations and contents'. In *Eye movements in reading: perceptual and language processes*, edited by Keith Rayner, pp. 53–61. New York and London: Academic Press
- Kosslyn, Stephen M. 1994. *Elements of graph design*. New York: W.H. Freeman and Company
- Kostelnick, Charles, and David R. Roberts. 1998. *Designing visual language: strategies for professional communicators*. Boston: Allyn and Bacon
- Kostelnick, Charles. 1995. 'Cultural adaptation and information design: two contrasting views'. *IEEE Transactions on Professional Communication*, vol. 38, no. 4, pp. 182–196
- Kowler, Eileen (editor). 1990. *Eye movements and their role in visual and cognitive processes*. Amsterdam and New York: Elsevier Science
- Krampen, Martin. 1983. 'Icons of the road'. Special issue of *Semiotica*, vol. 43, nos. 1/2
- Kravutske, Mary Eileen. 1994. 'The effect of serif versus sans serif typeface on reader comprehension and speed of reading'. PhD dissertation. Wayne State University
- Krull, Robert. 1997. 'What practitioners need to know to evaluate research'. *IEEE Transactions on Professional Communication*, vol. 40, no. 3, pp. 168–181
- Kuhn, Thomas S. [1962] 1970. *The structure of scientific revolutions*. 2nd edition. Chicago: University of Chicago Press
- Kunst, Robert John. 1972. 'Type styles as related to reading comprehension'. DEd dissertation. Arizona State University
- Kvernbekk, Tone. 1995. 'Erfaringstyranni eller teorityranni: et filosofisk perspektiv på praksis'. In *Profesjonsutdanning og forskning: FoU-perspektiver på praksisfeltet*, [edited by] Lærerutdanningsrådet, pp. 17–27. Oslo: Lærerutdanningsrådet
- Kyng, Morten, and Joan Greenbaum. 1991. *Design at work: cooperative design of computer systems*. Hillsdale, New Jersey: Lawrence Erlbaum

L

- LaCapra, Dominick. 1983. 'Rethinking intellectual history and reading texts'. In *Rethinking intellectual history: texts, contexts, language*, pp. 23–71. Ithaca: Cornell University Press. (The essay was first published in *History and Theory*, vol. 19, 1988, pp. 245–276. It was later reprinted with some changes in the volume above, as well as in *Modern European intellectual history: reappraisals and new perspectives*, edited by Dominick LaCapra and Steven L. Kaplan, pp. 47–85. Itacha: Cornell University Press)
- Lakatos, Imre. 1970. 'Falsification and the methodology of scientific research programs'. In *Criticism and the growth of knowledge*, edited by Imre Lakatos and Alan Musgrave, pp. 91–196. London: Cambridge University Press

- Landauer, Thomas K. 1991. 'Let's get real: a position paper on the role of cognitive psychology in the design of humanly useful and usable systems'. In *Designing interaction*, edited by John M. Carroll, pp. 60–73. Cambridge: Cambridge University Press
- Landauer, Thomas K. 1997. 'Behavioural research methods in human–computer interaction'. In *Handbook of human–computer interaction*, second, completely revised edition, edited by Martin G. Helander, Thomas K. Landauer and Prasad V. Prabhn, pp. 203–227. Amsterdam: Elsevier
- Lane, Barnett. 1966. 'Legibility depends on the reader'. *Print in Britain*, vol. 14, no. 8, p. 12
- [Lange, Rudi Wynand de]. 1992. 'The legibility of sans serif typefaces'. *Graphix*, December/January, 1993, pp. 15–16
- Lange, Rudi W. de, Hendry L. Esterhuizen, and Derek Beatty. 1993. 'Performance differences between Times and Helvetica in a reading task'. *Electronic Publishing*, vol. 6, no. 3, pp. 241–248
- Lange, Rudi Wynand de. 1993. 'The legibility of sans serif typefaces, an experimental and comparative study'. Master's Diploma dissertation. Bloemfontein: Technikon OFS
- Leahey, Thomas Hardy. 1992. *A history of psychology: main currents in psychological thought*. 3rd edn. Englewood Cliffs, New Jersey: Prentice Hall
- Leat, S.J., W. Li, and K. Epp. 1999. 'Crowding in central and eccentric vision: the effects of contour interaction and attention'. *Investigative Ophthalmology & Visual Science*, vol. 40, pp. 504–512
- Lee, Marshall. 1979. *Bookmaking: the illustrated guide to design / production / editing*. Second edition. New York: Bowker
- Lee, Vicki L. 1988. *Beyond behaviourism*. Hillsdale, New Jersey: Lawrence Erlbaum
- Legge, Gordon E., Denis G. Pelli, Gary S. Rubin, and Mary M. Schleske. 1985a. 'Psychophysics of reading – 1: Normal vision'. *Vision Research*, vol. 25, no. 2, pp. 239–252
- Legge, Gordon E., Gary S. Rubin, Denis G. Pelli, and Mary M. Schleske. 1985b. 'Psychophysics of reading – 2: Low vision'. *Vision Research*, vol. 25, no. 2, pp. 253–265
- Legge, Gordon E., Gary S. Rubin, and Andrew Luebker. 1987. 'Psychophysics of reading – 5: The role of contrast in normal vision'. *Vision Research*, vol. 27, no. 7, pp. 1165–1177
- Legge, Gordon E., Julie A. Ross, Kathleen T. Maxwell, and Andrew Luebker. 1989. 'Psychophysics of reading – 7: Comprehension in normal and low vision'. *Clinical Vision Science*, vol. 4, no. 1, pp. 51–60
- Legge, Gordon E., Sonia J. Ahn, Timothy S. Klitz, and Andrew Luebker. 1997. 'Psychophysics of reading – 16: The visual span in normal and low vision'. *Vision Research*, vol. 37, no. 14, pp. 1999–2010
- Legros, Lucien Alphonse. 1922. *A note on the legibility of printed matter*. Prepared for the information of the committee on type faces. London: His Majesty's Stationary Office

- Legros, Lucien Alphonse, and John Cameron Grant. 1916. *Typographical printing-surfaces: the technology and mechanism of their production*. London: Longmans, Green, and Co
- Lenze, J. 1989. 'Serif vs. sans serif type fonts: a comparison based on readers' comprehension'. (Not seen. Ref. in Kravutske 1994)
- Level, Jeff. 1989. 'Face to face with the daily news'. *Fine Print*, vol. 15, no. 1, pp. 23–30, plus a one-page bibliography on the back of a separately attached fold-out poster
- Levenston, E.A. 1992. *The stuff of literature: physical aspects of texts and their relation to literary meaning*. Albany: State University of New York Press
- Lewis, Clive. 1989. 'Typographic factors in reading'. PhD thesis. Lancashire Polytechnic, School of Psychology, Faculty of Science
- Lewis, Clive, and Peter Walker. 1989. 'Typographic influences on reading'. *British Journal of Psychology*, vol. 80, no. 2, pp. 241–257
- Licko, Zuzana. 1990. 'Typeface designs'. *Emigre*, no. 15, pp. 8–13
- Licko, Zuzana. 1996a. *Base 12/9*. Sacramento: Emigre. (Promotional folder with type specimens and discursive text.)
- [Licko, Zuzana]. 1996b. *A typeface designed by Zuzana Licko: based on the design of Baskerville*. Sacramento: Emigre. (Promotional pamphlet with type specimens and discursive text.)
- Long, John. 1996. 'Specifying relations between research and the design of human–computer interactions'. *International Journal of Human–Computer Studies*, vol. 44, pp. 875–920
- Long, Robert. 1995. 'David Kindersley: in love with letters'. *Serif: The Magazine of Type & Typography*, no. 3, 1995, pp. 32–37
- Lubell, Stephen. 1993. 'Ranged right, ranged left, justified and all that ... : a few thoughts on legibility, readability, typographic tradition and De gustibus non est disputandum'. *TypeLab Gazette*, ATypI (Association Typographique Internationale) conference, Antwerp, 1993, unpaginated, 4 tabloid sized pages
- Luckiesh, Matthew, and Frank K. Moss. 1937. 'The visibility of various type faces'. *Journal of the Franklin Institute*, no. 223, January, pp. 77–82
- Luckiesh, Matthew, and Frank K. Moss. 1938. 'Effects on leading on readability'. *Journal of Applied Psychology*, vol. 22, pp. 140–160
- Luckiesh, Matthew, and Frank K. Moss. 1942. *Reading as a visual task*. New York: Van Nostrand
- Lund, Ole. 1993. 'Description and differentiation of sans serif typefaces'. Postgraduate diploma dissertation. Department of Typography & Graphic Communication, The University of Reading
- Lund, Ole. 1995. Review article of *In black & white: an r & d report on typography and legibility*, by Kim Pedersen and Anders Kidmose. *Information Design Journal*, vol. 8, no. 1, pp. 91–95
- Lund, Ole. 1997. 'Why serifs are (still) important'. *Typography Papers*, no. 2, pp. 91–104
- Lund, Ole. 1997. Review article of *Type & layout: how typography and design can get your message across – or get in the way*, by Colin Wheildon. *Information Design Journal*, vol. 9, no. 1, pp. 74–77

M

- Macdonald-Ross, Michael. 1978. 'Graphics in text'. In *Review of research in education*, vol. 5, 1977, edited by Lee S. Shulman, pp. 49–85. Itacha, Illinois: F.E. Peacock Publishers, a publication of the American Educational Research Association
- Macdonald-Ross, Michael. 1989. 'Towards a graphic ecology'. In *Knowledge acquisition from text and pictures*, edited by Heinz Mandl and Joel R. Levin, pp. 145–154. Amsterdam: North Holland, Elsevier Science Publishers
- Macdonald-Ross, Michael. 1994. 'Print media, production of'. In *The international encyclopedia of education*, second edition, volume 8, pp. 4689–4694
- Macdonald-Ross, Michael, and Eleanor Smith. 1977. *Graphics in text: a bibliography*. IET monograph no. 6. Milton Keynes: Institute of Educational Technology, The Open University
- Macdonald-Ross, Michael, and Robert Waller. 1975. 'Criticism, alternatives and tests: a conceptual framework for improving typography'. *Programmed Learning and Educational Technology*, vol. 12, no. 2, pp. 75–83
- Mack, George E. 1979. 'Opinion/Commentary'. *Communication Arts*, vol. 21, pt. 2, May/June, pp. 96–97
- MacKenzie, Maureen. 1994. 'Our changing visual environment: questions and challenges'. *Communication News* (of the Communication Research Institute of Australia), vol. 7, no. 4, pp. 11–14
- Mackintosh, N.J. (ed). 1995. *Cyril Burt: fraud or framed?* Oxford: Oxford University Press
- Mackworth, N. 1965. 'Visual noise causes tunnel vision'. *Psychonomic Science*, no. 3, pp. 67–68
- Mansfield, J. Stephen, Gordon E. Legge, and Mark C. Bane. 1996. 'Psychophysics of reading – 15: Font effects in normal and low vision'. *Investigative Ophthalmology & Visual Science*, vol. 37, no. 8, pp. 1492–1501
- Manwaring, Tony, and Stephen Wood. 1984. 'The ghost in the machine: tacit skills in the labor process'. *Socialist Review*, no. 74, March/April, pp. 55–83
- Margolin, Victor. 1988. 'A decade of design history in the United States 1977–87'. *Journal of Design History*, vol. 1, no. 1, pp. 51–72
- Margolin, Victor. 1992. Design history or design studies: subject matter and methods'. *Design Studies*, vol. 13, no. 2, pp. 104–116. (Also published in *Design Issues*, vol. 11, no. 1, 1995, pp. 4–15)
- Margolin, Victor. 1998. 'Design studies: proposal for a new doctorate'. In *The education of a graphic designer*, edited by Steven Heller, pp. 163–170. New York: Allworth Press
- Marr, David. 1982. *Vision: a computational investigation into the human representation and processing of visual information*. San Francisco: W.H. Freeman
- Mason, John. 1994. 'British Airports Authority'. *Directions: newsletter of the Sign Design Society*, vol. 1, no. 6, pp. [1–3]
- Massin. 1970. *Letter and image*. Translated [from French] by Caroline Hillier and Vivienne Menkes. London: Studio Vista

- McClelland, J.L., and D.E. Rumelhart. 1981. 'An interactive model of context effects in letter perception: part 1: an account of findings'. *Psychological Review*, vol. 88, no. 5, pp. 375–407
- McCullough, Malcolm. 1996. *Abstracting craft: the practical digital hand*. Cambridge, Massachusetts: The MIT Press
- McGann, Jerome. 1988a. *The beauty of inflections: literary investigations in historical method and theory*. Oxford: Clarendon Press
- McGann, Jerome. 1988b. 'Theory of texts'. *London Review of Books*, vol. 10, no. 4, 18 February, pp. 20–21. (Review essay on D.F. McKenzie's *Bibliography and the sociology of texts: the Panizzi lectures 1985*)
- McGrew, Mac. 1993. *American metal typefaces of the twentieth century*. 2nd ed. New Castle, Delaware: Oak Knoll Books
- McLean, Ruari. 1980. *The Thames and Hudson manual of typography*. London: Thames and Hudson
- Mergenthaler Linotype. 1935. *The legibility of type*. Brooklyn, New York: Mergenthaler Linotype Company
- Michaels, Claire F., and Claudia Carello. 1981. *Direct perception*. Englewood Cliffs, New Jersey: Prentice-Hall
- Misanchuk, Earl R. 1989a. 'Learner/user preferences for fonts in microcomputer screen display'. *Canadian Journal of Educational Communication*, vol. 18, no. 3, pp. 193–205
- Misanchuk, Earl R. 1989b. 'Learner preferences for typeface (font) and leading in print materials'. ERIC no.: ED307854
- Michl, Jan. 1991. 'On the rumor of functional perfection'. *Pro Forma*, no. 2, pp. 67–81
- Michl, Jan. 1995. 'Form follows what?: the modernist notion of function as a carte blanche'. *Magazine of the Faculty of Architecture & Town Planning*, no. 10, pp. 31–20 [sic]
- Mijksenaar, Paul. 1997. *Visual function: an introduction to information design*. Rotterdam: 010 Publishers
- Ministry of Transport. 1962. *Traffic signs for motorways: final report of advisory committee*. London Her Majesty's Stationery Office. (Title on cover: *Motorway signs: final report of advisory committee on traffic signs for motorways*.) (The Anderson report)
- Ministry of Transport. 1963. *Report of the traffic signs committee: 18th April 1963*. London: Her Majesty's Stationery Office. (Title on cover: *Traffic signs 1963: report of the committee on traffic signs for all-purpose roads*.) (The Worboys report)
- Moede [1931/1932]. See *Futuraheft*. [1931/1932]
- Molander, Bengt. 1996. *Kunskap i handling*. Göteborg: Daidalos
- Moles, Abraham. [1958] 1966. *Information theory and esthetic perception*. Urbana and London: University of Illinois Press (Originally published in 1958 as *Théorie de l'information et perception esthétique*)
- Moles, Abraham. 1986. 'The legibility of the world: a project of graphic design'. *Design Issues*, vol. 3, no. 1, pp. 43–53. (Republished facsimile-like page by page in *Design discourse: history, theory, criticism*, edited by Victor Margolin, pp. 119–129. Chicago: University of Chicago Press, 1989)

- Mollerup, Per. [1997] 1998. *Marks of excellence: the history and taxonomy of trademarks*. Revised edition. London: Phaidon Press
- Monotype Recorder. 1964. 'Map design and typography', vol. 43, no. 1, pp. 42–52
- Moore, R.L., and A.W. Christie. 1963. 'Research on traffic signs'. In *Engineering for Traffic Conference*, July, 1963, p. 116
- Moriarty, Sandra Ernst, and Edward C. Scheiner. 1984. 'A study of close-set text type'. *Journal of Applied Psychology*, vol. 69, no. 4, pp. 700–702
- Morison, Stanley. 1996. *First principles of typography*. New edition. Introduction by Huib van Krimpen. Preface by David McKitterick. Leiden: Academic Press
- Morris, Robert A. 1988. 'Image processing aspects of type'. In *Document manipulation and typography*. Proceedings of the international conference on electronic publishing, document manipulation and typography, Nice, April 20–22 1988. Edited by J.C. van Vliet, pp. 139–155. Cambridge: Cambridge University Press
- Morris, R.A. 1989. 'Rendering digital type: a historical and economic view of technology'. *The Computer Journal*, vol. 32, no. 6, pp. 524–532
- Morris, R.A., K. Berry, K.A. Hargreaves, and D. Liarakis. 1991. 'How typeface variation and typographic scaling affect readability at small sizes'. In *Proceedings of the 7th International congress on advances in non-impact printing technologies*. Portland: Society for Imaging Science and Technology
- Morris, R.A., K. Berry, and K.A. Hargreaves. 1993. 'Towards quantification of the effects of typographic variation on readability'. *1993 SID international symposium: digest of technical papers*, edited by Jay Morreale, vol. 24, pp. 64–67. Playa del Rey, California: Society for Information Display
- Morris, R.A., R.D. Hersch, and A. Coimbra. 1998. 'Legibility of condensed perceptually-tuned grayscale fonts'. In *Electronic publishing, artistic imaging and digital typography*, edited by Roger D. Hersch, Jacques André, and Heather Brown, pp. 281–293. Berlin: Springer-Verlag
- Morrison, Gary R. 1986. 'Communicability of the emotional connotation of type'. *Educational Communication and Technology*, vol. 34, no. 4, pp. 235–244
- Morrison, Robert E., and Albrecht-Werner Inhoff. 1981. 'Visual factors and eye movements in reading'. *Visible Language*, vol. 15, no. 2, pp. 129–146
- Mosley, James. 1965. 'The nymph and the grot: the revival of the sanserif letter'. *Typographica*, no. 12, pp. 2–19
- Mosley, James. 1999. *The nymph and the grot: the revival of the sanserif letter*. London: Friends of the St. Bride Printing Library. (Updated version of Mosley 1964)
- Myers, Prue Wallis. 1983. 'Handwriting in English education'. *Visible Language*, vol. 17, no. 4, pp. 333–356

N

- Nagel, Ernest. 1961. *The structure of science: problems in the logic of scientific explanation*. London: Routledge & Kegan Paul

- Naiman, Avi C. 1991. *The use of grayscale for improved character presentation*. Dynamic graphics project technical memo DGP-91-01. Toronto: Computer systems research institute, University of Toronto. (Avi Naiman's PhD thesis)
- Nash, Walter. 1980. *Designs in prose: a study of compositional problems and methods*. London and New York: Longman (Extensively rewritten and updated as: Walter Nash and David Stacey. *Creating texts: an introduction to the study of composition*. London and New York: Longman, 1997)
- Nell, Victor. 1988. *Lost in a book: the psychology of reading for pleasure*. New Haven and London: Yale University Press
- Neman, Thomas Edward. 1968. 'The relative legibility of three cold type faces, three line lengths, three paper stocks, and the interaction of these three variables'. PhD dissertation. Indiana University
- Nes, F.L. van. 1991. 'Visual ergonomics of displays'. In *The man-machine interface*, edited by J.A.J. Roufs, pp. 70–82. Basingstoke: Macmillan Press
- Neue Deutsche Biographie. 1994. Siebzehnter Band: Melander–Moller. Berlin: Duncker & Humblot
- Nielsen, Jakob. 1993. *Usability engineering*. Boston: Academic Press
- Nolan, Carson Y. 1959. 'Readability of large types: a study of type sizes and type styles'. *International Journal of Education for the Blind*, vol. 9, December, pp. 41–44
- Norman, Donald A. 1988. *The psychology of everyday things*. New York: Basic Books (This book has since been published under the title: *The design of everyday things*)
- Norman, Donald A. 1995. 'On differences between research and practice'. *Ergonomics in Design*, April, pp. 35–36
- Norris, Christopher. 1995. 'Truth, science, and the growth of knowledge'. *New Left Review*, no. 210, pp. 105–123. (Also published in Christopher Norris, *Reclaiming truth: contribution to a critique of cultural relativism*, pp. 154–179. London: Lawrence & Wishart, 1996)
- Nunberg, Geoffrey. 1990. *The linguistics of punctuation*. Stanford: Center for the study of language and information
- Nyce, James M., and Jonas Löwgren. 1995. 'Toward foundational analysis in human-computer interaction'. In *The social and interactional dimensions of human-computer interfaces*, edited by Peter J. Thomas, pp. 37–47. Cambridge: Cambridge University Press

O

- Osborne, David J. 1987. *Ergonomics at work*. 2nd edn. Chichester: John Wiley
- OED. 1989. *The Oxford English dictionary*. 2nd ed. (20 vols). Oxford: Oxford University Press
- O'Regan, J. Kevin. 1990. 'Eye movements and reading'. In *Eye movements and their role in visual and cognitive processes*, edited by Eileen Kowler, pp. 395–453. Amsterdam and New York: Elsevier Science

- O'Regan, Kevin, Nicole Bismuth, Roger D. Hersch, and Alexandros Pappas. 1996. 'Legibility of perceptually-tuned grayscale fonts'. *International conference on image processing 1996*, vol. 1, pp. 537–540
- Östberg, Olov, Houshang Shahnava, and Rikard Stenberg. 1989. 'Legibility testing of visual display screens'. *Behaviour & Information Technology*, vol. 8, no. 2, pp. 145–153
- Outhwaite, William. 1987. *New philosophies of social science: realism, hermeneutics and critical theory*. New York: St. Martin's Press
- Ovink, G.W. 1938. *Legibility, atmosphere-value and forms of printing types*. Leiden: A.W. Sijthoff's Uitgeversmaatschappij

P

- Pace, Ann Jaffe. 1982. 'Analysing and describing the structure of text'. In *The technology of text: principles for structuring, designing, and displaying text*, [volume one], edited by David H. Jonassen, pp. 15–27. Englewood Cliffs, New Jersey: Educational Technology Publications
- Palmer, Carl. 1979–1981. 'Legibility is the cornerstone of typography', [which is the title of the first ten of sixteen articles on typography and legibility running in] *Graphic Arts Monthly*, [from May 1979 to June 1981]
- Pardoe, F.E. 1975. *John Baskerville of Birmingham: letter founder & printer*. London: Frederick Muller
- Paterson, Donald G., and Miles A. Tinker. 1932a. 'Studies of typographical factors influencing speed of reading. 8, Space between lines or leading'. *Journal of Applied Psychology*, vol. 16, pp. 388–397
- Paterson, Donald G., and Miles A. Tinker. 1932b. 'Studies of typographical factors influencing speed of reading. 10, Style of type face'. *Journal of Applied Psychology*, vol. 16, pp. 605–613
- Paterson, Donald G., and Miles A. Tinker. 1940a. *How to make type readable: a manual for typographers, printers and advertisers*. New York and London: Harper & Brothers
- Paterson, Donald G., and Miles A. Tinker. 1940b. 'Influence of line width on eye movements'. *Journal of Experimental Psychology*, vol. 27, pp. 572–577
- Paterson, D.G. and M. Tinker. 1947. 'The effect of typography upon the perceptual span in reading'. *American Journal of Psychology*, vol. 60, pp. 388–396
- Payne, Donald E. 1967. 'Readability of typewritten material: proportional versus standard spacing'. *Journal of Typographic Research*, vol. 1, no. 2, pp. 125–136
- Pedersen, Kim, and Anders Kidmose. 1993. *In black and white: an r & d report on typography and legibility*. Copenhagen: The Graphic College of Denmark. (Bilingual text in Danish and English)
- Pedhazur, Elazar J., and Liora Pedhazur Schmelkin. 1991. *Measurement, design, and analysis: an integrated approach*. Hillsdale, New Jersey: Lawrence Erlbaum Associates
- Peters, John. 1962. 'Sans-serifs and legibility'. Letter from a reader. *Monotype Newsletter*, no. 68, p. 12

- Phillips, Russel M. 1976. 'The interacting effects of letter style, letter stroke-width and letter size on the legibility of projected high contrast lettering'. EdD dissertation. Indiana University
- Pleasants, Nigel. 1996. 'Nothing is concealed: de-centering tacit knowledge and rules from social theory'. *Journal for the Theory of Social Behaviour*, vol. 26, no. 3, pp. 233–255
- Poffenberger, A.T.; and R.B. Franken. 1923. 'Appropriateness of type faces'. *Journal of Applied Psychology*, vol. 7, pp. 312–329
- Polanyi, Michael. 1958. *Personal knowledge: towards a post-critical philosophy*. London: Routledge & Kegan Paul; Chicago: University of Chicago Press
- Polanyi, Michael. 1966. *The tacit dimension*. London: Routledge and Kegan Paul
- Polkinghorne, Donald E. 1992. 'Postmodern epistemology of practice'. In *Psychology and postmodernism*, edited by Steinar Kvale, pp. 146–165. London: Sage
- Popper, Karl. 1963. *Conjectures and refutations: the growth of scientific knowledge*. London and Henley: Routledge and Kegan Paul
- Poulton, E.C. 1960. 'A note on printing to make comprehension easier'. *Ergonomics*, vol. 3, no. 3, pp. 245–248
- Poulton, E.C. 1965. 'Letter differentiation and rate of comprehension in reading'. *Journal of Applied Psychology*, vol. 49, no. 5, pp. 358–362
- Poulton, E.C. 1968. 'The measurement of legibility'. *Printing Technology*, vol. 12, no. 2, pp. 72–76
- Poynor, Rick. 1999. 'The great type debate rages on'. *Graphis*, no. 319, pp. [14], plus one page
- Prince, J.H. 1967. 'Printing for the visually handicapped'. *Journal of Typographic Research*, vol. 1, no. 1, pp. 31–47
- Pyke, R.L. 1926. *Report on the legibility of print*. Medical Research Council, Special report series, no. 110. London: His Majesty's Stationery Office

R

- Rainford, Paul. 1996. 'Motorway signs'. *Designing*, no. 44, p. 13
- Ralph, J.B. 1982. 'A geriatric visual concern: the need for publishing guidelines'. *Journal of the American Optometric Association*, vol. 53, pp. 43–50
- Rannem, Øyvinn. 1981. 'Leselighet: typografiens første prinsipper'. *Norsk Grafisk Tidsskrift*, November, pp. 6–9
- Rannem, Øyvinn. 1983. No title. (A comprehensive privately circulated bibliography on typography and legibility). Oslo
- Renear, Allen, Elli Mylonas, and David Durand. 1996. 'Refining our notion of what text really is: the problem of overlapping hierarchies'. In *Research in humanities computing 4*. Selected papers from the ALLC/ACH conference, Christ Church, Oxford, April 1992, pp. 263–280. Oxford: Clarendon Press
- Rayner, Keith. 1981. 'Visual cues in word recognition and reading: introduction'. *Visible Language*, vol. 15, no. 2, pp. 125–127
- Rayner, Keith (ed). 1983. *Eye movements in reading: perceptual and language processes*. New York and London: Academic Press

- Rayner, Keith, and Alexander Pollatsek. 1989. *The psychology of reading*. Englewood Cliffs, New Jersey: Prentice Hall
- Rayner, Keith, and Alexander Pollatsek. 1996. 'Reading unspaced text is not easy: comments on the implications of Epelboim *et al.*'s (1994) study for models of eye movement control in reading'. *Vision Research*, vol. 36, no. 3, pp. 461–465
- Reed, Edward S. 1988. *James J. Gibson and the psychology of perception*. New Haven and London: Yale University Press
- Regan, D., and X.H. Hong. 1994. 'Recognition and detection of texture-defined letters'. *Vision Research*, vol. 34, pp. 2403–2407
- Rehe, Rolf F. 1970. 'Psychological studies and their impact on modern typography'. *Inland Printer / American Litographer*, part one in vol. 164, no. 6, p. 53, and part two in vol. 165, no. 1, p. 66
- Rehe, Rolf F. 1971. 'Psychologische Studien und ihre Bedeutung für die Typografie von heute'. *Der Druckspiegel*, vol. 26, no. 3, p. 152
- Rehe, Rolf F. [1972] 1984. *Typography: how to make it most legible*. 5th edn. Carmel, Indiana: Design Research International
- Rehe, Rolf F. 1985. *Typography and design for newspapers*. Darmstadt: The International Association for Newspaper and Media Technology
- Reynolds, Linda. 1979a. 'The Graphic Information Research Unit: background and recent research'. *Visible Language*, vol. 13, no. 4, pp. 428–448
- Reynolds, L. 1979b. 'Legibility studies: their relevance to present day documentation methods'. *Journal of Documentation*, vol. 35, no. 4, pp. 307–340
- Reynolds, Linda. 1984. 'The legibility of printed scientific and technical information'. In *Information design: the design and evaluation of signs and printed material*, edited by Ronald Easterby and Harm Zwaga, pp. 187–208. Chichester: John Wiley
- Reynolds, Linda. 1988. 'Legibility'. *Baseline*, no. 10, pp. 26–29
- Richaudeau, François (ed.). 1984. *Reserches actuelles sur la lisibilité*. Paris: Editions Retz
- Rivlin, Christopher. 1987. 'A study of the relationship between visual perception and typographic organisation'. PhD thesis. Manchester Polytechnic
- Robinson, David Owen, Michael Abbamonte and Selby H. Evans. 1971. 'Why serifs are important: the perception of small print'. *Visible Language*, vol. 5, no. 4, pp. 353–359
- Rock, Irvin. 1984. *Perception*. New York: Scientific American Library
- Roethlein, Barbara Elizabeth. 1912. 'The relative legibility of different faces of printing types'. *American Journal of Psychology*, vol. 23, no. 1, pp. 1–36. (Also published in the series Publications of the Clark University Library, as vol. 3, no. 1. Worchester, Massachusetts: Clark University Press)
- Rögener, Stefan, Albert-Jan Pool, and Ursula Packhäuser. 1995. *Branding with type*. Edited by E.M. Ginger, translated by Stephanie Tripiet. Mountain View, California: Adobe Press. (German edition: *Typen machen Marken mächtig*. Hamburg: AdFinder, 1995)

- Rogers, Tim B. 1992. 'Antecedents of operationism: a case history in radical positivism'. In *Positivism in psychology: historical and contemporary problems*, edited by Charles W. Tolman, pp. 57–65. New York: Springer-Verlag
- Rolf, Bertil. 1991. *Profession, tradition och tyst kunskap: en studie i Michael Polanyis teori om den professionella kunskapens tysta dimension*. Sweden: Bokförlaget Nya Doxa
- Rollins, Carl Purington. 1942. 'Gilding the lily: in the designing of books there's no sin like complacency'. In *Bookmaking & kindred amenities*, edited by Earl Schenck Miers and Richard Ellis, pp. 21–31. New Brunswick, New Jersey: Rutgers University Press
- Rooum, Donald. 1981. 'Cyril Burt's "A psychological study of typography": a reappraisal'. *Typos: A Journal of Typography*, no. 4, pp. 37–40. London: College of Printing
- Rossum, Mark van. 1997. 'A new test of legibility'. *Quærendo*, vol. 27, no. 2, pp. 141–147
- Roth, Joel. 1969. 'Excerpt: typography that makes the reader work'. *Journal of Typographic Research*, vol. 3, no. 2, pp. 193–196
- Roth, Susan King. 1999. 'The state of design research'. *Design Issues*, vol. 15, no. 2, pp. 18–26
- Rowe, Camille L. 1982. 'The connotative dimensions of selected display typefaces'. *Information Design Journal*, vol. 3, no. 1, pp. 30–37
- Rubinstein, Richard. 1988. *Digital typography: an introduction to type and composition for computer system design*. Reading, Massachusetts: Addison-Wesley
- Ruder, Emil. 1957. 'Univers: eine Grotesk von Adrian Frutiger'. *Typografische Monatsblätter*, no. 5, 8 pp.
- Ruder, Emil. 1959. 'Univers: eine neue Grotesk von Adrian Frutiger / Univers: a new sans-serif type by Adrian Frutiger'. *Neue Grafik / New Graphic Design / Graphisme actuel*, no. 2, pp. 55–57
- Ruder, Emil. 1961. 'Univers and contemporary typography'. *Print in Britain*, no. 1, vol. 9, pp. 22–23
- Ruediger, W.C. 1907. 'The field of distinct vision: with reference to individual differences and their correlations'. *Archives of Psychology*, vol. 1, no. 5, pp. 1–68
- Runeson, Sverker. 1977. 'On the possibility of "smart" perceptual mechanisms'. *Scandinavian Journal of Psychology*, vol. 18, pp. 172–179
- Ryle, Gilbert. [1949] 1973. *The concept of mind*. Harmondsworth: Penguin Books

S

- Saenger, Paul. 1997. *Space between words: the origins of silent reading*. Stanford: Stanford University Press
- Samuel, Raphael. 1991/1992. 'Reading the signs' [part 1]; and 'Reading the signs: [Part] 2. Fact-grubbers and mind-readers'. *History Workshop*, respectively no. 32, 1991, pp. 88–109; and no. 33, 1992, pp. 220–251

- Salcedo, Rodolfo N., Hadley Read, James F. Evans, and Ada C. Kong. 1972. 'A broader look at legibility'. *Journalism Quarterly*, vol. 49, pt. 2, pp. 285–289, 295
- Sassoon, Rosemary. 1988. 'Joins in children's handwriting, and the effects of different models and teaching methods'. PhD thesis. Department of Typography & Graphic Communication, The University of Reading
- Sassoon, Rosemary. 1999. *Handwriting of the twentieth century*. London and New York: Routledge
- Schmidt, Nils. 1906. 'Några ord om huru böcker och tidningar ur ögon-hygienisk synpunkt böra tryckas'. Föredrag vid sv. graf. Fackskolefören. *Nordisk Boktryckarekonst*, vol. 7, no. 3, pp. 105–109
- Schön, Donald A. 1983. *The reflective practitioner: how professionals think in action*. London: Temple Smith. New York: Basic Books
- Schön, Donald A. 1987. *Educating the reflective practitioner: toward a new design for teaching and learning in the professions*. San Francisco: Jossey-Bass Publishers
- Schriver, Karen A. 1997. *Dynamics in document design: creating text for readers*. New York: John Wiley
- Schriver, Karen A. 1998. 'A review of "Building the bridge across the years and disciplines"'. *Information Design Journal*, vol. 9, no. 1, pp. 11–15
- Schulz-Anker, Erik. 1969. *Formanalyse und Dokumentation einer serifenlosen Linearschrift auf neuer Basis: Syntax Antiqua*. Frankfurt am Main: Stempel. (Also published in *Novum Gebrauchsgraphik*, no. 8, 1970, pp. 49–56; and in *Druck-Print*, no. 1, 1970, pp. 20–23)
- Schumacher, Gary M., and Robert Waller. 1985. 'Testing design alternatives: a comparison of procedures'. In *Designing usable texts*, edited by Thomas Duffy and Robert Waller, pp. 377–403. Orlando and London: Academic Press
- Scriven, M. 1967. 'The methodology of evaluation'. In *Perspectives of curriculum evaluation*, edited by R.W. Tyler, R. Gagné, and M. Scriven, pp. 39–83. Chicago: Rand McNally
- Searle, John. 1980. 'Minds, brains, and programs'. *Behavioral and Brain Sciences*, vol. 3, no. 3, pp. 417–424. (Republished on numerous occasions in collections and readers)
- Searle, John R. 1990. 'Cognitive science and the computer metaphor'. In *Artificial intelligence, culture and language: on education and work*, edited by Bo Göranzon and Magnus Florin, pp. 23–34. London: Springer-Verlag
- Searle, John R. 1992. *The rediscovery of the mind*. Cambridge, Massachusetts: The MIT Press
- Seybold, Jonathan. 1991. 'Adobe's "Multimaster" technology: breakthrough in type aesthetics'. *The Seybold Report on Desktop Publishing*, vol. 5, no. 7, pp. 3–7
- Shaw, Alison. 1969. *Print for partial sight: a report to the Library Association sub-committee on books for readers with defective sight*. London: The Library Association
- Shaw, Montague. 1989. *David Kindersley: his work and workshop*. Cambridge: Cardozo Kindersley Editions and Uitgeverij de Buitenkant

- Shaw, Paul. 1989. 'The Century family'. In *Fine Print on type: the best of Fine Print: a magazine on type and typography*, edited by Charles Bigelow, Paul Hayden Duensing and Linnea Gentry, pp. 46–49. San Francisco: Fine Print and Bedford Arts
- Shaw, Paul. 1996. 'Baskerville revisited'. *Print*, vol. 50, no. 6, pp. 28D, 28F, 115–116. (Review of the typeface family Mrs. Eaves, designed by Zuzana Licko, Emigre Fonts)
- Silver, N. Clayton, and Curt C. Braun. 1993. 'Perceived readability of warning labels with varied font sizes and styles'. *Safety Science*, volume 16, no. 5/6, pp. 615–625
- Silver, N. Clayton, Paul B. Kline, and Curt C. Braun. 1994. 'Type form variables: differences in perceived readability and perceived hazardousness'. *Proceedings of the Human Factors and Ergonomics Society 38th annual meeting – 1994*, pp. 821–825
- Simon, Herbert A. [1969] 1996. *The sciences of the artificial*. Third edition. Cambridge, Massachusetts: The MIT Press
- Skarzenski, Emily N. 1996. Book review of *Type & layout: how typography and design can get your message across – or get in the way*, by Colin Wheildon. *Technical Communication*, fourth quarter, pp. 424–426
- Skjervheim, Hans. [1957] 1996. 'Deltakar og tilskodar'. In *Deltakar og tilskodar og andre essays*, [second edition], pp. 71–87. Oslo: Aschehoug. (Published the very first time as a multiplied typewritten manuscript by the Department of Sociology at the University of Oslo, 1957)
- Skjervheim, Hans. [1963] 1996. 'Fenomenologi og psykologi'. In *Deltakar og tilskodar og andre essays*, [second edition], pp. 169–185. Oslo: Aschehoug. (Published the very first time in *Metapsykologi*, edited by Carl Erik Grenness and Steinar Kvale, Oslo, 1963)
- Sless, David. 1981. *Learning and visual communication*. London: Croom Helm; New York and Toronto: John Wiley
- Sless, David. 1994. 'What is information design?' In *Designing information for people: proceedings from the symposium*, edited by Robyn Penman and David Sless, pp. 1–16. Hackett: Communication Research Institute of Australia
- Sless, David. 1996. 'Better information presentation: satisfying consumers?'. *Visible Language*, vol. 30, no. 3, pp. 246–267
- Sless, David. 1997. 'Building the bridge across the years and the disciplines'. *Information Design Journal*, vol. 9, no. 1, pp. 3–10, 27–28
- Smedshammar, Hans, Kerstin Frenckner, Caroline Nordquist, and Steffan Romberger. 1990. *Läslighet på bildskärm hos några teckensnitt*. (English abstract: Legibility of four different typefaces on computer screens). TRITA-NA-P9022. Stockholm: Department of numerical analysis and computing science, Royal Institute of Technology
- Smith, Margaret M. 1983. 'Form and its relationship to content in the design of incunables'. PhD dissertation. University of Cambridge
- Smith, Margaret M. 1987. 'Printed foliation: forerunner to printed page-numbers?' *Gutenberg Jahrbuch*, 1987, pp. 54–70

- Smith, Margaret M. 1993. 'The pre-history of "small caps": from all caps to smaller capitals to small caps'. *Journal of the Printing Historical Society*, no. 22, pp. 79–106
- Smith, Margaret M. 1994. 'The design relationship between the manuscript and the incunable'. In *A millennium of the book: production, design & illustration in manuscript & print: 900–1900*, edited by Robin Myers and Michael Harris, pp. 23–44. Winchester: St Paul's Bibliographies. Delaware: Oak Knoll Press
- Smither, J.A., and C.C. Braun. 1994. 'Readability of prescription drug labels by older and younger adults'. *Journal of Clinical Psychology in Medical Settings*, vol. 1, no. 2, pp. 149–159
- Snyder, Gertrude, and Alan Peckolick. 1985. *Herb Lubalin: art director, graphic designer and typographer*. New York: American Showcase
- Solli, Susanna M. 1998. 'Bourdieu og konstruksjonen av forskningsobjektet'. In *Modernitet – refleksjoner og idébrytninger: en antologi*, edited by Elisabeth L'orange Fürst and Øystein Nilsen, pp. 235–253. Oslo: Cappelen Akademisk Forlag
- Sorg, J.A. 1985. 'An exploratory study of type face, type size, and color paper preferences among older adults'. Master thesis. Pennsylvania State University
- Southall, Richard. 1988. 'Visual structure and the transmission of meaning'. In *Document manipulation and typography*. Proceedings of the international conference on electronic publishing, document manipulation and typography, Nice (France) April 20–22 1998, edited by J.C. van Vliet, pp. 35–45. Cambridge: Cambridge University Press
- Southall, Richard. 1989. 'Interfaces between the designer and the document'. In *Structured documents*, edited by J. André, R. Furuta, and V. Quint, pp. 119–131. Cambridge: Cambridge University Press
- Southall, Richard. 1992. 'Presentation rules and rules of composition in the formatting of complex text'. In *EP92: proceedings of Electronic Publishing 1992*, edited by C. Vanoirbeek and G. Coray, pp. 275–290. Cambridge: Cambridge University Press
- Souttar, James. 1992. 'The voice of the establishment'. The Monotype conference 1992: typographic connections: a European conference about typographic history, theory and practice: proceedings. Queens' College, University of Cambridge, 1–3 September, 1992
- Spencer, Herbert. 1961. 'Mile-a-minute typography?'. *Typographica*, series 2, no. 4, pp. 2–28
- Spencer, Herbert. 1969. *The visible word*. 2nd edn. London: Lund Humphries / Royal College of Art
- Spencer, Herbert. 1974. 'Legibility research in information publishing'. *Penrose Annual*, vol. 67, pp. 153–162.
- Spencer, Herbert, Linda Reynolds, and Brian Coe. [1975] 1977. *The effect of image degradation and background noise on the legibility of text and numerals in four different typefaces*. [London]: Readability of Print Research Unit, Royal College of Art (revised 1977)

- Spiekermann, Erik. 1986. 'Post mortem: or how I once designed a typeface for Europe's biggest company'. *Baseline*, no. 7, pp. 6–7
- Spiekermann, Erik, and E.M. Ginger. 1993. *Stop stealing sheep & find out how type works*. Mountain View, California: Adobe Press
- Stahl, Albert F. 1989. 'The effects of type style upon readability'. PhD dissertation. Wayne State University
- Stake, Robert E. 1994. 'Case studies'. In *Handbook of qualitative research*, edited by Norman K. Denzin and Yvonne S. Lincoln, pp. 236–247. Thousand Oaks, California: Sage Publications
- Stamm, Beat. 1998. 'Visual TrueType: a graphical method for authoring font intelligence'. In *Electronic publishing, artistic imaging and digital typography*, edited by Roger D. Hersch, Jacques André, and Heather Brown, pp. 77–92. Berlin: Springer-Verlag
- Stanton, Neville (ed.). 1998. *Human factors in consumer products*. London: Taylor & Francis
- Steer, Vincent. No date [first edition 1934]. *Printing design and layout: the manual for printers, typographers and all designers and users of printing and advertising*. With a foreword by Beatrice Warde. London: Virtue and Company
- Stern, John A., Donna Boyer, and David Schroeder. 1994. 'Blink rate: a possible measure of fatigue'. *Human Factors*, vol. 36, no. 2, pp. 285–297
- Stiff, Paul. 1993a. 'Stop sitting around and start reading'. *Eye*, no. 11, vol. 3, pp. 4–5
- Stiff, Paul. 1993b. 'Graphic design, MetaDesign, and information design'. *Information Design Journal*, vol. 7, no. 1, pp. 41–46
- Stiff, Paul. 1994. 'Structuralists, stylists, and forgotten readers'. *Information Design Journal*, vol. 7, no. 3, pp. 227–241
- Stiff, Paul. 1995a. 'Design methods, cultural diversity, and the limits of designing'. *Information Design Journal*, vol. 8, no. 1, pp. 36–47
- Stiff, Paul. 1995b. 'Public graphics: the Lunteren symposium'. *Information Design Journal*, vol. 8, no. 1, pp. 64–71
- Stiff, Paul. 1996a. 'The end of line: a survey of unjustified typography'. *Information Design Journal*, vol. 8, no. 2, pp. 125–152
- Stiff, Paul. 1996b. 'Instructing the printer: what specification tells about typographic designing'. *Typography Papers*, no. 1, pp. 27–74
- Stone, Deborah Berman. 1997. 'The legibility of text on paper and laptop computer: a multivariable approach'. PhD dissertation. University of Maryland at College Park
- Stone, Deborah B.; Sylvia K. Fisher, and John Eliot. 1999. 'Adults' prior exposure to print as a predictor of the legibility of text on paper and laptop computer'. *Reading and Writing*, vol. 11, no. 1, pp. 1–28
- Suen, C.Y., and M.K. Komoda. 1986. 'Legibility of digital type-fonts and comprehension in reading'. In *Text processing and document manipulation*. Proceedings of the international conference, University of Nottingham, 14–16 April 1986, edited by J.C. van Vliet, pp. 178–187. Cambridge: Cambridge University Press on behalf of The British Computer Society

Sutherland, Sandra Wright. 1989. 'Miles Albert Tinker and the zone of optimal typography'. PhD dissertation. [Seattle]: University of Washington

T

- Tantillo, J., J. Dilorenzoais, and R.E. Mathisen. 1995. 'Quantifying perceived differences in type style: an exploratory study'. *Psychology & Marketing*, vol. 12, no. 5, pp. 447–457
- Taylor, Jeffrey Lynn. 1990. 'The effect of typeface on reading rates and the typeface preferences of individual readers'. PhD dissertation. Wayne State University
- Taylor, Conrad. 1998. 'Comments on David Sless's Schwarzenberg paper'. *Information Design Journal*, vol. 9, no. 1, pp. 23–25
- Tarr, John C. 1947. 'Legibility in printing'. *Schweizer Graphische Mitteilungen*, Jahrgang 66, Heft 3, pp. 97–100
- Tarr, John C. 1949. 'A critical discursus on type legibility', *Penrose Annual*, vol. 43, pp. 29–31
- Tempte, Nils. 1961. *Pedagogiska och typografiska anspråk på läroböckerna*. Nordisk Bokuke, Gausdal, September 1961
- Thompson, Evan. 1995. *Colour vision: a study in cognitive science and the philosophy of perception*. London and New York: Routledge
- Thompson, Martyn P. 1993. 'Reception theory and the interpretation of historical meaning'. *History and Theory*, vol. 32, no. 3, pp. 248–272
- Tinker, Miles A. 1944. 'Criteria for determining the readability of type faces', *Journal of Educational Psychology*, vol. 35, no. 7, pp. 385–396
- Tinker, Miles A. 1963. *Legibility of print*. Ames, Iowa: Iowa State University Press
- Tinker, Miles A. 1965. *Bases for effective reading*. Minneapolis: University of Minnesota Press
- Tinker, Miles A., and Donald G. Paterson. 1942. 'Reader preferences and typography'. *Journal of Applied Psychology*, vol. 26, pp. 38–40
- Tinker, Miles A., and Donald G. Paterson. 1949. 'What is legibility?' *Print*, vol. 6, no. 2, p. 61
- Tinker, Miles A., and Donald G. Paterson. 1955. 'The effect of typographical variations upon eye movement in reading'. *Journal of Educational Research*, vol. 49, pp. 171–183
- Tomaszewski, Roman. 1973. *Materiały do bibliografii na temat czytelności pism drukarskich (The materials of bibliography on the thema legibility of printing types)*. Poland. (Not seen; source: *Les sciences de l'écrit: encyclopédie internationale de bibliologie*, edited by Robert Estivals, Jean Meyriat, and François Richaudeau, p. 454. Paris: Retz, 1993)
- Tracy, Walter. 1986. *Letters of credit: a view of type design*. London: Gordon Fraser
- Tracy, Walter. 1988. *The typographic scene*. London: Gordon Fraser
- Turnbull, Arthur T., and Russel N. Baird. 1980. *The graphics of communication: typography, layout, design, production*. 4th ed. New York: Holt, Rinehart and Winston

- Turvey, M.T., R.E. Shaw, E.S. Reed, and W.M. Mace. 1981. 'Ecological laws of perceiving and acting: in reply to Fodor and Pylyshyn (1981)'. *Cognition*, vol. 9, pp. 237–304
- Twyman, Michael. 1970. *Printing 1770–1970: an illustrated history of its development and uses in England*. London: Eyre & Spottiswood. (New edition published by The British Library, 1998)
- Twyman, Michael. 1979. 'A schema for the study of graphic language'. In *Processing of visible language: volume 1*, edited by Paul A. Kolers, Merald E. Wrolstad and Herman Bouma, pp. 117–150. New York and London: Plenum Press. (Also published, partly rewritten, in *Media, knowledge and power*, edited by Oliver Boyd-Barrett and Peter Braham, pp. 201–225. London and Sydney: Croom Helm, in association with The Open University, 1987)
- Twyman, Michael. 1982. 'The graphic presentation of language'. *Information Design Journal*, vol. 3, no. 1, pp. 2–22
- Twyman, Michael. 1986. 'Articulating graphic language: a historical perspective'. In *Toward a new understanding of literacy*, edited by M.E. Wrolstad and D.F. Fisher, pp. 188–251. New York: Praeger
- Twyman, Michael. 1993. 'The bold idea: the use of bold-looking types in the nineteenth century'. *Journal of the Printing Historical Society*, no. 22, pp. 107–143
- Typografische Monatsblätter*, no. 1, 1961. (A special issue on the typeface family Univers)

U

- Uddike, Daniel Berkeley. [1922] 1962. *Printing types: their history, forms and use: a study in survivals*. Two volumes. Third edition. Cambridge, Massachusetts: Harvard University Press; London: Oxford University Press
- Uttal, William R. 1990. 'On some two-way barriers between models and mechanisms'. *Perception and Psychophysics*, vol. 48, no. 2, pp. 188–203

V

- Vagle, Wenche. 1995. Review article of *Skjermtekster: skriftkulturen og den elektroniske informasjonsteknologien*, edited by Ture Schwebs. *Norsk Medietidsskrift*, vol. 2, no. 1, pp. 165–170
- Valentine, Elizabeth R. 1982. *Conceptual issues in psychology*. London: George Allen & Unwin
- Vanderplas, James M., and Jean H. Vanderplas. 1980. 'Some factors affecting legibility of printed material for older adults'. *Perceptual and Motor Skills*, vol. 50, pp. 923–932

- Venezky, Richard L. 1984. 'The history of reading research'. In *Handbook of reading research* [vol. 1], edited by P. David Pearson, Rebecca Barr, Michael L. Kamil, and Peter Mosenthal, pp. 3–38. New York and London: Longman. (This article is based in part on an earlier article by the same author: 'Research on reading processes: a historical perspective', *American Psychologist*, vol. 32, no. 5, 1977, pp. 339–345)
- Vihma, Susann (ed.). 1990. *Semantic visions in design*. Proceedings from the symposium on design research and semiotics 17–18.5 1989, at the University of Industrial Arts, Helsinki. Helsinki: University of Industrial Arts
- Vihma, Susann (ed.) 1992. *Objects and images: studies in design and advertising*. Helsinki: University of Industrial Arts
- de Vinne, Theodore Low. 1899. *Plain printing types*. New York: The Century Co. (The whole title, including the introductory series title and a descriptive title of the volume, reads: *The practice of typography: a treatise on the processes of type-making, the point system, the names, sizes, styles and prices of: plain printing types*)
- Visible Language*, vol. 28, no. 3, and no. 4, 1994. (Special issues on graphic design historiography)

W

- Wade, Nicholas J., and Michael Swanston. 1991. *Visual perception: an introduction*. London and New York: Routledge
- Wakeman. [1999]. *Catalogue 43: the book arts*. Oxford: Frances Wakeman Books
- Walker Jr., Alvin (ed.). 1994. *Thesaurus of psychological index terms*. Twentieth anniversary 1974–1994. 7th ed. Washington, D.C.: The American Psychological Association
- Walker, John. 1989. *Design history and the history of design*. With a contribution by Judy Attfield. London: Pluto Press
- Waller, Robert. 1979. 'Functional information design: research and practice'. *Information Design Journal*, vol. 1, no. 1, pp. 43–50
- Waller, Robert H.W. 1980. 'Graphic aspects of complex texts: typography as macro-punctuation'. In *Processing of visible language: volume 2*, edited by Paul A. Kolers, Merald E. Wrolstad and Herman Bouma, pp. 241–253. New York: Plenum Press
- Waller, Robert. 1982. 'Text as diagram: using typography to improve access and understanding'. In *The technology of text: principles for structuring, designing, and displaying text* [volume one], edited by David H. Jonassen, pp. 137–166. Englewood Cliffs, New Jersey: Educational Technology Publications
- Waller, Robert H.W. 1985. 'Using typography to structure arguments: a critical analysis of some examples'. In *The technology of text: volume two: principles for structuring, designing, and displaying text*, edited by David H. Jonassen, pp. 105–125. Englewood Cliffs, New Jersey: Educational Technology Publications

- Waller, Robert. 1987. 'The typographic contribution to language: towards a model of typographic genres and their underlying structures'. PhD thesis. Department of Typography & Graphic Communication, University of Reading
- Waller, Robert. 1991. 'Typography and discourse'. In *Handbook of reading research*, vol. 2, edited by Rebecca Barr, Michael L. Kamil, Peter B. Mosenthal and P. David Pearson, pp. 341–380. New York: Longman
- Wacquant, Loïc J.D. 1993. 'Positivism'. In *The Blackwell dictionary of twentieth century social thought*, edited by William Outhwaite and Tom Bottomore, pp. 495–498. Oxford: Blackwell
- Warde, Beatrice. 1956. 'New light on typographic legibility'. *Penrose Annual*, vol. 50, pp. 51–55
- Watt, Roger. 1993. 'The visual analysis of pages of text'. In *Computers and typography*, compiled by Rosemary Sassoon, pp. 178–201. Oxford: Intellect Books
- Watts, Lynne, and John Nisbet. 1974. *Legibility in children's books: a review of research*. Windsor: NFER Publishing Company
- Webster, Helen Agnes. 1933. 'The influence of type face and paper surface on the legibility of print'. MA thesis. University of Minnesota. (Not seen)
- Webster, Helen A. and Miles A. Tinker. 1935. 'The influence of type face on the legibility of print'. *Journal of Applied Psychology*, vol. 19, February, pp. 43–52
- Wendt, Dirk. 1968. 'Semantic differentials of typefaces as a method of congeniality research'. *Journal of Typographic Research*, vol. 2, no. 1, pp. 3–25
- Wendt, Dirk. 1969. *Einflüsse von Schriftart (Bodoni vs. Futura), Schriftneigung und Fettigkeit auf die erzielbare Lesegeschwindigkeit mit einer Druckschrift*. Bericht Nr. 5, Untersuchungen zur Lesbarkeit von Druckschriften. Hamburg: Psychologisches Institut der Universität Hamburg
- Wendt, Dirk. 1970a. 'By what criteria are we to judge legibility?'. In *Typographic opportunities in the computer age*, pp. 42–46. Papers of the 11th congress of the Association Typographique Internationale, 1969, Prague. (Also published as 'Mehrdimensionale Kriterien der Lesbarkeit'. *Papier und Druck*, Fachausgabe Typographie, vol. 18, no. 12, 1969, pp. 261–264)
- Wendt, Dirk. 1970b. 'Probleme und Ergebnisse psychologischer Lesbarkeitforschung'. *Druck-Print*, vol. 107, no. 1, pp. 16–19
- Wendt, Dirk. 1971. 'Lesen und Lesbarkeit in Abhängigkeit von der Textanordnung'. *Papier und Druck*, vol. 20, no. 6, pp. 90–92
- Wendt, Dirk. 1972. 'Schriftgestaltung und experimental-psychologische Forschung'. *Druck Print*, no. 3, pp. 145–147
- Wendt, Dirk. 1994. 'Legibility'. In *Font technology: methods and tools*, by Peter Karow, pp. 271–306. Berlin: Springer-Verlag
- Wendt, Dirk, Wiebke Groggel, and Georg Gutschmidt. 1997. 'On the effectiveness of highlighting ads in telephone directories by color'. *Visible Language*, vol. 31, no. 3, pp. 326–337
- Weniger, Erich. 1990. *Ausgewählte Schriften: zur geisteswissenschaftlichen Pädagogik*. Weinheim: Verlag Julius Beltz

- Westinghouse. 1980. *Westinghouse product safety label handbook*. Trafford, Pennsylvania: Westinghouse Printing Division
- Whalley, Peter. 1993. 'An alternative rhetoric for hypertext'. In *Hypertext: a psychological perspective*, edited by C. McKnight, A. Dillon, and J. Richardson, pp. 7–18. New York and London: Ellis Horwood
- Wheatly, W.F. 1985. *Typeface analogue*. Arlington, Virginia: National Composition Association
- Wheildon, Colin. 1984. *Communicating, or just making pretty shapes? a study of validity – or otherwise – of some elements of typographic design*. Sydney: Newspaper Advertising Bureau of Australia
- Wheildon, Colin. 1995. *Type & layout: how typography and design can get your message across – or get in the way*. Edited and with an introduction by Mal Warwick. Berkeley, California: Strathmoor Press
- Whittemore, Irving C. 1948. 'What do you mean, – legibility?'. *Print*, vol. 5, no. 4, pp. 35–37
- Wijnholds, Aernout de Beaufort. 1996. *Using type: the typographer's craftsmanship and the ergonomist's research*. Literary [sic] review. Cognitive Ergonomics, Psychonomics Department: Utrecht University
- Wilkes, K.V. 1990. 'Modelling the mind'. In *Modelling the mind*, edited by K.A. Mohyeldin Said, W.H. Newton-Smith, R. Viale, and K.V. Wilkes, pp. 63–82. Oxford: Oxford University Press, Clarendon Press
- Wilkins, Arnold J. 1995. *Visual stress*. Oxford: Oxford University Press
- Williams, Mark Allen. 1990. 'Legibility of serif and sans serif type faces in computer displays'. MS thesis. Colorado State University.
- Williams, Thomas R., and Jan H. Spyridakis. 1992. 'Visual discriminability of headings in text'. *IEEE Transactions on Professional Communication*, vol. 35, no. 2, pp. 64–70
- Winograd, Terry, and Fernando Flores. 1986. *Understanding computers and cognition: a new foundation for design*. Norwood, New Jersey: Ablex. (Reprinted several times by Ablex, and republished as a paperback by Addison-Wesley; Reading, Massachusetts)
- Wright, Patricia. 1978. 'Feeding the information eaters: suggestions for integrating pure and applied research on language comprehension'. *Instructional Science*, 1978, vol. 7, no. 3, pp. 249–312
- Wright, Patricia. 1994. 'Enhancing the usability of written instructions'. In the conference proceedings of 'Public graphics: visual information for everyday use'; 26–30 September 1994, Lunteren, the Netherlands; edited by Harm J.G. Zwaga, Theo Boersema, and Henriëtte C.M. Hoonhout; pp. 1.1–1.18
- Wright, Patricia. 1998. 'Developments and growth in information design' *Information Design Journal*, vol. 9, no. 1, pp. 16–21

X

x-height. 1995. 'Boiled in oil', vol. 4, no. 2, pp. 14–15

Y

- Yager, Dean, and Robert Plass. 1995. 'Presentation of video-based text to low vision readers: rapid serial visual presentation vs. full-page reading'. *Investigative Ophthalmology & Visual Science*, vol. 36, p. S71 (Conference poster / meeting abstract)
- Yager, Dean, Kathy Aquilante, and Robert Plass. 1998. 'High and low luminance letters, acuity reserve, and font effects on reading speed'. *Vision Research*, vol. 8, no. 17, pp. 2527–2531

Z

- Zachrisson, Bror. 1965. *Studies in the legibility of printed text*. Stockholm: Almqvist & Wiksell
- Zachrisson, Bror. 1968. 'The ATypI legibility research committee'. *Journal of Typographic Research*, vol. 2, no. 3, pp. 271–277
- Zachrisson, Bror, and Hans Smedshammar. 1971. *Förundersökning av vissa typografiska variabler vid läsning med nedsatt synförmåga*. [Rapport no. 56]. Stockholm: Lärarhögskolan i Stockholm, Pedagogiska Institutionen
- Zachrisson, Bror, and Hans Smedshammar. 1973a. *Typografi för synsvaga: en översikt och bibliografi*. Rapport nr. 39. Stockholm: Grafiska Institutet
- [Zachrisson, Bror, and Hans Smedshammar]. 1973b. 'Typografi för synsvaga'. *Grafiskt Forum*, no. 6–7, p. 24
- Zakaluk, Beverly L., and S. Jay Samuels (eds). 1988. *Readability: its past, present and future*. Newark, Delaware: International Reading Association
- Zwahlen, H.T., M. Sunkara, and T. Schnell. 1995. 'Review of legibility relationships within the context of textual information presentation'. In *Human performance and safety in highway, traffic, and ITS systems*. Transportation research record, no. 1485, pp. 61–70. Washington, D.C.: National Research Council and National Academy Press

Colophon

This thesis is written and paginated as one single document in Microsoft Word on an Apple Macintosh computer. The main text is set in the serif typeface New Century Schoolbook. The title page, the headings and the running headline are set in variants of the sans serif typeface Stone Sans.